

Journal of Agricultural and Urban Entomology

Contents

Volume 23, No. 1
January 2006

RAMAZAN CANHILAL, GERALD R. CARNER, RANDALL P. GRIFFIN, D. MICHAEL JACKSON, AND DAVID R. ALVERSON – Life History of the Squash Vine Borer, <i>Melittia cucurbitae</i> (Harris) (Lepidoptera: Sesiidae) in South Carolina	1
DANIEL E. WALDSTEIN AND MARU K. KERING – Studies on Oriental Fruit Moth (Lepidoptera: Tortricidae) Pheromone Microcapsules Using Various Tree Fruit Species, Cultivars, and Application Methods	7
JAMES D. HANSEN, MILLIE L. HEIDT, AND PAULINE A. ANDERSON – Bin Sterilization to Prevent Reintroduction of Codling Moth	17
RAMAZAN CANHILAL AND GERALD R. CARNER – Efficacy of Entomopathogenic Nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) Against the Squash Vine Borer, <i>Melittia cucurbitae</i> (Lepidoptera: Sesiidae) in South Carolina.	27
ROBERTO LEZAMA-GUTIÉRREZ, JAIME MOLINE-OCCHOA, ALFONSO PESCADOR-RUBIO, EDELMIRA GALINDO-VELASCO, CÉSAR ANDRÉS ÁNGEL-SAHAGÚN, ALEJANDRO C. MICHEL-ACEVES, AND EZEQUIEL GONZÁLEZ-REYES – Efficacy of Steinernematid Nematodes (Rhabditida: Steinernematidae) on the Suppression of <i>Anastrepha ludens</i> (Diptera: Tephritidae) Larvae in Soil of Differing Textures: Laboratory and Field Trials	41
JOSE L. GONZALEZ-ANDUJAR, JOSE F. RAMIREZ-DAVILA, MARIA A. LOPEZ, AND RAFAEL OCETE – Spatial Distribution of <i>Jacobiasca lybica</i> (Bergenin & Zanon) (Homoptera: Cicadellidae) Egg Populations in an Irrigated Sherry Vineyard	51
Notice to Contributors	56

THE JOURNAL OF AGRICULTURAL AND URBAN ENTOMOLOGY

<http://entweb.clemson.edu/scesweb/jaue.html>

Volume 23 • Number 4 • October 2006

The *Journal of Agricultural and Urban Entomology* is published quarterly under the auspices of the South Carolina Entomological Society, Inc., at Clemson, SC. The *Journal* publishes contributions of original research concerning insects and other arthropods of agricultural and urban significance (including those affecting humans, livestock, poultry, and wildlife). The *Journal* is particularly dedicated to the timely publication of articles and notes pertaining to applied entomology, although it will accept suitable contributions of a fundamental nature related to agricultural and urban entomology.

A membership to the South Carolina Entomological Society is automatically accompanied by a subscription to the *Journal*. For information on the Society or *Journal*, contact our office: SCES, 3517 Flowering Oak Way, Mt. Pleasant, SC 29466 or see <http://entweb.clemson.edu/scesweb>.

For guidelines in preparing manuscripts for submission, contributing authors should consult "Instructions for Authors" on the Web site. Authors should submit **an original and three copies** of their manuscript typed double spaced to J. W. Austin, Center for Urban & Structural Entomology, Texas A&M University, Department of Entomology, 2143 TAMU, College Station, TX 77843-2143 USA. **An abstract, 3–10 key words, and a running head must be included with each manuscript.** All measurements should be given in metric or in metric and English units. A charge of \$55.00 per page will be made; photos, figures, tables, etc. are charged at an additional \$20.00 each. Excessive or complicated tables, sideways tables, oversize figures, or photos may be subject to an additional charge of \$5.00–\$20.00 per page depending on number, length, complexity, etc. Author's alterations on the galley will be charged at the rate of \$2.00 per line. (Page charges are subject to change without notice.)

For more information on prevailing page charges and charges for immediate publication, advertising rates, cost of reprints, etc., contact the Executive Director.

EDITORIAL COMMITTEE

Editor: J. W. Austin, Center for Urban & Structural Entomology, Texas A&M University, Department of Entomology, 2143 TAMU, College Station, TX 77843-2143 USA.

Executive Director: Randy McWhorter, Clemson University, 3517 Flowering Oak Way, Mt. Pleasant, SC 29466; phone 843.856.9411; fax 843.856.9411

Subject Editors:

J. K. Greene, Southeast Res. & Ext. Center, University of Arkansas CES, P.O. Box 3508, Monticello, AR 71656

W. O. Lamp, Dept. of Entomology, 4112 Plant Science Bldg., Univ. of Maryland, College Park, MD 23284-2012

J. W. McCreadie, Dept. of Biological Sciences, LSB 124, Univ. of South Alabama, Mobile, AL 36688-0002

T. G. Shanower, USDA-ARS Grain Marketing and Production Research Center, 1515 College Ave., Manhattan, KS 66502

C. E. Sorenson, Dept. of Entomology, North Carolina State Univ., Research Annex—West A, Campus Box 7630, Raleigh, NC 27695

D. K. Weaver, Dept. of Entomology, Montana State University, 333 Leon Johnson Hall, Box 173020, Bozeman, MT 59717-3020

NOTE: Statements in the *Journal of Agricultural and Urban Entomology*, whether fact or opinion, are the sole responsibility of the authors. Mention of a commercial product does not constitute endorsement by the South Carolina Entomological Society.

OFFICERS
SOUTH CAROLINA ENTOMOLOGICAL SOCIETY, INC.
2006

Executive Director: Randy McWhorter, 3517 Flowering Oak Way, Mt. Pleasant, SC 29466; phone 843.856.9411; fax 843.856.9411

Officers:

President: D. Michael Jackson, USDA/ARS/Veg. Lab., 2700 Savannah Highway, Charleston, SC 29414; phone 843.402.5308, fax 843.571.4654

VP/President Elect: Eric P. Benson, Clemson University, 105 Long Hall, Clemson, SC 29634-0365; phone 864.656.7860; fax 864.656.5065

Past President: Mike Hood, Clemson University, 305 Long Hall, Clemson, SC 29634; phone 864.656.0346; fax 864.656.5065

Secretary: Dwight Williams, Cypress Gardens, 3030 Cypress Gardens Rd., Moncks Corner, SC 29461; phone 843.553.0515; fax 843.569.0644

Historian: Mike Hood, Clemson University, 305 Long Hall, Clemson, SC 29634; phone 864.656.0346; fax 864.656.5065

Committee Chairs:

Auditing: Donald G. Manley, Pee Dee Research and Education Center, 2200 Pocket Rd., Florence, SC 29506-9706; phone 843.662.3526 ext. 231; fax 843.661.5676

Honors and Awards: Pat Zungoli, Clemson University, Urban Entomology Lab, 114 Long Hall, Clemson, SC 29634; phone 864.656.3137; fax 864.656.5065

Local Arrangements: J. Powell Smith, Edisto Research and Education Center, 64 Research Rd., Blackville, SC 29817, phone 803.284.3343 ext. 228; fax 803.284.3684

Long-Range Planning: Gerry Carner, Clemson University, 318 Long Hall, Clemson, SC 29634-0365; phone 864.656.5040; fax 864.656.5065

Membership: OPEN

Nomination: Merle Shepard, Coastal Research and Education Center, 2700 Savannah Highway, Charleston, SC 29414, phone 843.402-5393; fax 843.571.4654

Program: Eric P. Benson, Clemson University, 105 Long Hall, Clemson, SC 29634-0365; phone 864.656.7860; fax 864.656.5065

Publicity: Austin Jenkins, 406 Watts Ave., Greenville, SC 29601; phone 864.313.4233

Resolutions: Paula Mitchell, Winthrop University Biology Dept., Rock Hill, SC 29733; fax 803.323.3448

Life History of the Squash Vine Borer, *Melittia cucurbitae* (Harris) (Lepidoptera: Sesiidae) in South Carolina¹

Ramazan Canhilal, Gerald R. Carner,² Randall P. Griffin,²
D. Michael Jackson,³ and David R. Alverson²

Seyrani Agricultural Faculty, Erciyes University 38039 Melikgazi, Kayseri, Turkey

J. Agric. Urban Entomol. 23(1): 1–6 (Jan 2006)

ABSTRACT The life history of the squash vine borer (SVB), *Melittia cucurbitae* (Harris) (Lepidoptera: Sesiidae), was investigated in South Carolina. Biological parameters for SVB were determined in cages in a rearing room and outside. These parameters were used to construct a life table for SVB. Eggs hatched, on average, in 10.5 and 8.2 d in the rearing room and outside cages, respectively. Larval and pupal development times averaged 24.4 and 27.6 d in the rearing room, and 27.5 and 29.3 d in outside cages, respectively. Average total development time for immature stages, from egg to adult, was 62.1 d in the rearing room and 65 d in outside cages. Longevity of females averaged 5.2 and 3.9 d for the rearing room and the outside cages, respectively. Males lived, on average 3.1 and 3.0 d in the rearing room and the outside cages. Preoviposition time of females averaged about one day, and females continued to lay eggs for 3 d in the rearing room and 1.9 d in the outside cages. Fecundity averaged 42 and 40.5 eggs/female for the rearing room and the outside cages, respectively. Intrinsic rate of increase (r_m) was 0.035 for the rearing room and 0.029 for outside cages. Reproductive rate (R_o) was calculated to be 9.8 and 7.0 daughters per female for rearing room and outside cages, respectively. The mean period from birth of parents to birth of offspring (T) was calculated to be 65.2 d for the rearing room and 67.3 d for outside cages. SVB has 2 overlapping generations per year and a prolonged emergence pattern in South Carolina. The moths of the first generation emerge around mid-May and they emerge continuously about 4.0–4.5 months until the end of October even though individual adults live only 1–11 d. Larval activity continued until mid to late October when larvae made cocoons in the soil 2.5 (an inch) or 5 (two inches) cm deep to over-winter as larvae.

KEY WORDS Squash, squash vine borer, *Melittia cucurbitae*, life history, life table

The squash vine borer (SVB), *Melittia cucurbitae* (Harris) (Lepidoptera: Sesiidae), causes economic loss to members of the family Cucurbitaceae, especially squashes and pumpkins. Among all cucurbits, cultivars of *Cucurbitae maxima* (banana, buttercup, and hubbard) and *C. pepo* (crookneck, zucchini, and most

¹Accepted for publication 27 September 2006

²Department of Entomology, Clemson University, 113 Long Hall Box 340365, Clemson, South Carolina 29634-0365

³USDA Vegetable Laboratory, 2700 Savannah Hwy, Charleston, South Carolina 29414

pumpkins) are highly suitable hosts, and *C. mediate* and *C. ficifolia* (Mexican gourd) are intermediate in susceptibility. Cultivars of *C. moschata* (butternut) are the least satisfactory hosts (Howe & Rhodes 1972). The SVB has considerable economic significance in areas east of the Rocky Mountains from Canada to South America (Klun et al. 1990). Borers enter and feed in the stems of vines, and greenish-yellow excrement is pushed out of the entry hole on the stem. The pest can cause as much as 25% crop loss in commercial squash. Home plantings usually suffer greater losses than commercial plantings (Sorenson 1994).

The primary means of control for the SVB is chemical control; weekly sprays are needed for 2 months of the growing season. Once larvae have entered the stem, it is almost impossible to kill them with chemicals. Increasing concern for the impact of chemical insecticides on human health, effects on nontarget organisms, residues on crops, insect resistance, and inducing secondary pests have necessitated less pesticide usage under the concept of integrated pest management (IPM). Life-history studies are the first step and an important part in development of IPM practices. SVB has not been studied in South Carolina; therefore, our objective was to study the life history of SVB to provide information to use in development of IPM practices in South Carolina.

Materials and Methods

Maintaining of a laboratory colony. Techniques modified from Howe's (1950) method were used to maintain a laboratory colony of *M. cucurbitae*. The colony was started with eggs and larvae collected from squash in the field. The eggs were placed on small cubes of squash fruit in 30-mL (1 ounce) plastic Dixie® cups (Frenchtown, New Jersey). Upon hatching, the larvae bored into the cube and were allowed to feed for about a week. Cubes containing young larvae and field-collected larger larvae were then placed on larger squash cubes in the same cups. Cups were checked daily by removing the top portion or, if necessary, cutting into the fruit where the larvae had bored. If the squash cubes were nearly consumed or rotting, the larvae were transferred to fresh squash cubes similarly prepared. When the larvae were fully grown, some sand was placed at the bottom of the container to facilitate pupation. Upon maturity, the larvae entered the sand and constructed normal silken cocoons.

Life table studies of SVB. For determining biological parameters of SVB, 2–5 moths, <24-h old, were placed in a rearing cage (61 × 61 × 61 cm) containing 1–2 squash plants that were 35–40 cm in height. The plants were replaced every 1–2 d. An average of 70 plants were used per cage. The squash plants taken out were examined and the eggs were counted. Cages were either kept in a rearing room (25 ± 2°C, 65–70% humidity and a photoperiod of 16:8 [L:D] h), or outside under natural environmental conditions (14 May–24 July 1988) at the Entomology Cherry Farm, Clemson, South Carolina. After the larvae hatched, they were fed in squash fruit cubes as described previously in both laboratory and outdoor cages and monitored for pupation and adult emergence so that development time and mortality of eggs, larvae, and pupae could be determined. Dead adults were recorded, and their gender was identified. In this manner, information on pre-oviposition, oviposition and post oviposition periods and number of progeny for SVB were determined by making daily observations. These data were used to construct a life table for SVB.

Life tables were constructed for data from both the rearing room ($25 \pm 2^\circ\text{C}$) and outside conditions. The equation of Southwood (1978): $\sum l_x \cdot m_x \cdot e^{-r_m \cdot x} = 1$ was used where

- l_x = percentage survival at age x
- m_x = expected number of daughters per female at age x (female/female/day)
- e = natural log base
- r_m = intrinsic rate of increase
- x = age of females in days

The net reproductive rate (R_0) was calculated as the sum of the reproduction expectation ($\sum L_x m_x$) for each age group. The mean generation time (T) was calculated using the formula: $T = \log_e R_0 / r_m$ (Laing 1968).

Field observations for SVB. Yellow crookneck squash was sequentially planted in field plots at the beginning of May and June, and in mid-July at the Entomology Cherry Farm, 3.2 km South of the Clemson University Campus, Clemson, South Carolina, in 1997 and 1988 for field observations. Larval, pupal, and adult activities were observed in these squash plots.

Two-three plants in mature plantings were examined twice a week to observe larval and pupal activity. Holes made by larvae on the main stem were followed up by noting the frass pushed out by the larvae and/or cutting into larval galleries to check the larval activity. The activities of moths in the squash field were observed to determine the moths' fly pattern, what time of day they are active, and where they rest and lay eggs. It was made by slowly walking and sitting around the squash field about one hour early morning and one hour late afternoon just before it gets dark in a week for about one month in August when moth density is high.

Results and Discussion

Life table studies of SVB. SVB eggs hatched, on average, in 10.5 and 8.2 d after oviposition in the rearing room and outside cage, respectively (Table 1). Howe (1950) found that the incubation time of eggs under natural conditions was 8–9 d in New York, and Sorenson (1994) reported approximately a week in North Carolina. Welty (1991) and Klass (1995) reported that the incubation period of SVB eggs was 9–14 d in Ohio and 2 weeks in New York, respectively. In South Carolina, larval development time averaged 24.4 d in the rearing room and 27.5 d in outside cages (Table 1). Howe (1950), Klass (1995), Welty (1991), and Sorenson (1994) stated that the total larval development time of SVB averaged 30 d in New York, Ohio, and North Carolina, respectively. Pupal development time averaged 26.6 d in the rearing room and 29.3 d in outside cages (Table 1). Sorenson (1994) reported that pupal development time of SVB was 2–3 weeks in North Carolina under natural conditions. In South Carolina total development time for immature stages (when they were fed on squash fruit cubes), from egg laying to adult, was 62.1 d in the rearing room and 65 d in outside cages (Table 1). No mortality factor for immature stages was determined during rearing trials in rearing room and outside cages.

We found that SVB adults have a short life span; longevity of females averaged 5.2 d in the rearing room and 3.9 d in outside cages. Males lived, on average, 3.1 and 3.0 d in the rearing room and the outside cages, respectively. Preoviposition

Table 1. Mean and range of development times in days for immature stages of *Melittia cucurbitae* reared in a rearing room at 25 ± 2°C, 65–70% humidity, and a photoperiod of 16:8 (L:D) h, or in outside cages under natural environmental conditions.

	n	Rearing room	Outside cage
Egg	50	10.5 (9–11) ± 0.08	8.2 (7–9) ± 0.10
Larva1	20	4.6 (4–7) ± 0.18	5.2 (4–6) ± 0.13
Larva2	20	4.3 (4–6) ± 0.15	4.8 (4–7) ± 0.22
Larva3	20	4.7 (4–6) ± 0.17	5.3 (4–8) ± 0.27
Larva4	20	10.8 (7–12) ± 0.29	12.2 (8–13) ± 0.28
Total larva	40	24.4 (19–31) ± 0.58	27.5 (20–34) ± 0.65
Pupa	40	27.6 (24–34) ± 0.45	29.3 (22–52) ± 0.85
Total development time		62.5 (52–76)	65.0 (49–95)

time of females averaged about 1 d and females continued to lay eggs for 3 d in the rearing room and 1.9 d in the outside cages. Fecundity averaged 42 eggs per female in the rearing room and 40.5 eggs per female in the outside cages (Table 2). No information related to longevity and fecundity of SVB was found in the literature.

The calculation of the intrinsic rate of increase (r_m) was obtained by the equation: $\sum l_x \cdot m_x \cdot e^{-r_m \cdot x} = 1$. The value of r_m that made the sum closest to 1, by trial and error substitutions, was 0.035 for the rearing room and 0.029 for the outside cages. R_0 is the number of daughters that replace an average female. R_0 was calculated to be 9.8 for the rearing room and 7.0 for the outside cages. The mean period from birth of parents to birth of offspring (T) was calculated as 65.2 and 67.3 d for the rearing room and outside cages, respectively, by using the equation: $T = \log_e R_0 / r_m$ (Laing 1968).

Table 2. Longevity (days), fecundity (eggs/female), and range for adults of *Melittia cucurbitae* reared in the rearing room at 25 ± 2°C, 65–70% humidity, and a photoperiod of 16:8 (L:D) h, or in outside cages under natural conditions.

	n	Rearing room	Outside cage
Preoviposition	15	1.1 (1–3) ± 0.13	1.1 (1–2) ± 0.07
Oviposition	15	3.0 (1–5) ± 0.35	1.9 (1–4) ± 0.23
Postoviposition	15	1.1 (0.5–2) ± 0.12	1.0 (0.5–2) ± 0.10
Longevity of Female	15	5.2 (3–11) ± 0.59	3.9 (2–6) ± 0.25
Longevity of Male	15	3.1 (1–7) ± 0.35	3.00 (2–5) ± 0.21
Fecundity	15	42.0 (4–66) ± 4.21	40.5 (17–107) ± 5.37

Life tables provide important statistics for calculating the growth performance of a population (Birch 1948). Many factors such as temperature, moisture, and other components of weather, the quality and quantity of food, predators, pathogens, or other factors will determine the actual rate of increase of a population. Rate of increase will be greatest when the combinations of the factors are optimal. The r_m found for SVB is not the maximal value, but is the rate of increase obtained from the experimental conditions when competitive factors and predation were excluded. These parameters are important and valuable data to be used in developing biological control and IPM programs even they were obtained in experimental condition.

Field observations for SVB. Our observations in the field cages indicated that SVB has two generations per year in South Carolina. Jackson et al. (2005) stated that a pheromone trapping study of SVB from two locations in South Carolina over five seasons also showed that SVB has two overlapping generations per year and a prolonged emergence pattern in South Carolina. The moths of the first generation emerge around mid-May and they emerge continuously for approximately 4.0–4.5 months until the end of October (Jackson et al. 2005); even individual adults live only 1–11 days (Table 2). Sorenson (1994) reported that SVB has 2 generations per year in North Carolina. Howe (1950) and Welty (1991) stated that SVB has one and a partial second generation in New York and Ohio, respectively, whereas Gauthier (1991) and Ghidui (1987) found that SVB has a single generation in Connecticut and New Jersey.

Unlike most moths, the adults of SVB are active during the day. In the evening, they rest on leaves. They fly slowly in zigzag patterns around plants and lay eggs on the leaf stalks, on the upper and lower sides of leaves, on fruit buds, and often on the lower part of the main stem. After the larvae hatch, they bore into the main stem, feed, and pass through four instars. The larvae can be found in the fruit when population density is high. Fully-grown larvae enter into the soil to pupate. We found that larvae were present until mid to late October, before dropping to the soil and over-wintering as mature larva in cocoons 2.5–5-cm deep. Sorenson (1994) reported that SVB over-winters in the larval or pupal stage in North Carolina. Langevin (1996) found that they over-winter as pupae in New England. Klass (1995) and Welty (1991) stated that the larvae of SVB overwinter in the cocoons, changing to pupae the following spring in New York and Ohio, respectively.

Acknowledgments

We thank Jimmy Pilgrim for making soil preparation in the squash plots and Yayi Kusumah for helping plant squash plots. This is technical contribution number 4615 of the South Carolina Agricultural Experiment Station.

References Cited

- Birch, L. C. 1948.** The intrinsic rate of natural increase of an insect population. *J. Anim. Ecol.* 17: 15–26.
- Gauthier, N. L. 1991.** Squash vine borers on cucurbits. *Grower Vegetable and Fruit Newsletter*, The University of Connecticut Cooperative Extension System 91-6.
- Ghidui, G. M. 1987.** Squash vine borers. *Rutgers Cooperative Extension Leaflet FS 229.*

- Howe, W. L. 1950.** Biology and host relationship of squash vine borer. J. Econ. Entomol. 43: 480–483.
- Howe, W. L. & A. M. Rhodes. 1972.** Host relationship of squash vine borer, *Melittia cucurbitae* with species of Cucurbitae. Ann. Entomol. Sci. 66: 266–269.
- Jackson, D. M., R. Canhilal & G. R. Carner. 2005.** Trap monitoring squash vine borers in cucurbits. J. Agric. Urban Entomol. 22: 27–39.
- Klass, C. 1995.** Squash Vine Borer. Cornell University Extension Factsheet. <http://freedom.cce.cornel.edu/factsheets/home/insects/squash-vine-borer.html>. Accessed December 19, 2006.
- Klun, J. A., M. Schwarz, B. A. Leonardt & W. W. Cantelo. 1990.** Sex pheromone of the female squash vine borer. J. Entomol. Sci. 25: 64–72.
- Laing, J. E. 1968.** Life history and life table of *Phytoseilus persimilis* Athias-Henriot. Acarologica 10: 578–588.
- Langevin, D. 1996.** Cucurbit Family Plant Problems – Squash Vine Borer. http://www.backyardgardener.com/wcgp/problems/squash_borer.html. Accessed December 19, 2006
- Sorenson, K. A. 1994.** The Squash Vine Borer. North Carolina State University Extension Factsheet. <http://ipm-www.ncsu.edu/insect-notes/vegetables/veg>. Accessed December 19, 2006.
- Southwood, T. R. E. 1978.** Ecological methods, Chapman and Hall, London, 391 pp.
- Welty, C. 1991.** Squash Vine Borer. Ohio State University Extension Factsheet. <http://www.ces.ncsu.edu/depts/ent/notes/vegetables/veg20.html>. Accessed December 19, 2006.

Studies on Oriental Fruit Moth (Lepidoptera: Tortricidae) Pheromone Microcapsules Using Various Tree Fruit Species, Cultivars, and Application Methods¹

Daniel E. Waldstein and Maru K. Kering^{2,3}

Missouri State Fruit Experiment Station, Missouri State University,
Mountain Grove, Missouri 65711 USA

J. Agric. Urban Entomol. 23(1): 7–16 (Jan 2006)

ABSTRACT The number of microcapsules containing pheromone of the oriental fruit moth, *Grapholita molesta* (Busck), was determined on leaves from apple, Asian and European pear, peach, sweet and tart cherry, and plum trees. In addition, we compared the number of microcapsules on apple and peach leaves using a laboratory leaf dip application and a field airblast sprayer application. Finally, we compared the number of microcapsules on two apple and peach cultivars after airblast sprayer applications of microencapsulated pheromone. The bottom surface of ‘Gala’ apple leaves had microcapsule density of 0.47, the highest among the different tree fruits. The bottoms of ‘Shinko’ Asian pear had the lowest microcapsule density of 0.09 followed by the tops of ‘Bluebell’ plum and ‘Columbia’ sweet cherry leaves both with density of 0.10. The greater than 3-fold difference between the fruit leaves with the highest and lowest numbers of microcapsules may indicate the need for rate specific recommendations on different species of fruit trees. The number of microcapsules on apple and peach leaves treated with laboratory leaf dips was 8- to 60-fold greater than the number of microcapsules on field-treated leaves. Although laboratory treatments eliminated other potentially confounding field variables, they did not accurately represent microcapsule numbers on leaves treated with an airblast sprayer. There were significant differences in microcapsule abundance on ‘Delicious’ and ‘Gala’ apple leaves and ‘Encore’ and ‘Red Haven’ peach leaves. Trichome abundance and cuticle structure may be responsible for the differences in the number of microcapsules on various tree fruit species and cultivars.

KEY WORDS pheromone, microencapsulation, *Grapholita molesta*, apple, pear, peach, cherry, plum

Microencapsulated pheromones have been studied for management of pests of cotton (Campion et al. 1981, Critchley et al. 1983, Moawad et al. 1991), corn (Albajes et al. 2002), stored products (Prevett et al. 1989), and timber (Beroza et al. 1974). Most of this research has been conducted against fruit pests. Research on the field efficacy of microencapsulated pheromone has been conducted on pests

¹Accepted for publication 29 September 2006.

²Department of Agronomy, University of Missouri-Columbia, Columbia, Missouri, 65211, USA.

³Corresponding author: Maru K. Kering, 1-31 Agric. Bldg., University of Missouri, Columbia, Columbia, Missouri 65211, USA. E-mail: mkkqm2@mizzou.edu

of apples and peaches (Cardé et al. 1975, Trimble & Appleby 2004, Trimble et al. 2004, Kovanci et al. 2004), cranberries and blueberries (Polavarapu et al. 2001, Polavarapu et al. 2002), and grapes (Taschenberg & Roelofs 1976, Trimble et al. 2003).

Oriental fruit moth, *Grapholita molesta* (Busck) (Lepidoptera: Tortricidae) is one of the most widely studied fruit insect pests and has been successfully controlled using hand-applied pheromone dispensers (Jones et al. 1984, Pree et al. 1994, Trimble et al. 2001, Atanassov et al. 2002) and microencapsulated pheromone (Trimble et al. 2004, Kovanci et al. 2004). Oriental fruit moth is an ideal subject for mating disruption because of its high sensitivity and relatively long habituation to synthetic pheromone (Cardé et al. 1977, Charlton & Cardé 1981, Lofstedt et al. 1990, Figueredo & Baker 1992, Sanders & Lucuik 1996, Valeur & Lofstedt 1996, Rumbo & Vickers 1997).

Microcapsules containing *G. molesta* pheromone differ in their abundance on mature and immature apple leaves, one year old wood, and fruit (Waldstein & Gut 2003). In addition, leaf surface can be an important variable influencing the number of oriental fruit moth pheromone microcapsules (Waldstein & Gut 2003, Waldstein & Gut 2004). Little is known about the differences in microcapsule abundance on different types of fruit leaves. Because oriental fruit moth is a pest on a variety of fruit crops including apples, pears, peaches, nectarines, apricots, cherries, and plums (Howitt 1993), there is the potential for *G. molesta* microencapsulated pheromone to be used on these different fruit crops. Therefore, we compared the number of oriental fruit moth pheromone microcapsules on the top and bottom surfaces of apple, pear (Asian and European), peach, cherry (sweet and tart), and plum leaves.

Studies have been conducted on the number of oriental fruit moth pheromone microcapsules on apple leaves using laboratory leaf dip applications (Waldstein & Gut 2003, Waldstein & Gut 2004). In the field, the numbers of microcapsules on leaves may be substantially less than observed under laboratory conditions using the same concentration with a leaf dip application. Bioassay studies have demonstrated that higher concentrations of pesticides are necessary for sprayer applications under field conditions to achieve the same mortality as leaf dip treatments (Agnello et al. 1994, Horowitz et al. 1997, Stansly et al. 1998). Factors that influence spray coverage in the field (e.g., weather, sprayer type, tree size, planting density, and density of the leaf canopy; Johnstone 1985, Black & Bukovac 1996, Bohmont 1997) may be responsible for lower concentrations on foliage sprayed in the field than leaf dip treatments. The value of using a leaf dip application under laboratory conditions is that it eliminates these variables and allows for a relative comparison of microcapsule abundance. The value of using an airblast sprayer application is that it gives a more realistic depiction of microcapsule abundance in commercial orchards. Therefore, we compared these two application methods to gain a better understanding of the number of microcapsules on apple and peach leaves under laboratory and field conditions. In addition, the number of pheromone microcapsules on leaves from 'Gala' and 'Delicious' apple cultivars were compared after applications with an airblast sprayer to determine if cultivar differences existed.

Materials and Methods

Tree fruit comparison with laboratory leaf dips. Twenty-five mature leaves were collected from one to three fruit trees at the Missouri State Fruit

Experiment Station, Mountain Grove, Missouri on 16 June, 23 June, 30 June, and 9 July 2003. A total of 100 leaves from the four replicates were collected from the following cultivars: 'Gala' apple, 'Shinko' Asian pear, 'Duchess de Angloeme Bronze' European pear, 'Loring' peach, 'Columbia' sweet cherry, 'Montmorency' tart cherry, and 'Bluebell' plum trees. While we were still in the field, we placed the leaves in plastic bags and stored them in a lunchbox cooler to reduce moisture loss. The leaves then were brought into the laboratory and dipped for 5 sec in beakers containing 200 mL of water and 45.8 μ L of Checkmate® OFM-F microencapsulated pheromone for oriental fruit moth (22.06% Z-8 dodecenyl Acetate, 1.40% E-8 Dodecenyl Acetate, and 0.2% Z-8 Dodecenol, Suterra LLC, Bend, Oregon). The concentration of the solutions in beakers was equivalent to 86.6 mL of Checkmate® OFM-F in 378.5 L of water (2.93 fl oz Checkmate® OFM-F in 100 gal of water). After treatment with microencapsulated pheromone, the leaves were placed on waxed paper to air dry, and microcapsules were counted.

Laboratory versus field application. Leaves were collected from 'Gala' apple and 'Encore' and 'Red Haven' peach trees for the leaf dip application 1–3 d before airblast sprayer applications of microencapsulated pheromone. Leaves were brought into the laboratory and dipped for 5 sec in beakers containing 200 mL of water and 45.8 μ L of Checkmate® OFM-F (2.93 fl oz Checkmate® OFM-F in 100 gal of water) according to the method previously described for the different tree fruit leaves. Leaves were also removed 2–6 h after microencapsulated pheromone was applied with an airblast sprayer (FMC DP100, FMC Corporation, Jonesboro, Arkansas) at the labeled rate of 214.5 mL/ha (2.93 fl oz/acre; 49.5 g active ingredient/ha) in a spray volume of 153 L/ha (100 gal/acre). Microencapsulated pheromone was applied a total of six times using the airblast sprayer at 6- to 9-d intervals. Fifteen leaves per apple and peach tree on a total of three to four randomly selected trees were removed for both leaf dip and airblast sprayer applications. Leaves were selected from the third to the sixth leaf from the distal end of the terminal to decrease potential variability in leaf age. Leaves were collected in an even distribution around each tree at a height of one to two meters on the outer portion of the leaf canopy. A total of six replicates of leaves were sampled from apple and peach trees for both application types.

Cultivar comparison with field application. *Apple.* Checkmate OFM-F microencapsulated pheromone was applied a total of four times at 7–9-d intervals to an apple block with 'Gala' and 'Delicious' trees. The total size of the apple block was 1.05 ha (2.6 acres), and the 'Gala' and 'Delicious' trees comprised 0.40 ha (1.0 acres) of the total. Four of the sixteen trees in each row were 'Gala' and four of sixteen were 'Delicious' trees. The apple trees were 13-yr old with 4.6 m (15 ft) between trees and 6.1 m (20 ft) between rows.

Peach. Checkmate OFM-F microencapsulated pheromone was applied in a 0.89-ha (2.2 acres) block containing four rows of 'Encore' and four rows of 'Red Haven' peach trees. The peach trees were 14-yr old with 6.1 m (20 ft) between trees and 6.1 m (20 ft) between rows. A total of seven applications were made on one-half row of 'Encore' trees and eight applications were made on one-half row of 'Red Haven' trees. Microencapsulated pheromone was applied 6–15 d apart on the two peach cultivars. 'Encore' and 'Red Haven' were not sprayed and sampled on the same day because they were in separate rows and insufficient space was available to spray them separately and perform the objectives of a simultaneous study using microencapsulated pheromone in the same block. Fifteen leaves per

apple and peach tree on a total of three to four randomly selected trees were sampled 2–6 h after each microencapsulated pheromone application.

Microcapsule counts. A fluorescent dye (0.10%) was microencapsulated with the oriental fruit moth pheromone by Suterra chemists so that microcapsules could be counted using an UV light (Knight et al. 2004). Leaves were illuminated with an UV light (Spectroline Model BiB-150-P, 365 nm) to count the green fluorescing microcapsules. The entire leaf including both top and bottom surfaces were examined for microcapsules. Fluorescing microcapsules were counted with the naked eye protected by an UV visor. Because leaf surface area is a variable, a leaf area meter (LI-COR Model LI-3000) was used so that counts could be standardized to a per square centimeter surface area (microcapsule density).

Statistical analysis. The microcapsule density was normalized prior to analysis of variance (ANOVA) using a $\log(x + 1)$ transformation. Analysis was done using SAS program version 9.1 (SAS, 2004) to compare mean microcapsule density among fruit trees, laboratory and field treatments, field applications on ‘Gala’ and ‘Delicious’ apple trees, and field applications on ‘Encore’ and ‘Red Haven’ peach trees. Where significant differences occurred, means were separated using Fisher’s least significant difference (LSD) test.

Results

Tree fruit comparison with laboratory leaf dips. The bottom surfaces of apple (‘Gala’) leaves had a microcapsule density of 0.47, the highest density among the fruit leaf surfaces (Table 1). The bottoms of Asian pear (‘Shinko’) and peach leaves and the tops of plum (‘Bluebell’) and sweet cherry (‘Columbia’) leaves had the lowest number of microcapsules with densities of 0.09, 0.14, 0.10, and 0.10, respectively. There were significantly more microcapsules on the top than the bottom leaf surfaces of ‘Shinko’ Asian pear leaves. There were significantly more microcapsules on the bottom than the top leaf surfaces on ‘Gala’, ‘Columbia’, ‘Montmorency’, and ‘Bluebell’ apple, sweet cherry, tart cherry, and plum leaves, respectively. The number of microcapsules on the tops and bottoms of ‘Duchess de Angloeme Bronze’ European pear and ‘Loring’ peach leaves was not significantly different. The largest difference in leaf surfaces occurred with plum leaves. The number of microcapsules on the bottoms of plum leaves was 2.3-fold greater than the top leaf surfaces.

Laboratory versus field application. Apple and peach leaves treated with microencapsulated pheromone by laboratory leaf dips had significantly higher densities than by field applications using an airblast sprayer (Table 2). The greatest differences between the application types occurred on ‘Gala’ apple leaves where the average number of microcapsules on leaves exposed by laboratory treatments was 60-fold greater than the field-treated leaves. On peach leaves, the average number of microcapsules on laboratory treated leaves was 8-fold and 30-fold greater than field treated leaves for the ‘Encore’ and ‘Red Haven’ cultivars, respectively.

Cultivar comparison with field application. After treatment with an airblast sprayer, the bottom surfaces of ‘Gala’ apple leaves had a significantly greater microcapsule density than the ‘Delicious’ apple leaves (Table 3). On the tops of the ‘Gala’ and ‘Delicious’ apple leaves, however, there were no significant

Table 1. Mean ($\pm$ SE) microcapsule density on top and bottom surfaces of tree fruit leaves dipped in breakers containing 45.8 μ L Check-Mate® OFM-F microencapsulated pheromone in 200 mL of water (2.93 fl. oz./100 gal.)

Tree	Cultivar	Leaf surface	Mean ^a	SE
Apple	Gala	Top	0.28 b	0.006
Apple	Gala	Bottom	0.49 a	0.008
Asian Pear	Shinko	Top	0.16 ef	0.004
Asian Pear	Shinko	Bottom	0.09 h	0.003
European Pear	Duchess ^b	Top	0.21 cd	0.006
European Pear	Duchess ^b	Bottom	0.18 de	0.006
Peach	Loring	Top	0.15 ef	0.004
Peach	Loring	Bottom	0.14 fg	0.005
Sweet Cherry	Columbia	Top	0.10 gh	0.003
Sweet Cherry	Columbia	Bottom	0.17 def	0.003
Tart Cherry	Montmorency	Top	0.17 def	0.004
Tart Cherry	Montmorency	Bottom	0.27 b	0.006
Plum	Bluebell	Top	0.10 gh	0.003
Plum	Bluebell	Bottom	0.23 bc	0.005

$F = 7.6$; $df = 6, 21$; $P < 0.001$.

^aMeans followed by the same letter are not significantly different (Fisher's protected LSD, $P < 0.05$).

^bDuchess de Angloeme Bronze.

differences in microcapsule density. There were significantly higher numbers of microcapsules on the bottom of 'Gala' and 'Delicious' apple leaves than the top surfaces of the same leaves. The 'Encore' peach leaves had a significantly greater number of microcapsules on top and bottom leaf surfaces than the top and bottom surfaces of 'Red Haven' peach leaves. The number of microcapsules on the top and bottom surfaces of 'Encore' leaves was not significantly different. However, the number of microcapsules on the top of 'Red Haven' leaves was significantly greater than on the bottom of the same leaves.

Discussion

Similar to the results of a previous study (Waldstein & Gut 2003), leaf surface was an important variable affecting the microcapsule density. In addition to leaf surface, the species and cultivar of the fruit tree were additional variables that influenced the density of microcapsules on leaves. Differences in trichome abundance and cuticle structure have been proposed as factors responsible for differences in the number of pheromone microcapsules on tops and bottoms of leaves and immature and mature leaves (Waldstein & Gut 2003). Trichome abundance and cuticle structure may also have contributed to differences in the number of microcapsules among the different fruit leaves and cultivars. The greater than 3-fold difference between the fruit leaves with the highest and lowest numbers of microcapsules (i.e., 'Gala' and 'Shinko') may indicate the need for rate specific recommendations on different species of fruit trees. Further studies could be

Table 2. Mean ($\pm$ SE) microcapsule density on apple and peach leaves treated with CheckMate® OFM-F microencapsulated pheromone by leaf dips in breakers under laboratory conditions or in the field on tree with an airblast sprayer at a rate of 86.6 mL of Checkmate® OFM-F in 378.5 L of water (2.93 fl. oz./100 gal.), Mountain Grove, Missouri, 2003.

Tree	Cultivar	Treatment	Leaf surface	Mean ^a	SE
Apple	Gala	Laboratory	Top	1.89 b	0.070
		Field	Top	0.03 d	0.003
		Laboratory	Bottom	3.43 a	0.114
		Field	Bottom	0.06 d	0.004
Peach	Encore	Laboratory	Top	1.27 c	0.061
		Field	Top	0.14 d	0.012
		Laboratory	Bottom	1.02 c	0.068
		Field	Bottom	0.15 d	0.013
Peach	Red Haven	Laboratory	Top	1.91 b	0.145
		Field	Top	0.07 d	0.005
		Laboratory	Bottom	1.95 b	0.222
		Field	Bottom	0.06 d	0.004

$F = 52$; $df = 2, 2358$; $P < 0.001$.
^aMeans followed by the same letter are not significantly different (Fisher's protected LSD, $P < 0.05$).

Table 3. Mean ($\pm$ SE) microcapsule density on top and bottom 'Gala' and 'Delicious' apple leaves and 'Encore' and 'Red Haven' peach leaves after CheckMate® OFM-F microencapsulated pheromone was applied with an airblast sprayer at a rate of 86.6 mL of Checkmate® OFM-F in 378.5 L of water (2.93 fl. oz./100 gal.), Mountain Grove, Missouri, 2003.

Tree	Cultivar	Leaf surface	Mean ^a	SE
Apple	Gala	Top	0.021 e	0.002
		Bottom	0.076 a	0.007
Apple	Delicious	Top	0.015 e	0.002
		Bottom	0.062 b	0.005
Peach	Encore	Top	0.072 a	0.006
		Bottom	0.078 a	0.006
Peach	Red Haven	Top	0.052 c	0.003
		Bottom	0.038 d	0.002

$F = 7.05$; $df = 3, 18$; $P < 0.001$.
^aMeans followed by the same letter are not significantly different (Fisher's protected LSD, $P < 0.05$).

conducted to compare different fruit species using a field application method (e.g., airblast sprayer). However, factors such as tree size, leaf canopy density, and row spacing could be potentially confounding variables affecting the number of pheromone microcapsules on leaves.

The large differences between the number of microcapsules on laboratory-treated and field-treated leaves illustrate the distinction between these two application methods and environments. Laboratory leaf dips eliminate sprayer and environmental variables that can reduce coverage during airblast sprayer applications of microencapsulated pheromone. However, airblast sprayer applications are more realistic of a standard application in commercial orchards. As this study demonstrates, caution should be exercised when comparing leaf dip applications of microencapsulated pheromone in the laboratory to field applications using an airblast sprayer.

After an airblast sprayer application, only a fraction of the total number of pheromone microcapsules in a spray tank adheres to a leaf surface. Some of the microcapsules may come in contact with other tree tissues such as fruit, limbs, and the trunk, and some may contact nontarget plants on the orchard floor. The microcapsules that adhere to other tree tissues may still contribute to mating disruption, whereas, the microcapsules on the orchard floor may be less effective because oriental fruit moth mating occurs in the tree canopy. Knight & Larsen (2001) demonstrated that microencapsulated pheromone applied to the orchard floor was ineffective at disrupting mating of codling moth, *Cydia pomonella* L (Lepidoptera: Tortricidae). A previous study indicated that 1-yr-old woody tissue and fruit had a significantly higher microcapsules density than mature leaves (Waldstein & Gut 2003). However, that study was conducted using laboratory plant tissue dips with a different formulation of microencapsulated pheromone than the one used in this study (i.e., OFM MEC Phase III, 3M Canada company, London, Ontario, Canada). Further studies should be conducted to determine if other tree fruit tissues have higher microcapsule density than leaves after an airblast sprayer application of microencapsulated pheromone. Another potential topic of future studies is the contribution of oriental fruit moth mating disruption by microcapsules that adhere to nontarget plant tissues on the orchard floor.

Acknowledgments

We thank Jessica Bradshaw for her assistance in data collection, Tom Larsen and Kristen Ketner for technical assistance, and Suterra LLC and Missouri State University for funding assistance.

References Cited

- Agnello, A. M., W. H. Reissig & T. Harris. 1994.** Management of summer populations of European red mite (Acari: Tetranychidae) on apple with horticultural oil. *J. Econ. Entomol.* 87:148–161.
- Albajes, R., M. Knostantopoulou, O. Etchepare, M. Eizguirre, B. Frerot, A. Sans, F.**

- Krokos, A. Ameline & B. Mazomenos. 2002.** Mating disruption of the corn borer *Sesamia nonagrioides* (Lepidoptera: Noctuidae) using sprayable formulations of pheromone. *Crop Prot.* 21:217–225.
- Atanassov, A., P. W. Shearer, G. Hamilton & D. Polk. 2002.** Development and implementation of a reduced risk peach arthropod management program in New Jersey. *J. Econ. Entomol.* 95:803–812.
- Beroza, M., C. S. Hood, D. Trefrey, D. E. Leonard, E. F. Knipling, W. Klassen & L. J. Stevens. 1974.** Large field trial with microencapsulated sex pheromone to prevent mating of the gypsy moth. *J. Econ. Entomol.* 67:659–664.
- Black, B. & M. J. Bukovac. 1996.** Plant growth regulator application technology, Uptake and Action. pp. 44–50. *In* K. M. Maib, P. K. Andrews, G. A. Lang, and K. Mullinix [Eds.], *Tree fruit physiology: growth and development*. Good Fruit Grower, Yakima, Washington.
- Bohmont, B. L. 1997.** Pesticide application equipment. pp. 271–323. *In* B. L. Bohmont [Ed.], *The standard pesticide user's guide*, 4th ed. Prentice Hall, Upper Saddle River, New Jersey.
- Campion, D. G., L. J. McVeigh, P. Hunter-Jones, D. R. Hall, R. Lester, B. F. Nesbitt, G. J. Marrs & M. R. Alder. 1981.** Evaluation of microencapsulated formulations of pheromone components of the Egyptian cotton leafworm *Spodoptera littoralis* in Crete. pp. 253–265. *In* E. Mitchell [Ed.], *Management of insect pests with semiochemicals: concepts and practice*. Plenum Press, New York.
- Cardé, R. T., T. C. Baker & P. J. Castrovillo. 1977.** Disruption of sexual communication in *Laspeyresia pomonella* (codling moth) *Grapholitha molesta* (oriental fruit moth), and *G. prunivora* (lesser appleworm) with hollow fiber attractant sources. *Entomol. Exp. Applic.* 22:280–288.
- Cardé, R. T., K. Trammel & W. L. Roelofs. 1975.** Disruption of sex attraction of the redbanded leafroller (*Argyrotaenia velutinana*) with microencapsulated pheromone components [apples]. *Environ. Entomol.* 4:448–450.
- Charlton, R. & R. Cardé. 1981.** Comparing the effectiveness of sexual communications disruption in the oriental fruit moth (*Grapholitha molesta*) using different combinations and dosages of its pheromone blend. *J. Chem. Ecol.* 7: 501–507.
- Critchley, B. R., D. R. Hall, G. J. Marrs, M. M. Hosny & E. A. Nasr. 1983.** Control of pink bollworm, *Pectinophora gossypiella* (Saunders) (Lepidoptera: Gelechiidae), in Egypt by mating disruption using an aerially applied microencapsulated pheromone formulation. *Bull. Entomol. Res.* 73: 289–299.
- Figueredo, A. J. & T. C. Baker. 1992.** Reduction of the response to sex pheromone in the oriental fruit moth, *Grapholitha molesta* (Lepidoptera: Tortricidae) following successive pheromonal exposures. *J. Insect Behav.* 5: 347–363.
- Horowitz, A. R., Z. Mendelson & I. Ishaaya. 1997.** Effect of abamectin mixed with mineral oil on the sweetpotato whitefly (Homoptera: Aleyrodidae). *J. Econ. Entomol.* 90: 349–353.
- Howitt, A. J. 1993.** Common Tree Fruit Pests. Michigan State University Extension, NCR Extension Publication #63. East Lansing, Michigan.
- Johnstone, D. R. 1985.** Physics and meteorology. pp. 50–67. *In* P. T. Haskell [Ed.], *Pesticide application: principles and practice*. Clarendon Press, Oxford, England.
- Jones, E. L., G. H. L. Rothschild & R. A. Vickers. 1984.** Control of oriental fruit moth *Cydia molesta* (Busck) in peach orchards with commercial pheromone dispensers. pp. 82–87. *In* P. Bailey and D. Swincer [Eds.], *Proceedings of the Fourth Australian Applied Entomological Research Conference; Pest control: Recent advances and future prospects*.
- Knight, A. L. & T. E. Larsen. 2001.** The influence of spray application method on the efficacy of microencapsulated sprayable codling moth pheromone. *Proc. Western Orchard Pest Manage. Conf.* 75: 139–140.
- Knight, A. L., T. E. Larsen & K. C. Ketner. 2004.** Rainfastness of a microencapsulated

- sex pheromone formulation for codling moth (Lepidoptera: Tortricidae). J. Econ. Entomol. 97:1987–1992.
- Kovanci, O. B., J. F. Walgenback & G. G. Kennedy. 2004.** Evaluation of extended-season mating disruption of the Oriental fruit moth *Grapholitha molesta* (Busck) (Lep., Tortricidae) in apples. J. Appl. Entomol. 128:664–669.
- Lofstedt, C., N. J. Vickers & T. C. Baker. 1990.** Courtship, pheromone titre and determination of the male mating success in the oriental fruit moth, *Grapholitha molesta* (Lepidoptera: Tortricidae). Entomologia Generalis. 15: 121–125.
- Moawad, G., A. A. Khidr, M. Zaki, B. R. Critchley, L. J. McVeigh & D. G. Campion. 1991.** Large-scale use of hollow fibre and microencapsulated pink bollworm pheromone formulations integrated with conventional insecticides for the control of the cotton pest complex in Egypt. Trop. Pest Management. 37: 10–16.
- Polavarapu, S., G. Lonergan, H. Peng & K. Neilsen. 2001.** Potential for mating disruption of *Sparganothis sulfureana* (Lepidoptera: Tortricidae) in cranberries. J. Econ. Entomol. 94:658–665.
- Polavarapu, S., M. Wicki, K. Vogel, G. Lonergan & K. Nielsen. 2002.** Disruption of sexual communication of oriental beetles (Coleoptera: Scarabaeidae) with a microencapsulated formulation of sex pheromone components in blueberries and ornamental nurseries. Environ. Entomol. 6:1268–1275.
- Pree, D. J., R. M. Trimble, K. J. Whitty & P. M. Vickers. 1994.** Control of oriental fruit moth by mating disruption using sex pheromone in the Niagara Peninsula, Ontario. Can. Entomol. 126:1287–1299.
- Prevett, P. F., F. P. Benton, D. R. Hall, R. J. Hodges & R. Serodio. 1989.** Suppression of mating in *Ephestia cautella* (Walker) (Lepidoptera: Phycitidae) using microencapsulated formulations of synthetic sex pheromone. J. Stored Prod. Res. 25:147–154.
- Rumbo, E. R. & R. A. Vickers. 1997.** Prolonged adaptation as possible mating disruption mechanism in oriental fruit moth, *Cydia molesta* L. (Lepidoptera: Tortricidae). J. Chem. Ecol. 23:445–457.
- Sanders, C. J. & G. S. Lucuik. 1996.** Disruption of male oriental fruit moth to calling females in a wind tunnel by different concentrations of synthetic pheromone. J. Chem. Ecol. 22:1971–1986.
- SAS Institute. 2004.** The SAS system, version 9.1. SAS Institute Inc., Cary, North Carolina, USA.
- Stansly, P. A., T. X. Liu & C. S. Vavrina. 1998.** Response of *Bemisia argentifolii* (Homoptera: Aleyrodidae) to imidacloprid under greenhouse, field, and laboratory conditions. J. Econ. Entomol. 91:686–692.
- Taschenberg, E. F. & W. L. Roelofs. 1976.** Pheromone communication disruption of the grape berry moth (*Paralobesia viteana*) with microencapsulated and hollow fiber systems. Environ. Entomol. 4:688–691.
- Trimble, R. M. & M. E. Appleby. 2004.** Comparison of efficacy of programs using insecticide and insecticide plus mating disruption for controlling the obliquebanded leafroller in apple (Lepidoptera: Tortricidae). J. Econ. Entomol. 97:518–524.
- Trimble, R. M., D. J. Pree & N. J. Carter. 2001.** Integrated control of oriental fruit moth (Lepidoptera: Tortricidae) in peach orchards using insecticide and mating disruption. J. Econ. Entomol. 94:476–485.
- Trimble, R. M., D. J. Pree, E. S. Barszcz & N. J. Carter. 2004.** Comparison of a sprayable pheromone formulation and two hand-applied pheromone dispensers for use in the integrated control of oriental fruit moth (Lepidoptera: Tortricidae). J. Econ. Entomol. 97:482–489.
- Trimble, R. M., P. M. Vickers, K. E. Nielsen & G. Barinshteyn. 2003.** Sprayable pheromone for controlling the North American grape berry moth by mating disruption. Agric. For. Entomol. 5:263–268.
- Valeur, P. G. & C. Lofstedt. 1996.** Behavior of male oriental fruit moth, *Grapholitha*

molesta, in overlapping sex pheromone plumes in a wind tunnel. Entomol. Exp. Applic. 79: 51–59.

Waldstein, D. E. & L. J. Gut. 2003. Comparison of microcapsule density with various apple tissues and formulations of oriental fruit moth (Lepidoptera: Tortricidae) sprayable pheromone. J. Econ. Entomol. 96:58–63.

Waldstein, D. E. & L. J. Gut. 2004. Effects of rain and sunlight on oriental fruit moth (Lepidoptera: Tortricidae) pheromone microcapsules applied to apple foliage. J. Agric. & Urban Entomol. 21:117–128.

Bin Sterilization to Prevent Reintroduction of Codling Moth¹

James D. Hansen, Millie L. Heidt, and Pauline A. Anderson

USDA-ARS Yakima Agricultural Research Laboratory, 5230 Konnowac Pass Road,
Wapato, Washington 98951

J. Agric. Urban Entomol. 23(1): 17–26 (Jan 2006)

ABSTRACT An important source of reinfestation of codling moth, *Cydia pomonella* L. (Lepidoptera: Tortricidae), is the return of fruit bins containing diapausing larvae. Laboratory tests, conducted to determine efficacious temperatures of hot water baths to prevent adult emergence, found baths at 80°C for ≥ 30 sec were effective and adding a surfactant to the water bath slightly improved efficacy. Validation tests conducted at a packing house, using an 80°C submersion bath for bins, resulted in $< 0.2\%$ survival. This procedure should be an important component for field control of codling moth.

KEY WORDS *Cydia pomonella*, codling moth, heat, apple bins, water baths

Considerable advancements have been made in controlling codling moth, *Cydia pomonella* L. (Lepidoptera: Tortricidae), by using environmentally friendly methods, such as encouraging natural enemies and promoting mating disruption (Judd et al. 1997, Calkins 1998, Calkins et al. 2000). The result has been fewer applications of insecticides, which reduce costs to the growers and reduce environmental contamination. However, a major problem with these programs is the reintroduction of codling moths to commercial orchards by returning empty fruit bins containing overwintering (or diapausing) larvae (Proverbs & Newton 1975, Higbee et al. 1999, 2001), with Higbee et al. (2001) finding an average of 12.2 emerged adults from wooden bins. The efficacy of these control programs has been compromised by the reintroduction of codling moth from the bins to the orchards when empty bins are brought back to the field with diapausing larvae attached to or hidden in them. If diapausing codling moth could be eliminated from these containers, these innovative control strategies would be more effective.

One procedure that may be applicable is to use hot water dips for treating larvae-contaminated bins. Diapausing codling moth larvae are the most heat resistant life stage, although direct exposure in air for 2 min at 52°C will immediately kill all treated individuals (Wang et al. 2004). Prevention of adult emergence through long-term (or chronic) mortality is an effective mitigation measure. The effect of heat need not be immediate and the water temperature does not have to be 52°C to prevent emergence, which would reduce potential energy costs.

In this research, the level of chronic mortality of diapausing larvae caused by submersion in hot water was determined by temperature and exposure duration.

¹Accepted for publication 13 October 2006.

Treated diapausing larvae must pass through the pupal stage, a physiological bottleneck, before becoming adults. Intense heat exposure may interfere with normal insect development, resulting in chronic mortality. Adding surfactants, which aid treatment efficacy by breaking down the surface tension of water and allowing heat to reach previously protected areas, was also examined. The goal of this research was to provide recommendations for treating bins at the packing house by using hot water dips to prevent adult moth emergence.

Materials and Methods

Experimental design. Diapausing codling moth larvae were obtained from an established colony at the USDA-ARS Yakima Agricultural Research Laboratory (YARL) in Wapato, Washington (Hansen & Anderson 2006). Diapausing larvae spin cocoons within cells of corrugated cardboard after development under reduced temperature and photoperiod. Diapausing larvae are generally pale in comparison with the pinkish normal larvae, have thicker cocoons, and are less active.

Initial tests were done by directly submersing the strips, on average ($\pm$ SD) containing 26 ($\pm$ 12) diapausing larvae, into baths with 0.01% food grade defoamer (Columbia Technical Assoc., Wenatchee, Washington), the same material currently used in water systems in packing houses. Water was heated using a 6-quart immersible cooker (15.2-cm deep $\times$ 23.5-cm diameter) (Model No. 060000; National Presto Industries, Inc., Eau Claire, Wisconsin), and the heat treatments were: no water (control), 25°, 50°, 60°, 70°, or 80°C for 20 sec, followed by air cooling.

Because diapausing larvae frequently overwinter in cracks and corners of wood bins, treatment efficacy was determined by simulating this environment by placing the cardboard strips with larvae between two wooden strips (3.5 $\times$ 17.5 $\times$ 0.7 cm) during treatment. Each strip contained on average 22 ($\pm$ 7) diapausing larvae.

To test if reduced water surface tension would improve treatment efficiency, three food grade surfactants (Ivanhoe Industries Inc., Mundelein, Illinois) were tested separately: "Surf 1" = a polydimethyl siloxane emulsion with silicone defoamer, "Surf 2" = similar to the proceeding material but with higher concentration of active ingredients, "Surf 3" = a polydimethyl silicone emulsion along (exact formulations protected by proprietary agreement). Surfactant concentrations varied from 0.01–1%.

Initial tests with diapausing larvae between wooden strips were done with 0.01% Surf 1 for 20 sec submersion in baths at 25°, 50°, 60°, 70°, or 80°C. The next set of tests used 0.1% Surf 1 in the same baths, but at 20, 30, or 40 sec submersions. Water only at 25°C and no submersion served as controls. There were five replicates for each treatment.

To examine if different surfactant formulations would affect efficacy, diapausing larvae between wood strips were submersed in hot water baths with the three silicone surfactants in either at 25°C (control), 60°, or 70°C for 20, 30, or 40 sec. There were four replicates for each treatment.

In two studies (Study #1 and Study #2), each with five replicates, acute and chronic survival was determined for diapausing larvae submerged between wooden strips in 1% Surf 3 for 30 sec at 25°, 50°, 60°, 70°, and 80°C. No submersion and submersion in deionized water at 25°C for 30 sec were the controls.

Another series of tests compared survival with submersion durations between 5 and 30 sec using 50°, 60°, or 70°C baths with no surfactant with deionized water or with 3% Surf 3. Each treatment was replicated five times.

Larval survival was evaluated by inspection the day following treatment. Larval and pupal mortalities were recorded after dissecting the cardboard strips. Percent acute survival was calculated by the number of pupae divided by the total number of treated larvae, multiplied by 100%. Adult survival was measured by emergence from treated larvae held in specially designed plastic chambers at room temperature; these chambers were similar to those already in use in the codling moth rearing colony. Percent chronic survival was determined by the number of adult moths and the total number of treated larvae. There were five replicates for each treatment.

Packinghouse validation. Tests were conducted at a commercially built facility (Van Doren Sales, Incorp., Wenatchee, Washington) for sterilizing wooden bins. Empty bins move along a track, are submerged for 20 sec in an 80°C water bath under an enclosed structure, and then emerge on an open track. It takes about 2.5 min to complete the cycle.

To test efficacy, about 40 diapausing larvae in cardboard strips were placed between strips of wood, as described earlier. Using a screen sleeve, these were attached to three locations inside of a bin: the bottom, middle, and top. The same procedure was used when Surf 3 was added to the water bath to make a 0.03% mixture. Two types of controls were used: diapausing larvae between wood strips submerged in water at room temperature (25°) and the same with no submersion. Five replicates were used for each of these treatments. The total number of treated diapausing larvae were: 807 in air (control), 662 in 25°C water, 656 in 80°C water, and 669 in 80°C water with surfactant.

Statistical analysis. Data were managed and univariate statistics calculated by using Microsoft® Excel 2002 Spreadsheet (Microsoft Corp., Redmond, Washington). Data were analyzed using SAS® (SAS Institute 1996). Because of the possible lack of homogeneity among the variances, nonparametric tests were used to determine significant differences by first arranging data by PROC RANK, then performing PROC GLM. This approach is equivalent to a Wilcoxon rank sum test for two samples and the Kruskal-Wallis *k*-sample test for more than two samples (Zolman 1993). Paired comparisons were done by determining the difference between the two variables and using the T and PRT options in PROC Means.

Results

Direct submersion. Acute (larval) survival was not significantly different among the dry control and submersion at room temperature in water or in a bath with 0.01% defoamer (Table 1). However, larval survival decreased (mean $\pm$ SE = $7.4 \pm 6.0\%$) with the 50°C treatment, and no larvae survived the higher temperatures.

Treatment using wood strips. Chronic (adult) survival decreased with 30-sec bath temperatures (Table 2), particularly between 60° and 70°C (Surf 1: $F = 6.22$; $df = 1, 8$; $P < 0.05$; Surf 2: $F = 4.50$; $df = 1, 8$; $P > 0.05$; Surf 3: $F = 19.20$; $df = 1, 8$; $P < 0.01$) 70° and 80°C (Surf 1: $F = 14.80$; $df = 1, 8$; $P < 0.01$; Surf 2: $F = 19.20$; $df = 1, 8$; $P < 0.01$; Surf 3: $F = 4.60$; $df = 1, 8$; $P > 0.05$) with Surf 3 consistently having the lowest value. Larval survival was not significantly

Table 1. Average acute percent survival of diapausing codling moth larvae submerged in various baths foe 20 sec; five replicates per treatment, at YARL, 2004.

Treatment	Temp. °C	Treated larvae		% Live	
		Avg. ±	SE	Avg. ±	SE
Dry ^a	25	33.4	2.3	93.2	1.2
Water	25	37.8	4.6	94.4	1.4
0.01% defoamer	25	35.8	4.9	95.3	2.8
0.01% defoamer	50	37.0	1.5	7.4	6.0
0.01% defoamer	60	37.6	5.4	0.0	0.0
0.01% defoamer	70	35.0	0.9	0.0	0.0
0.01% defoamer	80	36.4	6.0	0.0	0.0

^aNo submersion.

different among the three surfactants at 30 sec in the 60° baths ($F = 0.17$; $df = 2, 10$; $P > 0.05$), 70° ($F = 3.23$; $df = 2, 10$; $P > 0.05$), or 80°C ($F = 0.56$; $df = 2, 10$; $P > 0.05$). The 0.1% and 0.01% Surf 1 formulations were not significantly different for 20 sec at 80°C ($F = 2.51$; $df = 1, 8$; $P > 0.05$), but were between 20 and 30 sec for the 0.1% Surf bath at 80°C ($F = 6.36$; $df = 1, 8$; $P < 0.05$).

Because previous tests suggested that Surf 3 was the most efficient formulation, further tests on surfactants were only done with this material on diapausing larvae. Acute survival was also examined to determine which life stage was affected by the heat treatments. Study 2 tended to have higher survival for all treatments, including the dry control and water at room temperature, than the first study (Table 3). Adult survival declined rapidly from 60° to 70° for a 30 sec bath of 1% Surf (combined mean for both replicates: 60°C = 31.3%, 70°C = 5.2%). There were no significant differences in adult emergence for 30 sec in a 1% Surf bath between 70° and 80°C ($F = 2.60$; $df = 3, 16$; $P > 0.05$).

To determine a descriptive mathematical model on efficacy, average acute and chronic survivorships was plotted for water baths at 25°, 50°, 60°, or 70°C from 5–30 sec in deionized water or 3% Surf baths (Figs. 1, 2), but no distinct pattern was detected. Pupation was little effected with time in either the water or in the Surf 3 baths at 25°C (Fig. 2). Larvae tended to tolerate the surfactant baths better than the water baths for temperatures of 50° and 60°C, but survival declined with exposures at 70°C. Furthermore, paired t -tests showed significant differences ($P < 0.05$) between pupation and adult emergence within the same replicate for all exposures for the dry control and all heated baths in both water and surfactant, except for 30 sec of water at 70°C, which indicated extensive chronic mortality after treatment. In neither water nor surfactant bath at 25°C was adult survival significantly different among time periods (water: $F = 4.10$; $df = 1, 28$; $P > 0.05$; and 3% Surf: $F = 3.22$; $df = 1, 28$; $P > 0.05$). At 30 sec, there was no significant difference in adult survival in exposures among the temperatures in the water bath ($F = 2.47$; $df = 2, 12$; $P > 0.05$), but differences were found in the 3% Surf ($F = 9.65$; $df = 2, 12$; $P < 0.01$), with the 70°C bath producing the lowest adult survival.

Table 2. Average percent chronic survival of diapausing codling moth larvae ($n = 22 \pm 7$) submerged between two wooden strips into various water baths; five replicates for each treatment, YARL, 2004–2005.

Chem.	Concentration	Temp. °C	% Chronic survival time (sec)					
			20		30		40	
			Avg. $\pm$	SE	Avg. $\pm$	SE	Avg. $\pm$	SE
Dry ^a							99.1	0.8
Water		25	95.6	1.1	96.7	1.3	95.2	1.2
Surf 1	0.01%	25	99.2	0.8				
		50	95.3	1.2				
		60	97.5	1.0				
		70	66.5	8.0				
		80	64.3	6.1				
	0.10%	25	99.0	0.8	91.3	4.7	94.5	1.9
		50	94.6	1.3	95.0	1.3	95.7	1.6
		60	95.6	1.6	82.0	3.6	86.3	4.2
		70	74.6	11.4	59.5	8.2	65.5	6.0
		80	42.6	9.4	11.7	7.8	11.3	7.3
Surf 2	1.00%	25			100.0	0.0		
		60			89.0	3.7		
		70			57.1	12.5		
		80			15.8	4.0		
Surf 3	1.00%	25			100.0	0.0		
		60			90.9	1.5		
		70			24.2	6.7		
		80			8.0	5.3		

^aNo submersion.

Packinghouse validation. Adult survival in the untreated controls was 64.7% and in the 25° bath was 45.1%. Only one adult emerged from the treated larvae in each of the hot water baths, with an adult survival of < 0.2%. Location of larvae on the bin did not affect survival.

Discussion

Diapause is a type of dormancy, characterized by an altered physiological state that allows an insect to survive adverse environmental conditions (Brown 1991). In the codling moth, diapause is induced by decreasing day length, lower temperatures, increased crowding, and limited food supply (Riedl 1983). Diapause in codling moth allows the fifth instar (the last larval stage) to survive winter where cocoons are spun under loose bark, in litter at the base of trees, or in fruit bins left in the orchard, and then continue normal development in the spring with adult emergence (Bloem et al. 1999). Thus, diapausing larvae are conditioned to toler-

Table 3. Acute and chronic survival of diapausing codling moth larvae (average $\pm$ SEM/replicate, Study 1 = 46.4 ± 2.1 ; Study 2 = 62.3 ± 1.1) submerged between wood strips into various baths for 30 sec, YARL, 2005–2006.

Treatment	Temperature (°C)	% Acute survival study				% Chronic survival study			
		1		2		1		2	
		Mean	$\pm$ SEM	Mean	$\pm$ SEM	Mean	$\pm$ SEM	Mean	$\pm$ SEM
Dry	25	82.8	12.3	94.2	1.2	43.2	4.4	67.7	3.7
Water	25	93.7	0.9	92.8	1.8	59.1	3.2	64.5	2.7
1% Surf 3	25	95.0	1.6	^b	^b	50.8	5.9	^b	^b
1% Surf 3	50	88.0	3.4	91.1	2.2	33.2	4.8	51.8	3.4
1% Surf 3	60	57.5	9.5	78.8	3.2	18.1	6.9	44.5	4.5
1% Surf 3	70	14.7	6.0	13.9	2.1	2.5	1.7	7.8	2.3
1% Surf 3	80	3.9	2.6	5.0	2.0	2.4	1.6	2.0	1.0

^a* = 0.05, ** = 0.01, ns = not significantly different, using Wilcoxon rank sum tests conducted between studies for same treatment.

^bNot conducted.

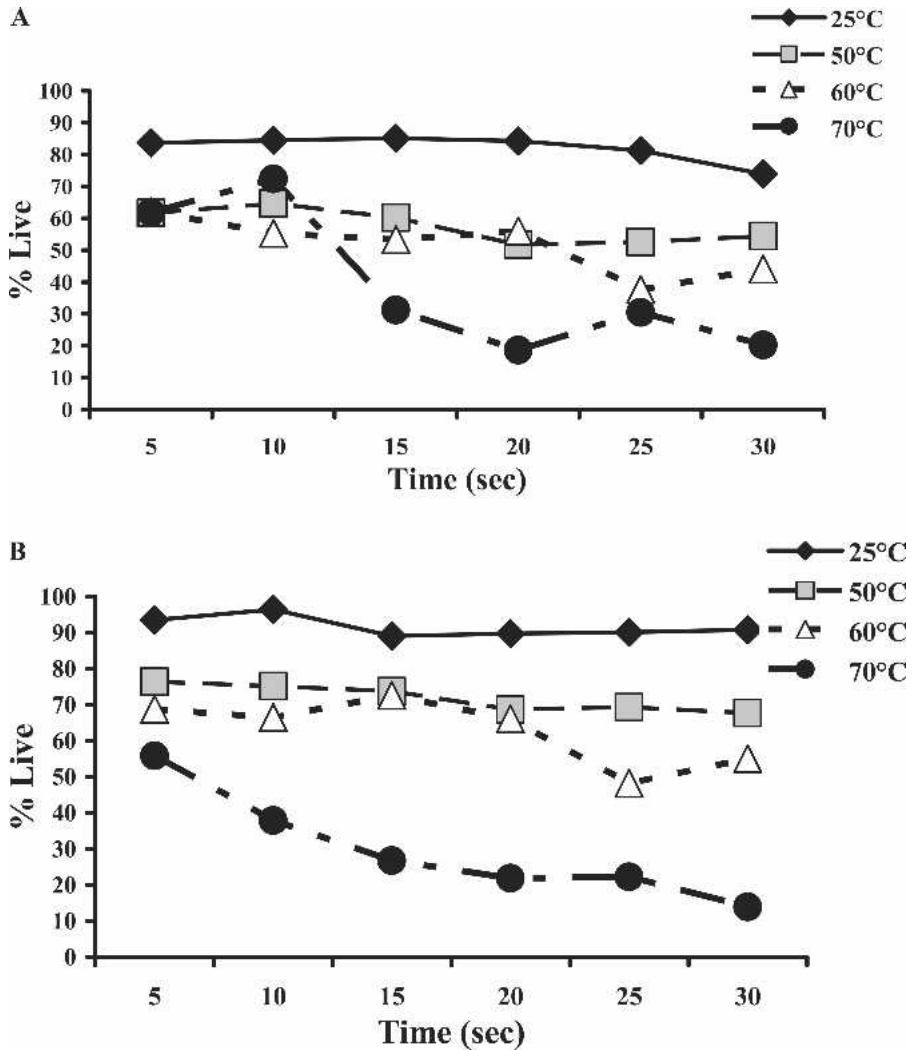


Fig. 1. Average acute survival of about 25 diapausing codling moth larvae, among five replicates, submerged between two wooden strips for 5-sec intervals to 30 sec in 25°, 50°, 60°, or 70°C baths at YARL, 2005–2006: A, water bath; B, bath with 3% Surf 3.

ate more extreme temperatures, including heat (Wang et al. 2004), than nondiapausing mature larvae.

Various methods have been proposed to control diapausing codling moth larvae in fruit bins. Proverbs & Newton (1975) suggested methyl bromide fumigation, but the use of this chemical has been severely restricted by international agreements (Hansen et al. 2004). Cossentine et al. (2004) recommended fumigating empty fruit bins with carbon dioxide, but obtained only 60% mortality of

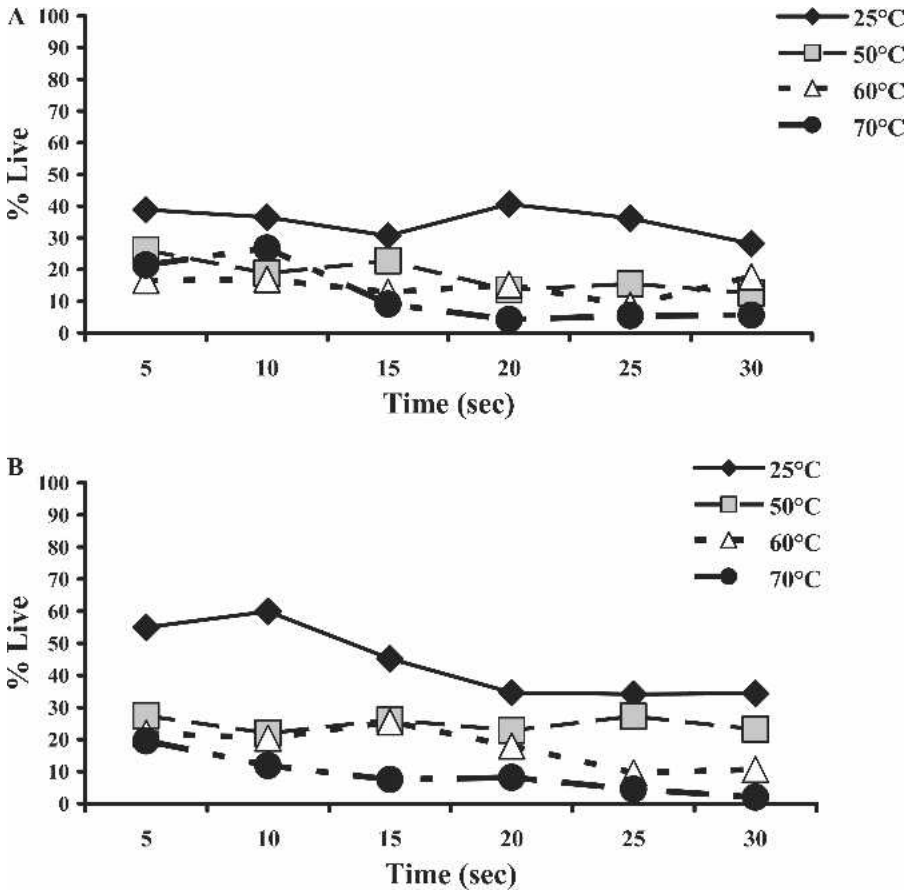


Fig. 2. Average chronic survival of about 25 diapausing codling moth larvae, among five replicates, submerged between two wooden strips for 5-sec intervals to 30 sec in 25°, 50°, 60°, or 70°C baths at YARL, 2005–2006: A, water bath; B, bath with 3% Surf 3.

diapausing codling moth larvae after six days of 40% carbon dioxide at 15°C (59°F). Higbee et al. (2001) covered bin stacks with clear plastic and observed earlier adult emergence than with uncovered bins. Lacey & Chauvin (1999) submerged fruit bins containing diapausing codling moth larvae for 1 min into suspensions containing entomopathogenic nematodes and obtained > 98% larval mortality with ≥ 50 infectious nematodes per ml. Cossentine et al. (2002) performed a similar treatment and observed > 80% codling moth larval mortality.

Our studies demonstrate that diapausing larvae can be controlled by submersion in hot water. Acute survival decreases with increased temperature, indicating immediate effects on larvae. Also, our data show significant mortality among diapausing larvae, even without a treatment. Adding a surfactant may aid efficacy in the low range of treatment temperatures, but is not necessary at the

higher ones. When applying a heat treatment, time is not a limiting factor because it is difficult to move bins within short periods. In our packing house test, total bin submersion was only 20 sec, but the effective time was much longer because the material did not rapidly cool after treatment. Furthermore, chronic mortality allows some flexibility in selecting treatment temperatures.

Diapausing codling moth larvae in fruit bins can be effectively controlled by hot water submersion. Eliminating the source of codling moths for possible reintroduction in orchards should aid in pest management and reduce costs. Furthermore, treatment temperatures of 80°C can provide pathogen control, another important benefit.

Acknowledgments

The authors thank Walt Hough and Auvil Fruit (Orlando, Washington) for the packing house validation, Kaylin Phillips for laboratory assistance, and Eugene Kupferman and Chang-Lin Xiao (Washington State University—Wenatchee) for their advice and support. P. J. Landolt and S. F. Garczynski (USDA-ARS, Wapato) and M. K. Hennessey and Ned Jones (USDA-APHIS, Raleigh) reviewed earlier versions of this manuscript.

References Cited

- Bloem, S., K. A. Bloem & C. O. Calkins. 1999. Is it possible to use mass-reared or field-collected diapaused codling moth larvae, *Cydia pomonella*, (Lepidoptera: Tortricidae), to predict spring biofix? J. Entomol. Soc. Brit. Col. 96: 111–117.
- Brown, J. J. 1991. Diapause. In: L. P. S. Van Der Geest and H. H. Evenhuis (Ed.), Tortricid pests, their biology, natural enemies and control. Elsevier, New York. Pp. 175–185.
- Calkins, C. O. 1998. Review of the codling moth areawide suppression program in the western United States. J. Agric. Entomol. 15: 327–333.
- Calkins, C. O., A. L. Knight, G. Richardson & K. A. Bloem. 2000. Area-wide population suppression of codling moth. IV: K. Tan (Ed.), Area-wide control of fruit flies and other insect pests. Penerbit Univ. Sains Malaysia, Pinang, Malaysia. Pp. 215–219.
- Cossentine, J. E., L. B. Jensen & L. Moys. 2002. Fruit bins washed with *Steinernema carpocapsae* (Rhabditida: Steinernematidae) to control *Cydia pomonella* (Lepidoptera: Tortricidae). Biocon. Sci. Technol. 12: 251–258.
- Cossentine, J. E., P. L. Scholberg, L. B. J. Jensen, K. E. Bedford & T. C. Shephard. 2004. Fumigation of empty fruit bins with carbon dioxide to control diapausing codling moth larvae and *Penicillium expansum* Link. Ex Thom spores. HortScience 39: 429–432.
- Hansen, J. D. & P. A. Anderson. 2006. Mass rearing codling moths: Improvements and modifications. J. Entomol. Soc. Brit. Col. 103: 33–36.
- Hansen, J. D., S. R. Drake, M. L. Heidt, M. A. Watkins, J. Tang & S. Wang. 2004. Radio frequency treatments for postharvest codling moth control in fresh apples. Horttechnology 14: 533–537.
- Higbee, B. S., C. O. Calkins & C. Temple. 1999. Larval infestation of harvest bins by codling moth. Good Fruit Grower 50: 26–31.
- Higbee, B. S., C. O. Calkins & C. Temple. 2001. Overwintering of codling moth (Lepidoptera: Tortricidae) larvae in apple harvest bins and subsequent moth emergence. J. Econ. Entomol. 94: 1511–1517.
- Judd, G. J. R., M. G. T. Gardiner & D. R. Thomson. 1997. Control of codling moth in organically-managed apple orchards by combining pheromone-mediated mating disruption, post-harvest fruit removal and tree banding. Entomol. Exp. Appl. 83: 137–146.
- Lacey, L. A. & R. L. Chauvin. 1999. Entomopathogenic nematodes for control of diapausing codling moth (Lepidoptera: Tortricidae) in fruit bins. J. Econ. Entomol. 92: 104–109.

- Proverbs, M. D. & J. R. Newton. 1975.** Codling moth control by sterile insect release: importance of fruit and fruit containers as a source of reinfestation. *J. Entomol. Soc. Brit. Col.* 72: 6–9.
- Riedl, H. 1983.** Analysis of codling moth phenology in relation to latitude, climate and food availability. *In*: V. K. Brown and I. Hodek (Ed.). *Diapause and life cycle strategies in insects*. Dr. W. Junk Publ., Boston. Pp. 233–252.
- SAS Institute. 1996.** The SAS System for Windows: Release 6.12. Cary, North Carolina.
- Wang, S., X. Yin, J. Tang & J. D. Hansen. 2004.** Thermal resistance of different life stages of codling moth (Lepidoptera: Tortricidae). *J. Stored Prod. Res.* 40: 565–574.
- Zolman, J. F. 1993.** *Biostatistics: Experimental design and statistical inference*. 343 p. Oxford Univ. Press, New York.
-

Efficacy of Entomopathogenic Nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) Against the Squash Vine Borer, *Melittia cucurbitae* (Lepidoptera: Sesiidae) in South Carolina¹

Ramazan Canhilal and Gerald R. Carner²

Seyrani Agricultural Faculty, Erciyes University 38039 Melikgazi, Kayseri, Turkey

J. Agric. Urban Entomol. 23(1): 27–39 (Jan 2006)

ABSTRACT Laboratory bioassays and field experiments were conducted to study the effects of entomopathogenic nematodes on squash vine borer (SVB), *Melittia cucurbitae* (Harris) (Lepidoptera: Sesiidae). In the laboratory, SVB larvae were exposed to *Steinernema riobrave* TX strain, *S. feltiae* SN strain, *S. carpocapsae* All strain, *S. carpocapsae* Sal strain, *Heterorhabditis bacteriophora* Hb strain, and *H. sp.* Hbl strain. *Steinernema riobrave* was the best-performing nematode (overall), producing earlier and greater rates of mortality than the other strains. Species and strains of *Steinernema* were more effective than either heterorhabditid nematodes. Field experiments were carried out on yellow crookneck squash, *Cucurbita pepo*, in six trials over the course of 3 years (1997–99) in Clemson, South Carolina. The same nematodes as used in lab bioassays except *S. carpocapsae* Sal strain and *H. bacteriophora* Hb strain were compared to an insecticide standard, endosulfan (Phaser), and an untreated check. Two application methods were employed: soil and stem spray, and injection into the main stem. Infection of SVB larvae in the main stem of squash plants ranged from 19–61% in 1997, 9–27% in 1998, and 0–2% in 1999 trials. In the second trial of 1997, the injection treatment did not differ from the untreated check through the entire trial. However, the soil and stem sprays of both nematode species provided control equal to that of endosulfan. In the first trial of 1998, both *S. carpocapsae* and *S. feltiae* were equal to endosulfan in minimizing damage. As in 1997, the injection method did not perform well. In the first trial of 1999, *S. carpocapsae* and *S. riobrave* were not statistically different from the untreated check. In second trial, *S. riobrave* differed significantly from the untreated check and provided the same protection as endosulfan. Steinernematid nematodes thus may be promising biological control agents against SVB.

KEY WORDS Squash vine borer, *Melittia cucurbitae*, biological control, entomopathogenic nematodes, Steinernematidae, Heterorhabditidae

The squash vine borer (SVB), *Melittia cucurbitae* (Harris) (Lepidoptera: Sesiidae) is an important pest of Cucurbitae, especially squash and pumpkin. It causes considerable economic losses in areas east of the Rocky Mountains from

¹Accepted for publication 31 October 2006.

²Department of Entomology, Clemson University 113 Long Hall Box 340365 Clemson, South Carolina 29634-0365

Canada to South America (Klun et al. 1990, Jackson et al. 2005). Borers enter and feed in the stems of vines and greenish-yellow sawdust-type excrement is pushed from the hole onto the surface of the stem. The pest can cause as much as 25% crop loss in commercial squash, and home plantings usually suffer greater losses than commercial growing (Sorensen 1994, Jasinski & Welty 2005).

The main control method for SVB involves the application of chemical insecticides; weekly sprays are needed for 2 months of the growing season. Once larvae have entered into the stem, they are protected from contact with conventional chemical insecticides by the frass, which plugs the feeding tunnels. Increasing concern for human safety, effects on nontarget organisms, residues on crops, insect resistance to insecticide, and induction of secondary pests have created a need for alternative control strategies such as biological control.

Predation and parasitism levels for eggs, larvae, and pupae of the SVB usually are quite low so few biological control agents are available for development against this pest (Sorensen & Baker 1983). Entomopathogenic nematodes are one group that might be effective. These nematodes, which have been used to control insects since the 1930s (Smart 1995), are important biological control agents of soil-inhabiting insects such as the Japanese beetle, mole crickets, and root weevils (Gaugler 1981, Georgis & Poinar 1984, Klein 1990). They also have been used successfully against above-ground insects in cryptic habitats, i.e., navel orange worm, the codling moth and the artichoke plume moth (Begley 1990, Kaya 1990). Nematodes have shown promise for control of boring insects such as the currant borer, *Synanthedon tipuliformis* (Clerk) (Lepidoptera: Sesiidae) (Bedding & Miller 1981, Ralph 1981, Vreditelyami et al. 1992), the crown borer, *Pennisetia marginata* (Harris) (Lepidoptera: Sesiidae) (Capinera et al. 1986) and the clearwing moth, *S. culiciformis* (Lepidoptera: Sesiidae) on alder trees (Kaya 1988).

No published information is available on the use of entomopathogenic nematodes against SVB, except the results of a field efficacy trial using foliar sprays of *Steinernema carpocapsae* (Boucher & Nixon 1995). Our study is the first report to compare different species and strains of entomopathogenic nematodes against SVB in the laboratory and field. Our objectives were (1) to assess selected steinernematid and heterorhabditid nematodes against SVB in the laboratory to serve as a basis for their potential use in biological control programs and (2) to evaluate the field efficacy of the promising ones against this pest.

Materials and Methods

Laboratory bioassays. *Steinernema carpocapsae* All, *S. carpocapsae* Sal, *S. feltiae* SN, *Heterorhabditis bacteriophora* Hb, and *H. sp.* Hbl strain were obtained from Integrated BioControl Systems, Inc. (Aurora, Indiana) and *S. riobrave* TX strain was provided by Thermo Trilogy Corporation (Columbia, Maryland). The method described by Woodring & Kaya (1988) was used to culture the nematodes in late stage larvae of *Galleria mellonella* L. (Lepidoptera: Pyralidae). A modified White Trap (White 1929), consisting of a folded 11-cm filter paper (3 mm in depth after folding) in a Petri dish (100 × 15 mm) with 15–20 mL of distilled water, was used to collect the infective juveniles (IJs) that emerged after 7–10 d. These IJs were stored at 8–10°C in tissue culture flasks for 20–30 d.

A modified method from Howe (1950) was used to maintain a laboratory colony of SVB. The colony was started with eggs and larvae collected from field grown

squash. The eggs were placed on small cubes of squash fruit in 30-mL (1-oz.) plastic Dixie cups. Upon hatching, the larvae bored into the cubes, and were allowed to feed 3–4 d. Then, the cubes containing young larvae and field-collected larger larvae were placed into larger squash cubes in the same cups. If the squash cubes were nearly consumed or seriously decayed, the larvae were transferred to fresh squash cubes. When larvae were fully grown, some sand [2.5 cm (1 inch) in depth] was placed in the bottom of the container before pupation occurred. Upon maturity, the larvae entered the soil and constructed the normal silken cocoons. Newly-emerged moths were transferred to 61 × 61 × 61 cm aluminum frame rearing cages (2–5 adults/cage) and squash plants were placed in the cages for oviposition sites.

Bioassays were conducted by a method modified from Miller (1989). Unautoclaved sand (30 g) was placed in 30-ml plastic Dixie cups and moistened with 2 ml of distilled water. Nematode dosages (50, 100, and 200 IJs / cup [= 0.31, 0.62, and 1.24 billion/hectare] in 1 mL of water) were applied to the sand. One third- to fourth-stage larva was placed in each cup. The cups were incubated at $25 \pm 2^\circ\text{C}$ and a photoperiod of 16:8 (L:D) h. Four replicates of 5 cups each were tested per strain and, as untreated controls, 4 replicates were tested without nematodes. Larval mortality was recorded every 24 h for 4 d. All data were analyzed using the analysis of variance procedure and the treatment means were compared using Fisher's LSD comparison test (SAS Institute 1997).

Field efficacy trials. *Plot design.* Yellow crookneck squash, *Cucurbita pepo* L. 'Dixie,' was seeded directly in the field at the beginning of May and June of 1997 and 1998 and at the beginning of May in 1999 at the Cherry Farm, Clemson University, Clemson, South Carolina. Plots consisted of two 6.10-m (20 ft) rows, 1.52-m (5 ft) apart with plants spaced 1.22-m (4 ft) apart. Plots were arranged in a randomized complete block design with four replicates in each test.

Application methods. The promising nematodes from laboratory bioassays, *S. carpocapsae* All strain, *S. feltiae* SN strain, *S. riobrave* TX strain, and *H. sp.* Hbl strain were applied as a soil and stem spray and also were injected into the main stem of squash plants. The viability of the nematodes was checked by examining under a binocular microscope before applications. Nematodes were kept in a cooler at 10–15°C during transportation and application. For the soil and stem spray, suspensions of IJs were sprayed on the soil surface around the crown and on the base of the main stem with a CO₂-powered backpack sprayer using a TX 10 hollow cone tip on a one nozzle boom; the sprayer delivered 42.9 l (11.33 gal) finished spray/acre at 4.22 kg/cm² (60 psi). Squash plots were watered just before application and every other day to provide enough moisture for survival of the nematodes. For the injection method, IJs of *S. carpocapsae* were injected by a 1–10-mL plastic syringe into the main stem of squash plants at 1–4 different places at the rate of 14,000–234,000 IJs/plant, depending on plant size (Table 1).

Nematodes were applied in the late afternoon to reduce adverse effects of high temperatures, especially in the case of soil applications. Humidity and temperature at application times ranged between 36–68.5% and 26–29.1°C in the 1997 trials, 31.4–67.4% and 27.5–31.4°C in the 1998 trials, and 38.3–84.6% and 27.8–30.8°C in the 1999 trials, respectively.

Table 1. Average aliquot injected in to squash stems and number of injections made based on stem diameter.

Diameter of main stem (mm)	Aliquot injected (mL)	No. of injections
<7	0.7	1
10	1.4	2
15	4.6	2
20	7.6	3
25	10.0	3
>30	11.7	4

Evaluation methods. Percent infection of SVB larvae (%) by nematodes was assessed by counting the dead infected and live larvae in the main stem at the end of each trial. Dead infected larvae were put on the White Traps and the water in the petri dish was checked for IJs during a 10- to 12-d period to confirm nematode infection.

To evaluate damage by SVB in 1997, the holes made by SVB on the main stem were counted ca. every 5 days after each application. In 1998 and 1999, a different damage rating system was used. After each application, the total area damaged was measured in mm² for each main stem. This gave a better estimate of the degree of damage because holes would merge as feeding by SVB progressed. In 1999, in addition to measuring the damaged area, the main stems were split and live larvae in each main stem were counted.

Statistical analysis. The SAS general linear model procedure was used to evaluate data converted by square root transformation (SAS Institute 1997). Differences ($P = 0.05$) among means were analyzed using the analysis of variance procedure (ANOVA), and the Fisher's Protected Least Significant Difference test was used for all pairwise multiple comparison procedures (SAS Institute 1997).

1997 trials. Three trials were conducted in 1997. In the first trial, *Steinernema carpocapsae* and *Heterorhabditis* sp. were applied as stem and soil sprays at the rate of 2.5 billion IJs/hectare. Only *S. carpocapsae* was injected. Also included were an insecticide standard, endosulfan (Phaser), and an untreated check. Two applications were made on 26 June and 9 July, and damage evaluated on 1, 8, and 14 July. In the second trial, the same treatments were applied three times (31 July, 11 and 23 August), and damage was evaluated on 5, 9, 16, 22, and 27 August. In the third trial, the only treatment was a soil and stem spray of *S. feltiae* applied once (26 September). This was compared with an untreated check. The only evaluation method used in this trial was the percent infestation determination.

1998 trials. In the first 1998 trial, *S. carpocapsae* and *S. feltiae* were applied as soil and stem sprays at the rate of 5 billion IJs/hectare. Only *S. carpocapsae* was injected. Three treatments were made on 15, 25 June and 6 July, and five evaluations were made on 19, 24, and 30 June, and 5 and 10 July. In the second trial, *S. feltiae* and *S. riobrave* were applied three times (July 24, August 4 and 14) as soil and stem sprays at the rate of 5 billion IJs/hectare. No injection method was used in this trial. Damage evaluations were made on 28 July, 3, 8, 12, and 18 August. An insecticide standard and untreated check were included in both 1998

Table 2. Percent mortality of third to fourth stage SVB larvae exposed to various species and strains of entomopathogenic nematodes at 1, 2, 3, and 4 d after inoculation (laboratory bioassay at 25 ± 2°C), Clemson, South Carolina, 1997.

Nematodes ^a	50 infective juveniles/larva ^b				100 infective juveniles/larva ^b				200 infective juveniles/larva ^b			
	1 st d	2 nd d	3 rd d	4 th d	1 st d	2 nd d	3 rd d	4 th d	1 st d	2 nd d	3 rd d	4 th d
<i>S. riobrave</i>	0	50 a	70 a	70 a	0	80 a	85 a	85 a	0	80 a	95 a	95 a
<i>S.f.</i> SN st.	0	35 ab	50 ab	60 ab	10	40 b	55 b	65 ab	15	60 ab	80 a	85 a
<i>S.c.</i> All st.	0	30 ab	40 bc	55 ab	10	35 b	65 ab	65 ab	0	50 bc	90 a	90 a
<i>S.c.</i> Sal st.	0	15 bc	40 bc	40 bc	0	20 bc	50 b	60 b	0	30 dc	70 ab	80 a
<i>H.b.</i> Hb st.	0	0 c	15 cd	15 cd	0	10 c	20 c	3 c	0	20 d	50 b	50 b
<i>H.sp.</i> Hbl st.	0	0 c	0 d	0 d	0	0 c	0 c	0 d	0	10 d	20 c	30 b

^a*S. riobrave* = *Steinernema riobrave* TX strain, *S.f.* SN st. = *Steinernema feltiae* SN strain, *S.c.* All st. = *Steinernema carpocapsae* All strain, *S.c.* Sal st. = *Steinernema carpocapsae* Sal strain, *H.b.* Hb st. = *Heterorhabditis bacteriophora* Hb strain, and *H.sp.* Hbl st. = *Heterorhabditis* sp. Hbl strain.

^bMeans within a column followed by the same letter are not significantly different at 0.05 (LSD).

trials. After the second application in the 1st trial in 1998, soil samples were taken from soil and stem application plots of *S. carpocapsae* and *S. feltiae* daily for 10 days. Five late-stage *Galleria mellonella* (L.) larvae were buried in these soil samples in a 500-mL plastic container to check survivability of nematodes in the soil (Bedding & Akhurst 1975).

1999 trials. In the first trial, *S. feltiae* and *S. riobrave* were applied three times (3, 16, and 27 June) as soil and stem sprays at the rate of 5 billion IJs/hectare. Damage was evaluated on 7, 11, 20, 27 June, and 2 July. In the second trial, only *S. riobrave* was applied at the same rate on July 30, August 8 and 18. Damage was evaluated on 4, 10, 15, and 22 August. Insecticide standard and untreated checks were included in both trials.

Results and Discussion

Laboratory bioassays. There was no mortality in untreated controls. All dead larvae in the treatments showed typical symptoms of nematode infection. Very low mortality occurred during the first day after treatment. However, after 2 d, enough mortality had occurred that comparisons could be made. In general, mortality rates increased from day 2 to day 4 (Table 2).

The best-performing nematode overall was *S. riobrave* TX strain. It caused earlier and higher mortality than most of the other strains. For example, at 100 IJs/larva, it caused significantly higher ($P < 0.05$) mortality (80%) after 2 days than any other nematode strain ($df = 5$; $F = 12.32$; $P = 0.001$) (Table 2).

The next-best-performing strains after the *S. riobrave* TX strain were *S. feltiae* SN strain and *S. carpocapsae* All strain. They produced significantly lower ($P < 0.05$) mortality than *S. riobrave* in only 2 of 9 and 3 of 9 data sets, respectively. For instance, *S. feltiae* differed from *S. riobrave* at only the 100 IJs/larva rate (2 and 3 d readings) ($df = 5$; $F = 12.32$; $P = 0.001$, $df = 5$; $F = 11.26$; $P = 0.001$).

Steinernema carpocapsae Sal strain was less effective than the other species and strains of *Steinernema*. It differed from *S. riobrave* in 7 of the 9 data sets. Both strains of heterorhabditid nematodes were poor performers against SVB. They both differed from *S. riobrave* in all nine data sets.

Neither heterorhabditid strain was effective against SVB. The steinernematid nematodes, *S. riobrave* TX strain, *S. feltiae* SN strain, and *S. carpocapsae* All strain appeared promising for SVB control in laboratory bioassays.

In the literature, there was no record related to efficacy of entomopathogenic nematodes against SVB in laboratory. Williams et al. (2002) found that contrary to our results, heterorhabditid nematodes were more effective than steinernematids against one of the boring insects, grape root borer *Vitacea polistiformis* (Lepidoptera: Sesiidae) in laboratory and greenhouse bioassays. *Heterorhabditis bacteriophora* (GPS11 strain) produced the highest rate of infection at 92% followed by *H. zealandica*, which had an infection rate of 86% whereas *S. carpocapsae* produced an infection rate of 69%, with the remaining *Steinernema* species producing zero to low mortality.

Field efficacy trials. **1997 trials.** In the first trial, there was no statistical difference among treatments attributable to low population density and high variance between replications except for the second count after the first application, in which only endosulfan differed from the untreated check ($df = 4$; $F = 2.28$; $P = 0.121$) (Table 3). Larval infection rates ranged from 19% to 40%,

Table 3. Damage by SVB in plots treated with nematodes, Clemson, South Carolina, 1997.

Treatments ^b	Mean no. of holes/plant									
	First trial ^a				Second trial ^a				Third trial	
	1 st appl.		2 nd appl.		1 st appl.		2 nd appl.		1 st appl.	
	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count
	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count
Sc injection	0.12 a	0.37 ab	0.25 a	0.12 ab	0.02 a	0.77 ab	0.159 a	2.40 ab		
Sc soil appl.	0.04 a	0.37 ab	0.09 a	0.01 b	0.00 a	0.18 c	0.74 b	1.64 c		
H. sp. soil appl.	0.18 a	0.86 a	0.67 a	0.01 b	0.01 a	0.31 bc	0.69 b	1.93 bc		
Endosulfan	0.00 a	0.00 b	0.00 a	0.02 b	0.01 a	0.17 c	0.31 b	0.85 d		
Untreated	0.28 a	0.98 a	0.66 a	0.64 a	0.01 a	1.19 a	1.80 a	3.10 a		

^aMeans in a column followed by the same letter are not significantly different ($P = 0.05$, LSD).

^bSc injection = *Steinernema carpocapsae* injection, Sc soil appl. = *Steinernema carpocapsae* soil application, and H. sp. soil appl. = *Heterorhabditis* sp. soil application.

Table 4. Infection of SVB larvae (%) in the main stem of squash treated with entomopathogenic nematodes in field trials during 1997, 1998, and 1999, Clemson, South Carolina.

Treatments*	1997			1998		1999	
	1 st trial	2 nd trial	3 rd trial	1 st trial	2 nd trial	1 st trial	2 nd trial
Sc soil	41.0 a	11.7 a	—	9.6 a	—	3.2 a	—
Sc inj.	39.7 a	13.0 a	—	27.0 a	—	—	—
Sf soil	—	—	60.8 a	9.0 a	24.0 a	—	—
Sr soil	—	—	—	—	0.0 b	2.0 a	1.0 a
H.sp. soil	19.2 a	1.2 b	—	—	—	—	—
Untreated	0.0 b	0.0 b	0.0 b	0.0 b	0.0 b	0.0 a	0.0 a

*Sc soil = *Steinernema carpocapsae* soil application, Sc inj. = *Steinernema carpocapsae* injection, Sf soil = *S. feltiae* soil application, Sr soil = *S. riobrave* soil application, and H. sp. soil = *Heterorhabditis* sp. soil application.

confirming that nematodes were causing mortality in the larval population (Table 4).

In the second trial, pest pressure was higher and all treatments, except the injection treatment, differed significantly from the check by the second count after the first application ($df = 4$; $F = 4.10$; $P = 0.025$) (Table 3). The injection treatment did not differ from the check through the entire trial. However, the soil and stem sprays of both nematode species provided control equivalent to that of endosulfan except in the final count after the third application. Nematode infection rates were somewhat lower in the second trial, and infection by *Heterorhabditis* was significantly lower than for *S. carpocapsae* ($df = 3$; $F = 24.17$; $P = 0.0001$) (Table 4). In the third trial, the infection rate for *S. feltiae* (61%) was the highest in any of the trials (Table 4).

1998 trials. In the first trial, although SVB pressure was quite high at the beginning of the test, infestation levels were lower in the check plots than other treatments when treatments began. Because of this, none of the treatments differed significantly from the check until the second count after the second application (Table 5). On this date (12 August), only *S. carpocapsae* sprays and endosulfan were significantly better than the check ($df = 4$; $F = 3.99$; $P = 0.027$). By the final count after the third application, however, both *S. carpocapsae* and *S. feltiae* were equal to endosulfan in reducing damage ($df = 4$; $F = 4.82$; $P = 0.015$). As in 1997, the injection method did not perform well. Infection levels in this trial ranged from 9–27% with no significant differences between nematode treatments ($df = 3$; $F = 5.23$; $P = 0.023$) (Table 4).

In the second trial, all the treatments on all dates had statistically less damage than the untreated check. *S. riobrave* provided the same control as endosulfan throughout the test (Table 5).

Nematode activity in the soil samples taken during the first trial is presented in Table 6. Results confirm that IJs of both *S. carpocapsae* and *S. feltiae* remained active and infective in the soil throughout the 10 d period.

1999 trials. In the first trial, nematode treatments protected squash plants from SVB damage early in the test. As pest pressure increased, this protection

Table 5. Mean areas of damage per plant by SVB on the main stem of squash in plots treated with nematodes, Clemson, South Carolina, 1998.

Mean areas (mm ²) of damage per plant											
First trial ^a						Second trial ^a					
1 st appl.		2 nd appl.		3 rd appl.		1 st appl.		2 nd appl.		3 rd appl.	
1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count	1 st count	2 nd count
Treatments											
Sc injection	19.1 ab	70.9 a	85.7 a	101.6 ab	319.7 ab	—	—	—	—	—	—
Sc soil appl.	3.2 b	26.0 a	45.8 a	65.0 bc	149.3 b	—	—	—	—	—	—
Sf soil appl.	2.7 b	42.6 a	70.2 a	84.5 ab	108.2 b	15.6 b	75.2 b	182.0 b	486.6 b	912.0 b	912.0 b
Sr soil appl.	—	—	—	—	—	14.7 b	38.6 b	129.7 b	198.8 bc	490.2 bc	490.2 bc
Endosulfan	18.8 ab	33.3 a	26.5 a	25.8 c	95.8 b	6.9 b	54.6 b	68.9 b	70.4 c	192.7 c	192.7 c
Untreated	1.1 b	80.1 a	103.2 a	153.0 a	602.7 a	50.0 a	453.7 a	858.5 a	1007.4 a	2775.2 a	2775.2 a

^aMeans in a column followed by the same letter are not significantly different ($P = 0.05$, LSD).

^bSc injection = *Steinernema carpocapsae* injection, Sc soil appl. = *Steinernema carpocapsae* soil application, Sf soil appl. = *Steinernema feltiae* soil application, and Sr soil appl. = *Steinernema riobrave* soil application.

Table 6. Mortality of wax moth larvae exposed to soil samples taken from nematode-treated plots after the second application of the first 1998 trial, Clemson. South Carolina.

	Days after treatment ^a																			
	1		2		3		4		5		6		7		8		9		10	
	d	l	d	l	d	l	d	l	d	l	d	l	d	l	d	l	d	l	d	l
Nematodes																				
<i>S. carpocapsae</i>	4	1	1	4	4	1	2	3	1	4	0	5	3	2	0	5	1	4	2	3
<i>S. feltiae</i>	4	1	1	4	4	1	0	5	1	4	2	3	2	3	2	3	0	5	1	4

^ad = dead larva caused by nematodes, l = live larva.

broke down, and by end of the test, only endosulfan differed significantly from the untreated check ($df = 3$; $F = 5.80$; $P = 0.017$) (Table 7). Larval counts at the end of this trial, however, showed that nematode treatments significantly reduced larval numbers ($df = 3$; $F = 5.67$; $P = 0.018$). Infection levels were quite low in this trial (Table 4), suggesting that test conditions were not favorable for survival of nematodes.

In the second trial, no significant differences were apparent among treatments until the final count after the third application; at that time, *S. riobrave* treatment differed significantly from the check and provided the same protection as endosulfan ($df = 2$; $F = 2.64$; $P = 0.031$) (Table 7). Larval numbers, however, did not differ significantly from the check ($df = 2$; $F = 2.52$; $P = 0.16$) (Table 7), and again, the infection rate was quite low (Table 4).

Boucher & Nixon (1995) indicated that foliar sprays of *S. carpocapsae* were not effective in reducing SVB damage. On the other hand, Kuepper & Thomas (2001) stated that on a small scale, gardeners sometimes could choose to inject liquid suspensions of predatory nematodes into the stem bases of squash plants using a syringe against SVB. Successful results were also obtained for control of boring insects by nematodes. Bedding & Miller (1981) found that spraying of *S. feltiae* (Filipjev) eliminated the currant borer, *S. tipuliformis* (Clerk) from black currant cuttings. Suspensions of *S. feltiae* applied at a rate of 60,000 infective juveniles (IJs) per plant to the soil at the base of raspberry clumps gave good control of the crown borer, *P. marginata* (Harris) (Capinera et al. 1986). Bark sprays of *S. carpocapsae* (Weiser) caused 77–84% mortality of the clearwing moth, *S. culiciformis* on alder trees (Kaya 1988).

In conclusion, stem and soil application of steinernematid nematodes, *S. carpocapsae* All strain, *S. feltiae* SN strain, and *S. riobrave* TX strain were successful in reducing SVB damage. The use of these nematodes especially in organic squash production will be a valuable alternative to chemical insecticides. Further tests need to be conducted to determine optimum timing and rates for this pest.

Acknowledgments

We thank Professor Randall Griffin for sharing his unique field experience and use of his research materials, Dr. David Alverson for use of his laboratory, Jimmy Pilgrim for soil

Table 7. Mean areas of damage per plant on the main stem of squash by SVB and mean no. larvae of SVB in the main stem of squash in plots treated with nematodes, Clemson, South Carolina, 1999.

Mean areas of damage (mm ² /plant) ^a												
First trial						Second trial						Mean no. larvae ^c
1 st appl.			2 nd appl.			1 st appl.			2 nd appl.			1 st trial
1 st count	2 nd count		1 st count	2 nd count		1 st count	2 nd count		1 st count	2 nd count		3 rd appl.
Trts ^b	1 st count	2 nd count	1 st count	2 nd count	3 rd appl.	1 st count	2 nd count	3 rd appl.	1 st count	2 nd count	3 rd appl.	3 rd appl.
Scs	0.0 b	0.0 b	1.9 bc	17.3 ab	40.6 a	—	—	—	—	—	1.0 b	—
Srs	0.0 b	0.0 b	3.1 ab	29.4 a	24.0 ab	1.2 a	13.2 a	10.1 a	13.8 a	110.3 bc	1.2 b	0.3 a
End	0.0 b	0.0 b	0.0 c	0.0 b	0.0 b	0.8 a	11.3 a	14.1 a	14.4 a	87.5 c	0.0 b	0.0 a
Untr	1.8 a	3.3 a	9.5 a	56.6 a	109.8 a	3.6 a	14.2 a	16.7 a	47.6 a	317.6 a	7.1 a	0.5 a

^aMeans in a column followed by the same letter are not significantly different ($P = 0.05$, LSD).
^bTrts = Treatments, Scs = *Steinernema carpocapsae* soil application, Srs = *Steinernema riobrave* soil application, End = Endosufan, and Untr = Untreated.

preparation in the squash plots, Yayi Kusumah for helping plant squash plots, Dr. David Shapiro of Integrated BioControl Systems and Ms. Peggy Mandanas of Termo TRilogy for providing us entomopathogenic nematodes, and Dr. James Rieck for advice on statistical analysis of our data. This is technical contribution number 4621 of the South Carolina Agricultural Experiment Station.

References Cited

- Bedding, R. A. & R. J. Akhurst. 1975.** A simple technique for the detection of insect parasitic rhabditid nematodes in soil. *Nematologica* 21: 109–116.
- Bedding, R. A. & L. A. Miller. 1981.** Disinfesting black currant cuttings of *Synanthedon tipuliformis*, using the insect parasitic nematode, *Neoapectana bibionis*. *J. Entomol. Sci.* 10: 449–453.
- Begley, J. W. 1990.** Efficacy against insects in habitats other than soil, pp. 215–231. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic Nematodes in Biological Control*. CRC, Boca Raton, Florida.
- Boucher, J. & G. Nixon. 1995.** Evaluation of alternative insecticides for squash vine borer on summer squash. *Arthropod Management Tests* 21: 175.
- Capinera, J. L., W. Cranshaw & H. G. Huges. 1986.** Suppression of raspberry crown borer, *Pennisetia marginata* (Harris) (Lepidoptera: Sesiidae) with soil application of *Steinernema feltiae* (Rhabditida: Steinernematidae). *J. Invertebr. Pathol.* 48:257–258.
- Gaugler, R. 1981.** Biological control potential of neoapectanid nematodes. *J. Nematol.* 13:241–249.
- Georgis, R. & G. O. Poinar. 1984.** Field control of the strawberry root weevil, *Nemocestes incomptus*, by neoapectanid nematodes (Steinernematidae: Nematoda). *J. Invertebr. Pathol.* 43:103–131.
- Howe, W. L. 1950.** Biology and host relationship of squash vine borer. *J. Econ. Entomol.* 43:480–483.
- Jackson, D. M., R. Canhilal & G. R. Carner. 2005.** Trap monitoring squash vine borers in cucurbits. *J. Agric. Urban Entomol.* 22:27–39.
- Jasinski, J. & C. Welty. 2005.** Squash bugs and squash vine borer on the rise. Ohio State University Extension Vegetable Crops. VegNet Vol. 12, No. 14. <http://www.ag.ohio-state.edu/~vegnet/news/vnew1405.htm>. Accessed December 19, 2006.
- Kaya, H. K. 1988.** Princes from todes. *Am. Nurseryman*. 168:63, 65–69.
- Kaya, H. K. 1990.** Soil ecology, pp. 93–115. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic nematodes in biological control*. CRC, Boca Raton, Florida.
- Klein, M. G. 1990.** Efficacy against soil-inhabiting insect pests, pp. 195–214. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic nematodes in biological control*. CRC, Boca Raton, Florida.
- Klun, J. A., M. Schwarz, B. A. Leonardt & W. W. Cantelo. 1990.** Sex pheromone of the female squash vine borer. *J. Entomol. Sci.* 25: 64–72.
- Kuepper, G. & R. Thomas. 2001.** Organic control of squash vine borer. National Sustainable Agriculture Information Service. <http://www.attra.org/attra-pub/squashbore.html>. Accessed December 19, 2006.
- Miller, R. W. 1989.** Novel pathogenicity assessment technique for *Steinernema* and *Heterorhabditis* entomopathogenic nematodes. *J. Nematol.* 21: 574.
- Ralph, W. 1981.** Nematodes for insect control. *Rural Research* (No. 113): 11–14.
- SAS Institute. 1997.** SAS/STAT software, release 6.12. SAS Institute, Cary, North Carolina.
- Smart Jr., G. C. 1995.** Entomopathogenic nematodes for the biological control of insects. *J. Nematol.* 27(4 Suppl):529–534.
- Sorensen, K. A. 1994.** The Squash Vine Borer. North Carolina State University Extension Factsheet. www.ces.ncsu.edu/depts/ent/notes/Vegetables/veg20.html. Accessed December 19, 2006.

- Sorensen, K. A. & J. R. Baker. 1983.** Insect and Related Pests of Vegetables: Some Important, Common, and Potential Pests in the Southeastern United States. Pub. AG-295. North Carolina State University, Raleigh, North Carolina 173 p.
- Vreditelyami, S. V., V. P. C. Smorodiny & A. S. Zainalov. 1992.** Attempts to increase efficiency of control measures against the principle pests in black currant nurseries, pp. 17–20. Thesis Russian Academy of Agricultural Sciences, Moscow, Russia.
- White, G. F. 1929.** A method for obtaining infective nematode larvae from cultures. Science 66: 302–303.
- Williams, R. N., D. S. Fickle, P. S. Grewal & J. R. Meyer. 2002.** Assessing the potential of entomopathogenic nematodes to control the grape root borer *Vitacea polistiformis* (Lepidoptera: Sesiidae) through laboratory and greenhouse bioassays. Biocontrol Sci.Technol. 12: 35–42.
- Woodring, J. L. & H. K. Kaya. 1988.** Steinernematid and Heterorhabditid Nematodes: A handbook of Biology and Techniques. South Coop. Ser. Bull. 331, Arkansas Agric. Exp. Stat., Fayetteville, Arkansas.
-

Efficacy of Steinernematid Nematodes (Rhabditida: Steinernematidae) on the Suppression of *Anastrepha ludens* (Diptera: Tephritidae) Larvae in Soil of Differing Textures: Laboratory and Field Trials¹

Roberto Lezama-Gutiérrez,² Jaime Molina-Ochoa, Alfonso Pescador-Rubio,³
Edelmira Galindo-Velasco,⁴ César Andrés Ángel-Sahagún,⁴
Alejandro C. Michel-Aceves,⁵ and Ezequiel González-Reyes⁶

Facultad de Ciencias Biológicas y Agropecuarias, Universidad de Colima. Apartado Postal No. 36,
Tecomán, Colima C. P. 28100, México

J. Agric. Urban Entomol. 23(1): 41–49 (Jan 2006)

ABSTRACT The efficacy of the entomopathogenic nematodes, *Steinernema carpocapsae* (Weiser 1955) All strain and *S. riobrave* (Cabanillas, Poinar & Raulston), against last instar of *Anastrepha ludens* (Loew) was determined in different soil types. Larvae were exposed to entomopathogenic nematodes in sandy loam, loam, and clay texture soils. Higher rates of larval mortality were observed in sandy loam and loam than in clay. *S. carpocapsae* showed LC₅₀ values of 35.15 and 18.32 infective juveniles (IJs)/cm² in sandy loam and loam soils, respectively. The nematode *S. riobrave* showed LC₅₀ values of 14.55 and 10.31 IJs/ cm² in sandy loam and loam, respectively. In field conditions of sandy loam soil, *S. carpocapsae* reduced adult emergence by 64%, whereas *S. riobrave* reduced adult emergence by only 14% compared with the control using the higher IJ concentration. These results suggest that soil type is a critical factor that must be taken into account when selecting the nematode species and planning fruit fly biological control strategies.

KEY WORDS Entomopathogenic nematodes, Mexican fruit fly, biological control, mango, Mexico, *Steinernema carpocapsae*, *Steinernema riobrave*.

Fruit flies of the genus *Anastrepha* (Schiner) (Diptera: Tephritidae) comprise about 180 species (Norrbom & Kim 1988). Several species significantly affect commercially grown fruit in the American tropics and subtropics (Steck et al. 1990). The Mexican fruit fly, *A. ludens* (Loew), is one of seven species that cause the greatest economic damage (Aluja 1994). This species, as well as *A. obliqua*

¹Accepted for publication 16 November 2006.

²Corresponding author: Roberto Lezama-Gutiérrez, Facultad de Ciencias Biológicas y Agropecuarias Universidad de Colima, Apartado Postal No. 36, Tecomán, Colima, C. P. 28100, México. E-mail: rlezama@ucol.mx, jmolina@ucol.mx

³Centro Universitario de Investigación y Desarrollo Agropecuario. Apartado Postal No. 36, Tecomán, Colima C. P. 28100, México.

⁴Facultad de Medicina Veterinaria y Zootecnia, Universidad de Colima. Apartado Postal No. 36, Tecomán, Colima C. P. 28100, México.

⁵Colegio Superior Agropecuario del Estado de Guerrero (CSAEGRO). Av. Guerrero No. 81, 1^{er} piso, C.P. 40000, Iguala, Guerrero, México. E-Mail: amichelaceves@yahoo.com.mx

⁶Facultad de Biología. Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacán, México.

(Macquart), account for 96.6% of all captured adults in commercial mango orchards in the Soconusco Region, Chiapas, Mexico (Aluja et al. 1996). Because mango production plays an important role in the economy of many tropical countries, including Mexico as the world's leading mango exporter (Zahler 1990), it is important to improve current control methods.

Chemical control is known to be an effective strategy for controlling fruit fly populations (Aluja 1994); however, the side effects on nontarget organisms, groundwater contamination, residues on food crops, and the development of insect resistance to chemicals have forced the industry to focus on the development of alternative control measures.

Alternative control measures, including entomopathogenic nematodes such as *Steinernema* species, have been recommended (Gaugler 1981, Georgis 1992). Poinar & Hislop (1981) first suggested their use as a suppression method for fruit flies. Entomopathogenic nematodes in the Steinernematidae and Heterorhabditidae have shown promise for the biological control of fruit flies. Third instar *A. ludens* are susceptible to infection by infective juveniles (IJs) during the period when they exit the host fruit and burrow into the soil for pupation. Once the infested fruit is on the ground for 3 days, the larvae leave the host fruit over a 2-day period.

Lindegren & Vail (1986) evaluated the susceptibility of the Mediterranean fruit fly, melon fly, and the Oriental fruit fly to the entomopathogenic nematode *Steinernema feltiae* (Filipjev) in laboratory tests. In addition, Lindegren et al. (1990) conducted field experiments to evaluate the susceptibility of *Ceratitis capitata* (Weidemann) to *S. feltiae* concluding that this nematode was an alternative for a biological control program. Beavers & Calkins (1984) evaluated the susceptibility of *A. suspensa* (Loew) to steinernematid and heterorhabditid nematodes under laboratory conditions, and found that larvae and adults were highly susceptible.

Several reports indicate that the soil environment affects the efficacy of entomopathogenic nematodes in suppressing pests inhabiting soil (Gaugler 1988; Choo & Kaya 1991). Moreover, the application of nematodes has produced discouraging or inconsistent results in field conditions because of a poor understanding of nematode virulence in different soil types (Gaugler 1988). Therefore, knowledge on the behavior of a specific nematode species in the soil environment is essential to developing an effective biological control program (Kaya 1990). Overall, entomopathogenic nematodes are more infective in sandy and sandy loam soils than in clay soils (Georgis & Poinar 1983a, 1983b, Molyneux & Bedding 1984, Kung et al. 1990, Barbercheck & Kaya 1991); however, at present, no research has been done on the evaluation of entomopathogenic nematode species against *A. ludens* larvae in different soil types under laboratory and field conditions.

The objective of our study was to evaluate the mortality of *A. ludens* larvae caused by the entomopathogenic nematodes *S. carpocapsae* All strain and *S. riobrave* in three different soil types and their subsequent effectiveness in reducing the emergence of adults in field conditions.

Materials and Methods

Three soil types, sandy loam, loam, and clay, were collected from mango orchards from the State of Colima, Mexico. Each soil type was air dried and sieved

through a 2 mm-pore sieve (Molyneux & Bedding 1984, Kung et al. 1991). The pH and organic matter content (%) of the soils were 7.1 and 1.2% for sandy loam soil (clay 4.84%, silt 19.28%, and sand 75.88%), 7.01 and 2.3% for loamy soil (clay 16.84%, silt 34.92%, sand 48.24%), and pH 7.6 and 1.4% for clay soil (clay 73.40%, silt 19.28%, and 7.32%).

The *A. ludens* larvae were supplied by the Mass Rearing Laboratory of the Moscamed Program (SAGARPA-USDA) from Metapa de Domínguez, Chiapas, México. The entomopathogenic nematode *S. carpocapsae* (All strain) was obtained through the courtesy of Dr. Harry K. Kaya, Department of Nematology and Entomology, University of California, Davis, USA, and *S. riobrave* was supplied through Dr. Grover C. Smart, Entomology and Nematology Department, University of Florida, Gainesville, USA. IJs of both nematodes were produced in the last-instar of the greater wax moth, *Galleria mellonella* (L.) (Lepidoptera: Pyralidae), following the procedures of Dutky et al. (1964) and Kaya & Stock (1997). Stock suspensions of IJs were stored in 0.1% formaldehyde solution at 5°C for 2 weeks before the tests were conducted (Kung et al. 1990).

Before the assays, viability was confirmed by observing IJ activity (rapid wiggling) in a five-drop (eye dropper) aliquot from the liquid suspension (Buhler & Gibb 1994). In addition, IJ infectivity was confirmed by placing 100 IJs in a Petri dish (100 × 15 mm) lined with a moistened 9-cm no. 1 Whatman filter paper; ten *G. mellonella* larvae were then added to each dish. This was replicated four times for each nematode species (Forschler & Gardner 1991).

Laboratory trials. Twenty four 1000-ml plastic cups, 15-cm in diameter, containing 300-g of sterilized soil were prepared from each soil type (Lindgren & Vail 1986). The soil in each cup was moistened with sterile distilled water after equilibration for 4–5 days at 25°C; then, a field capacity was attained adding 31.8, 63.0, and 35.3 mL of sterile distilled water to the sand, clay, and loam soil types, respectively. Groups of four cups were used for each variable to be studied; each cup was inoculated with 1 mL of an IJ suspension containing 0, 1000, 2000, 3000, 4000, or 5000 IJs/mL applied on the soil surface, representing concentrations of 0, 5.66, 11.32, 16.98, 22.64, or 28.30 IJs/cm². After IJ application, 100 third instars of *A. ludens* were placed on the soil surface. Each cup was covered with its plastic cover provided with 40 ventilation holes and incubated in the dark at 25°C under laboratory conditions.

A factorial design with 36 treatments was established; factor A had two levels representing two IJ species; factor B had three levels representing three soil types, and factor C had five levels representing five nematode concentrations and water-treated controls. Each treatment was replicated four times.

To confirm nematode infection, 6, 12, and 24 d after the initial larval exposure to IJs, dead larvae and pupae were collected by hand from the surface sieved and individually placed on a moistened No. 1 Whatman filter paper in a Petri dish. Larvae and pupae were incubated until emergence of IJs or dissected to verify nematode parasitism. The remaining pupae were held until adult emergence or dissected to verify the absence or presence of nematodes. Pupae without nematodes were considered as not infected.

Cumulative mortality percent data were transformed by the arcsine transformation before the analysis of variance (Sokal & Rohlf 1995), and means were separated by the Tukey's test ($P = 0.05$; SAS Institute 1997). LC_{50} values were calculated for each nematode species and soil type; values were not considered to

be different if the 95% fiducial limits (FL) overlapped (Finney 1971). Percentage of mortality was adjusted with Abbott's (1925) formula. Because the mortality percentages in the clay type were lower than 15%, the LC_{50} values were not calculated.

Field trials. Six mini plots ($30 \times 30 \times 17$ cm) were used for each nematode species and control and placed in a mango orchard with loamy soil. The orchard was irrigated until field capacity one day before setting up the experiments. On the experimental areas the litter was removed. In each experimental area a suspension of 200 mL containing the nematodes was applied with a hand sprayer using a dose of 250 IJs/cm². 200 ml of sterile distilled water was applied to the controls. After the nematode applications, 100 third instars were placed in each area and covered with a wood cage ($30 \times 30 \times 17$ cm) with mosquito net on the top to capture emerging Mexican fruit fly adults. The borders of the cages were covered with soil to support them. Three treatments, the *S. riobrave*, *S. carpocapsae*, and the control, with 6 replicates per treatment were established. Starting with the first application, and every 48 h until adult emergence, the soil was sampled for moisture that varied from 8.65–7.53%. The number of emerged adults were counted and subjected to ANOVA, and means were separated using Tukey's test ($P = 0.05$; SAS Institute 1997).

Results

The entomopathogenic nematodes *S. carpocapsae* and *S. riobrave* were able to kill the third instars of *A. ludens* in the 3 different soil types tested in our study. The analysis of variance showed highly significant differences for nematode species ($F = 31.7$; $df = 1, 108$; $P = 0.0001$), soil type ($F = 326.3$; $df = 2, 108$; $P = 0.0001$), IJ concentrations ($F = 88.5$; $df = 5, 108$; $P = 0.0001$), interaction nematode species*soil type ($F = 14.15$; $df = 2, 108$; $P = 0.0017$), interaction soil type*concentration ($F = 14.1$; $df = 10, 108$; $P = 0.0001$); however, nonsignificant differences were found for the interaction IJ*concentration ($F = 2.1$; $df = 5, 108$; $P = 0.08$) and the interaction IJ*concentration*soil type ($F = 0.971$; $df = 10, 108$; $P = 0.47$), $n = 144$, $r^2 = 0.81$, $CV\% = 34.43$, whereas *S. riobrave* caused higher larval mortality (MSD = 3.37) than *S. carpocapsae* (Sc mean = 26.1, $n = 72$); *S. riobrave* (Sr mean = 32.6, $n = 72$). Means were those exhibited by the IJ species in all concentrations and all soil types.

Soil type had a significant effect on the ability of the nematodes to kill *A. ludens* larvae. Comparing the IJ species, *S. riobrave* caused higher rates of mortality in sandy and sandy loam soil than *S. carpocapsae* even when they exhibited low mortalities. Both IJ species exhibited similar effects on *A. ludens* larvae in clay soil, as evidenced by the significant nematode * soil type interaction.

As expected, *A. ludens* mortality increased with nematode concentration (Table 1); however, in clay the larval mortality was reduced at all *S. carpocapsae* concentrations used, ranging from 7.25% to 12.00% at 5.66 and 28.30 IJ/cm², respectively. This response was also observed with the *S. riobrave* concentrations ranging because 4.75% to 14.00% with 5.66 and 28.30 IJ/cm², respectively (Table 1).

Steinernema riobrave LC_{50} values indicated that this species is nearly 2.5 times more virulent than *S. carpocapsae* in sandy loam type, and 1.8 times more

Table 1. Mortality rates of third instar and pupae, adult emergence and LC₅₀ values estimated for *S. carpocapsae* and *S. riobrave* under different soil types and IJ concentrations.

<i>Steinernema carpocapsae</i>						<i>Steinernema riobrave</i>					
Soil type	IJ Conc.	% Mortality	% Adult emergence	LC ₅₀	95% Fiducial limits	Soil type	IJ Conc.	% Mortality	% Adult emergence	LC ₅₀	95% Fiducial limits
Sandy loam	0	0.0 c	95	35.15	23.27–53.09	Sandy loam	0	0.0 d	95	14.55	10.32–20.50
	5.66	19.00 bc	80				5.66	41.24 c	57		
	11.32	29.75 b	67				11.32	44.25 c	55		
	16.98	29.75 b	68				16.98	46.00 bc	52		
	22.64	39.5 ab	60				22.64	57.25 ab	41		
Loam	28.30	51.25 a	46			Loam	28.30	63.25a	35		
	0	0.0 c	98	18.32	11.93–28.12		0	0.0 c	89	10.31	7.21–14.75
	5.66	38.75 b	60				5.66	39.25 b	59		
	11.32	43.00 ab	55				11.32	55.75 ab	43		
	16.98	46.25 ab	52				16.98	56.25 ab	41		
Clay	22.64	50.75 ab	49			Clay	22.64	62.00 a	37		
	28.30	58.50 a	40				28.30	63.25 a	36		
	0	0.0 a	99	Not estimated	Not estimated		0	0.0 b	92	Not estimated	Not estimated
	5.66	7.25 a	85				5.66	4.75 ab	91		
	11.32	8.00 a	90				11.32	6.00 ab	89		
	16.98	10.75 a	87				16.98	7.50 ab	90		
	22.64	10.75 a	86				22.64	13.75 ab	85		
	28.30	12.00 a	85				28.30	14.00 a	80		

Means within columns followed by the same letter, between each soil type or IJ concentration are not significantly different (Tukey's test, $P = 0.05$). LC₅₀ values expressed as the number of IJ/cm².

virulent than *S. carpocapsae* in loamy type. Within species, texture differences were only evident with respect to clay soil, where mortality rates were less than 15% and their LC_{50} values were not estimated (Table 1).

Under field conditions, *S. carpocapsae* reduced adult emergence by 64.1%, whereas *S. riobrave* reduced adult emergence by only 14.0% compared with the controls. The adult emergence exhibited by the control and *S. riobrave* treatments was not significantly different. In field conditions, statistical differences were obtained between treatments ($F = 24.5$; $df = 2, 15$; $P = 0.0001$, $r^2 = 0.77$, $CV\% = 15.86$).

Discussion

Our hypothesis was that soil type and texture affects the infectivity of entomopathogenic nematodes against *A. ludens* larvae on soil surfaces. Our results confirmed this hypothesis and corroborate the reports of Georgis & Poinar (1983a, 1983b), Gaugler (1988), and Choo & Kaya (1991). Mortality rates in clay soil were severely reduced. A direct relationship between mortality rates and concentration of nematodes was observed in sandy loam and loam soil, and in addition, a direct relationship between reduction of adult emergence and nematode concentration was also observed. These findings corroborate the reports by Molyneux & Bedding (1984) on the larvae of sheep blowfly, *Lucilia cuprina* (Weidemann) (Diptera: Calliphoridae); those of Kung et al. (1991) and Choo & Kaya (1991), on *Galleria mellonella*, and Eidt & Thurston (1995) on wireworms, where the soil type affected the infectivity of entomopathogenic nematodes.

Our study showed that the *S. carpocapsae* and *S. riobrave* differed in activity on third instar of *A. ludens* in the laboratory. The LC_{50} values showed that *S. riobrave* was about twice as efficacious as *S. carpocapsae* in both soil types (sandy and sandy loam).

The number and size of soil pores also could play a diminishing role on the reduction of cumulative percent mortality because the space availability and water films are diminished in the clay type, affecting the movement of the nematodes in the soil (Barbercheck & Kaya 1991, Gouge et al. 2000). There is a clear correlation between soil type and porosity, even when *S. carpocapsae* remains near to the inoculation point when applied to soil as reported by Moyle & Kaya (1981). The nematode *S. carpocapsae* was more efficacious in loam texture than in sandy and clay soils. Eben & Barbercheck (1997) observed that *S. carpocapsae* was more efficient in sandy loam soil than sand. However, *S. riobrave* exhibited similar larvicidal activity in both soil textures (sandy loam and loam soils). Foraging behavior of IJs can affect their efficacy. Some species are ambushers (e.g., *S. carpocapsae*, and *S. scapterisci*) that tend to remain near the soil surface and attach to and infect mobile hosts at the soil-litter interface (Campbell & Gaugler 1993). Other species (i.e., *S. glaseri* Steiner and *Heterorhabditis bacteriophora* Poinar) are cruisers that have an active searching strategy and are more effective against less mobile insects in the soil (Lewis et al. 1992, Campbell & Gaugler 1997). Most nematode species are situated between these two extremes, placing them as intermediate foraging strategists (e.g., *S. riobrave*, and *S. feltiae*; Campbell & Gaugler 1997). An explanation of the higher mortality rates obtained with *S. riobrave* could be the result of its host seeking behavior. This behavior provides the nematodes with the capability to search for host insects (Kaya & Gaugler 1993).

Gazit et al. (2000) discussed that a biological pest control system based on entomopathogenic nematodes requires not only an effective strain of nematode but also one that can maintain its infectivity in the soil under the wide range of environmental conditions that the pest inhabits. Considering this, our results demonstrated that *S. carpocapsae* was able to reduce about 64% of adult emergence under field conditions, whereas *S. riobrave* reduced only 14%. This finding demonstrates that field environmental conditions could affect the behavior and infectivity of a nematode species such as *S. riobrave* that exhibited high larvicidal activity in the laboratory but had reduced its infectivity under field conditions. Gazit et al. (2000) mentioned that highest mortalities in *C. capitata* were achieved in small and large arenas with rates higher than 100 IJs/cm² in this study we used 250 IJs/cm². Possibly, other conditions affected the movement of *S. riobrave* more than *S. carpocapsae*. It might be argued that IJ vigor could be a factor in explaining poor field efficacy in the field trial this could be true for *S. riobrave* however it can not explain our results with *S. carpocapsae*, where IJ viability was excellent.

The effectiveness of *S. riobrave* and *S. carpocapsae* to infect *A. ludens* larvae in the sandy loam and loam soils are important because most mango orchards are established in sandy loam and loam soil in the tropics. The application of these nematodes in the dropping area of mango canopies for controlling *A. ludens* larvae could reduce the next generations by suppressing the last instars and emerging adults. We encourage the utilization of entomopathogenic nematodes under mango trees as a promising technique for the control of fruit flies. Gazit et al. (2000) observed that infectivity of nematodes in soil decreased within 10 d and was completely lost after 14 d. To discriminate the effectiveness of the nematode species in soils including repeated applications, more studies are required.

This is the first report evaluating the virulence of entomopathogenic nematodes on *A. ludens* larvae in three different soils under both laboratory and field conditions. Our data further demonstrate that soil type is a critical factor that must be taken into account when selecting nematode species and planning biological control strategies for fruit flies if entomopathogenic nematodes are to be effectively used.

Acknowledgments

We thank Mass Rearing Laboratory of the Program Moscamed (SAGARPA-USDA) from Metapa de Domínguez, Chiapas, Mexico for supplying the *A. ludens* larvae for the experiments, as well as to Dr. Harry K. Kaya and Dr. Grover C. Smart, Jr. for supplying the nematode strains. We thank Dr. John J. Hamm, Orville Marti, Jr., (USDA-ARS-IBPMRL, Tifton, Georgia), and Dr. E. A. Heinrichs (Department of Entomology, University of Nebraska Lincoln) for critical reviews of the manuscript and Héctor Enrique Ascencio Jaramillo and Arthur Edwards Block (Universidad de Colima, Colima, Mexico) for helping us to review the manuscript.

References Cited

- Abbott, W. S. 1925.** A method for computing the effectiveness of an insecticide. *J. Econ. Entomol.* 18: 265–267.
- Aluja, M. 1994.** Bionomics and management of *Anastrepha*. *Annu. Rev. Entomol.* 39: 155–178.

- Aluja, M., H. Celedonio-Hurtado, P. Liedo, M. Cabrera, F. Castillo, J. Guillen & F. Rios. 1996.** Seasonal population fluctuations and ecological implications for management of *Anastrepha* fruit flies (Diptera: Tephritidae) in commercial mango orchards in Southern Mexico. *J. Econ. Entomol.* 89: 654–667.
- Barbercheck, M. E. & H. K. Kaya. 1991.** Effect of host condition and soil texture on host finding by the entomogenous nematodes *Heterorhabditis bacteriophora* (Rhabditida: Heterorhabditidae) and *Steinernema carpocapsae* (Rhabditidae: Steinernematidae). *Environ. Entomol.* 20: 538–589.
- Beavers, J. B. & C. O. Calkins. 1984.** Susceptibility of *Anastrepha suspensa* (Diptera: Tephritidae) to steinernematid and heterorhabditid nematodes in laboratory studies. *Environ. Entomol.* 13: 137–139.
- Buhler, W. G. & T. J. Gibb. 1994.** Persistence of *Steinernema carpocapsae* and *S. glaseri* (Rhabditida: Steinernematidae) as measured by their control of black cutworm (Lepidoptera: Noctuidae) larvae in bentgrass. *J. Econ. Entomol.* 87: 638–642.
- Campbell, J. F. & R. Gaugler. 1993.** Nictation behaviour and its implications in the host search strategies of entomopathogenic nematodes (Heterorhabditidae and Steinernematidae). *Behaviour* 126: 155–169.
- Campbell, J. F. & R. Gaugler. 1997.** Inter-specific variation in entomopathogenic nematode foraging strategy: Dichotomy or variation along a continuum? *Fund. Appl. Nematol.* 20: 393–398.
- Choo, H. Y. & H. K. Kaya. 1991.** Influence of soil texture and presence of roots on host finding by *Heterorhabditis bacteriophora*. *J. Invertebr. Pathol.* 58: 279–280.
- Dutky, S. R., J. V. Thompson & G. E. Cantwell. 1964.** A technique for the mass propagation of the DD-136 nematode. *J. Insect Pathol.* 6: 417–422.
- Eben, A., & M. E. Barbercheck. 1997.** Host plant and substrate effects on mortality of southern corn rootworm from entomopathogenic nematodes. *Biological Control* 8: 89–96.
- Eidt, D. C. & G. S. Thurston. 1995.** Physical deterrents to infection by entomopathogenic nematodes in wireworms (Coleoptera: Elateridae) and other soil insects. *Can. Entomol.* 127: 423–429.
- Finney, D. J. 1971.** Probit analysis. 3rd ed. Cambridge University Press, Cambridge, England.
- Forschler, B. T. & W. A. Gardner. 1991.** Field efficacy and persistence of entomogenous nematodes in the management of white grubs (Coleoptera: Scarabaeidae) in turf and pasture. *J. Econ. Entomol.* 84: 1454–1459.
- Gaugler, R. 1981.** Biological control potential of neoaplectanid nematodes. *J. Nematol.* 13: 241–249.
- Gaugler, R. 1988.** Ecological considerations in the biological control of soil-inhabiting insects with entomopathogenic nematodes. *Agric. Ecosyst. Environ.* 24: 351–360.
- Gazit, Y., Y. Rössler & I. Glazer. 2000.** Evaluation of entomopathogenic nematodes for the control of Mediterranean fruit fly (Diptera: Tephritidae). *Biocontrol Sci. Technol.* 10: 157–164.
- Georgis, R. 1992.** Present and future prospects for entomopathogenic nematode products. *Biocontrol Sci. Technol.* 2: 83–99.
- Georgis, R. & G. O. Poinar Jr. 1983a.** Effect of soil texture on the distribution and infectivity of *Neoaplectana carpocapsae* (Nematoda: Steinernematidae). *J. Nematol.* 15: 308–315.
- Georgis, R. & G. O. Poinar Jr. 1983b.** Effect of soil texture on the distribution and infectivity of *Neoaplectana glaseri* (Nematoda: Steinernematidae). *J. Nematol.* 15: 329–332.
- Gouge, D. H., K. A. Smith, L. L. Lee & T. J. Henneberry. 2000.** Effect of soil depth and moisture on the vertical distribution of *Steinernema riobrave* (Nematoda: Steinernematidae). *J. Nematol.* 32: 223–228.

- Kaya, H. K. 1990.** Soil ecology. In: *Entomopathogenic Nematodes in Biological Control* (Gaugler, R. and Kaya, H. K., Eds.), pp. 93-115, CRC Press, Boca Raton, Florida, USA.
- Kaya, H. K. & R. Gaugler. 1993.** Entomopathogenic nematodes. *Annu. Rev. Entomol.* 38: 181-206.
- Kaya, H. K. & S. P. Stock. 1997.** Techniques in insect nematology. In *Manual of techniques in insect pathology*, L. A. Lacey (ed.). Biological Techniques Series, Academic Press, San Diego, California, p. 281-324.
- Kung, S. P., R. Gaugler & H. K. Kaya. 1990.** Soil type and entomopathogenic nematode persistence. *J. Invertebr. Pathol.* 55: 401-406.
- Kung, S. P., R. Gaugler & H. K. Kaya. 1991.** Effects of soil temperature, moisture, and relative humidity on entomopathogenic nematode persistence. *J. Invertebr. Pathol.* 57: 242-249.
- Lewis, E. E., R. Gaugler & R. Harrison. 1992.** Entomopathogenic nematode host finding: Response to contact cues by cruise and ambush foragers. *Parasitology* 105: 309-315.
- Lindgren, J. E. & P. V. Vail. 1986.** Susceptibility of Mediterranean fruit fly, melon fly, and Oriental fruit fly (Diptera: Tephritidae) to the entomogenous nematode *Steinernema feltiae* in laboratory tests. *Environ. Entomol.* 15: 465-468.
- Lindgren, J. E., T. T. Wong & D. O. Mcinnis. 1990.** Response of Mediterranean fruit fly (Diptera: Tephritidae) to the entomogenous nematode *Steinernema feltiae* in field tests in Hawaii. *Environ. Entomol.* 19: 383-386.
- Molyneux, A. S. & R. A. Bedding. 1984.** Influence of soil texture and moisture on the infectivity of *Heterorhabditis* sp. D1 and *Steinernema glaseri* for larvae of the sheep blowfly, *Lucila cuprina*. *Nematologica* 30: 358-365.
- Moyle, P. L. & H. K. Kaya. 1981.** Dispersal and infectivity of the entomopathogenic nematode, *Neoaplectana carpocapsae* Weiser (Rhabditida: Steinernematidae), in sand. *J. Nematol.* 13: 419-421.
- Norrbom, A. L. & K. C. Kim. 1988.** A list of the reported host plants of the species of *Anastrepha* (Diptera: Tephritidae). U.S. Dept. Agric., APHIS-PPQ, Hyattsville, Maryland. 114 p.
- Poinar Jr., G. O. & R. G. Hislop. 1981.** Mortality of Mediterranean fruit fly adults (*Ceratitis capitata*) from parasitic nematodes (*Neoaplectana* and *Heterorhabditis* spp.). *IRCS Med. Sci. Microbiol. Parasitol. Infect. Dis.* 9: 641.
- Sokal, R. R. & F. J. Rohlf. 1995.** Biometry. Freeman, New York, USA, pp. 887.
- SAS Institute. 1997.** SAS user's guide, statistics, version 5. SAS Institute, Cary, North Carolina.
- Steck, G. J., L. E. Carroll, H. Celedonio-Hurtado & J. Guillen-Aguilar. 1990.** Methods for identification of *Anastrepha* larvae (Diptera: Tephritidae), and key to 13 species. *Proc. Entomol. Soc. Wash.* 92: 333-346.
- Zahler, P. M. 1990.** Moscas-das-frutas em três pomares do Distrito Federal: levantamento de espécies e flutuação populacional. *Ciência e Cultura (Brasil)* 42: 177-182.
-

NOTE

Spatial Distribution of *Jacobiasca lybica* (Bergenin & Zanon) (Homoptera: Cicadellidae) Egg Populations in an Irrigated Sherry Vineyard¹

Jose L. Gonzalez-Andujar,² Jose F. Ramirez-Davila,³ Maria A. Lopez,⁴
and Rafael Ocete⁴

Departamento de Protección de Cultivos. Instituto de Agricultura Sostenible (CSIC), Apdo. 4084.
14080 Cordoba, Spain.

J. Agric. Urban Entomol. 23(1): 51–55 (Jan 2006)

Jacobiasca lybica (Berg. & Zanon) (Homoptera: Cicadellidae), the green leaf hopper, is the most serious phytosanitary problem in vineyards in the southern regions of the Iberian Peninsula, producing serious economic damage (Araújo 1988, Quartau et al. 1989, Quartau & Rebelo 1992). Damage is usually very serious because it affects not only the quality and quantity of grapes but also because the plant nutrient reserves for the following season are severely curtailed. Adults and nymphs feed almost continuously, piercing and sucking the cell contents in all mesophyll cells. Attacked leaves eventually change color and become scorched from the lack of sap movement. The leaves frequently curl toward the undersides. After very severe attacks, leaves drop to the ground, leaving the grapes exposed to the sun and thus causing “scalding” of the berries. Despite the high incidence of *J. lybica* infestations, at present little is known of its spatial distribution and biology because of its new pest status in Spanish vineyards (Toledo 1992, Ramirez-Davila 2003). Previous studies in Italy (Delrio et al. 2001) suggested that the nymphs of *J. lybica* follow an aggregated pattern in vineyard. However, they don't give any indication about the egg distributions.

Spatial distribution of pest populations is an important aspect of population biology because it is a result of the interaction between individuals of the species and their habitat (Sevacherian & Stern 1972). Knowledge of the distribution allows a better understanding of the relationship between an insect and its environment and provides basic information for interpreting spatial dynamics, designing efficient sampling programs for population estimation, and pest management (Harcourt 1961, Sevacherian & Stern 1972), and the development of

¹Accepted for publication 11 December 2006.

²Corresponding author: José L. Gonzalez-Andujar Protección de cultivo Instituto de Agricultura Sostenible (CSIC) Apdo. 4084 14080 Cordoba, Spain. E-mail: andujar@cica.es

³Facultad de Geografía, Universidad Autónoma del estado de México, Cerro de Coatepec, C.U., 50110 Toluca, México.

⁴Laboratorio de Entomología Aplicada. Universidad de Sevilla, Facultad de Biología, Avda. Reina Mercedes, 6, 41012 Sevilla, Spain.

population models (Croft & Hoyt 1983). Methods that are commonly used to describe spatial distribution of arthropod populations have been summarized by Southwood (1978). Several estimates based on sample mean and variances are used as indices for aggregation (Pielou 1977) and the dispersion parameter k for the negative binomial distribution (Southwood 1978).

The objective of this study was to describe the spatial pattern of the eggs of *J. lybica* using four statistical distributions and dispersion indices. The study was conducted at Jerez de la Frontera (Cádiz, Spain, 36°31'N, 5°27'W), for two consecutive seasons (2001 and 2002). Samples were collected during the months of higher insect activity (June-October) (Toledo 1992) and comprised 126 sample units (plants) in each month. At each plant, the number of eggs was directly counted by visual inspection of ten random leaves (López 1997). The plants were marked and the surveys were performed always on the same plants.

Spatial distribution of a population can be determined by working out different dispersion parameters. In this study, the distribution of leaf hopper in vineyard was known by calculating the following dispersion parameters:

- (1) The index of dispersion (variance to mean ratio) is based on the equality of variance and mean in a Poisson series. If the sample is in agreement with a theoretical Poisson series, we would expect this ratio to be equal to 1. A value of zero implies regular distribution; whereas a value greater than one implies aggregate or contagious distribution.
- (2) David and Moore's index (or Index of Clumping) (David & Moore 1954) is given by the formula

$$IC = \frac{S^2}{\bar{X}} - 1$$

Here, values greater than zero indicate aggregation, if IC is equal, a 0 indicates random pattern, and when IC is -1, there is maximum regularity.

- (3) The reciprocal of k of negative binomial, is another measure of clumping, using the reciprocal of k i.e. $1/k$ (Southwood 1978). Values for $1/k$, which were the distribution can be random ($1/k = 0$), aggregated ($1/k > 0$), and uniform ($1/k < 0$).
- (4) The negative binomial distribution (NBD) is probably the most commonly used probability distribution for aggregated populations (Pielou 1977). The probability of finding x individuals in a sampling unit, that is, P_x , where $x = 0, 1, 2, \dots, n$ individuals, is given by

$$P_x = \frac{(x+k-1)!}{x!(k-1)!} \left[\frac{k}{k+\bar{x}} \right]^k \left[\frac{\bar{x}}{k+\bar{x}} \right]^x$$

Where $\bar{x}$ is the sample mean and the k parameter defines the shape of the NBD and serves as a general indicator of aggregation, with the smaller values of k indicating increased aggregation (Southwood 1978). NBD was fitted to the count of eggs data for each month, and goodness of fit was examined with a χ^2 test. The program MLP (Maximum Likelihood Program) (Ross 1987) was used to fit distribution, using maximum likelihood estimation.

Stability of parameter k (k_c) for the NBD was assessed using the regression procedures presented by Bliss & Owen (1958) (also see Elliot 1979, Feng & No-

wierski 1992). A stable k indicates that the level of clumping is a fairly constant characteristic of a species. Estimates of k_c were made using Elliot's (1979) techniques, which estimates k_c by regressing two statistics, $y' [= s^2 - \bar{x}]$ on $x' [= \bar{x} - (s^2/N)]$, where $\bar{x}$ was the mean, s^2 the variance and N the number samples, and k_c was defined by $k_c = 1/\text{slope}$. The adequacy of this k_c estimate was previously evaluated using a regression analysis of $1/k$ on $\bar{x}$: a trend between these variables discredits the suitability of a single k_c (Bliss & Owen 1958, Southwood 1978).

The general linear model procedure (GLM) of SAS (SAS Institute 1999) was used to estimate the linear regressions. Student's t -tests were used to determine if the slopes (b) of the regression lines were significant (Sokal & Rohlf 1981).

Table 1 shows that the variance is greater than the corresponding mean values for all the months, suggesting the aggregative nature of the oviposition of leaf hopper. The index of dispersion (Table 2) shows that the values are more than one for all the months, also suggesting aggregation of *J. lybica* population in the field.

The results presented in Table 2 also show that the observed David and Moore's index value for all the months were greater than the value of maximum regularity (-1) and randomness (0), indicating the contagious aggregation.

In this study the values of $1/k$ (Reciprocal of k of negative binomial) were more than zero for all the months where it was possible to calculate it, ranging between 0.31 and 3.57 (Table 2) thereby suggesting aggregation of eggs populations. Reddy et al. (1993) used this test to establish the aggregated distribution of leaf hopper in rice.

The negative binomial distribution fitted well for 7 of the 10 cases studied. There was insufficient information to fit distributions or test goodness of fit for 2 of the cases, and in 1 case NBD was rejected ($P = 0.05$).

The stability of k is necessary to develop parametric sequential sampling methods (Nyrop et al. 1989) because k is used to determine the stopping criterion that indicates the minimum number of samples for which the estimate of mean density has met a specific level of precision. There is not a relationship between $1/k$ and $\bar{x}$ ($r^2 = 0.08$; $P = 0.52$) and, therefore, the calculation of a common k (k_c) is

Table 1. Monthly values of mean and variance for *J. lybica*.

Date	Mean	Variance
2001		
29-06	0.35	0.41
07-08	0.08	0.09
30-08	0.53	1.13
27-09	0.92	1.58
26-10	0.69	0.80
2002		
27-06	0.82	1.36
25-07	1.38	1.74
30-08	1.56	1.89
27-09	1.66	2.27
28-10	0.61	0.84

Table 2. Dispersion indices and negative binomial distribution.

Date	Dispersion index	David & Moore's index	Binomial negative ^a	K	1/k
2001					
29-06	1.17	0.37	s	2.15	0.46
07-08	1.12	0.12	ns	—	—
30-08	2.09	1.13	s	0.28	3.57
27-09	1.70	0.72	s	1.03	0.97
26-10	1.15	0.16	s	3.19	0.31
2002					
27-06	1.64	0.66	s	1.00	1.00
25-07	1.25	0.26	NA	—	—
30-08	1.21	0.21	NA	—	—
27-09	1.36	0.36	s	3.17	0.31
28-10	1.37	0.37	s	5.21	0.19

^as = significant at P = 0.05; ns = not significant; NA = not possible to fit negative binomial.

adequate. The estimates of k_c were 0.80. So, the hypothesis of a common or stable k_c was not rejected. The stability of k implies that the grade of aggregation of the eggs population was relatively constant over two years, despite variation in sample means. Faleiro et al. (2002) used this test to confirm the stability of k of red palm weevil distribution in coconut throughout the time.

This research provides the basis for understanding the spatial distribution of *J. lybica* in vineyards. The results presented and discussed document that green leaf hopper egg populations in vineyards are aggregated and follow the negative binomial distribution pattern. Management decisions are most feasible when decisions are based on knowledge of insect distributions. Thus, the practical significance of knowledge of frequency distribution may include accurate estimation of green leaf hopper threshold densities and implementation of efficient insect scouting programs (Pedigo & Buntin 1994). These results will provide information to develop scouting programs for *J. lybica* in Spanish vineyards.

References Cited

- Araújo, J. A. 1988. Panorâmica da viticultura alentejana, pp. 1-6. In I Simpósio de Vitivinicultura, Evora. Portugal. 150 pp.
- Bliss, C. L. & A. R. G. Owen. 1958. Negative binomial distributions with a common k . Biometrika 45:37–58.
- Croft, B. A. & S. C. Hoyt. 1983. Integrated management of insect pests of pome and stone fruits. Wiley, New York, 250 pp.
- David, F. N. & P. G. Moore. 1954. Note on contagious distributions in plant populations. Ann. Bot. (Lond.) 18:47–53.
- Delrio, G., A. Lentini & G. Serra. 2001. Spatial distribution and sampling of *J. lybica* on grapevine. IOBC WRPS Bulletin 24:211–216.
- Elliot, J. M. 1979. Some methods for statistical analysis of samples of benthic invertebrates. Freshwater Biological Association. Scientific Publication #25. UK, 160 pp.

- Faleiro, J. R., J. A. Kumar & P. A. Rangnekar. 2002.** Spatial distribution of red palm weevil *Rhynchophorus ferrugineus* Olic. (Coleoptera: Curculionidae) in coconut plantations. *Crop Prot.* 21:171–176.
- Feng, M. G. & R. M. Nowierski. 1992.** Variation in spatial patterns of the Russian wheat aphid (Homoptera: Aphididae) among small grains in the northwestern United States. *Environ. Entomol.* 21:1029–1034.
- Harcourt, D. G. 1961.** Design of a sampling plan for studies on the population dynamics of the diamond-back moth, *Plutella maculipennis* (Curt.) (Lepidoptera: Plutellidae). *Canadian Entomol.* 93:820–831.
- López, M. A. 1997.** Incidencia de *Kaloterme flavicollis* (Fabr.) (Isoptera, Kalotermitidae) en el Marco del Jerez / Ensayos de Técnicas Blandas de Control sobre Plagas del Viñedo. PhD Thesis. Univ. de Sevilla, 477 pp.
- Nyrop, J. P., A. M. Agnello, J. Kovach & W. H. Reissig. 1989.** Binomial sequential classification sampling plans for European red mite (Acari: Tetranychidae) with special reference to performance criteria. *J. Econ. Entomol.* 82:482–490.
- Pedigo, L. P. & G. D. Buntin. 1994.** Handbook of sampling methods for arthropods in agriculture. CRC Press, Boca Raton, Florida, 714 pp.
- Pielou, E. C. 1977.** Mathematical ecology. Wiley. New York, 385 pp.
- Quartau, J. A. & M. T. Rebelo. 1992.** Estudos preliminares sobre os cicadelídeos que constituem pragas das vinhas em Portugal (Homoptera: Cicadellidae). *Bol. San. Veg. Plagas* 18:407–417.
- Quartau, J. A., A. Francony & G. Andre. 1989.** *Jacobiasca lybica* (Bergening & Zanon) (Homoptera: Cicadellidae, Typhlocybinae) a new leafhopper infesting vineyards in Southern Portugal. *Bolm Soc. Port. Ento.* 114:129–133.
- Ramirez-Davila, J. F. 2003.** Modelización y generación de mapas de la distribución espacial de los estadios preimaginales del mosquito verde *Jacobiasca lybica* (Bergenin & Zanon) (Homoptera, Cicadellidae) en viñedos y su aplicación en la agricultura de precisión. PhD Dissertation. Universidad de Sevilla. Sevilla, 355 pp.
- Reddy, C. N., T. V. K. Singh, D. P. R. Reddy & T. R. Goud. 1993.** Spatial distribution of rice leaf and plant hoppers. *Indian J. Entomol.* 55:1–10.
- Ross, G. J. S. 1987.** Maximum Likelihood Program. Rothamsted Experimental Station, Harpenden. UK.
- SAS Institute. 1999.** SAS version 8.02. SAS Institute, Cary, North Carolina.
- Sevacherian, V. & V. M. Stern. 1972.** Spatial distribution patterns of *Lygus* bugs in California cotton fields. *Environ. Entomol.* 1:695–704.
- Sokal, R. R. & F. J. Rohlf. 1981.** Biometry, 2nd ed. Freeman, San Francisco, California, 832 pp.
- Southwood, T. R. E. 1978.** Ecological Methods. 2nd ed. Wesley-Halsted, New York, 524 pp.
- Toledo, J. 1992.** Mosquito verde (*Empoasca spp.*), pp 67–70. In Grupo trabajo de la Vid [Eds]. Los parásitos de la vid. M.A.P.A & Mundi-Prensa. Madrid, Spain, 393 pp.
-

NOTICE TO CONTRIBUTORS

The Editorial Committee of *Journal of Agricultural Entomology* considers the timely publication and documentation of research endeavors critical to the advancement of agricultural and urban entomology. As such, JAUE places a high priority on prompt processing of submitted works, as well as prompt publication of accepted works. However, the overall turnaround time is to a large extent in the hands of authors making revisions.

In light of this, the Committee requests that authors return revised manuscripts to their subject editor within three months of receipt. Manuscripts held for revisions for six months or more will, at the discretion of the editor, be subject to a 10% surcharge over the normal cost of publication.

The Committee recognizes that extenuating circumstances arise, and will consider special arrangements for authors in such situations.

Journal of Agricultural and Urban Entomology

Contents

 Volume 23, No. 2

 April 2006

- ZHENGYAN WANG, JIANCHU MO, AND SHIMIAO ZHANG – Laboratory and Field Evaluations of Potential Human Host Odors for *Aedes albopictus* Skuse (Diptera: Culicidae) 57
- SURENDRA K. DARA AND PAUL J. SEMTNER – Within-Plant Distribution of *Pandora neoaphidis* (Zygomycetes: Entomophthorales) in Populations of the Tobacco-Feeding Form of *Myzus persicae* (Homoptera: Aphididae) on Flue-Cured Tobacco 65
- D. MICHAEL JACKSON AND JANICE R. BOHAC – Survival and Growth of *Diabrotica balteata* Larvae on Insect-Resistant Sweetpotato Genotypes 77
- W. PAUL WILLIAMS, PAUL M. BUCKLEY, AND CHRISTOPHER A. DAVES – Identifying Resistance in Corn to Southwestern Corn Borer (Lepidoptera: Crambidae), Fall Armyworm (Lepidoptera: Noctuidae), and Corn Earworm (Lepidoptera: Noctuidae) 87
- W. PAUL WILLIAMS, GARY L. WINDHAM, PAUL M. BUCKLEY, AND CHRISTOPHER A. DAVES – Aflatoxin Accumulation in Corn Hybrids Infested at Different Growth Stages with Southwestern Corn Borer (Lepidoptera: Crambidae) 97
- JULIANE M. DEACUTIS, CHERYL A. LEICHTER, ALEC C. GERRY, DONALD A. RUTZ, WES D. WATSON, CHRISTOPHER J. GEDEN, AND JEFFREY G. SCOTT – Susceptibility of Field Collected House Flies to Spinosad Before and After a Season of Use 105

Laboratory and Field Evaluations of Potential Human Host Odors for *Aedes albopictus* Skuse (Diptera: Culicidae)¹

Zhengyan Wang, Jianchu Mo, and Shimiao Zhang

Urban Entomology Research Center, College of Agriculture and Biotechnology, Zhejiang University, Zhejiang 310029, China

J. Agric. Urban Entomol. 23(2): 57–64 (April 2006)

ABSTRACT Behavioral responses of female *Aedes albopictus* Skuse (Diptera: Culicidae) to ethyl butyrate, L-lactic acid, dimethyl disulfide, and hexanoic acid over a range of concentrations (0.01–100 mg/L) were investigated in a Y-tube olfactometer. The results indicated that 55–81% of female mosquitoes were activated to fly in all these tests. Ethyl butyrate, hexanoic acid and dimethyl disulfide had significantly greater levels of activation than L-lactic acid ($P < 0.05$), but activation was not significantly different between concentrations (0.01, 0.1, 1, 10, 100 mg/L; $P = 0.80$). Treatment of 0.01 mg/L ethyl butyrate, 0.1 mg/L L-lactic acid, and 0.01 mg/L dimethyl disulfide were significantly more attractive than control ($P < 0.05$). Field evaluation of ethyl butyrate, or L-lactic acid, 1-octen-3-ol, ammonia, 6-methyl-5-hepten-2-one, hexanoic acid, dimethyl disulfide, and their blends over a range of concentrations (0.1–1000 mg/L) for mosquito traps showed that the odors tested attracted many fewer mosquitoes (*Culex* and *Aedes*) compared with the control (a live mouse).

KEY WORDS *Aedes albopictus*, human skin volatile, activation, attraction, trap

Native to Asia, *Aedes albopictus* Skuse (Diptera: Culicidae) has been reported in more than 25 countries on the five continents outside its natural region since the end of the 1970s. It is an important vector of several arboviruses, including dengue, yellow fever, and diverse types of encephalitis. This mosquito is widely distributed in China and causes prevalent occurrence of dengue in some local areas, such as the cities of Guangzhou and Fuzhou.

Common approaches to control mosquitoes are the use of chemical insecticides for area-wide mosquito abatement and repellents for personal protection. However, because of insecticide resistance in mosquitoes and social concerns about environmental pollution, there has been an increasing emphasis on the development of alternative mosquito control technologies, such as the use of attractant-baited traps for mass trapping. Odors that attract mosquitoes could be useful in the development of new trapping methods (Meijerink & van Loon 1999, Silva et al. 2005).

¹Accepted for publication 13 December 2006

Many compounds that emanate from human skin and breath play an important role in the attraction of female mosquitoes to the host at a relatively close range (Gillies 1980, Schreck et al. 1990, Takken 1991). L-lactic acid is only slightly effective by itself but acts synergistically with other compounds from human skin for *Anopheles gambiae* Giles *sensu stricto* (Dekker et al. 2002). Ammonia is attractive to *Aedes aegypti* L. when presented together with lactic acid (Geier et al. 1999a). Hexanoic acid elicits responses of the carboxylic acid excited cells of *An. gambiae* in an electrophysiological study (Meijerink & van Loon 1999). Dimethyl disulfide is detected in human skin emanations (Bernier et al. 2000), and it synergistically attracts female *Ae. aegypti* with L-lactic acid, but the role of this disulfide in mosquito attraction process is not thoroughly understood (Bernier et al. 2003). Bernier et al. (2002) found that 6-methyl-5-hepten-2-one is more abundant on the host that attracts a greater percentage of *Ae. aegypti*. Certain combinations of chemicals synergistically attract some species more than others, although each attractant is capable of attracting mosquitoes on its own.

Recent studies on mosquito behaviors to human odors have focused mostly on *Ae. aegypti* and *An. gambiae*. The work reported here comprises laboratory and field studies, and the purpose is to evaluate the response of female *Ae. albopictus* to seven synthetic chemicals that have been reported as attractants of other mosquitoes, individually or as blends over a wide range of concentrations.

Materials and Methods

Mosquitoes. *Ae. albopictus* mosquitoes were obtained from the colony maintained at Shanghai Institute of Entomology, Chinese Academy of Sciences. The insects were maintained at 25°C and a L:D 14:10 h photoperiod without simulated dusk. Female and male adults were kept together in a gauze cage (35 × 35 × 35 cm) at 70% relative humidity and provided with diet of 5% sucrose solution on cotton. Females were routinely offered a mouse as bloodmeal from the fifth day after emergence. Eggs were laid on wet tissue paper, hatched in water trays, and the larvae were fed with rat food. Pupae were collected into cups and placed in cages for adults to emerge. For olfactometer tests, we used nonblood-fed, 5- to 7-d-old female *Ae. albopictus*.

Test odors. Dimethyl disulfide (98%), hexanoic acid (98%), 1-octen-3-ol (98%), L-lactic acid (85–90% w/w aqueous solution), and 6-methyl-5-hepten-2-one (96%) were purchased from Fluka Company (Buchs, Switzerland); ammonia (25–28% w/w aqueous solution) from Changzheng Chemical Product Company (Hangzhou, China); ethyl butyrate (98%) from Medicine Group Company of Shanghai (Shanghai, China).

Ethyl butyrate (0.117 mL), dimethyl disulfide (0.096 mL), hexanoic acid (0.110 mL), 1-octen-3-ol (0.122 mL), and 6-methyl-5-hepten-2-one (0.122 mL) were dissolved in 2 mL of absolute alcohol and then diluted with distilled water to a series of concentrations (0.01, 0.1, 1, 10, 100, and 1000 mg/L). Ammonia and L-lactic acid were directly diluted with distilled water to the same concentrations.

Experimental set-up and testing procedures. We used a modified dual-choice olfactometer of Plexiglas® (Geier & Boeckh 1999). A constant air stream was purified with activated charcoal filter and adjusted to 70 ± 5% relative humidity and 26 ± 1°C. The wind speed was 0.2 m/sec in the arms and 0.4 m/sec in the downwind arm. A 40-W light bulb was provided as overhead illumination. An

hour before each experiment, 10–25 female mosquitoes were lured from the holding cage into a release cage and left in the olfactometer air stream for 20 min to acclimatize. A piece of filter paper (6.2 cm diameter) was dipped into odor solutions for 5 sec for bioassay. Filter paper treated with distilled water served as control. The treated filter paper was placed in one of the arms, and the control paper was placed in the other arm for 60 sec, then the test was started by slowly opening the release cage. During the 5-min test period, mosquitoes could enter either upwind arm. After each test the upwind arms and the release cage were closed and the mosquitoes in each arm of the olfactometer were counted. Female mosquitoes were exposed to ethyl butyrate, dimethyl disulfide, hexanoic acid, and L-lactic acid in concentrations ranging from 0.01–100 mg/L. Each treatment test was replicated eight times. To eliminate environmental influence, test and control arm were exchanged every four replicates. The olfactometer was cleaned with absolute alcohol and dried with a hair drier before each test. To avoid possible side effects such as learning or habituation, each group of mosquitoes was tested only once. Tests were conducted during 17.00–22.00 h everyday.

Finally, we examined the field trapping effect of seven chemicals over concentrations ranging from 0.1 to 1000 mg/L or blends that were mixed with equal volumes of its components. A vial (8 mL) containing 5 mL of solution was suspended in the middle of a gauze cage (35 × 35 × 35 cm). To trap and prevent mosquitoes from escaping out of the cage once trapped, four paper funnels, tapering from 9 cm to 3 cm, were inserted in the middle of four walls of the cages, respectively. The traps were placed 10 m apart in the field. Each treatment was replicated three times. A live mouse wrapped in gauze served as control bait. The trial was implemented in a grove between 17.00 h and 7.00 h the next morning from April to July 2006.

Data analysis. The olfactometer results were expressed in terms of three parameters (Bosch et al. 2000): (i) the percentage of mosquitoes found outside of the release cage after 5 min was taken as a measure of activation; (ii) the percentage of mosquitoes trapped in the treated arm was a measure for attraction to the treatment; and (iii) the percentage of mosquitoes in the control arm measured the response to pure air. All these data were arcsin transformed for statistical analysis. Data of activation were analyzed with a two-way analysis of variance (ANOVA) and least significant difference (LSD) to sort out differences between treatments. The relative attractiveness of the treatment and the control was analyzed with a one-sample *t*-test (Dekker et al. 2002). Data of attraction of the same compound were analyzed by a one-way ANOVA and LSD to sort out differences between five concentrations. The field results were analyzed by a one-way ANOVA and LSD for comparison of treatment. All data were analyzed by means of DPS[®] software (Tang & Feng 2002).

Results

Effects of odors on female mosquitoes in olfactometer. When any compound was tested, 55–81% of the mosquitoes left the release chamber (Fig. 1). Activation was significantly different between four compounds (ANOVA: $F = 4.354$, $df = 3,140$, $P = 0.006$), and ethyl butyrate, hexanoic acid and dimethyl disulfide had significantly higher activation than L-lactic acid ($P < 0.001$, $P =$

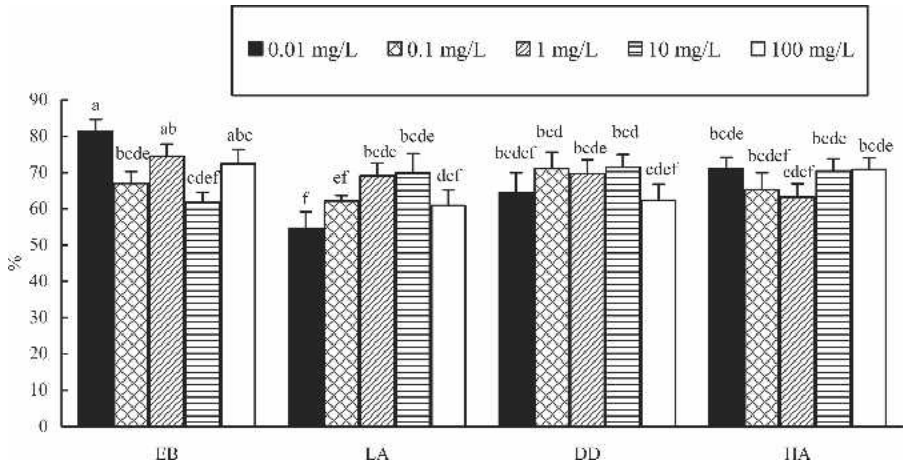


Fig. 1. Female *Aedes albopictus* activation by ethyl butyrate (EB), L-lactic acid (LA), dimethyl disulfide (DD), hexanoic acid (HA) over different concentrations. Bars with no letters in common above are significantly different at $P < 0.05$. Means of eight replicates $\pm$ SE.

0.033, $P = 0.038$). Activation was not significantly different between concentrations (0.01, 0.1, 1, 10, 100 mg/L; ANOVA: $F = 0.411$, $df = 4, 140$, $P = 0.800$).

Compared with the control, 0.01 mg/L ethyl butyrate (t -test: $t = 2.731$, $df = 7$, $P = 0.029$), 0.1 mg/L L-lactic acid (t -test: $t = 6.902$, $df = 7$, $P < 0.001$), and 0.01 mg/L dimethyl disulfide (t -test: $t = 2.648$, $df = 7$, $P = 0.033$) solution had significantly greater attractiveness to mosquitoes than distilled water and 100 mg/L dimethyl disulfide solution also had greater attraction than distilled water, although not significantly (t -test: $t = 2.315$, $df = 7$, $P = 0.054$; Fig. 2). These results suggest that they could attract *Ae. albopictus* females at the level of the aforementioned concentration. Attractiveness to mosquitoes was significantly different between five concentrations of ethyl butyrate (ANOVA: $F = 3.952$, $df = 4, 35$, $P = 0.009$) but was not significantly different between five concentrations of dimethyl disulfide (ANOVA: $F = 0.813$, $df = 4, 35$, $P = 0.525$), hexanoic acid (ANOVA: $F = 2.564$, $df = 4, 35$, $P = 0.055$), or L-lactic acid (ANOVA: $F = 0.104$, $df = 4, 35$, $P = 0.980$).

Attraction of odors on mosquitoes in field. Compared with a positive control (a live mouse), ethyl butyrate, L-lactic acid, 1-octen-3-ol, ammonia, 6-methyl-5-hepten-2-one, dimethyl disulfide, and hexanoic acid at the concentrations from 0.1 to 1000 mg/L attracted many fewer mosquitoes (*Culex* and *Aedes*) into the field traps. Blending L-lactic acid or ammonia with any other compound did not increase the attraction of mosquitoes over that of the single compound tested at the respective concentration (Table 1). These data indicated the synergist was not observed for blends of L-lactic acid or ammonia with the tested compounds.

Discussion

Similar to methyl butyrate in chemical structure, which is a tentatively identified component from human emanations (Bernier et al. 2000), ethyl butyrate

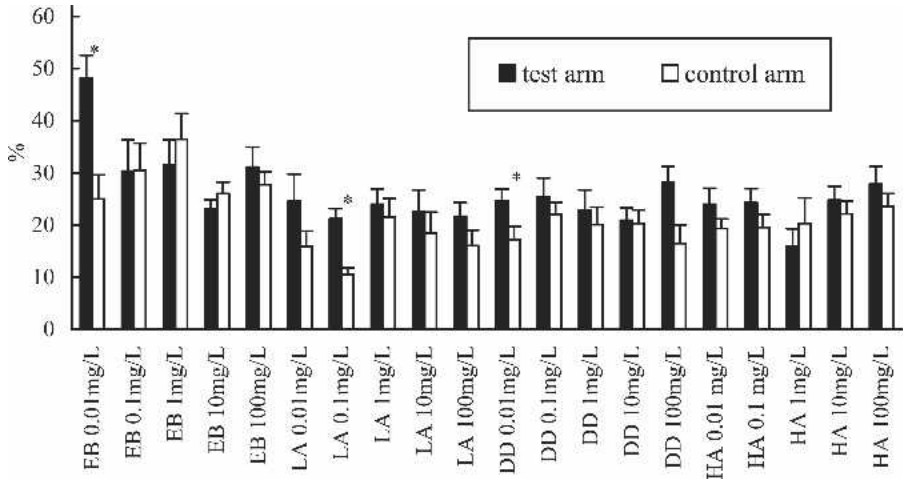


Fig. 2. Proportion of female *Aedes albopictus* caught in the two upwind arms (test and control) of the olfactometers. Test arms are added with ethyl butyrate (EB), or L-lactic acid (LA), dimethyl disulfide (DD), and hexanoic acid (HA) at different concentrations. Significant difference at $P < 0.05$ indicated by *. Means of eight replicates $\pm$ SE.

was tested in this study. Bernier et al. (2003) found that the blend of L-lactic acid with dimethyl disulfide was synergistic to attract female *Ae. aegypti*. In our olfactometer, the 0.01 mg/L ethyl butyrate solution, 0.01 and 100 mg/L dimethyl disulfide solutions produced greater attractiveness for *Ae. albopictus* than control. It also could be concluded that dimethyl disulfide exhibits some attractiveness for mosquitoes over a wide range of concentration. However, field results showed that the combination of ethyl butyrate or dimethyl disulfide with L-lactic acid was no more attractive to mosquitoes than either compound without L-lactic acid, at the respective concentrations. However, it is possible that different ratios of some compounds or all of them could still be promising attractants for *Ae. albopictus* and must be useful in the development of new trapping methods for control of mosquitoes.

L-lactic acid is a well-known attractant for *Ae. aegypti* (Acree et al. 1968). It also plays a significant role in host-finding by female *An. gambiae* (Dekker et al. 2002). Geier et al. (1999b) found the most effective lactic acid concentration was 1.7 nmol/L (153 ng/L), resulting in 20% upwind-flying mosquitoes in the homogeneous plume. In contrast to these reports suggesting when L-lactic acid attracts mosquitoes, Shirai et al. (2001) found that, compared with water-treated control skin, L-lactic acid repelled *Ae. albopictus* using human and mouse skin and the number of mosquito alightments decreased at increasing concentrations of L-lactic acid. Attraction of lactic acid to mosquitoes varied between 16% and 27% in our study, which was consistent with earlier findings of Bosch et al. (2000), and 0.1 mg/L L-lactic acid showed significant attractiveness when compared with the negative control in an olfactometer test. These findings imply that the attractiveness of L-lactic acid to mosquitoes is related to its concentration.

Table 1. Field attraction of mosquitoes to putative human odors (Mean $\pm$ SE for three replicates).

Component I ^a	Concentration of Component I (mg/L)					Component II ^b	Component III ^c	Control ^d
	0.1	1	10	100	1000			
EB	0.3 ± 0.3	0.0	2.7 ± 1.7	0.3 ± 0.3	1.3 ± 0.9			7.0 ± 3.1
LA	0.0	2.0 ± 0.0	0.0	0.7 ± 0.7	0.3 ± 0.3			7.7 ± 4.4
Octenol	0.7 ± 0.3	0.0	0.0	0.0	0.7 ± 0.3			3.0 ± 0.6
AM	0.3 ± 0.3	0.3 ± 0.3	0.3 ± 0.3	1.0 ± 0.6	0.3 ± 0.3			9.3 ± 4.3
MHO	0.7 ± 0.3	1.0 ± 0.0	0.0	1.0 ± 0.6	0.3 ± 0.3			16.6 ± 4.3*
HA	0.0	0.0	0.7 ± 0.7	0.0	0.0			30.0 ± 1.2*
DD	0.3 ± 0.3	0.0	1.0 ± 1.0	0.7 ± 0.7	0.0			22.0 ± 1.7*
Octenol	0.3 ± 0.3	0.7 ± 0.3	1.0 ± 0.6	0.0	0.0		AM	9.3 ± 3.5
EB	0.0	0.0	0.0	0.0	0.0	LA		9.7 ± 2.3*
Octenol	0.0	0.0	0.0	0.3 ± 0.3	0.0	LA		10.0 ± 1.5*
AM	0.0	0.0	0.0	0.0	0.3 ± 0.3	LA		15.7 ± 3.5*
MHO	0.0	0.0	0.0	0.0	0.0	LA		16.0 ± 2.5*
HA	0.0	0.0	0.3 ± 0.3	0.0	0.0	LA		27.7 ± 5.9*
DD	0.0	0.3 ± 0.3	0.3 ± 0.3	0.3 ± 0.3	0.3 ± 0.3	LA		23.7 ± 8.1*
HA	0.0	1.0 ± 1.0	0.0	0.0	1.3 ± 0.3	LA	AM	37.7 ± 11.7*

^aEB, ethyl butyrate; LA, L-lactic acid; Octenol, 1-octen-3-ol; AM, ammonia; MHO, 6-methyl-5-hepten-2-one; HA, hexanoic acid; DD, dimethyl disulfide.^bComponent II was 100 mg/L LA solution.^cComponent III was 100 mg/L AM solution.^dControl was a live mouse.*Significant difference existed at $P < 0.05$ level in the same line.

Geier et al. (1999a) found that the addition of 0.17 mg/L ammonia doubled the percentage of mosquitoes attracted to lactic acid alone, but greater concentrations did not further raise the attractiveness, and 0.017–1.7 mg/L ammonia alone was not attractive to mosquitoes compared with controls. Therefore, they concluded that attractive effects of certain compounds can be discovered in a bioassay only in combination with lactic acid. Bosch et al. (2000) showed a mixture of ammonia, lactic acid, propanoic acid, and valeric acid was the most attractive artificial odor blend for *Ae. aegypti* to date, with attraction levels that almost reached that of human skin washings. But L-lactic acid and ammonia showed no synergistic effect in trapping mosquitoes to other components in field test. Although Bosch et al. (2000) pointed out that the attractiveness of lactic acid was significantly augmented when combined with hexanoic acid, the hexanoic acid showed no significant attraction of mosquitoes in olfactometer and field tests even with the addition of L-lactic acid. Vale (1980) even found hexanoic acid was repellent for tsetse fly when used in an odor-baited trap. Our tests also indicated that when in combination with 1-octen-3-ol or lactic acid and hexanoic acid, ammonia was not attractive for *Ae. albopictus* in field.

Results from our field tests showed that all compounds attracted fewer mosquitoes than the control, but there was some minimum attractiveness of the chemicals tested. The concentration, ratio, and composition of chemical attractants in a blend can directly affect the efficacy of mosquitoes captured by a baited trap (Silva et al. 2005). It is conceivable that the composition, concentrations, and proportions of the synthetic compounds tested so far are not optimal. At the same time, host-seeking behavior of mosquitoes in the field also could be influenced by the mosquito physiological states such as sucrose-feeding, blood-feeding, oviposition, age, crowding and the environmental factors such as light, heat, windspeed, and moisture (van Dyk 2002). These factors also would undoubtedly influence the attractiveness of the test compounds for mosquitoes in field, especially single or simple blends, which only had very slight attraction of mosquitoes. In addition, the collection efficiency of mosquitoes can be strongly influenced by trap design. Silva et al. (2005) reported the attractive effects of a blend of acetone, L-lactic acid and dimethyl disulfide on *Ae. aegypti* varied among the trap designs under controlled laboratory conditions. Therefore, it may be important to study more factors in the design of mosquito traps to promote their efficiency of luring and trapping mosquitoes.

Acknowledgment

This work was a part of the project “Study on the Mechanism of Human Volatiles Attractions to *Culex pipiens pallens*” supported by the Natural Science Foundation of China.

References Cited

- Acree Jr., F., R. B. Turner, H. K. Gouck, M. Beroza & N. Smith. 1968. L-Lactic acid: a mosquito attractant isolated from humans. *Science* 161: 1346–1347.
- Bernier, U. R., D. L. Kline, D. R. Barnard, C. E. Schreck & R. A. Yost. 2000. Analysis of human skin emanations by gas chromatography/mass spectrometry. 2. Identification

- of volatile compounds that are candidate attractants for the yellow fever mosquito (*Aedes aegypti*). *Anal. Chem.* 72: 747–756.
- Bernier, U. R., D. L. Kline, C. E. Schreck, R. A. Yost & D. R. Barnard. 2002.** Chemical analysis of human skin emanations: comparison of volatiles from humans that differ in attraction of *Aedes aegypti* (Diptera: Culicidae). *J. Am. Mosq. Control Assoc.* 18: 186–195.
- Bernier, U. R., D. L. Kline, K. H. Posey, M. M. Booth, R. A. Yost & D. R. Barnard. 2003.** Synergistic attraction of *Aedes aegypti* (L.) to binary blends of L-lactic acid and acetone, dichloromethane, or dimethyl disulfide. *J. Med. Entomol.* 40: 653–656.
- Bosch, O. J., M. Geier & J. Boeckh. 2000.** Contribution of fatty acids to olfactory host finding of female *Aedes aegypti*. *Chem. Senses* 25: 323–330.
- Dekker, T., B. Steib, R. T. Cardé & M. Geier. 2002.** L-lactic acid: a human-signifying host cue for the anthropophilic mosquito *Anopheles gambiae*. *Med. Vet. Entomol.* 16: 91–98.
- Geier, M. & J. Boeckh. 1999.** A new Y-tube olfactometer for mosquitoes to measure the attractiveness of host odours. *Entomol. Exp. Appl.* 92: 9–19.
- Geier, M., O. J. Bosch & J. Boeckh. 1999a.** Ammonia as an attractive component of host odour for the yellow fever mosquito, *Aedes aegypti*. *Chem. Senses* 24: 647–653.
- Geier, M., O. J. Bosch & J. Boeckh. 1999b.** Influence of odour plume structure on upwind flight of mosquitoes towards hosts. *J. Exp. Biol.* 202: 1639–1648.
- Gillies, M. T. 1980.** The role of carbon dioxide in host-finding by mosquitoes (Diptera: Culicidae): a review. *Bull. Entomol. Res.* 70: 525–532.
- Meijerink, J. & J. J. A. van Loon. 1999.** Sensitivities of antennal olfactory neurons of the malaria mosquito, *Anopheles gambiae*, to carboxylic acids. *J. Insect Physiol.* 45: 365–373.
- Schreck, C. E., D. L. Kline & D. A. Carlson. 1990.** Mosquito attraction to substances from the skin of different humans. *J. Am. Mosq. Control Assoc.* 6: 406–410.
- Shirai, Y., K. Kamimura, T. Seki & M. Morohashi. 2001.** L-lactic acid as a mosquito (Diptera: Culicidae) repellent on human and mouse skin. *J. Med. Entomol.* 38: 51–54.
- Silva, I. M., Á. E. Eiras, D. L. Kline & U. R. Bernier. 2005.** Laboratory evaluation of mosquito traps baited with a synthetic human odor blend to capture *Aedes aegypti*. *J. Am. Mosq. Control Assoc.* 21: 229–233.
- Takken, W. 1991.** The role of olfaction in host-seeking of mosquitoes: a review. *Insect Sci. Appl.* 12: 287–295.
- Tang, Q. Y. & M. G. Feng. 2002.** DPS Data Processing System for Practical Statistics (DPS software, Version 6.50, May, 2002). Science Press, Beijing, 33–92.
- Vale, G. A. 1980.** Field studies on the responses of tsetse flies (Glossinidae) and other Diptera to carbon dioxide, acetone and other chemicals. *Bull. Entomol. Res.* 70: 563–570.
- van Dyk, J. 2002.** Mosquito Host-Seeking: A Partial Review. <http://www.ent.iastate.edu/dept/research/vandyk/hostseek.html>, pp. 1–20. Accessed on 5 April 2006.
-

Within-Plant Distribution of *Pandora neoaphidis* (Zygomycetes: Entomophthorales) in Populations of the Tobacco-Feeding Form of *Myzus persicae* (Homoptera: Aphididae) on Flue-Cured Tobacco¹

Surendra K. Dara² and Paul J. Semtner

Southern Piedmont Agricultural Research and Extension Center, Virginia Polytechnic Institute and State University, Blackstone, Virginia 23824 USA

J. Agric. Urban Entomol. 23(2): 65–76 (April 2006)

ABSTRACT The entomophthoralean, *Pandora neoaphidis* (Remaudière & Hennebert) Humber, is a common pathogen of the tobacco-adapted form of the green peach aphid (tobacco aphid), *Myzus persicae* (Sulzer) (Homoptera: Aphididae), a major pest of tobacco. This study assessed the within-plant distribution of *P. neoaphidis* in populations of the tobacco aphid on flue-cured tobacco, *Nicotiana tabacum* L., in Virginia during 1994 and 1995. Aphid populations and levels of mycosis were compared among three leaf positions, the bottom, middle, and top thirds of plants in preflowering and flowering growth stages. Each year, aphid populations and proportions of mycosis caused by *P. neoaphidis* were usually highest in the top third of the plant and lowest on the bottom third. However, mycosed aphids were more evenly distributed when the epizootics were in decline. Preflowering plants had significantly higher proportions of mycosed aphids than flowering plants in August when aphid populations were declining. Epizootics of *P. neoaphidis* occurred when temperatures were moderate (highs < 32°C) and weather was wet in late July and early August 1994 and late June and early July 1995. Hot weather (average highs > 35°C) in mid to late July 1995 was associated with a reduced incidence of the pathogen. Compared with the bottom leaf position, the levels of mycosis on the top and middle leaves were more highly correlated with the level of mycosis on the entire plant each year, which suggests that sampling the upper third of the plant for healthy aphids and those infected by *P. neoaphidis* gives a good indication of aphid populations and the levels of mycosed aphids on the entire plant.

KEY WORDS *Myzus nicotianae*, green peach aphid, tobacco aphid, within-plant distribution, *Nicotiana tabacum*, flue-cured tobacco

The tobacco-feeding form of the green peach aphid, *Myzus persicae* (Sulzer) (Homoptera: Aphididae) [= tobacco aphid, *M. nicotianae* Blackman], is a major pest of tobacco in North America (Cheng & Hanlon 1985, Blackman 1987, Lampert 1989, Reed & Semtner 1992). Heavy infestations of the aphid may reduce tobacco yield and value by 10–30% (Reed & Semtner 1992) and cause serious

¹Accepted for publication 9 March 2007

²Current address: Certis USA, 720 5th St, Wasco, California 93280

reductions in leaf quality (Mistic & Clark 1979). A red morph of the tobacco aphid was first observed in 1985 on tobacco, *Nicotiana tabacum* L., and quickly replaced the green morph as the dominant aphid (McPherson 1989, Harlow et al. 1991, Reed & Semtner 1992). The distribution of these aphids varies within the tobacco plant canopy. Webster et al. (1983) and Lampert (1989) reported that densities of the green morph are highest on leaves in the upper third of tobacco plants, but populations are highest on the middle third.

The entomophthoralean, *Pandora neoaphidis* (Remaudière & Hennebert) Humber [= *Erynia neoaphidis* Remaudière & Hennebert] is the most common and dominant pathogen of various aphids on a wide range of host plants (Wilding & Brady 1984, Feng et al. 1991, Kish et al. 1994, McLeod et al. 1998). Yu et al. (1995) reported that *P. neoaphidis* infects the tobacco aphid in the United States. In addition, unidentified fungal pathogens cause epizootics in aphids on tobacco in Greece (Lykouressis & Mentzos 1995). Frequent epizootics of another entomophthoralean, *Neozygites fresenii* (Nowakowski) Batko, result in significant reductions in cotton aphid, *Aphis gossypii* Glover, populations on cotton, often reducing the need for chemical control of the aphid in southeastern United States (Sánchez-Pena 1993, Steinkraus et al. 1995, Wells et al. 2000, Jones et al. 2003).

A better understanding of where aphids with mycosis occur on tobacco plants may aid in the development of techniques for using naturally occurring *P. neoaphidis* to manage aphid infestations on organically grown tobacco, low-residue tobacco, and for managing insecticide-resistant aphids. Information on within-plant distribution of *P. neoaphidis* and its host will provide a better understanding of how they fit into the tobacco ecosystem and aid in the development of methods for predicting epizootics. This study was conducted to determine the within-plant distribution of *P. neoaphidis* in populations of *M. persicae* on flue-cured tobacco in the preflowering and flowering growth stages.

Materials and Methods

Flue-cured tobacco 'K-326' was transplanted on 17 May 1994 (0.08 ha) and 16 May 1995 (0.06 ha) at the Virginia Tech Southern Piedmont Agricultural Research and Extension Center, Blackstone, Virginia. Plant spacing was 122 cm between rows and 56 cm between plants within rows. Standard agronomic practices were followed for flue-cured tobacco production in Virginia (Reed et al. 2005) except that plants were not treated for aphid or sucker control or harvested and the tobacco was not topped (terminal floral portion not removed) because aphid populations often decline after topping (Lampert 1989).

Observations were made at weekly intervals on four dates each year, from 27 July to 15 August 1994 and from 12 July to 2 August 1995. Counts were started after weekly samples indicated that *P. neoaphidis* was present in aphid populations on >50% of the sample plants in the test plots. Aphid populations and mycosis caused by *P. neoaphidis* were monitored on tobacco in two growth stages: preflowering (vegetative to flower buds) and flowering (at least 50% of buds flowering). The two stages occurred simultaneously in the test plots throughout the observation periods. The numbers of healthy aphids and those with mycosis were counted on each leaf of randomly selected plants. Aphids with mycosis included both fungus-killed and live fungus-infected aphids. Fungus-killed aphids turn golden-brown whereas live fungus-infected aphids appear pale (Wilding & Brady

1984). Aphids without symptoms were classified as healthy. On each date and growth stage, 10 and 7 plants were sampled in 1994 and 1995, respectively. Healthy, dead, and infected aphids were counted on each leaf of the sample plants beginning with the bottom harvestable leaf and proceeding up the plant to the bud or inflorescence. Counts were made in the morning when sporulating cadavers were easiest to see. Before analysis, leaves from each plant were grouped into three leaf positions each containing about six leaves and the numbers of aphids and levels of *P. neoaphidis* infection among the leaf positions were analyzed and compared. Temperature and rainfall data were obtained from a weather station (OmniData International Inc., Logan, Utah) located about 0.5 km from the test fields.

Identification of *P. neoaphidis*. On each sampling date, conidia from cadavers on 5–10 plants in each field were collected on cover slips and examined in lactophenol at 400X under a phase-contrast microscope. All cadavers examined were identified as *P. neoaphidis* based on the descriptions of Remaudière & Henebert (1980), Wilding & Brady (1984), and Humber (1989). The primary characteristics of *P. neoaphidis* included the shape of the conidia, the absence of resting spores and the shapes of the cystidia and rhizoids (Humber 1989).

Statistical analysis. PROC MIXED (SAS Institute 2000) was used to analyze data for leaf position and growth stage within date and year for total aphids and the percentage of aphids with mycosis caused by *P. neoaphidis*. The random effects were replication and the interactions of replication with the fixed effects. To normalize data aphid counts were transformed to $\text{Log}_{10}(x + 1)$ and percentage data for mycosis was transformed with an arcsine, square root transformation before analysis. Significantly different means within dates and stages were separated by Tukey-Kramer honestly significant difference mean separation test ($P < 0.05$; SAS Institute 2000). In addition, Pearson correlation was used to determine the relationship of percentage mycosis for each leaf position with the percentage of mycosis for the entire plant (SAS Institute 2000).

Results

In 1994, tobacco aphid populations peaked at >8,000 aphids per plant on 18 July and averaged >3,500 aphids per plant on 27 July and 1 August (first and second observation dates), decreasing to 700 aphids per plant on 15 August, the fourth observation date (Fig. 1A and B). In 1995, aphid populations peaked between 12 and 19 July (Fig. 2A and B). Populations for the two growth stages averaged >13,000 aphids per plant on 12 and 19 July and decreased to 1027 and 183 aphids per plant on preflowering and flowering stages, respectively, on 2 August. Aphid populations over the four sample dates in the 1995 study were 3.2-fold higher than in 1994 and most of this difference was related to observations starting 15 days earlier in 1995 than in 1994, but *P. neoaphidis* appeared earlier in 1995 than in 1994. Aphid populations were similar for both growth stages on 26 July 1995 and 27 July 1994 and for the full flower stages for the 1 August 1994 and 2 August 1995 sampling dates.

Pandora neoaphidis was first observed on 18 July 1994 and peaked at over 80% mycosis on the upper leaves on 8 August (Fig. 1C and D). In 1995, infected aphids were first observed on 1 July and the epizootic peaked at 24–28% mycosis on the upper leaves on 12 July (Fig. 2C and D).

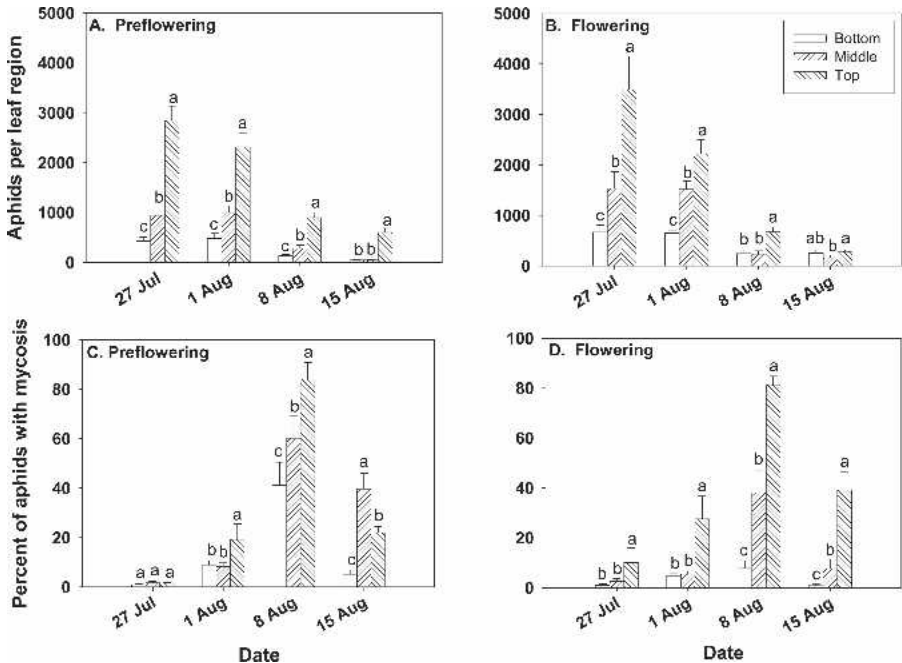


Fig. 1. Within-plant distribution of the tobacco aphid (A and B) and the percentage of aphids with mycosis caused by *P. neoaphidis* (C and D) among three leaf positions on preflowering and flowering plants, 1994. Bars with same letters within each date were not significantly different ($P < 0.05$).

Aphid Distribution. Leaf position had highly significant effects ($F = 102.35$, $df = 2, 18$, $P < 0.0001$ for 1994; $F = 348.63$, $df = 2, 12$, $P < 0.0001$ for 1995) on tobacco aphid distribution throughout the study each year (Figs. 1A, B and 2A, B). The highest aphid populations usually occurred on the top third of the plant. The differences among the two growth stages were not significant ($t = 0.58$, $df = 1, 15$, $P < 0.4450$). However, within dates, aphid distribution varied between stages with significantly higher populations on preflowering plants on 26 July ($t = 6.88$, $df = 2, 12$, $P = 0.0394$) and 2 August ($t = 5.44$, $df = 2, 12$, $P = 0.0584$) 1995 when 92–98% of the aphids were on the top third of the plant (Figs. 1B and 2A, B).

Each year, aphid populations increased from the bottom to the top of both preflowering and flowering plants except on 15 August 1994 on the flowering plants where populations were similar on bottom and top leaf positions and on 12 July 1995 on both stages of plants where populations were similar on the middle and upper leaf positions (Figs. 1A, B and 2A, B).

Distribution of *P. neoaphidis*. Leaf position had a significant effect on the incidence of *P. neoaphidis* mycosis in tobacco aphid populations (Figs. 1C, D and 2C, D). The highest levels of aphid mycosis occurred on the top leaves on 1 ($F = 9.08$ $df = 2, 18$, $P \leq 0.0005$) and 8 August ($F = 40.66$, $df = 2, 18$, $P \leq 0.0001$) 1994, and 12 July 1995 ($F = 7.05$, $df = 2, 12$, $P = 0.0031$). On 15 August 1994,

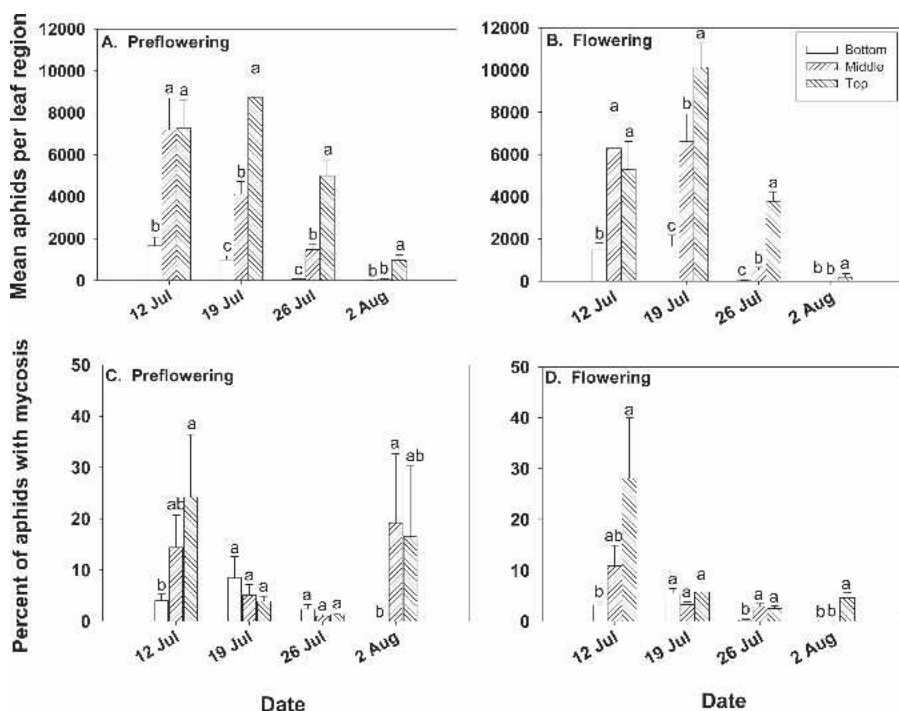


Fig. 2. Within-plant distribution of the tobacco aphid (A and B) and the percentage of mycosis caused by *P. neoaphidis* (C and D) among three leaf positions on plants in preflowering and flowering growth stages, 1995. Bars with same letters within each date were not significantly different ($P < 0.05$).

the incidence of *P. neoaphidis* was highest on the top third of flowering plants ($F = 33.78$, $df = 2, 18$, $P \leq 0.0001$) and the middle third of preflowering plants ($F = 25.43$, $df = 2, 12$, $P \leq 0.0001$; Fig. 1C, D) and on both stages on 26 July 1995 ($F = 6.13$, $df = 2, 12$, $P = 0.0058$) and on 2 August 1995 ($F = 5.33$, $df = 2, 12$, $P = 0.0104$; Fig. 2C, D). The stage of plant development also affected the levels of mycosis with the highest percentages of mycosed aphids on preflowering plants on 8 August 1994 ($t = 18.53$, $df = 1, 18$, $P \leq 0.0001$). On 2 August 1995, no *P. neoaphidis* was observed on the middle and bottom leaf positions of flowering plants or on the bottom third of preflowering plants. Aphid populations on the middle and bottom leaves of each growth stage made up <8% of the aphids on the entire plant on 2 August.

1994. *Pandora neoaphidis* was first observed in *M. persicae* populations on 18 July. Mycosis was generally most prevalent in aphids on the top third of the plant, particularly when the epizootic was increasing or at its peak on 1 and 8 August (Fig. 1C, D). On the top third of the plant, mycosis caused by *P. neoaphidis* increased from 1.5% on preflowering plants and 10% on flowering plants on 27 July to peaks of 84% on preflowering plants and 78% on flowering plants on 8

August (Fig. 1C, D). On 15 August, levels of mycosis on the top leaf position had declined to 22% on preflowering plants and 39% on flowering plants (Fig. 1C, D). Several factors probably contributed to the decline in mycosis between 8 and 15 August. These included warm, dry conditions (Fig. 2A) that are less favorable for *P. neoaphidis* sporulation (Yu et al. 1995) and germination (Shah et al. 2002), the development of new leaves above those with infected aphids, and reductions in aphid populations caused by predators and a decline in the quality of the host plant. In 1994, daily maximum and minimum temperatures averaged 33.0°C and 21.2°C, respectively between 13 and 27 July, and rainfall was moderate (29 mm) from 21–27 July (Fig. 3A). During the epizootic between 27 July and 15 August, the temperatures were relatively cool with maximum and minimum temperatures averaging 31.0 and 17.5°C, respectively and total rainfall was <10 mm. This cool weather provided conditions for sporulation and germination that might have helped the pathogen spread rapidly through the aphid populations. Low temperatures of 12°C and highs of 28–30°C occurring on 6–8 August might have contributed to the high levels of sporulation and symptoms of mycosis on 8 August.

1995. Very low levels of *P. neoaphidis* were first observed on 1 July. Mycosis on the top leaf position peaked at 24–28% on 12 July (first observation date), declined to 1.3–2.5% on 26 July, and increased to 4.6–16.5% on 2 August (Fig. 2C, D). The distribution of *P. neoaphidis* was less distinct than in 1994 with the percentages of aphids with mycosis being significantly higher on the top two-thirds and one-third of the plant on 12 July and 2 August, respectively (Fig. 2C, D). On 19 and 26 July, mycosis levels were similar for the three leaf positions (Fig. 2C, D). The epizootic peaked much earlier in the season and levels of mycosed aphids were much lower than in 1994. Aphid populations built up rapidly and extremely wet weather and moderate temperatures in late June and early July were ideal for the development of an epizootic of *P. neoaphidis* (Fig. 3B). Between 21 and 30 June, rainfall totaled 108 mm and maximum and minimum temperatures averaged 30.8 and 19.7°C, respectively. Between 1 July and when the mycosis peaked on 12 July, maximum and minimum temperatures averaged 31.9 and 18.1°C, respectively and total rainfall was >100 mm including 47 mm on 11 July. Although this relatively cool, moist weather was associated with the establishment of the pathogen, high temperatures (daily maxima 33–38°C) and dry weather after 12 July coupled with a general decrease in aphid populations after 18 July might have contributed to the rapid decline in infection levels between 12 and 26 July. Lower temperatures, moderate rainfall, high humidity and leaf wetness after 26 July might have contributed to the increase in the percentages of mycosis despite low aphid populations on 2 August.

The percentage of aphids with mycosis on the upper and middle thirds of the plant were highly correlated ($r > 0.60$) with the percentage of mycosed aphids on the whole plant having 12 of the 16 date by stage combinations for the top third and 11 of 15 date by stage combinations for middle third with an $r > 0.60$ (Table 1). This study indicates that samples from the upper and middle leaves give the best estimate of the incidence of *P. neoaphidis* in the tobacco-feeding form of the tobacco aphid in the field. Although the incidence of *P. neoaphidis* was higher on the lower leaves on one occasion in this study, mycosis in aphids on the upper

Table 1. Correlation coefficients (r) and probability levels for the incidence of *P. neoaphidis* by leaf position by whole plant average, Blackstone, Virginia, 1994–1995.

Year	Plant stage	Leaf position	27 July		1 August		8 August		15 August	
			r	P > F	r	P > F	r	P > F	r	P > F
1994	Flowering	Bottom	0.6664	0.0013	0.4150	0.0688	0.4160	0.0681	0.5310	0.0160
		Middle	0.7499	0.0001	0.8521	0.0001	0.6590	0.0016	0.5841	0.0068
		Top	0.8812	0.0001	0.8384	0.0001	0.4411	0.0516	0.8220	0.0001
	Preflowering	Bottom	-0.0653	0.7844	0.1410	0.5540	0.6350	0.0026	0.2442	0.2995
		Middle	0.8169	0.0001	0.7113	0.0004	0.7612	0.0001	0.6649	0.0014
		Top	0.9371	0.0001	0.9243	0.0001	0.9434	0.0001	0.3872	0.1097
1995	Flowering	Bottom Middle Top	12 July		19 July		26 July		2 August	
			r	P > F	r	P > F	r	P > F	r	P > F
			0.3348	0.4190	0.1832	0.5307	-0.2331	0.4225	—	—
	Preflowering	Bottom Middle Top	0.7561	0.0018	0.6255	0.0167	0.3199	0.2649	—	—
			0.9814	0.0001	0.9645	0.0001	0.6968	0.0056	0.4338	0.1212
			0.7090	0.0045	0.7947	0.0007	0.0840	0.7755	—	—
			0.8057	0.0005	0.6550	0.0110	0.1325	0.6517	0.5483	0.0120
			0.9915	0.0001	0.7938	0.0007	0.4188	0.1361	0.9979	0.0001

leaves, where aphid populations are usually higher, represented the best estimate of the level of *P. neoaphidis* infection.

Discussion

Although maximum temperatures in the tobacco-producing region of Virginia exceed levels that favor *P. neoaphidis*, minimum temperatures and high humidities during the nights are usually favorable for sporulation and germination. Yu et al. (1995) reported that 10–25°C is optimum temperature for sporulation by *P. neoaphidis*, but no sporulation occurs at 30°C. FuRong et al. (2005) reported that 25°C was optimum for germination by *P. neoaphidis* on *M. persicae* and that the developmental time between inoculation and sporulation was 2.9–3.9 days depending on the temperature. Shah et al. (2002) found less sporulation in environments with temperatures fluctuating between 18 and 30°C than for those changing between 18 and 20 or 25°C. In our study, daily field temperatures generally ranged from 20 and 32°C with extremes from 12–34°C in 1994 and between 20 and 36°C in 1995 with extremes of 15–38°C (Fig. 3A, B). Freestanding water or humidities >93% also favor sporulation by the pathogen (Yu et al. 1995, Shah et al. 2002) although FuRong et al. (2005) reported successful germination of *P. neoaphidis* conidia at 85% RH in studies on *M. persicae*.

The within-plant distribution of *P. neoaphidis* appeared to be influenced by the weather conditions and whether the epizootic was increasing or declining. Although aphid populations were higher in 1995 than in 1994, the level of mycosis was more evenly distributed among the leaf positions possibly because the pathogen occurred at low levels. Our observations were similar to those for the levels of unidentified fungal pathogens in *M. nicotianae* in Greece, where the levels of mycosis varied between the upper and lower halves of tobacco plants when the disease incidence was low (Lykouressis & Mentzos 1995). When the disease was at its peak, aphids on the upper half of the plant had significantly higher infection levels than those on the bottom half.

In general, levels of mycosis caused by *P. neoaphidis* were similar on both preflowering and flowering plants when the epizootics were at low levels. However, higher levels of mycosis occurred on preflowering plants than on flowering plants when the epizootics were increasing or peaking in 1994 (Fig. 1C, D). Brobyn et al. (1985) found that conidia of *P. neoaphidis* were viable for longer periods on lower leaves than on upper leaves of field beans (*Vicia faba* L.). This was attributed to the protection of lower leaves from harmful radiation that inactivates the conidia and from rainfall that washes conidia off the leaves. In addition, the bottom leaves of tobacco stay wet from dew later in the morning than do the upper leaves (Norse 1971). Therefore, one would expect higher infection levels on the bottom leaf position because microclimatic conditions are more favorable for the pathogen to germinate and sporulate. Prevalence of *N. fresenii* was also higher in *A. gossypii* in the lower parts of cotton canopy than in higher parts (O'Brien et al. 1993, Hollingsworth et al. 1995). However, that was not the case in this study.

Greater percentages of mycosed aphids probably occurred on the top leaves for several reasons. First, the much higher aphid densities on the upper leaves are more favorable for epizootics than the more sparsely populated lower leaves.

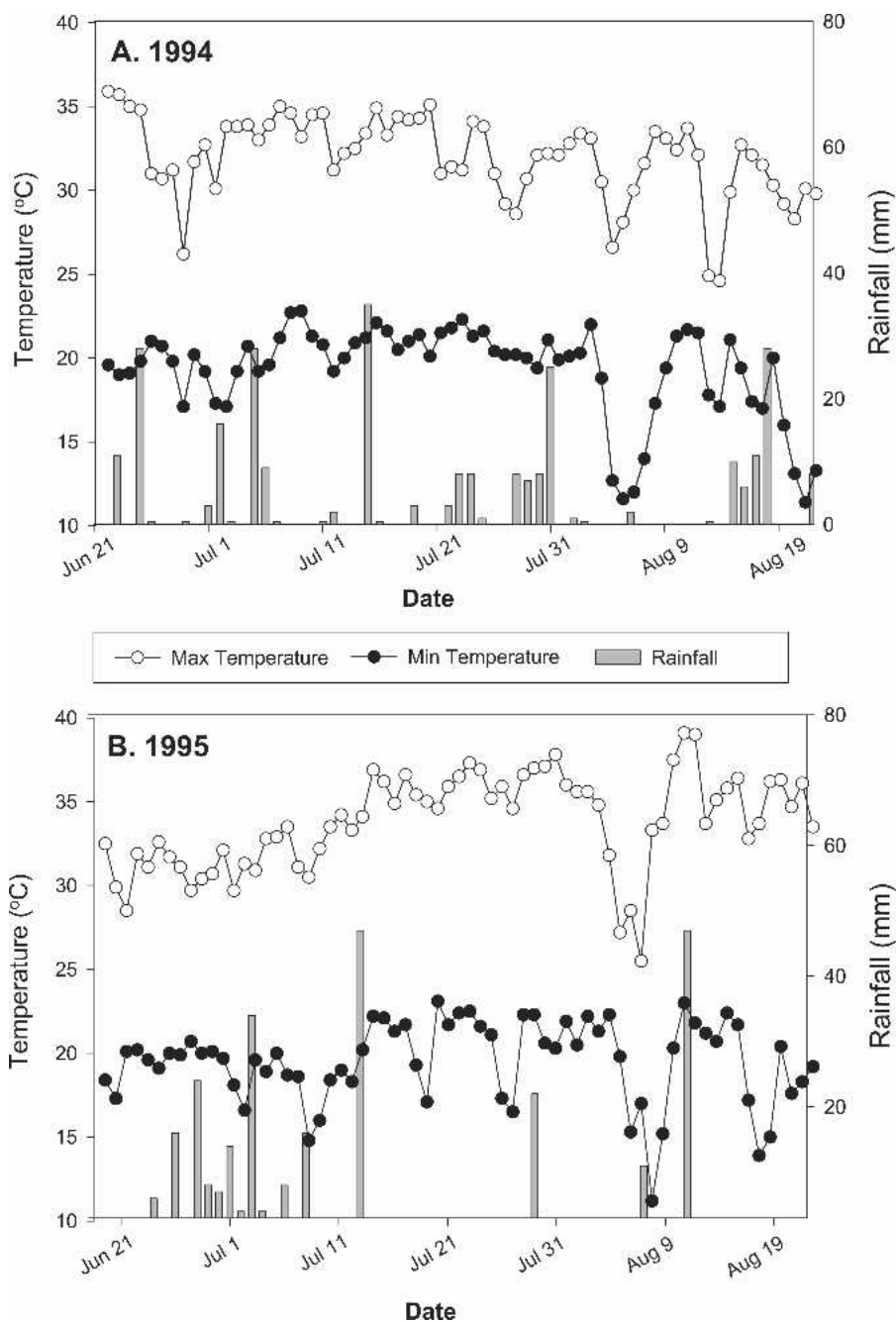


Fig. 3. Maximum and minimum temperatures and daily rainfall, Southern Piedmont AREC, Blackstone, Virginia, 20 June to 25 August 1994 (A) and 1995 (B).

Second, alate aphids infected with the pathogen and migrating in from other fields are more likely to land on the upper leaves and initiate an epizootic (Chen & Feng 2004). Third, wind currents may carry more airborne inoculum (conidia) to the upper leaves than the lower leaves. Fourth, because most aphids feed on the abaxial side of the leaf, those on the upper leaves are probably exposed to more airborne fungal inoculum than aphids feeding on the more protected undersides of the bottom leaves (Webster et al. 1983, Lampert 1989).

Thurston (1961) reported that aphid populations are greater on the upper leaves than on the middle or lower leaves of most tobacco cultivars because the nicotine levels are lowest on the youngest upper leaves. Aphids first increase on the lower leaves and then move up the plant as the leaves mature. In addition, Brown et al. (1995) demonstrated that green leaf volatiles from tobacco inhibit germination of *P. neoaphidis* spores until they encounter a potential aphid host. Differences in the leaf surface chemistry among leaf positions may influence the pathogen because several of these chemicals are toxic to some phytopathogenic fungi (Severson et al. 1985, Kennedy et al. 1992). Diterpene compounds isolated from *Nicotiana* spp. inhibit germination and development of several phytopathogenic fungi, including *Erysiphe cichoracearum* DC (tobacco powdery mildew pathogen) and *Peronospora tabacina* Adam (blue mold) (Bailey et al. 1974). Conidial germination of the entomopathogenic fungus, *Metarhizium anisopliae* (Metschnikoff) Sorokin, and its virulence to the mustard beetle, *Phaedon cochleariae* Fabricius, was increased by the leaf extracts of cruciferous plants (Inyang et al. 1999). The overall difference in the leaf surface chemistry of preflowering and flowering plants might have influenced the levels of mycosis in aphids on plants in these two stages of development. However, Court (1982) found that duvatrienediol content increases toward the top leaf positions in flue-cured tobacco at flowering and just before topping.

Because the percentage of mycosis in aphid populations was generally higher in upper leaf positions in our study, duvatrienediols probably had little influence on inoculum of *P. neoaphidis*. Higher numbers of aphids in upper leaf positions also seem to be an important factor that might have promoted the spread of the inoculum when weather and host conditions were favorable. Feng et al. (1991) reported that the development of entomophthoralean infections in cereal aphids on wheat were more associated with host density than with temperature and rainfall. Similarly, infections of *P. neoaphidis* and another fungal pathogen, *Schizolachnus piniradiatae* (Davidson) were associated with high levels of inoculum and high densities of their aphid hosts (Wilding 1975, Soper & MacLeod 1981). Our results also suggest that, within each tobacco plant, higher infection levels, in general, were associated with higher aphid populations in the upper parts of the plant when epizootic was at or near its peak. A study based on the measurement of inoculum levels and microenvironmental conditions at different leaf positions may provide further insight on the interrelation among levels of infection and of inoculum, and microenvironmental differences among leaf positions.

Acknowledgment

We are grateful for the financial support provided by R. J. Reynolds Tobacco Company and the Virginia Bright Flue-Cured Tobacco Board.

References Cited

- Bailey, J. A., G. G. Vincent & R. S. Burden. 1974. Diterpenes from *Nicotiana glutinosa* and their effect on fungal growth. *J. Gen. Microbiol.* 85: 57–64.
- Blackman, R. L. 1987. Morphological discrimination of a tobacco-feeding form from *Myzus persicae* (Sulzer) (Homoptera: Aphididae), and a key to New World *Myzus* (*Nectarosiphon*) species. *Bull. Entomol. Res.* 77: 713–730.
- Brobyn, P. J., N. Wilding & S. J. Clark. 1985. The persistence of infectivity of conidia of the aphid pathogen *Erynia neoaphidis* on leaves in the field. *Ann. Appl. Biol.* 107: 365–376.
- Brown, G. C., G. L. Prochaska, D. F. Hilderbrand, G. L. Nordin & D. M. Jackson. 1995. Green leaf volatiles inhibit conidial germination of the entomopathogen, *Pandora neoaphidis* (Entomophthorales: Entomophthoraceae). *Environ. Entomol.* 24: 1637–1643.
- Chen, C. & M. Feng. 2004. *Sitobion avenae* alate infected by *Pandora neoaphidis*: their flight ability, post-flight colonization, and mycosis transmission to progeny colonies. *J. Invert. Path.* 86: 117–123.
- Cheng, H. H. & J. J. Hanlon. 1985. Effects of green peach aphid, *Myzus persicae* (Sulzer), on yield and quality of flue-cured tobacco in Ontario. *Tob. Sci.* 29: 144–148.
- Court, W. A. 1982. Factors affecting the concentration of duvatrienediols of flue-cured tobacco. *Tob. Sci.* 24: 40–43.
- Feng, M., J. B. Johnson & S. E. Halbert. 1991. Natural control of cereal aphids (Homoptera: Aphididae) by entomopathogenic fungi (Zygomycetes: Entomophthorales) and parasitoids (Hymenoptera: Braconidae and Encyrtidae) on irrigated spring wheat in southwestern Idaho. *Environ. Entomol.* 20: 1699–1710.
- FuRong, G., L. YaHong & L. ZhengVue. 2005. Effect of temperature on virulence of *Pandora neoaphidis* against *Myzus persicae*. *Plant Protection* 31: 61–64.
- Harlow, C. D., S. P. Southern & E. P. Lampert. 1991. Geographic distribution of two color forms, carboxylesterase activity, and chromosome configuration of the tobacco aphid (Homoptera: Aphididae) in North Carolina. *J. Econ. Entomol.* 84: 1175–1179.
- Hollingsworth, R. G., D. C. Steinkraus & R. W. McNew. 1995. Sampling to predict fungal epizootics in cotton aphid (Homoptera: Aphididae). *Environ. Entomol.* 24: 1414–1421.
- Humber, R. A. 1989. Synopsis of a revised classification for the Entomophthorales (Zygomycotina). *Mycotaxon* 34: 441–460.
- Inyang, E. N., T. M. Butt, A. Beckett & S. Archer. 1999. The effect of crucifer epicuticular waxes and leaf extracts on the germination and virulence of *Metarhizium anisopliae* conidia. *Mycol. Res.* 103: 419–426.
- Jones, R. H., B. R. Leonard & R. D. Bagwell. 2003. Fungus helps control Louisiana cotton aphids. *Louisiana Agriculture Magazine* 46: 24–25.
- Kennedy, B. S., M. T. Nielsen, R. F. Severson, V. A. Sisson, M. K. Stephenson & D. M. Jackson. 1992. Leaf surface chemicals from *Nicotiana* affecting germination of *Peronospora tabacina* (Adam) sporangia. *J. Chem. Ecol.* 18: 1467–1479.
- Kish, L. P., I. Majchrowicz & K. D. Biever. 1994. Prevalence of natural fungal mortality of green peach aphid (Homoptera: Aphididae) on potatoes and nonsolanaceous hosts in Washington and Idaho. *Environ. Entomol.* 23: 1326–1330.
- Lampert, E. P. 1989. Seasonal abundance and within-plant distribution of aphids (Homoptera: Aphididae) on flue-cured tobacco. *J. Econ. Entomol.* 82: 114–118.
- Lykouressis, D. P. & G. V. Mentzos. 1995. Effects of biological control agents and insecticides on the population development of *Myzus nicotianae* Blackman (Homoptera: Aphididae) on tobacco. *Agric. Ecosyst. Environ.* 52: 57–64.
- McLeod, P. J., D. C. Steinkraus, J. C. Correll & T. E. Morelock. 1998. Prevalence of *Erynia neoaphidis* (Entomophthorales: Entomophthoraceae) infections of green peach aphid (Homoptera: Aphididae) on spinach in the Arkansas River Valley. *Environ. Entomol.* 27: 796–800.

- McPherson, R. M. 1989.** Seasonal abundance of red and green morphs of the tobacco aphid (Homoptera: Aphididae) on flue-cured tobacco in Georgia. *J. Entomol. Sci.* 24: 531–538.
- Mistic, W. J. & G. B. Clark. 1979.** Green peach aphid injury to flue-cured tobacco leaves. *Tob. Sci.* 23: 23–24.
- Norse, D. 1971.** Lesion and epidemic development of *Alternaria longipes* (Ell. & Ev.) Mason on tobacco. *Ann. Appl. Biol.* 69: 105–123.
- O'Brien, P. J., M. B. Stoetzel, R. C. Navasero & J. B. Graves. 1993.** Field biology studies of the cotton aphid, *Aphis gossypii* Glover. *Southwestern Entomol.* 18: 25–35.
- Reed, T. D. & P. J. Semtner. 1992.** Effects of tobacco aphid (Homoptera: Aphididae) populations on flue-cured tobacco production. *J. Econ. Entomol.* 85: 1963–1971.
- Reed, T. D., C. S. Johnson, P. J. Semtner & C. A. Wilkinson. 2005.** 2006 flue-cured tobacco production guide. Virginia Coop. Ext. Serv. Publ. 436-048. 117 p.
- Remaudière, G. & G. L. Hennebert. 1980.** Revision systematique de *Entomophthora aphidis* Hoffm. In Fres. Description de deux nouveaux pathogenes d'aphides. *Mycotaxon* 11: 269–321.
- Sánchez-Pena, S. R. 1993.** Entomogenous fungi associated with the cotton aphid in the Texas high plains. *Southwestern Entomol.* 18: 69–71.
- SAS Institute. 2000.** SAS User's Guide: Statistics, release 8.2. SAS Institute, Inc., Cary, North Carolina.
- Severson, R. F., A. W. Johnson & D. M. Jackson. 1985.** Cuticular constituents of tobacco: factors affecting their production and their role in insect and disease resistance and smoke quality. *Rec. Adv. Tob. Sci.* 11: 105–174.
- Shah, P. A., M. Aebi & U. Tuor. 2002.** Effects of constant and fluctuating temperatures on sporulation and infection by the aphid-pathogenic fungus *Pandora neoaphidis*. *Entomol. Exp. Appl.* 103: 257–266.
- Soper, R. S. & D. M. MacLeod. 1981.** Descriptive epizootiology of an aphid mycosis. *USDA Tech. Bull.* 1632.
- Steinkraus, D. C., R. G. Hollingsworth & P. H. Slaymaker. 1995.** Prevalence of *Neozygites fresenii* (Entomophthorales: Neozygitaceae) on cotton aphids (Homoptera: Aphididae) in Arkansas cotton. *Environ. Entomol.* 24: 465–474.
- Thurston, R. 1961.** Resistance in *Nicotiana* to the green peach aphid and some other tobacco insect pests. *J. Econ. Entomol.* 54: 946–949.
- Webster, J. A., R. Thurston & R. P. Hoxie. 1983.** Green peach aphid distribution on burley tobacco in the field. *Tob. Sci.* 27: 41–44.
- Wells, M. L., R. M. McPherson, J. R. Ruberson & G. A. Herzog. 2000.** Effect of fungicide application on activity of *Neozygites fresenii* (Entomophthorales: Neozygitaceae) and cotton aphid (Homoptera: Aphididae) suppression. *J. Econ. Entomol.* 93: 1118–1126.
- Wilding, N. 1975.** *Entomophthora* species infecting pea aphids. *Trans. Royal Entomol. Soc.* 127: 171–183.
- Wilding, N. & B. L. Brady. 1984.** *Erynia neoaphidis*. CMI Desc. of Pathogenic Fungi and Bacteria, No. 815. The Commonwealth Mycological Institute, Kew, Surrey, England.
- Yu, Z., G. L. Nordin, G. C. Brown & D. M. Jackson. 1995.** Studies on *Pandora neoaphidis* (Entomophthorales: Entomophthoraceae) infectious to the red morph of tobacco aphid (Homoptera: Aphididae). *Environ. Entomol.* 24: 962–966.
-

Survival and Growth of *Diabrotica balteata* Larvae on Insect-Resistant Sweetpotato Genotypes^{1,2}

D. Michael Jackson and Janice R. Bohac

USDA, ARS, U.S. Vegetable Laboratory, Charleston, South Carolina 29414 USA

J. Agric. Urban Entomol. 23(2): 77–86 (April 2006)

ABSTRACT Production of sweetpotatoes is severely limited by several insect pests, and new pest-management approaches for this crop are needed. A host plant resistance research program typically depends on reliable bioassay procedures to streamline evaluation of germplasm. Thus, a laboratory bioassay procedure was developed for larvae of the banded cucumber beetle, *Diabrotica balteata* LeConte (Coleoptera: Chrysomelidae). Plugs (0.9 cm diam.) of sweetpotato peel (periderm plus cortex), cortex, or stele were placed into microcentrifuge tubes (1.5 mL) containing 1.0 mL of water–agar to prevent desiccation. One second instar, *D. balteata*, was added to each microcentrifuge tube and held at 25°C for 12 d. Surviving larvae were weighed. The peels and stele of five sweetpotato genotypes ('Beauregard', 'Charleston Scarlet', 'Liberty', 'Regal', and SC1149–19) were evaluated. *Diabrotica* larvae grew larger when they were fed stele from any of the sweetpotato genotypes than when they were fed peels of any genotype. Larval growth was not different among genotypes for any of the stele treatments. However, larval growth on the peels of the resistant genotypes ('Liberty', 'Regal', and 'Charleston Scarlet') was significantly lower than for the susceptible cultivars ('Beauregard' or SC1149–19). Larvae did not survive well or gain much weight when they were starved or fed only the scraped periderm of 'Regal.' There was no difference in the inhibition of larval growth by the cortex or the peel of 'Regal' roots. In general, survival effects were not as dramatic as differences in weight gain. These bioassays were consistent with field results, indicating that these techniques could be useful for evaluating pest resistance in sweetpotato genotypes for *Diabrotica* larvae and other insect species.

KEY WORDS host plant resistance, sweetpotato, *Ipomoea*, *Diabrotica*, bioassay

Several soil-inhabiting insect pests cause economic damage to the edible roots of sweetpotato, *Ipomoea batatas* (L.) Lam. (Convolvulaceae) (Cuthbert 1967, Schalk & Jones 1985). Control of these pests with insecticides is difficult, unreliable, expensive, and may lead to environmental or safety concerns. Thus, there is a need for integrated pest management (IPM) techniques that reduce the im-

¹Accepted for publication 19 March 2007.

²Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture.

pect of pesticides in sweetpotato production (Talekar 1988, Lawrence et al. 2005). Host plant resistance is one such insect control technique that could result in a more environmentally friendly IPM approach in sweetpotato production (Jansson & Raman 1991, Jackson et al. 2002).

The spotted cucumber beetle (or southern corn rootworm), *Diabrotica undecimpunctata howardi* Barber, and the banded cucumber beetle, *D. balteata* LeConte, are two of the most important vegetable pests in the Western Hemisphere (McKinlay 1992, Capinera 2001). These species belong in the highly polyphagous *fucata* species group (Coleoptera: Chrysomelidae: Galerucinae: Luperini: Diabroticina) and are nondiapausing and multivoltine (Branson & Krysan 1981). They have wide host ranges (Saba 1970, Krysan & Miller 1986), and are among the most damaging insect pests of sweetpotato in the southeastern United States (Cuthbert 1967, Chalfant et al. 1990). The spotted cucumber beetle is a more widespread pest than the banded cucumber beetle that is found only in the southern United States and further south (Pitre & Kantack 1962, Krysan & Miller 1986).

Because insect pests are seldom found when sweetpotato roots are dug, it is difficult to determine whether root injury was caused by wireworms (W) (Coleoptera: Elateridae), cucumber beetles (*Diabrotica* sp. [D]), or flea beetles (*Systena* sp. [S]) (Coleoptera: Chrysomelidae), so damage by this coleopteran pest complex is often lumped into a single category called WDS (Cuthbert & Davis 1971, Schalk et al. 1991). WDS damage can be caused by one or more of these pests, including a combination of cucumber beetle species. However, it was shown that the level of host resistance to the WDS complex could be significantly increased after only four generations of mass selection in randomly crossing populations of sweetpotato genotypes (Cuthbert & Jones 1972). Using these recurrent mass selection techniques (Jones et al. 1986) and a rating system based on a weighted index (Cuthbert & Jones 1972), several improved sweetpotato varieties and advanced breeding lines with resistance to the WDS complex have been developed and released by the USDA, ARS at the U. S. Vegetable Laboratory (USVL), Charleston, South Carolina (Schalk et al. 1991, Collins & Hall 1992, Bohac et al. 2000, 2002, 2003, Bohac & Jackson 2005).

A host plant resistance research program typically depends on reliable bioassay procedures to streamline evaluation of germplasm (Smith et al. 1994). Also, a well-designed bioassay can be critical to advancing an understanding of the mechanisms of host plant resistance to insects in a particular crop. Thus, for the study described herein, we developed a laboratory bioassay to evaluate the survival and development of banded cucumber beetle larvae on storage roots of resistant and susceptible sweetpotato genotypes. Previously, we developed a laboratory bioassay for measuring resistance of sweetpotato genotypes to adult *Diabrotica* species (Jackson & Bohac 2007).

Materials and Methods

Five sweetpotato genotypes were evaluated in these experiments, including two insect-susceptible, orange-fleshed cultivars ('Beauregard' and SC1149-19), two insect-resistant, orange-fleshed cultivars ('Charleston Scarlet' and 'Regal'), and a dry-fleshed cultivar ('Liberty'). All of the sweetpotato entries except 'Beauregard' were from the USDA, ARS breeding program and were developed using

recurrent or mass selection techniques (Jones 1965, Jones et al. 1986). Controls consisted of treatments in which cucumber beetles were fed sprouting wheat (Schalk 1986). 'Charleston Scarlet' and 'Regal' have high levels of resistance to the WDS complex in the field (Jones et al. 1985, Bohac & Jackson 2005, USDA 2005). 'Liberty' has low-to-moderate levels of resistance to the WDS complex (Bohac et al. 2003). 'Beauregard' and SC1149-19 are susceptible to most soil insect pests, including WDS (Rolston et al. 1987, Collins & Hall 1992, USDA 2005).

Storage roots used for these experiments were grown in field plots in Bamberg County, South Carolina, 2003, using standard production practices, except that no insecticides were used. After harvest, the roots were cured at about 45°C for 7–10 d then stored at 15–17°C until needed for the bioassays. Undamaged roots of each genotype were selected for bioassays. While being careful not to scratch the periderm, roots were washed in a 0.1% solution of Tween® 20 (ICI Americas Inc., Bridgewater, New Jersey) nonionic detergent, rinsed with de-ionized water, and allowed to air-dry.

Pieces of unblemished sweetpotato peel (defined here as the periderm plus cortex with stele removed) were then cut from the sweetpotato roots using a sharp knife. Because the thickness of periderm and cortex vary among sweetpotato genotypes (Schalk et al. 1986b), care was taken to ensure that the periderm was not damaged and that no stele portion of the sweetpotato root was left attached to the peel samples. Then, plugs (0.9 cm diam.) were taken from each peel using a #5 cork-hole borer. Plugs of stele, 3–5 mm thick, also were obtained from these sweetpotato roots. Plugs of peel or stele were soaked for 30 min in a 1.5% solution of captan (Captan® Fungicide 50WP, Southern Agricultural Insecticides, Inc., Palmetto, Florida) to prevent growth of fungi. A preliminary experiment showed that there were no measurable effects on survival or feeding behavior of cucumber beetle larvae that were fed either root plugs that had been soaked for 30 min in a captan solution compared with larvae fed untreated root plugs. However, on the untreated root plugs, there often was excessive growth of fungi. The pieces of peel or stele were then allowed to air-dry for 3–4 h. During this drying period, several layers of the outermost parenchyma cells desiccate followed by a gradual suberization of the parenchyma cells below the desiccated cells (McClure 1960, Walter & Schadel 1983). Within a few days, a lignified wound-periderm layer, 3–7 cells thick, forms beneath the suberized cells (Artschwager & Starrett 1931, Morris & Mann 1955).

Each piece of sweetpotato was placed, periderm-side up, into an autoclaved 1.5-mL graduated microcentrifuge tube (clear copolymer polypropylene, Product No. 1415-2500, USA Scientific, Ocala, Florida) that was 20 mm in diameter, 40 mm in height, and had a flat cap. To help prevent desiccation, each microcentrifuge tube was filled with 1.0 mL of water-agar prepared by adding 3.75 g of agar (Fisher Scientific Worldwide, Hampton, New Hampshire) to 250 mL of boiling water. The sweetpotato plug was pushed firmly onto the water-agar. The sweetpotato plug was slightly smaller in diameter than the inside of the microcentrifuge tube. These sweetpotato pieces tended to swell slightly and would become dislodged if they initially fit too tightly. Additional water-agar was placed around the edge of the plug with a pipette to prevent insect larvae from getting underneath it and tunneling into the water-agar below. Care was taken to ensure that only periderm was exposed, and that the edges of the cut pieces were completely covered by the water-agar, which was then allowed to dry. Thus, the insects were

forced to initially feed through the unblemished periderm before reaching the cortex of the sweetpotato peel.

A laboratory colony of *D. balteata* was originally started from adults collected from cucurbit fields at the USVL, Charleston, South Carolina. The colony has been maintained continuously for more than 25 years at the USVL (Schalk 1986, Schalk & Peterson 1990). Rearing procedures require germinating wheat sprouts to support the first two instars, but the remainder of the larval development is on a commercial artificial diet (wheat germ and casein base with no antibiotics; product no. F9760B, Bio-Serv, Frenchtown, New Jersey). Adults also were fed this dry artificial diet. More than 100 *D. balteata* adults that had been collected from sweetpotato fields at the USVL were added to the laboratory colony in 2002.

Every 2 days, cheese-cloth egg pads were collected from the *D. balteata* colony and incubated inside plastic bags containing sprouted wheat. Larval cohorts of a uniform size were collected from the sprouted wheat and used for each replication of an experiment. One second instar *D. balteata* was added to each microcentrifuge tube using a wet brush, and a small hole was punched into the top with a probe to provide ventilation. Twenty microcentrifuge tubes per treatment were set up for each replicate of the two bioassay experiments described below. Tubes were put into 24-hole racks, and the racks were placed in an environmental growth chamber at 25°C for 12 d. Then, surviving larvae were weighed to five decimal places using an A&D GR-202 Semi Micro Balance (A&D Company, Limited, Tokyo, Japan; accurate to 0.01 mg). Initial weights of larvae were too small to be weighed accurately as individuals. Therefore, 5 groups of 10 larvae were weighed at the beginning of each replication of the experiments, but because these larvae could have been damaged during handling, they were not used for the bioassay experiments. The laboratory bench, brushes, core-hole borer, and other handling equipment were frequently sterilized with 70% ethanol during the bioassay procedures.

Data for weight gain and survival were subjected to analysis of variance (ANOVA). When there were significant treatment effects, means were separated by Fisher' least significant difference at the 5% probability level (PROC GLM, SAS Institute 1989).

Experiment 1. Treatments evaluated in this experiment in 2004 were: (i) sprouted wheat, (ii) stele of SC1149–19, (iii) peel of SC1149–19, (iv) peel of 'Beauregard', (v) peel of 'Regal', (vi) peel of 'Charleston Scarlet', and (vii) peel of 'Liberty'. Sprouting wheat (Schalk 1986) was used as a control treatment. In a preliminary test, larvae did not survive on wheat sprouts that had been removed from the wheat kernel, probably because the sprouts decomposed rapidly. Thereafter, two whole wheat kernels with new sprouts were placed in each microcentrifuge tube. Also, two layers of autoclaved paper toweling were placed on the water-agar surface before the wheat was put inside the tubes. This experiment was repeated eight times, with each time repetition being treated as a replication for data analysis (= 160 larvae per treatment).

Experiment 2. Roots from only four of the aforementioned sweetpotato genotypes were evaluated in the second experiment, as the moderately resistant 'Liberty' was deleted. Treatments were: (i) sprouted wheat, (ii) stele of SC1149–19, (iii) peel of SC1149–19, (iv) stele of 'Beauregard', (v) peel of 'Beauregard', (vi) stele of 'Charleston Scarlet', (vii) peel of 'Charleston Scarlet', (viii) stele of 'Regal', (ix) peel of 'Regal', (x) cortex of 'Regal', (xi) periderm of 'Regal', and (xii) starved

larvae. The periderm treatment was made by scraping the periderm from the unblemished surface of whole sweetpotatoes with a scalpel (Peterson et al. 1998, Jackson & Peterson 2000, Harrison et al. 2003a). Periderm from about 4 cm² was placed into each microcentrifuge tube directly on a piece of autoclaved paper towel. The cortex treatment was prepared by first scraping off the periderm from a whole root with a scalpel and then preparing the sweetpotato plug as described above. Starved larvae were simply placed on a piece of autoclaved paper towel on top of the water-agar. This experiment was repeated 13 times, with each time repetition being treated as a replication for data analysis (= 260 larvae per treatment).

Field evaluations (1997–2004). The five sweetpotato entries used in the first bioassay experiment were grown in 16 replicated field experiments at the USVL during 1997–2004. Each sweetpotato entry was planted in 3–4 replications of single row, 25-plant plots arranged in a randomized complete block experimental design. Local production practices were followed, except that no insecticides were applied. When rainfall was not adequate during the growing season, supplemental overhead irrigation was applied. Plots were harvested in the fall when roots had reached maturity.

After harvest, all individual roots were scored for insect damage by previously published procedures (Schalk et al. 1991, Lawrence et al. 1999, Jackson et al. 1999, 2002). The severity index for the WDS complex was calculated by averaging the rating given to each root (1 = 1–5 holes or scars, 2 = 6–10 holes or scars, 4 = >10 holes or scars). Because there were significant yearly fluctuations in the levels of WDS injury and because not every sweetpotato genotype was grown in each experiment, WDS severity index data were weighted by multiplying each data point by a weighting factor calculated as a proportion of the average WDS rating for that experiment against the average for all experiments during the eight years of evaluations. A combined data set for all sweetpotato genotypes was subjected to ANOVA. Means were separated by Fisher's least significant difference at the 5% probability level (PROC GLM, SAS Institute 1989).

Results

Experiment 1. For the first experiment, there were highly significant treatment effects for weight gain ($F = 547.5$; $df = 6, 672$; $P < 0.0001$) and survival ($F = 6.14$; $df = 6, 1150$; $P < 0.0001$) of *D. balteata* larvae. Larvae gained nearly four times as much weight on the sprouted wheat control as they did on any other treatment (Table 1). Larvae had significantly higher weight gain on the stele of SC1149–19 than on any of the peel treatments. Larvae also grew larger on the peels of the susceptible genotypes (SC1149–19 and 'Beauregard') than on the peels of the resistant genotypes ('Charleston Scarlet', 'Liberty', and 'Regal'; Table 1). There was an average loss of weight for larvae on the peel of 'Regal', but survival on 'Regal' peel was no lower than for other peel treatments. Survival effects were not as dramatic as differences in weight gain, and survival of larvae on the sprouted wheat control was not significantly different from any of the treatments except for the peel of 'Charleston Scarlet.'

Experiment 2. For experiment two, ANOVA indicated that there were highly significant treatment effects for weight gain ($F = 811.1$; $df = 11, 1476$; $P < 0.0001$) and survival ($F = 71.5$; $df = 11, 2964$; $P < 0.0001$) of *D. balteata*

Table 1. Weight gain and survival of banded cucumber beetle larvae in a laboratory bioassay, experiment 1, Charleston, South Carolina, 2004.

Treatment	Average weight gain (mg $\pm$ SE) ^a	Survival (% $\pm$ SE)
Sprouted wheat	8.03 $\pm$ 0.37 a	62.5 $\pm$ 4.0 ab
SC1149-19 stele	2.09 $\pm$ 0.19 b	70.8 $\pm$ 4.7 a
SC1149-19 peel	0.57 $\pm$ 0.07 c	60.4 $\pm$ 3.5 b
'Beauregard' peel	0.48 $\pm$ 0.07 cd	71.4 $\pm$ 3.3 a
'Charleston Scarlet' peel	0.18 $\pm$ 0.05 de	46.9 $\pm$ 3.6 c
'Liberty' peel	0.09 $\pm$ 0.05 e	57.8 $\pm$ 3.6 b
'Regal' peel	-0.01 $\pm$ 0.03 e	53.6 $\pm$ 3.6 bc

Means within columns followed by a common letter are not significantly different ($P = 0.05$, LSD) (SAS 1989).

^aAverage initial weight of larvae was 0.59 mg as determined by weighing 5 groups of 10 larvae before each of the eight times this experiment was repeated.

larvae. Larvae grew larger when they were fed stele from any of the sweetpotato genotypes than when they were fed the peels of any of the genotypes (Table 2). Larval growth differed little among the stele treatments of any of the genotypes. Similar to the first experiment, larvae also grew larger on the peels of the susceptible genotypes (SC1149-19 and 'Beauregard') than on the peels of the resistant genotypes ('Charleston Scarlet' and 'Regal'). Starved larvae, those that were fed 'Regal' periderm alone, and larvae fed 'Regal' peel all lost weight. Larvae that were fed 'Regal' cortex gained little weight, which was not significantly different from 'Regal' peel (Table 2).

Larval survival was highest on the sprouted wheat control and the stele from the resistant genotypes. The stele from 'Beauregard' and SC1149-19 degraded faster than the stele from 'Charleston Scarlet' and 'Regal', which contributed to larval mortality. Larval mortality was not significantly different among the peel treatments. Larvae survived poorly when starved or fed 'Regal' periderm alone (Table 2).

Field evaluations (1997-2004). There were highly significant treatment effects for WDS severity index ($F = 59.6$; $df = 4, 119$; $P < 0.0001$) and percent uninfested roots ($F = 102.4$; $df = 4, 119$; $P < 0.0001$) in the field. These measurements of WDS injury (Table 3), which encompass damage by cucumber beetle larvae, were consistent with results of the larval bioassay, indicating that these techniques could be useful for evaluating pest resistance in sweetpotato genotypes to *Diabrotica* species.

Discussion

In this study, *D. balteata* larvae survived well on the sprouted wheat control; however, they performed poorly on the peels of the insect-resistant sweetpotato cultivars. In our bioassay procedure, larvae were required to feed through the periderm to reach the cortex of the peel treatments. *D. balteata* larvae gained

Table 2. Weight gain and survival of banded cucumber beetle larvae in a laboratory bioassay, experiment 2, Charleston, South Carolina, 2004–2005.

Treatment	Average weight gain (mg $\pm$ SE) ^a	Survival (% $\pm$ SE)
Sprouted wheat	8.39 $\pm$ 0.91 a	84.1 $\pm$ 2.0 a
'Regal' stele	0.92 $\pm$ 0.06 b	71.5 $\pm$ 2.8 b
'Beauregard' stele	0.82 $\pm$ 0.01 bc	39.2 $\pm$ 3.0 d
SC1149-19 stele	0.73 $\pm$ 0.08 c	50.7 $\pm$ 3.1 c
'Charleston Scarlet' stele	0.69 $\pm$ 0.06 c	67.3 $\pm$ 2.9 b
SC1149-19 peel	0.31 $\pm$ 0.04 d	57.3 $\pm$ 3.1 c
'Beauregard' peel	0.31 $\pm$ 0.05 d	66.2 $\pm$ 2.9 bc
'Regal' cortex	0.11 $\pm$ 0.01 de	54.2 $\pm$ 3.1 c
'Charleston Scarlet' peel	0.03 $\pm$ 0.03 e	57.3 $\pm$ 3.1 c
'Regal' peel	-0.01 $\pm$ 0.03 e	55.0 $\pm$ 3.1 c
'Regal' periderm	-0.08 $\pm$ 0.02 e	6.7 $\pm$ 1.0 e
Starved larvae	-0.10 $\pm$ 0.01 e	19.6 $\pm$ 2.5 e

Means within columns followed by a common letter are not significantly different ($P = 0.05$, LSD) (SAS 1989).

^aAverage initial weight of larvae was 0.55 mg as determined by weighing 5 groups of 10 larvae before each of the 13 times this experiment was repeated.

Table 3. Average WDS severity index and percent uninfested roots for five sweetpotato genotypes from field trials, Charleston, South Carolina, 1997–2005.

Sweetpotato genotype	WDS index ($\pm$ SE) ^a	Percent uninfested roots ($\pm$ SE)
SC1149-19	1.16 $\pm$ 0.10 a	11.7 $\pm$ 2.1 a
'Beauregard'	0.97 $\pm$ 0.09 b	25.2 $\pm$ 3.1 b
'Liberty'	0.39 $\pm$ 0.03 c	51.1 $\pm$ 4.2 c
'Regal'	0.40 $\pm$ 0.05 c	59.8 $\pm$ 4.0 d
'Charleston Scarlet'	0.31 $\pm$ 0.05 c	64.5 $\pm$ 4.3 d

Means within columns followed by a common letter are not significantly different ($P = 0.05$, LSD) (SAS 1989).

^aAverage from 16 field experiments, 1997–2005 at the U.S. Vegetable Laboratory, Charleston, SC. WDS (wireworm + *Diabrotica* spp. + *Systema* spp.) severity index: 1 = 1–5 scars, 2 = 6–10 scars, 4 = >10 scars, averaged over all roots; maximum score = 4.0.

little or no weight on 'Regal' periderm alone, 'Regal' cortex alone, or whole 'Regal' peels (periderm plus cortex). Larval survival was significantly lower on the periderm, cortex, and peel of 'Regal' roots than it was on 'Regal' stele. Survival on the scraped periderm treatment was significantly lower than any other treatment, except for starved larvae. These results are not surprising because antibiotic

components from the periderm and cortex have been identified from these genotypes (Cuthbert & Davis 1971, Schalk et al. 1986a, Peterson et al. 1998, 2003, 2005, Jackson & Peterson 2000, Harrison et al. 2003a, 2003b). However, this is the first report of a laboratory bioassay for *D. balteata* larvae on the isolated individual components of sweetpotato roots. It should be noted that weight gain of *D. balteata* larvae on sprouted wheat was much higher than on the stele of any of the sweetpotato genotypes, suggesting that additional resistance factors also may be present in sweetpotato stele.

Although Cuthbert & Jones (1972) first showed that the level of resistance to the WDS complex could be increased incrementally using recurrent selection, they did not attempt to simultaneously select for other desirable characteristics. Jones (1990) reported that sweetpotato germplasm could be developed that was resistant to several insect pests and disease pathogens, using a systematic approach. Progress toward improved sweetpotato cultivars must be done through simultaneous evaluation procedures for all desirable characteristics, including pest resistance (Jones et al. 1986). As part of the evaluation procedure, each new sweetpotato genotype must be assessed to see if it has retained its resistance to insect pests. Schalk & Jones (1982) developed a field evaluation technique for banded cucumber beetles in the field. However, this technique requires large, replicated field plots. The results of the present study demonstrate that a laboratory bioassay with *D. balteata* larvae could be useful as part of the evaluation process for resistance to *Diabrotica* species.

Acknowledgments

We thank John Fender, Jennifer Cook, and Louise Cauthen, U. S. Vegetable Laboratory, Charleston, South Carolina, for their excellent technical assistance. This research was supported in part by a grant from the McKnight Foundation, Collaborative Crop Research Program.

References Cited

- Artschwager, E. & R. C. Starrett. 1931.** Suberization and wound-periderm formation in sweetpotato and gladiolus as affected by temperature and relative humidity. *J. Agric. Res.* 43: 353–364.
- Bohac, J. R. & D. M. Jackson. 2005.** Notice of release of ‘Charleston Scarlet’, a very sweet, orange fleshed sweetpotato, with high resistance to insects. USDA, ARS, Washington, DC.
- Bohac, J. R., D. M. Jackson, J. D. Mueller, M. Sullivan & P. D. Dukes. 2003.** Notice of release of “Liberty,” a multiple pest resistant, dry fleshed sweetpotato cultivar. USDA, ARS, 15 September 2003, Washington, DC.
- Bohac, J. R., D. M. Jackson, J. D. Mueller, P. D. Dukes & J. M. Schalk. 2000.** Notice of release of ‘Patriot’, an insect-resistant, copper-rose skinned, orange-fleshed sweetpotato cultivar. USDA, ARS, 19 December 2000, Washington, DC.
- Bohac, J. R., D. M. Jackson, P. D. Dukes & J. D. Mueller. 2002.** ‘Ruddy’: a multiple-pest-resistant sweetpotato. *HortScience* 37: 993–994.
- Branson, T. F. & J. L. Krysan. 1981.** Feeding and oviposition behavior and life cycle strategies of *Diabrotica*: an evolutionary view with implications for pest management USA. *Environ. Entomol.* 10: 826–831.
- Capinera, J. L. 2001** Handbook of Vegetable Pests. Academic Press, New York.

- Chalfant, R. B., R. K. Jansson, D. R. Seal & J. M. Schalk. 1990.** Ecology and management of sweet potato insects. *Annu. Rev. Entomol.* 35: 157–180.
- Collins, W. W. & M. R. Hall. 1992.** Sweetpotato breeding and genetics, germplasm and cultivar development, pp. 16–28. *In* A. Jones & J. C. Bouwkamp [Eds.], Fifty years of cooperative sweetpotato research. South. Coop. Ser. Bull. No. 369.
- Cuthbert Jr., F. P. 1967.** Insects affecting sweet potatoes. USDA Agric. Handb. No. 329.
- Cuthbert Jr., F. P. & A. Jones. 1972.** Resistance in sweet potatoes to Coleoptera increased by recurrent selection. *J. Econ. Entomol.* 65: 1655–1658.
- Cuthbert Jr., F. P. & B. W. Davis. 1971.** Factors associated with insect resistance in sweet potatoes. *J. Econ. Entomol.* 64: 713–717.
- Harrison Jr., H. F., J. K. Peterson, D. M. Jackson & M. E. Snook. 2003a.** Periderm resin glycoside contents of sweetpotato, *Ipomoea batatas* (L.) Lam. clones and their biological activity. *Allelopathy J.* 12: 53–60.
- Harrison Jr., H. F., J. K. Peterson, M. E. Snook, J. R. Bohac & D. M. Jackson. 2003b.** Quantitation and potential biological activity of caffeic acid in sweetpotato [*Ipomoea batatas* (L.) Lam] storage roots. *J. Agric. Food Chem.* 51: 2943–2948.
- Jackson, D. M. & J. K. Peterson. 2000.** Sublethal effects of resin glycosides from sweetpotato skins on larvae of the diamondback moth. *J. Econ. Entomol.* 93: 388–393.
- Jackson, D. M. & J. R. Bohac. 2007.** Resistance of sweetpotato genotypes to adult *Diabrotica* beetles. *J. Econ. Entomol.* 100: 566–572.
- Jackson, D. M., J. R. Bohac, J. Lawrence & J. D. Mueller. 1999.** Multiple insect resistance in dry-fleshed sweet potato breeding lines for the USA and Caribbean, pp. 274–280. *In* Progress in IPM CRSP Research, Proceedings of the Third IPM CRSP Symposium, 15–18 May 1998, Virginia Tech Univ., Blacksburg, Virginia.
- Jackson, D. M., J. R. Bohac, K. M. Dalip, J. Lawrence, D. Clarke Harris, L. McComie, J. Gore, D. McGlashan, P. Chung, S. Edwards, S. Tolin & C. Edwards. 2002.** Integrated Pest Management of Sweetpotato in the Caribbean, pp. 143–154. *In* T. Ames [Ed.], Proceedings of the First International Conference on Sweetpotato Food and Health for the Future, 26–30 November 2001, Lima, Peru. *Acta Horticulturae* No. 583.
- Jansson, R. K. & K. V. Raman. 1991.** Sweet Potato Pest Management: A Global Perspective. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Jones, A. 1965.** A proposed breeding procedure for sweetpotato. *Crop Sci.* 5: 191–192.
- Jones, A. 1990.** Breeding sweetpotato for resistance to multiple insect pests. *HortScience* 25: 1177–1178.
- Jones, A., P. D. Dukes & J. M. Schalk. 1986.** Sweet potato breeding, pp. 1–35. *In* M. J. Bassett [Ed.], Breeding Vegetable Crops. AVI Publ. Co., Westport, Connecticut.
- Jones, A., P. D. Dukes, J. M. Schalk, M. G. Hamilton, M. A. Mullen, R. A. Baumgardner, D. R. Paterson & T. E. Boswell. 1985.** ‘Regal’ sweet potato. *HortScience* 20: 781–782.
- Krysan, J. L. & T. A. Miller. 1986.** Methods for the Study of Pest *Diabrotica*. Springer Verlag, New York, 260 pp.
- Lawrence, J., D. M. Jackson & J. R. Bohac. 1999.** Evaluation of USDA sweetpotato breeding lines as a potential tactic for managing sweet potato pests in the Caribbean and USA, pp. 298–304. *In* Progress in IPM CRSP Research, Proceedings of the Third IPM CRSP Symposium, 15–18 May 1998, Virginia Tech Univ., Blacksburg, Virginia.
- Lawrence, J., S. Tolin, C. Edwards, S. Fleischer, D. M. Jackson, D. Clarke-Harris, S. McDonald, K. Dalip & P. Chung. 2005.** Developing IPM packages in the Caribbean, pp. 95–119. *In* G. W. Norton, E. A. Heinrichs, G. C. Luther & M. E. Irwin [Eds.], Globalizing Integrated Pest Management: A Participatory Research Process. Blackwell Publ., Ames, Iowa, 338 pp.
- McClure, T. T. 1960.** Chlorogenic acid accumulation and wound healing in sweet potato roots. *Am. J. Bot.* 47: 277–280.
- McKinlay, R. G. 1992.** Vegetable Crop Pests. CRC Press, Inc., Boca Raton, Florida.

- Morris, L. L. & L. K. Mann. 1955.** Wound healing, keeping quality, and compositional changes during curing and storage of sweet potatoes. *Hilgardia* 24: 143–183.
- Peterson, J. K., H. F. Harrison Jr. & A. E. Muckenfuss. 1998.** Sweetpotato [*Ipomoea batatas* (L.)] resin glycosides: Evidence of antibiosis effects in the diamondback moth *Plutella xylostella* L. (Lepidoptera: Plutellidae). *Allelopathy J.* 5: 43–52.
- Peterson, J. K., H. F. Harrison Jr., D. M. Jackson & M. E. Snook. 2003.** Biological activities and contents of scopolin and scopoletin in sweetpotato clones. *HortScience* 38: 1129–1133.
- Peterson, J. K., H. F. Harrison Jr., M. E. Snook & D. M. Jackson. 2005.** Chlorogenic acid content in sweetpotato germplasm: possible role in disease and pest resistance. *Allelopathy J.* 16: 239–249.
- Pitre, H. N. & E. J. Kantack. 1962.** Biology of the banded cucumber beetle, *Diabrotica balteata*, in Louisiana. *J. Econ. Entomol.* 55: 904–906.
- Rolston, L. H., C. A. Clark, J. M. Cannon, W. M. Randell, E. G. Riley, P. W. Wilson & M. L. Robbins. 1987.** 'Beauregard' sweetpotato. *HortScience* 22: 1338–1339.
- Saba, F. 1970.** Host plant spectrum and temperature limitation of *Diabrotica balteata*. *Can. Entomol.* 102: 684–691.
- SAS Institute. 1989.** SAS/STAT User's Guide, Version 6, Fourth Edition, Volumes 1 and 2. SAS Institute, Inc., Cary, North Carolina.
- Schalk, J. M. 1986.** Rearing and handling of *Diabrotica balteata*, pp. 49–56. In: J. L. Krysan & T. A. Miller [Eds.], *Methods for the study of pest Diabrotica*. Springer Verlag, New York, 260 pp.
- Schalk, J. M. & A. Jones. 1982.** Methods to evaluate sweet potatoes for resistance to the banded cucumber beetle in the field. *J. Econ. Entomol.* 75: 76–79.
- Schalk, J. M. & A. Jones. 1985.** Major insect pests, pp. 59–78. In: J. C. Bouwkamp [Ed.], *Sweet Potato Products: A Natural Source for the Tropics*. CRC Press, Boca Raton, Florida.
- Schalk, J. M. & J. K. Peterson. 1990.** A meridic diet for banded cucumber beetle larvae (*Diabrotica balteata* LeConte). *J. Agric. Entomol.* 7: 333–336.
- Schalk, J. M., A. Jones & P. D. Dukes. 1986a.** Factors associated with resistance in recently developed sweet potato cultivars and germplasm to the banded cucumber beetle, *Diabrotica balteata* Leconte. *J. Agric. Entomol.* 3: 329–334.
- Schalk, J. M., A. F. Jones, P. D. Dukes & J. K. Peterson. 1991.** Approaches to the control of multiple insect problems in sweet potato in the southern United States, pp. 283–301. In: R. K. Jansson & K. V. Raman [Eds.], *Sweet potato pest management: A global perspective*. Westview Press, Boulder, Colorado.
- Schalk, J. M., J. K. Peterson, A. Jones, P. D. Dukes & W. M. Walter Jr. 1986b.** The anatomy of sweet potato periderm and its relationship to wireworm, *Diabrotica*, *Systema* resistance. *J. Agric. Entomol.* 3: 350–356.
- Smith, C. M., Z. R. Khan & M. D. Pathak. 1994.** *Techniques for Evaluating Insect Resistance in Crop Plants*. Lewis Publ., CRC Press, Boca Raton, Florida.
- Talekar, N. S. 1988.** How to control sweetpotato weevil: A practical IPM approach. AVRDC Publ. 88-292, Taiwan, Republic of China, 6 pp.
- [USDA] U.S. Department of Agriculture. 2005.** USDA, ARS, National Genetic Resources Program. Germplasm Resources Information Network (GRIN). National Germplasm Resources Laboratory, Beltsville, Maryland. Available at <http://www.ars-grin.gov/cgi-bin/npgs/html/croplist.pl>, accessed 24 November 2006.
- Walter Jr., W. M. & W. E. Schadel. 1983.** Structure and composition of normal skin (periderm) and wound tissue from cured sweet potatoes. *J. Am. Soc. Hortic. Sci.* 108: 909–914.

Identifying Resistance in Corn to Southwestern Corn Borer (Lepidoptera: Crambidae), Fall Armyworm (Lepidoptera: Noctuidae), and Corn Earworm (Lepidoptera: Noctuidae)¹

W. Paul Williams, Paul M. Buckley, and Christopher A. Daves

USDA-ARS Corn Host Plant Resistance Research Unit, Box 9555, Mississippi State, Mississippi 39762 USA

J. Agric. Urban Entomol. 23(2): 87–95 (April 2006)

ABSTRACT Southwestern corn borer, *Diatraea grandiosella* Dyar (Lepidoptera: Crambidae); fall armyworm, *Spodoptera frugiperda* (J.E. Smith) (Lepidoptera: Noctuidae); and corn earworm, *Helicoverpa zea* Boddie (Lepidoptera: Noctuidae), are major insect pests of corn, *Zea mays* L., in the southern United States. Corn hybrids produced from resistant germplasm were evaluated together with Bt and nonBt commercial hybrids in field tests and laboratory bioassays for resistance to leaf and ear damage by southwestern corn borer, fall armyworm, and corn earworm. The Bt hybrids, Mp496 × Mp716 and Mp704 × Mp716, exhibited resistance to leaf feeding damage by southwestern corn borer and fall armyworm. The Bt hybrids sustained significantly less southwestern corn borer ear damage than their nonBt versions. Those genotypes expressing resistance to leaf feeding also showed resistance, as indicated by reduced larval growth or survival in laboratory bioassays based on lyophilized leaf tissue, to southwestern corn borer, fall armyworm, and corn earworm. Only the Bt hybrids expressed resistance when larval growth was evaluated in bioassays based on lyophilized husk tissues. Using both field tests and laboratory bioassays to assess resistance to these insects will enhance efforts to identify useful germplasm and determine factors contributing to resistance.

KEY WORDS *Bacillus thuringiensis*, corn, Crambidae, *Diatraea grandiosella*, *Helicoverpa zea*, Lepidoptera, Noctuidae, *Spodoptera frugiperda*, *Zea mays*

Southwestern corn borer, *Diatraea grandiosella* Dyar (Lepidoptera: Crambidae); fall armyworm, *Spodoptera frugiperda* (J.E. Smith) (Lepidoptera: Noctuidae); and corn earworm, *Helicoverpa zea* Boddie (Lepidoptera: Noctuidae), are major lepidopteran pests of corn, *Zea mays* L., in the southern United States. Each of these insects attack corn plants in both the vegetative and reproductive stages of development. Because plant resistance generally is considered a desirable means of reducing losses to these insects, research programs designed to

¹Accepted for publication 20 March 2007

identify and develop resistant germplasm have been established at several locations (Overman 1989; Smith et al. 1989; Widstrom 1989; Williams & Davis 1997).

The USDA-ARS Corn Host Plant Resistance Research Unit at Mississippi State University has developed and released corn germplasm lines with resistance to leaf damage caused by feeding of southwestern corn borer and fall armyworm (Scott & Davis 1981; Williams & Davis 1982, 1984, 2000, 2002). Selection for resistance to leaf feeding was accomplished by infesting plants at the 7- to 8-leaf stage of growth with 30 neonate larvae per plant and visually rating damage 14 d later.

Selection for resistance to southwestern corn borer damage after anthesis was first attempted by infesting plants after silk emergence and measuring stalk tunnels and counting entry and exit holes in the stalk (Davis & Williams 1983, 1997). Resistance to southwestern corn borer was also determined by visually rating damage to husks and leaf sheaths and counting and weighing surviving larvae (Williams et al. 1998a).

Laboratory bioassays were developed to investigate resistance to leaf feeding by southwestern corn borer and fall armyworm (Williams et al. 1990). Laboratory diets were prepared from finely ground, lyophilized leaf tissue. Larvae were weighed 10–14 d later. Weights of both southwestern corn borer and fall armyworm larvae in the bioassays were positively correlated with leaf feeding damage sustained in field tests. Laboratory bioassays also were used to assess southwestern corn borer growth on diets prepared from lyophilized husk tissue (Williams & Buckley 1996). Larvae that fed on husks of corn inbred lines Mp305 or T202 tended to weigh less than those fed on husks of other genotypes.

Genetically engineered corn hybrids expressing genes that encode proteins having the property of those found in the bacterium *Bacillus thuringiensis* Berliner (Bt) provided new opportunities for insect control (Boulter 1993). Transgenic Bt hybrids exhibited high levels of resistance to southwestern corn borer in both the field and laboratory (Williams et al. 1997, 1998b). The same hybrids exhibited substantial levels of resistance to fall armyworm as well.

This investigation was undertaken to assess resistance potential by (i) evaluating leaf and ear damage from southwestern corn borer, fall armyworm, and corn earworm sustained by corn hybrids in field tests; (ii) determining larval growth on laboratory diets prepared from leaf and husk tissue; and (iii) comparing results of field tests and laboratory bioassays for the three insect species.

Materials and Methods

Nine corn hybrids were grown at the R.R. Foil Plant Science Research Center at Mississippi State University, Mississippi State, Mississippi, in 2005. The hybrids included two pairs of nonBt and transgenic Bt commercial hybrids, Pioneer brand 3223 and 31B13 and DEKALB brand DK679 and DK679 BTY, respectively. Two hybrids, Ab24E × SC229 and Mp313E × Mo18w, were selected for their susceptibility to leaf feeding by southwestern corn borer and fall armyworm. The remaining three hybrids, Mp305 × Mp496, Mp496 × Mp716, and Mp704 × Mp716, were included because their inbred parents exhibit varying degrees of resistance to southwestern corn borer and fall armyworm leaf feeding (Williams & Davis 1997).

Leaf feeding damage by southwestern corn borer, fall armyworm, and corn

earworm was evaluated in separate experiments planted 22 April 2005. Hybrids were planted in single-row plots with 20 plants. Rows were approximately 4 m long and spaced 0.97 m apart. Within each experiment, hybrids were arranged in a randomized complete block design with three replications. On 31 May, plants at the 7-leaf stage of growth in each experiment were infested with 30 neonates each of the appropriate insect: southwestern corn borer, fall armyworm, or corn earworm. Larvae were obtained from laboratory colonies maintained by the USDA-ARS Corn Host Plant Resistance Research Unit, Mississippi State, Mississippi (Davis & Williams 1997). Larvae were mixed with corn cob grits and dispensed into the plant whorl using a mechanical applicator (Wiseman et al. 1980). Leaf feeding damage was visually rated 14 d after infestation.

The visual ratings of damage were based on the number and size of lesions (Williams et al. 1989). Southwestern corn borer leaf feeding was visually rated on the following scale: 0 = no visible leaf damage; 1 = small amount of pin-hole damage; 2 = shot-hole damage on 1–3 leaves; 3 = shot-hole damage on 4–6 leaves; 4 = elongated lesions (<25 mm) on 1–3 leaves; 5 = elongated lesions (<25 mm) on 4–7 leaves; 6 = elongated lesions (25–35 mm) on 3–5 leaves; 7 = long lesions (>35 mm) on 50% of the leaves; 8 = long lesions (>35 mm) on 70% of the leaves; and 9 = long lesions (>35 mm) on most leaves.

Because fall armyworm feeding causes a somewhat different pattern of damage, the rating scale for fall armyworm damage was defined as follows: 0 = no visible leaf damage; 1 = only pin-hole damage on leaves; 2 = pin-hole and shot-hole damage to leaves; 3 = small elongated lesions (5–10 mm) on 1–3 leaves; 4 = midsized lesions (10–30 mm) on 4–7 leaves; 5 = large elongated lesions (>30 mm) or small portions eaten on 3–5 leaves; 6 = elongated lesions (>30 mm) and large portions eaten on 3–5 leaves; 7 = elongated lesions (>30 mm) and large portions eaten on 50% of leaves; 8 = elongated lesions (>30 mm) and large portions eaten on 70% of leaves; and 9 = leaves destroyed on 70% of leaves.

Ear damage resulting from southwestern corn borer, fall armyworm, and corn earworm was made separately in three experiments planted 25 April. These experiments were similar in design to those previously described. Plants were infested with 30 neonates approximately 7 d after mid silk, the date when silks had emerged from the top ear shoots of 50% of the plants in a plot. For southwestern corn borer and fall armyworm, the larvae-corn cob grits mixture was placed in the axil of the top-ear leaf and in the axil of the next higher leaf. For corn earworm, the mixture was placed in the axil of the top-ear leaf and on the silks. Approximately 15 larvae were placed at each site.

Ear damage was visually rated after harvest on 10 randomly selected ears from each plot on the following scale: 1 = no damage to any ears; 2 = tip (<30 mm) damage to 1–3 ears; 3 = tip damage to 4–6 ears; 4 = tip damage to 7 or more ears and damage below ear tips to 1–3 kernels of 1–3 ears; 5 = tip damage to 7 or more ears and damage to 1–3 kernels below tips of 4–6 ears; 6 = ear tip damage to 7–10 ears and damage to 1–4 kernels below tips of 7–10 ears; 7 = ear tip damage to 7–10 ears and 4–6 kernels destroyed on 4–6 ears; 8 = ear tip damage to all ears and 4–6 kernels below tips destroyed on 7–8 ears; and 9 = ear tip damage to all ears and 5 or more kernels destroyed below tips of 9–10 ears.

Twenty additional rows of each hybrid were planted on 25 April to provide leaf and husk tissue for laboratory bioassays. On June 7, plant whorls were collected from 10 rows of each hybrid. The whorls were trimmed to approximately 15 cm in

length so that the yellow-green portion of the leaves was retained. The yellow-green portion of the leaf is the primary larval feeding site and the portion of the leaf where resistance is most strongly expressed (Williams et al. 1998c, Davis et al. 1999). The remaining 10 rows of each hybrid provided husks for the bioassays. Husks were removed from ear shoots from 3–7 d after silk emergence. The plant tissue, either leaves or husks, was placed in plastic freezer bags and frozen at -18°C . The frozen tissue was later lyophilized and ground to a fine powder using a laboratory mill with a 1-mm mesh screen.

Laboratory bioassays were conducted following the procedures described by Williams & Buckley (1992, 1996). Test diets were prepared for each hybrid by combining 250 ml distilled water, 2400 mg agar, 12.5 mg gentamicin sulfate, 132 mg sorbic acid, and 528 mg ascorbic acid. The mixture was heated to 82°C while stirring. Either 10 g of lyophilized leaf tissue or 15 g of lyophilized husk tissue was then added to the mixture. The mixture was dispensed in 10-mL aliquots into 30-mL plastic diet cups.

A test for southwestern corn borer was initiated on 28 March 2006 with 20 cups each of leaf and husk diets prepared for each of the nine hybrids. Each cup was infested with one neonate and covered with a paper board lid. Cups for both the leaf and husk bioassays were arranged in a randomized complete block design with five replications of four cups per replication. The cups were placed in an environmental chamber maintained at 28°C with a photoperiod of 12:12 (L:D). Surviving larvae were counted and weighed after 21 d. Growth of corn earworm and fall armyworm was assessed in similar experiments initiated on 30 March and 4 April 2006, respectively.

Table 1. Leaf feeding damage sustained by corn hybrids infested with 30 southwestern corn borer (SWCB) or fall armyworm (FAW) larvae per plant in field test at Mississippi State, Mississippi, 2005.

Hybrid	Leaf damage (mean $\pm$ SE) ^a	
	SWCB ^b	FAW ^b
DK679	9.0 $\pm$ 0.0a	8.7 $\pm$ 0.6a
Ab24E $\times$ SC229	8.3 $\pm$ 0.6ab	9.0 $\pm$ 0.0a
3223	8.0 $\pm$ 1.0b	8.7 $\pm$ 0.6a
Mp313E $\times$ Mo18w	8.0 $\pm$ 0.0b	8.3 $\pm$ 0.6ab
Mp305 $\times$ Mp496	6.7 $\pm$ 0.6c	7.7 $\pm$ 0.6bc
Mp496 $\times$ Mp716	5.7 $\pm$ 0.6d	7.0 $\pm$ 0.0c
Mp704 $\times$ Mp716	4.7 $\pm$ 0.6e	5.7 $\pm$ 0.6d
31B13	1.0 $\pm$ 0.0f	2.3 $\pm$ 0.6e
DK679BTY	1.0 $\pm$ 0.0f	1.3 $\pm$ 0.6f
LSD (0.05)	0.9	0.9
$F_{8, 16}$	97.29	85.39
P	0.001	0.001

^aLeaf feeding damage was visually rated 14 d after infestation on a scale of 0 = no damage, to 9 = extensive damage.

^bValues in a column followed by the same letter do not differ at $P = 0.05$ (Fisher's Protected LSD).

Table 2. Ear damage sustained by corn hybrids infested with southwestern corn borer (SWCB), fall armyworm (FAW), or corn earworm (CEW) at Mississippi State, Mississippi, 2005.

Hybrid	Leaf damage (mean $\pm$ SE) ^a		
	SWCB ^b	FAW ^b	CEW
DK679	6.6 $\pm$ 2.1a	5.7 $\pm$ 1.3a	7.1 $\pm$ 0.6
Ab24E $\times$ SC229	5.7 $\pm$ 0.5ab	5.4 $\pm$ 0.9ab	5.7 $\pm$ 1.0
3223	5.5 $\pm$ 1.1ab	3.5 $\pm$ 1.2cd	6.9 $\pm$ 0.6
Mp313E $\times$ Mo18w	5.0 $\pm$ 1.8abcd	4.0 $\pm$ 0.9bcd	5.9 $\pm$ 0.9
Mp305 $\times$ Mp496	4.5 $\pm$ 1.3abcd	2.9 $\pm$ 1.0cd	6.9 $\pm$ 0.2
Mp496 $\times$ Mp716	4.1 $\pm$ 2.1bcd	4.1 $\pm$ 0.8bcd	5.3 $\pm$ 2.4
Mp704 $\times$ Mp716	5.4 $\pm$ 1.4abc	6.1 $\pm$ 2.0a	6.3 $\pm$ 2.0
31B13	2.8 $\pm$ 0.8d	4.1 $\pm$ 0.5bcd	6.6 $\pm$ 1.7
DK679BTY	3.1 $\pm$ 0.6cd	4.5 $\pm$ 0.9abc	7.6 $\pm$ 0.5
LSD (0.05)	2.4	1.7	—
$F_{8, 16}$	2.52	3.43	1.14
P	0.05	0.02	0.39

^aEar damage was visually rated after harvest on a scale of 1 = no damage, to 9 = heavy damage.

^bValues in a column followed by the same letter do not differ at $P = 0.05$ (Fisher's Protected LSD).

All data were analyzed by analysis of variance (SAS Institute 1987). Means were compared by Fisher's protected least significant difference (LSD) test (Steel & Torrie 1980).

Results and Discussion

The transgenic Bt hybrids sustained only minor pinhole-type damage when infested at the whorl stage of growth with southwestern corn borer (Table 1). Mp496 $\times$ Mp716 and Mp704 $\times$ Mp716 exhibited a moderate level of resistance. These two crosses were produced by crossing two inbred lines that were developed and released as sources of resistance to southwestern corn borer. Mp305 $\times$ Mp496, a cross between a susceptible and resistant inbred line, sustained more leaf damage than the crosses between two resistant lines, but not as much as the susceptible hybrids, DK679 and Ab24E $\times$ SC229.

Leaf damage caused by fall armyworm feeding was highly correlated with southwestern corn borer leaf feeding damage ($r^2 = 0.97$, $P < 0.01$). The Bt hybrid 31B13, however, sustained heavier damage from fall armyworm than southwestern corn borer. The Bt hybrids exhibited the least leaf feeding damage, but among the nonBt hybrids Mp704 $\times$ Mp716 again exhibited the lowest level of leaf feeding damage. Because corn earworm larvae failed to become established on the whorl stage plants, no data are available on resistance to leaf feeding. Others have had similar difficulties with larval establishment when infesting whorl stage plants with corn earworm although leaf feeding from naturally occurring corn earworm infestations is observed in field plots (F.M. Davis, personal communication).

All hybrids sustained ear damage whether infested with southwestern corn

Table 3. Mean weights of southwestern corn borer (SWCB), fall armyworm (FAW), and corn earworm (CEW) larvae reared on laboratory diets containing lyophilized leaf tissue of corn hybrids, Mississippi State, Mississippi, 2006.

Hybrid	Leaf damage (mean $\pm$ SE) ^a mg ^a		
	SWCB ^b	FAW ^b	CEW ^b
DK679	79 $\pm$ 6a	369 $\pm$ 7a	234 $\pm$ 27a
Ab24E $\times$ SC229	80 $\pm$ 6a	359 $\pm$ 34a	179 $\pm$ 34a
3223	75 $\pm$ 7a	332 $\pm$ 92ab	210 $\pm$ 41a
Mp313E $\times$ Mo18w	74 $\pm$ 7a	268 $\pm$ 79c	232 $\pm$ 56b
Mp305 $\times$ Mp496	80 $\pm$ 9a	285 $\pm$ 34bc	102 $\pm$ 54c
Mp496 $\times$ Mp716	33 $\pm$ 5b	61 $\pm$ 37d	33 $\pm$ 23d
Mp704 $\times$ Mp716	27 $\pm$ 7b	49 $\pm$ 49d	41 $\pm$ 27d
31B13	— ^c	16 $\pm$ 5d	— ^c
DK679BTY	— ^c	11 $\pm$ 4d	— ^c
LSD (0.05)	9	60	47
<i>F</i> _{6, 24}	54.73	—	28.92
<i>F</i> _{8 29}	—	56.36	— ^c
<i>P</i>	0.001	0.001	0.001

^aLarvae were weighed after feeding for 21 d on laboratory diets.
^bMeans in a column followed by the same letter do not differ at *P* = 0.05 (Fisher's Protected LSD).
^cNo larvae survived.

borer, fall armyworm, or corn earworm (Table 2). When infested with southwestern corn borer larvae, the transgenic Bt hybrids, DK679BTY and 31B13, sustained significantly less ear damage than on their nontransgenic versions, DK679 and 3223. Mp305 $\times$ Mp496 and Mp496 $\times$ Mp716 also sustained moderate ear damage and did not differ from the Bt hybrids. Williams et al. (1998b) reported that southwestern corn borer larval survival and growth were reduced on ears of corn hybrids produced from parental inbred lines Mp305 and Mp496.

There were significant differences among hybrids for ear damage caused by fall armyworm. The differences in damage sustained by the Bt hybrids and their nontransgenic versions were, however, not statistically significant. The beneficial effects expressed by Bt hybrids in reducing ear damage by southwestern corn borer were not realized when the hybrids were infested with fall armyworm. The hybrids Mp305 $\times$ Mp496 and Mp496 $\times$ Mp716, which exhibited resistance to southwestern corn borer, also exhibited moderate resistance to fall armyworm.

Hybrids infested with corn earworm exhibited higher ear damage ratings compared with the other insects. The damage ratings ranged from 5.3–7.6 for hybrids infested with corn earworm, but the differences among hybrids were not significant.

When larval growth was evaluated in laboratory bioassays with diets prepared from lyophilized leaf tissue, none of the southwestern corn borer or corn earworm larvae survived on the Bt hybrids (Table 3). Fall armyworm larvae survived, but their weight was markedly reduced. For all three species, larval weight was

Table 4. Mean weights of southwestern corn borer (SWCB), fall armyworm (FAW), and corn earworm (CEW) larvae reared on laboratory diets containing lyophilized husk tissue of corn hybrids, Mississippi State, Mississippi, 2006.

Hybrid	Leaf damage (mean $\pm$ SE) ^a mg ^a		
	SWCB ^b	FAW ^b	CEW ^b
DK679	29 $\pm$ 2bcd	13 $\pm$ 2abc	16 $\pm$ 4b
Ab24E $\times$ SC229	36 $\pm$ 9b	14 $\pm$ 6ab	20 $\pm$ 3b
3223	25 $\pm$ 3cd	14 $\pm$ 7ab	24 $\pm$ 5b
Mp313E $\times$ Mo18w	44 $\pm$ 3a	21 $\pm$ 7a	44 $\pm$ 22a
Mp305 $\times$ Mp496	32 $\pm$ 4bc	8 $\pm$ 1bc	16 $\pm$ 4b
Mp496 $\times$ Mp716	22 $\pm$ 3d	10 $\pm$ 4bc	14 $\pm$ 3bd
Mp704 $\times$ Mp716	29 $\pm$ 6bcd	19 $\pm$ 13a	23 $\pm$ 5bd
31B13	— ^c	4 $\pm$ 1c	— ^c
DK679BTY	— ^c	— ^c	— ^c
LSD (0.05)	7	9	47
<i>F</i> _{6, 24}	8.18	—	6.43
<i>F</i> _{8, 29}	—	3.09	—
<i>P</i>	0.001	0.017	0.001

^aLarvae were weighed after feeding for 21 d on laboratory diets.^bMeans in a column followed by the same letter do not differ at *P* = 0.05 (Fisher's Protected LSD).^cNo larvae survived.

significantly lower for Mp496 $\times$ Mp716 and Mp704 $\times$ Mp716. These are the same two nonBt hybrids that sustained the lowest levels of leaf feedings damage from southwestern corn borer and fall armyworm in the field tests (Table 1).

Larval growth was substantially less on diets prepared from husks tissue than on diets prepared from leaf tissue (Table 4). As with leaf tissue, neither southwestern corn borer nor corn earworm larvae survived on husk tissue of the Bt hybrids. There were no surviving fall armyworm on the husk tissue of DK679BTY, and the larvae recovered from 31B13 were small. The heaviest southwestern corn borer, fall armyworm, and corn earworm larvae were those that fed on husk tissue of Mp313E $\times$ Mo18w. This was not anticipated based on the ear damage ratings for the hybrids, and there is no obvious explanation for the heavier weights on Mp313E $\times$ Mo18w.

Bioassays that reflect differences in levels of resistance in the field are useful in investigations of the factors associated with resistance. Southwestern corn borer and fall armyworm growth on lyophilized leaf tissue in these bioassays was lowest on the Bt hybrids that also showed the highest level of resistance in the field. Reduced larval growth and lower damage ratings also were observed on the resistant corn hybrids developed through conventional breeding. Although the field evaluation of corn earworm resistance was unsuccessful, corn earworm growth on lyophilized leaf tissue was reduced on the same hybrids that exhibited resistance to southwestern corn borer and fall armyworm in the field, suggesting that those factors associated with resistance to one insect may also be associated

with resistance to the other insects and that corn earworm leaf feeding resistance can be evaluated with bioassays.

Although larval survival in the husk bioassays would appear to indicate that the Bt hybrids have high levels of resistance to all insects, the ear damage ratings do not reflect this except for southwestern corn borer. Corn earworm damage was great on all hybrids, including the Bt hybrids. One of the limitations of the husk bioassays is that, although southwestern corn borer and fall armyworm larvae feed on husks before feeding on developing kernels, corn earworm larvae generally feed on silks before feeding on developing kernels. Bioassays for corn earworm that include silks and kernels might yield results more predictive of actual field resistance than assays that use only husk tissue. The results of this investigation indicate not only a need for improved bioassays, but also a need for corn hybrids with higher levels of resistance to ear damage caused by fall armyworm or corn earworm feeding.

Acknowledgments

Appreciation is extended to Michael N. Alpe, Gerald A. Matthews, and Susan H. Wolf for excellent technical assistance. This article reports the results of research only. Mention of a proprietary product does not constitute an endorsement or a recommendation for use by USDA or Mississippi State University. This article is a contribution of the Corn Host Plant Resistance Research Unit, Agricultural Research Service, U.S. Department of Agriculture, in cooperation with the Mississippi Agricultural and Forestry Experiment Station. It is published with the approval of both agencies as Journal No. J-11037 of the Mississippi Agricultural and Forestry Experiment Station.

References Cited

- Boulter, D. 1993.** Insect pest control by copying nature using genetically engineered crops. *Phytochemistry* 34: 1453–1466.
- Davis, F. M. & W. P. Williams. 1983.** Second-generation southwestern corn borer: ear and stalk damage to susceptible and resistant maize. *J. Econ. Entomol.* 76: 507–509.
- Davis, F. M. & W. P. Williams. 1997.** A new technique for evaluating southwestern corn borer damage to post-anthesis maize, pp. 189–194. *In* J. A. Mihm [Ed.], *Proceedings of international symposium: Insect resistant maize—recent advances and utilization*. CIMMYT, Mexico, D. F., Mexico, 302 pp.
- Davis, F. M., W. P. Williams, Y. M. Chang, G. T. Baker & P. A. Hedin. 1999.** Differential growth of fall armyworm larvae (Lepidoptera: Noctuidae) reared on three phenotypic regions of whorl leaves from a resistant and a susceptible maize hybrid. *Fla. Entomol.* 82: 248–254.
- Overman, J. L. 1989.** A maize breeding program for development of hybrids with resistance to multiple species of leaf-feeding and stalk-boring Lepidoptera, pp. 235–243. *In* *Toward insect resistance maize for the third world. Proceedings international symposium on methodologies for developing host plant resistance to maize insects*. CIMMYT, Mexico, D. F., Mexico, 327 pp.
- SAS Institute. 1987.** SAS/STAT guide for personal computers. 7th ed. SAS Inst. Inc. Cary, North Carolina.
- Scott, G. E. & F. M. Davis. 1981.** Registration of Mp496 inbred line of maize. *Crop Sci.* 21: 353.
- Smith, M. E., J. A. Mihm & D. C. Jewell. 1989.** Breeding for multiple resistance to temperate, subtropical, and tropical maize insect pests at CIMMYT, pp. 222–234. *In* *Toward insect resistance for the third world. Proceedings international symposium on methodologies for developing host plant resistance to maize insects*. CIMMYT, Mexico, D. F., Mexico, 327 pp.

- Steel, R. D. G. & J. H. Torrie. 1980.** Principles and procedures of statistics. McGraw-Hill, New York, 633 pp.
- Widstrom, N. W. 1989.** Breeding methodology to increase resistance in maize to corn earworm, fall armyworm, and maize weevil. pp. 211–221. *In* Toward insect resistance for the third world. Proceedings of international symposium on methodologies for developing host plant resistance to maize insects. CIMMYT, Mexico, D.F., Mexico. 327 pp.
- Williams, W. P. & P. M. Buckley. 1992.** Growth of fall armyworm (Lepidoptera: Noctuidae) larvae in resistant and susceptible corn. *J. Econ. Entomol.* 85: 2039–2042.
- Williams, W. P. & P. M. Buckley. 1996.** Southwestern corn borer growth on laboratory diets containing lyophilized corn husks. *Crop Sci.* 36: 362–364.
- Williams, W. P. & F. M. Davis. 1982.** Registration of Mp704 germplasm line of maize. *Crop Sci.* 22: 1269–1270.
- Williams, W. P. & F. M. Davis. 1984.** Registration of Mp705, Mp706, and Mp707 germplasm lines of maize. *Crop Sci.* 24: 1217.
- Williams, W. P. & F. M. Davis. 1997.** Maize germplasm with resistance to southwestern corn borer and fall armyworm, pp. 226–229. *In* J.A. Mihm [Ed.], Proceedings of international symposium: Insect resistant maize – recent advances and utilization. CIMMYT, Mexico, D. F., Mexico, 302 pp.
- Williams, W. P. & F. M. Davis. 2000.** Registration of maize germplasms Mp713 and Mp714. *Crop Sci.* 40: 584.
- Williams, W. P. & F. M. Davis. 2002.** Registration of maize germplasm line Mp716. *Crop Sci.* 42: 671–672.
- Williams, W. P., P. M. Buckley & F. M. Davis. 1989.** Combining ability for resistance in corn to fall armyworm and southwestern corn borer. *Crop Sci.* 29: 913–915.
- Williams, W. P., P. M. Buckley, P. A. Hedin & F. M. Davis. 1990.** Laboratory bioassay for resistance in corn to fall armyworm (Lepidoptera: Noctuidae) and southwestern corn borer (Lepidoptera: Pyralidae). *J. Econ. Entomol.* 83: 1578–1581.
- Williams, W. P., J. B. Sagers, J. A. Hanten, F. M. Davis & P. M. Buckley. 1997.** Transgenic corn evaluated for resistance to fall armyworm and southwestern corn borer. *Crop Sci.* 37: 957–962.
- Williams, W. P., P. M. Buckley, J. B. Sagers & J. A. Hanten. 1998a.** Evaluation of transgenic corn for resistance to corn earworm (Lepidoptera: Noctuidae), fall armyworm (Lepidoptera: Noctuidae), and southwestern corn borer (Lepidoptera: Crambidae) in a laboratory bioassay. *J. Agric. Entomol.* 15: 105–112.
- Williams, W. P., F. M. Davis & P. M. Buckley. 1998b.** Resistance to southwestern corn borer in corn after anthesis. *Crop Sci.* 38: 1514–1517.
- Williams, W. P., F. M. Davis, P. M. Buckley, P. A. Hedin, G. T. Baker & D. S. Luthe. 1998c.** Factors associated with resistance to fall armyworm (Lepidoptera: Noctuidae) southwestern corn borer (Lepidoptera: Crambidae) in corn at different vegetative stages. *J. Econ. Entomol.* 91: 1471–1480.
- Wiseman, B. R., F. M. Davis & J. E. Campbell. 1980.** Mechanical infestation device used in fall armyworm plant resistance programs. *Fla. Entomol.* 63: 425–432.
-

Aflatoxin Accumulation in Corn Hybrids Infested at Different Growth Stages with Southwestern Corn Borer (Lepidoptera: Crambidae)¹

W. Paul Williams, Gary L. Windham, Paul M. Buckley, and
Christopher A. Daves

USDA-ARS Corn Host Plant Resistance Research Unit, Mississippi State, Mississippi 39762 USA

J. Agric. Urban Entomol. 23(2): 97–103 (April 2006)

ABSTRACT Aflatoxin is a potent toxin produced by the fungus *Aspergillus flavus*. Contamination of corn, *Zea mays* L., with aflatoxin greatly reduces the value of corn grain and is a major impediment to profitable corn production in the South. Infestation of developing corn ears with southwestern corn borer, *Diatraea grandiosella* Dyar (Lepidoptera: Crambidae), increases the likelihood of aflatoxin contamination. In this investigation, six corn hybrids with different levels of resistance to southwestern corn borer damage and aflatoxin contamination were evaluated for damage by southwestern corn borer and aflatoxin contamination after infestation with southwestern corn borer at 7-d intervals from 35–84 d after planting and inoculation with an *A. flavus* conidial suspension. Transgenic (Bt) hybrids exhibited excellent resistance to southwestern corn borer and less aflatoxin contamination than conventional versions of the same hybrids. A conventional hybrid with intermediate resistance to southwestern corn borer also exhibited relatively low levels of aflatoxin contamination. A conventional hybrid selected for resistance to aflatoxin contamination was heavily damaged by southwestern corn borer, but exhibited little aflatoxin contamination. Hybrids that have resistance to both southwestern corn borer and aflatoxin contamination appear to have the greatest potential for effectively reducing losses to aflatoxin in the South. Damage by southwestern corn borer to plants in both the vegetative and reproductive stages of growth can increase aflatoxin contamination in susceptible corn hybrids.

KEY WORDS Aflatoxin, *Aspergillus flavus*, *Diatraea grandiosella*, Lepidoptera, Crambidae, *Zea mays*, host plant resistance

Southwestern corn borer, *Diatraea grandiosella* Dyar (Lepidoptera: Crambidae), is a major pest of corn, *Zea mays* L., in the southern United States. Because this insect completes more than one generation (three in Mississippi) each year, plants are attacked at different stages of growth. When plants in the whorl stage of growth are infested with southwestern corn borer, larvae feed in the whorl of the plant for about 10 d and then tunnel into the stalk. When plants are infested after anthesis, larvae feed first on the husks, ears, and ear shoots and then tunnel into the stalk (Davis et al. 1972). Yield losses result not only from

¹Accepted for publication 28 March 2007

reduced grain production but also from dropped ears and broken stalks that impede efficient harvesting. Feeding by the insects also provides entrance for fungi into the ears. The presence of fungi can further reduce the quality and value of the grain.

In experiments in which plants were manually infested with southwestern corn borer, yield losses exceeded 50% when infestations were made during the whorl stage of growth (Williams & Davis 1990). When plants were infested at anthesis, yield losses exceeded 20% (Williams et al. 1983). Younger plants tend to sustain greater damage from southwestern corn borer. When plants were infested weekly from 4–8 wk after planting, the greatest yield loss was sustained by plants infested at 4 wk after planting, but yields at all ages of plants at infestation were significantly lower than yields of noninfested plants (Williams & Davis 1985). Heights at maturity of plants infested at 4 wk after planting were significantly less than noninfested plants. Plant heights increased linearly with each week's delay in infestation from 4 to 8 wk.

Damage to developing ears caused by feeding of southwestern corn borer has been shown to be associated with high levels of aflatoxin accumulation in harvested corn grain (Williams et al. 2002a, 2002b, 2003). Aflatoxin, a naturally occurring toxin produced by the fungus *Aspergillus flavus* Link: Fries is a potent carcinogen (Castegnaro & McGregor 1998, Pittet 1998). Aflatoxin contamination of grain is frequently associated with high temperatures and drought. The presence of aflatoxin in grain reduces its value and marketability.

Host plant resistance generally is considered a desirable way to reduce aflatoxin contamination of corn. Growing hybrids with resistance to *A. flavus* infection and aflatoxin accumulation has been demonstrated to substantially reduce aflatoxin contamination (Williams et al. 2003). When ears were manually infested with southwestern corn borer larvae and inoculated with *A. flavus*, resistance to southwestern corn borer, in either transgenic hybrids or conventionally bred hybrids, was also effective in reducing aflatoxin contamination (Williams et al. 2002c, 2005).

In previous investigations of the association between southwestern corn borer damage and aflatoxin accumulation, infestations were made when plants were in the reproductive growth stages. Those investigations focused on damage to developing ears. In this investigation, southwestern corn borer infestations were begun while plants were in the vegetative stages of growth so that the effects of damage at earlier growth stages could also be ascertained. The first objective of this investigation was to determine the effects of southwestern corn borer infestations from 35–84 d after planting on mature plant height of corn hybrids exhibiting varying levels of resistance to leaf feeding damage. The second objective was to compare aflatoxin accumulation among hybrids infested with southwestern corn borer from 35–84 d after planting and among noninfested hybrids.

Materials and Methods

Six hybrids were included in the investigation: Pioneer Brand 3223 and DK 687 are conventional, nontransgenic commercial hybrids; Pioneer Brand 31B13 and DK 6870 are their transgenic versions and express insecticidal proteins (Cry1Ab) isolated from *Bacillus thuringiensis* (Bt); Mp704 × Mp707 is a cross between two inbred lines that were selected for resistance to leaf feeding damage

by southwestern corn borer (Williams & Davis 1982, 1984); and Mp313E $\times$ Mp717 is a cross between two inbred lines selected for resistance to *A. flavus* infection and aflatoxin contamination (Scott & Zummo 1990, Williams & Windham 2006). The hybrids were planted in single-row plots approximately 4 m long with 1 m between rows on 18 April 2003, 21 April 2004, and 22 April 2005. Plots were over planted and thinned to 20 plants. Standard corn production practices were followed.

The experimental design was a randomized complete block with five replications with a split-plot arrangement of treatments and hybrids. Treatments, which consisted of infestation with southwestern corn borer larvae at 35, 42, 49, 56, 63, 70, 77, or 84 d after planting and a noninfested check, were assigned to main plots. Hybrids were assigned to subplots. Within main plots, plants were infested with 30 southwestern corn borer neonates mixed with corn cob grits at the designated time using a mechanical larval dispenser (Wiseman et al. 1980). The larvae were obtained from a laboratory colony maintained by USDA-ARS, Mississippi State, Mississippi. When the earlier infestations were made, larvae were placed in the plant whorls. At the later stages of development as tassels and ear shoots emerged, larvae were placed in the axil of the top-ear leaf. Approximately 7 d after initiation of silk emergence, all plants were inoculated five times at 7-d intervals with a suspension containing 9×10^7 *A. flavus* conidia per milliliter in distilled water, to which a spreader sticker was added (Hi-Yield Chemical Co., Bonham, Texas). The suspension was applied to ears and ear shoots using a backpack sprayer (Solo, Newport News, Virginia). Inoculum was prepared using *A. flavus* isolate NRRL3357 as described by Windham & Williams (1999).

Leaf feeding damage sustained by plants infested with southwestern corn borer 35 d after planting was visually rated 14 d after infestation on a scale of 0, no damage, to 9, extensive damage (Williams et al. 1989). At maturity, heights of 10 plants per plot were measured. Grain from all plots was harvested, dried, and ground using a Romer mill (Union, Missouri). Aflatoxin contamination was determined for each plot using the Vicam Aflatest (Watertown, Massachusetts).

Data were analyzed using the SAS general linear models procedure (SAS Institute 2001). Means were compared by Fisher's Protected Least Significant Difference (LSD) at $P = 0.05$ (Steel & Torrie 1980). Data for aflatoxin contamination were transformed before analysis of variance by adding 1 and taking the logarithm of each number [$\ln(y+1)$]. Aflatoxin data are reported as the antilogarithm of the transformed mean.

Results and Discussion

The relative level of resistance to southwestern corn borer of the six hybrids is indicated by the visual leaf feeding damage ratings (Table 1). Both hybrids ($F = 782.56$; $df = 5, 20$; $P < 0.001$) and years ($F = 96.37$; $df = 2, 8$; $P < 0.001$) were significant sources of variation. Mp313E $\times$ Mp717 sustained the heaviest damage overall; however, in 2003 and 2005, differences between Mp313E $\times$ Mp717 and the two conventional commercial hybrids were not significant. Both of the transgenic Bt hybrids sustained only minor damage each year. Although Mp704 $\times$ Mp707 sustained significantly more damage than the Bt hybrids, it sustained significantly less leaf damage than the three susceptible hybrids each year.

Table 1. Leaf feeding damage sustained by six corn hybrids infested with southwestern corn borer larvae 35 d after planting in 2003, 2004, and 2005 at Mississippi State, Mississippi.

Hybrid	Mean $\pm$ SE leaf feeding damage rating ^a			
	2003	2004	2005	Mean
3223	7.8 $\pm$ 0.5	5.6 $\pm$ 0.5	8.6 $\pm$ 0.5	7.3 $\pm$ 1.4
DK 687	7.9 $\pm$ 0.2	5.0 $\pm$ 1.2	8.8 $\pm$ 0.4	7.2 $\pm$ 1.8
31B13	1.1 $\pm$ 0.1	1.2 $\pm$ 0.4	1.4 $\pm$ 0.5	1.2 $\pm$ 0.4
DK 6870	1.3 $\pm$ 0.2	1.0 $\pm$ 0.0	1.4 $\pm$ 0.5	1.2 $\pm$ 0.4
Mp704 $\times$ Mp707	5.4 $\pm$ 0.4	3.4 $\pm$ 0.5	6.0 $\pm$ 0.7	4.9 $\pm$ 1.3
Mp313E $\times$ Mp717	8.2 $\pm$ 0.2	6.8 $\pm$ 0.8	8.8 $\pm$ 0.4	7.9 $\pm$ 1.0
LSD (0.05)	0.4	0.9	0.7	0.4

^aEach plant was infested with 30 neonates 35 d after planting. Damage was visually rated 14 d after infestation on a scale of 0 (no damage) to 9 (extensive damage).

Another measure of susceptibility to southwestern corn borer is reflected in a reduction in plant height. Infestation times ($F = 14.10$; $df = 8, 32$; $P < 0.001$), hybrids ($F = 165.78$; $df = 5, 180$; $P < 0.001$), and infestation times $\times$ hybrids ($F = 6.96$; $df = 40, 180$; $P < 0.001$) were significant sources of variation. The three hybrids that sustained the highest level of leaf feeding damage exhibited significant reductions in plant height when infested at 35, 42, or 49 d after planting (Table 2). When infested at 35 d after planting, 3223 and DK 687 were reduced in plant height by 29 and 24%, respectively. Mp313E $\times$ Mp717, the tallest of the hybrids, sustained a 13% reduction in height when infested at 35 d after planting. The height of Mp313E $\times$ Mp717 also was significantly ($P < 0.05$) reduced when plants were infested at 42 and 49 d after planting. Mp704 $\times$ Mp707 was also reduced in height by 13% when infested at 35 d, but its height was not significantly reduced at other infestation dates. This gives further evidence of its resistance to southwestern corn borer. Plant height of the two Bt hybrids was not reduced at any date of infestation. Both plant heights and leaf feeding damage ratings are indicative of the excellent resistance to southwestern corn borer that these hybrids possess.

Aflatoxin contamination differed among years ($F = 89.52$; $df = 8, 40$; $P < 0.0001$) and among hybrids ($F = 80.67$; $df = 2, 8$; $P < 0.0001$). Overall aflatoxin accumulation was lowest in 2003 (12 ng/g). The mean aflatoxin accumulation for all infestation times was less than 1 ng/g for DK 6870, Mp704 $\times$ Mp707, and Mp313E $\times$ Mp717 that year. Aflatoxin accumulation was highest for 3223 (28 ng/g). Overall aflatoxin accumulation was higher in 2004 than in 2003 and varied from 3 ng/g for DK 6870–127 for 3223. Aflatoxin was highest in 2005 when it ranged from 11–221 ng/g for Mp313E $\times$ Mp717 and DK687, respectively.

Aflatoxin contamination of the six hybrids after inoculation with *A. flavus* and infestation with southwestern corn borer at different stages of plant development are given in Table 3. Differences among times of infestation with southwestern corn borer were significant ($F = 4.28$; $df = 8, 32$; $P < 0.001$). Neither hybrids $\times$ infestation times ($F = 2.136$; $df = 40, 180$; $P < 0.345$) nor years $\times$ infestation

Table 2. Plant heights of six corn hybrids infested with southwestern corn borer larvae 35 to 84 d after planting in 2003, 2004, and 2005 at Mississippi State, Mississippi.

Infestation: days after planting	Mean $\pm$ SE plant height (cm)					
	3223 ^a	DK 687	31B13	DK 6870	Mp704 $\times$ Mp707	Mp313E $\times$ Mp717
35 ^b	160 $\pm$ 36	170 $\pm$ 38	225 $\pm$ 62	218 $\pm$ 29	203 $\pm$ 37	228 $\pm$ 41
42	173 $\pm$ 23	186 $\pm$ 26	211 $\pm$ 29	224 $\pm$ 27	217 $\pm$ 29	240 $\pm$ 25
49	190 $\pm$ 34	195 $\pm$ 28	218 $\pm$ 30	221 $\pm$ 30	221 $\pm$ 43	243 $\pm$ 33
56	208 $\pm$ 40	213 $\pm$ 32	219 $\pm$ 31	223 $\pm$ 34	219 $\pm$ 28	262 $\pm$ 34
63	221 $\pm$ 44	218 $\pm$ 34	215 $\pm$ 28	217 $\pm$ 39	222 $\pm$ 33	261 $\pm$ 46
70	226 $\pm$ 34	225 $\pm$ 29	217 $\pm$ 32	224 $\pm$ 32	224 $\pm$ 32	268 $\pm$ 37
77	225 $\pm$ 36	222 $\pm$ 30	218 $\pm$ 36	219 $\pm$ 36	213 $\pm$ 39	265 $\pm$ 43
84	230 $\pm$ 37	226 $\pm$ 32	220 $\pm$ 37	231 $\pm$ 29	235 $\pm$ 79	274 $\pm$ 40
Uninfested	226 $\pm$ 35	224 $\pm$ 31	215 $\pm$ 28	224 $\pm$ 29	232 $\pm$ 18	264 $\pm$ 36

^aFor comparisons of infestation times within hybrids (within columns): LSD (0.05) = 19.

^bFor comparisons of hybrids within one infestation time (across rows): LSD (0.05) = 12.

times ($F = 0.96$; $df = 16, 64$; $P < 0.5127$) were significant. Mp313E $\times$ Mp717, a cross of two germplasm lines selected for resistance to aflatoxin accumulation, exhibited relatively low levels of aflatoxin contamination at all dates of southwestern corn borer infestation. When hybrids were infested with southwestern corn borer at 35, 42, 49, 70, 77, and 84 d after planting, hybrids 3223 and DK 687 had significantly greater levels of contamination than the other hybrids.

The levels of aflatoxin accumulation did not differ between Mp313E $\times$ Mp717 and DK 6870 at any date of infestation. Mp704 $\times$ Mp707 did not differ in aflatoxin accumulation from DK 6870 at any time of infestation although its level of aflatoxin accumulation was higher than Mp313E $\times$ Mp717 when infestations were made at 35, 42, and 84 d. Although Mp313E $\times$ Mp717 was heavily damaged by southwestern corn borer, its inherent resistance to aflatoxin accumulation was sufficient to provide protection from aflatoxin contamination.

A comparison of the levels of aflatoxin contamination of conventional hybrids 3223 and DK 687 and their transgenic versions, 31B13 and DK 6870, indicates the value of resistance to southwestern corn borer in reducing aflatoxin in environments where the insect is found. The relatively low levels of aflatoxin accumulation in Mp704 $\times$ Mp707 provide further evidence of the value of resistance to southwestern corn borer in reducing contamination. Both types of resistance derived from another species as with the Bt hybrids and naturally occurring genetic resistance as in Mp704 $\times$ Mp707 could prove valuable in reducing aflatoxin contamination. In areas in which the acreage planted to Bt hybrids is restricted, naturally occurring genetic resistance such as that found in Mp704 $\times$ Mp707 could be especially useful. Although resistance to either aflatoxin accumulation per se or resistance to southwestern corn borer damage resulted in relatively low levels of aflatoxin contamination in this investigation, combining the two types of resistance could prove valuable in years or locations in which the potential for

Table 3. Aflatoxin accumulation in six corn hybrids infested with southwestern corn borer larvae 35 to 84 d after planting in 2003, 2004, and 2005 at Mississippi State, Mississippi.

Infestation: days after planting	Mean aflatoxin (ng/g) and transformed means [$\ln(y + 1)$] ^a						
	3223 ^c	DK 687	31B13	DK 6870	Mp704 × Mp707	Mp313E × Mp717	Mean ^d
35 ^b	37 (3.64)	64 (4.17)	9 (2.26)	3 (1.26)	8 (2.22)	2 (1.09)	10 (2.38)
42	84 (4.44)	108 (4.69)	22 (3.16)	5 (1.77)	13 (2.63)	2 (1.07)	17 (2.89)
49	101 (4.62)	50 (3.93)	11 (2.53)	4 (1.51)	6 (1.93)	2 (1.19)	18 (2.64)
56	57 (4.06)	18 (2.92)	7 (2.09)	4 (1.66)	8 (2.21)	4 (1.52)	10 (2.41)
63	84 (4.44)	27 (3.35)	8 (2.18)	2 (1.23)	6 (1.90)	7 (2.07)	12 (2.57)
70	37 (3.62)	38 (3.67)	12 (2.57)	5 (1.77)	8 (2.17)	3 (1.33)	12 (2.54)
77	23 (2.19)	28 (3.38)	14 (2.72)	3 (1.50)	6 (2.01)	2 (1.01)	9 (2.30)
84	107 (4.68)	32 (3.49)	6 (1.97)	4 (1.47)	9 (2.35)	1 (0.77)	11 (2.47)
Uninfested	39 (3.69)	12 (2.55)	6 (1.91)	3 (1.31)	4 (1.56)	2 (1.00)	7 (2.10)
Mean ^e	57 (4.07)	33 (3.53)	10 (2.36)	4 (1.51)	8 (2.15)	2 (1.25)	

^aValues for aflatoxin in contamination were transformed [$\ln(y + 1)$] before analysis. The transformed values are in parenthesis following the non-transformed means in each column.

^bFor comparison among infestation times within hybrids (within columns) of the transformed values: LSD (0.05) = 0.83.

^cFor comparison among hybrids within infestation times (across rows) of the transformed values: LSD (0.05) = 1.01.

^dFor comparison of means of means for infestation times across hybrids of the transformed values: LSD (0.05) = 0.34.

^eFor comparisons of means for hybrids across infestation times of the transformed values: LSD (0.05) = 0.42.

high levels of contamination is high. The potential for high levels of aflatoxin is especially high when corn is grown under drought stress (Windham & Williams 1999). Combining resistance to drought stress with resistance to southwestern corn borer damage and aflatoxin accumulation in corn hybrids could further reduce losses to aflatoxin.

Acknowledgments

We gratefully acknowledge the excellent technical assistance provided by M. N. Alpe, G. A. Matthews, L. T. Owens, and S. H. Wolf. The manuscript is a contribution of the USDA-ARS Corn Host Plant Resistance Research Unit in cooperation with the Mississippi Agricultural and Forestry Experiment Station. It is published as Journal No. J-11027 of the Miss. Agric. and Forestry Exp. Stn. Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture.

References Cited

- Castegnaro, M. & D. McGregor. 1998.** Carcinogenic risk assessment of mycotoxins. *Rev. Med. Vet.* 149: 671–678.
- Davis, F. M., C. A. Henderson & G. E. Scott. 1972.** Movements and feeding of southwestern corn borer on two stages of corn growth. *J. Econ. Entomol.* 65: 519–521.

- Pittet, A. 1998.** Natural occurrence of mycotoxins in foods and feeds—an updated review. *Rev. Med. Vet.* 149: 479–492.
- SAS Institute. 2001.** SAS/STAT user's guide, version 8. Cary, North Carolina.
- Scott, G. E. & N. Zummo. 1990.** Registration of Mp313E parental line of maize. *Crop Sci.* 30: 1378.
- Steel, R. D. G. & J. H. Torrie. 1980.** Principles and procedures of statistics. McGraw-Hill, New York.
- Williams, W. P. & F. M. Davis. 1982.** Registration of Mp704 germplasm line of maize. *Crop Sci.* 22: 1269–1270.
- Williams, W. P. & F. M. Davis. 1984.** Registration of Mp705, Mp706, and Mp707 germplasm line of maize. *Crop Sci.* 24: 1217.
- Williams, W. P. & F. M. Davis. 1985.** Southwestern corn borer damage to maize: influence of plant age at time of larval infestation. *Agric. Ecosyst. Environ.* 12: 201–205.
- Williams, W. P. & F. M. Davis. 1990.** Response of corn to artificial infestation with fall armyworm and southwestern corn borer larvae. *Southwestern Entomol.* 15: 163–166.
- Williams, W. P. & G. L. Windham. 2006.** Registration of Mp717 germplasm line of maize. *Crop Sci.* 46: 1407–1408.
- Williams, W. P., P. M. Buckley & F. M. Davis. 1989.** Combining ability for resistance to fall armyworm and southwestern corn borer. *Crop Sci.* 29: 913–915.
- Williams, W. P., P. M. Buckley & G. L. Windham. 2002a.** Southwestern corn borer (Lepidoptera: Crambidae) damage and aflatoxin accumulation in maize. *J. Econ. Entomol.* 95: 1049–1053.
- Williams, W. P., F. M. Davis & G. E. Scott. 1983.** Second-brood southwestern corn borer infestation levels and their effect on corn. *Agron. J.* 75: 132–134.
- Williams, W. P., F. M. Davis, G. L. Windham & P. M. Buckley. 2002b.** Southwestern corn borer damage and aflatoxin accumulation in a diallel cross of maize. *J. Genet. Breed.* 56: 165–169.
- Williams, W. P., F. M. Davis, G. L. Windham & P. M. Buckley. 2003.** Aflatoxin accumulation in maize after inoculation with *Aspergillus flavus* and infestation with southwestern corn borer. *J. Genet. Breed.* 57: 365–370.
- Williams, W. P., G. L. Windham, P. M. Buckley & C. A. Daves. 2002c.** Aflatoxin accumulation in conventional and transgenic corn hybrids infested with southwestern corn borer (Lepidoptera: Crambidae). *J. Agric. Urban Entomol.* 19: 227–236.
- Williams, W. P., G. L. Windham, P. M. Buckley & J. M. Perkins. 2005.** Southwestern corn borer damage and aflatoxin accumulation in conventional and transgenic corn hybrids. *Field Crops Res.* 91: 329–336.
- Windham, G. L. & W. P. Williams. 1999.** Aflatoxin accumulation in commercial corn hybrids in 1998. *Miss. Agric. Forestry Exp. Stn. Res. Rep.* 22: 8.
- Wiseman, B. R., F. M. Davis & J. E. Campbell. 1980.** Mechanical infestation device used in fall armyworm plant resistance programs. *Fla. Entomol.* 63: 426–432.
-

Susceptibility of Field Collected House Flies to Spinosad Before and After a Season of Use¹

Juliane M. Deacutis, Cheryl A. Leichter, Alec C. Gerry,² Donald A. Rutz, Wes D. Watson,³ Christopher J. Geden,⁴ and Jeffrey G. Scott

Department of Entomology, Comstock Hall, Cornell University, Ithaca, New York 14853 USA

J. Agric. Urban Entomol. 23(2): 105–110 (April 2006)

ABSTRACT Spinosad is a new and highly promising insecticide with a novel mode of action, and it was first available for house fly control in the United States in 2005. To maintain the effectiveness of this new insecticide, it will be important to monitor populations for the evolution of resistance. We compared three bioassay methods (topical application, residual exposure, and feeding) using laboratory strains of spinosad susceptible and resistant adult house flies. Topical application was the most reliable and efficient method for detection of spinosad-resistant individuals. Using this bioassay we evaluated house flies that were collected before spinosad use (in 2004 and 2005). We found significant differences in survival between dairies, suggesting there was inherent variability between house fly populations prior to spinosad use. We also evaluated field collected house flies from before and after one season of spinosad use (2005). There was no evidence spinosad resistance was evolving after one season of spraying. Uses and limitations of monitoring spinosad resistance by this method are discussed.

KEY WORDS insecticide resistance monitoring, *Musca domestica*, dairy, poultry

House flies, *Musca domestica* L. (Diptera: Muscidae), are major pests in and around dairy, poultry, and hog facilities. Given that resistance to organophosphate and pyrethroid insecticides in house flies in the United States is widespread (Scott et al. 1989, Scott et al. 2000, Kaufman & Rutz 2001, Kaufman et al. 2001, Darbro & Mullens 2004), there is an urgent need for new insecticides that are effective against this pest.

Spinosad is a new and highly promising insecticide, derived from the soil actinomycete *Saccharopolyspora spinosa*. In 2005, spinosad was made available for control of house flies in the United States. Spinosad acts at the nicotinic acetylcholine receptor and has efficacy against a wide range of insects, including house flies (Bret et al. 1997, Scott 1998, Salgado & Sparks 2005). Recently, a strain of house flies (NYSPINR) that had high levels of resistance to spinosad was

¹Accepted for publication 7 May 2007

²Department of Entomology, University of California, Riverside, California 92521

³Department of Entomology, North Carolina State University, Raleigh, North Carolina 27695

⁴USDA, ARS, Gainesville, Florida 32607

selected (in the laboratory) from field collected house flies (Shono & Scott 2003), indicating that there was potential for the evolution of resistance after repeated use of spinosad in the field. Isolation of the NYSPINR strain required only 10 generations of selection, suggesting resistance might be able to evolve in as little as one season. However, there is no information about the baseline susceptibility of field populations of house flies to spinosad, and no method in place for resistance monitoring.

The goals of this study were to identify an effective bioassay method for detection of spinosad resistant house flies, survey for baseline susceptibility to spinosad in field collected house flies (i.e. determine variability between populations) in 2004 and 2005, and to determine if we could detect any increase in the frequency of resistant individuals at three dairies in California and three dairies in New York after a season of spinosad use (2005).

Methods and Materials

House flies. Two reference (laboratory) strains of house flies were used for comparison of topical, residual, and feeding bioassays. CS (Hamm et al. 2005) is an insecticide susceptible strain and NYSPINR is a spinosad-resistant strain (Shono & Scott 2003). To produce flies that were heterozygous for spinosad resistance (for bioassays) we crossed NYSPINR females and susceptible aabys (Hamm et al. 2005) males and *en masse*.

In 2004 (prior to spinosad use), six different strains of house flies (*Musca domestica*) were collected at various dairy (Alachua County, Florida, Schuyler and Tompkins Counties, New York), poultry (Sullivan and Wayne Counties, New York), and hog (Wake County, North Carolina) facilities in the Eastern United States. The levels of resistance to permethrin, cyfluthrin, pyrethrins, dimethoate, tetrachlorvinphos, and methomyl in house flies from the Schuyler County dairy and Wayne County poultry facility have been previously reported (Scott et al. 2000, Kaufman et al. 2001).

In 2005, house flies were collected from four dairies (P, M, H, and C) in Tompkins County New York and from four dairies in San Diego (DV), San Bernardino (AM and BJ), and Riverside (BS) Counties, California. These facilities were chosen because they were willing to participate in this study, they were within collecting distance, and because they represented two geographically distant regions (California and New York). Two collections were taken from each dairy. The first collection was made before spinosad was used. A second collection was taken at the end of the season ("postseason"), but while flies were still abundant. In New York, the dairies applied up to six applications of spinosad, except for the H dairy that served as our no spinosad control. In California, dairies applied spinosad four to five times, except for the DV dairy, which served as our no spinosad control.

House fly larvae were reared on medium containing 2.3 L of water, 0.5 kg of calf manna (Manna Pro Corp, St. Louis, Missouri), 90 g of bird and reptile little wood chips (Northeastern Products Corp, Warnersburg, New York), 0.8 kg of wheat bran (Agway; Ithaca, New York), and 50 g of dried active baker's yeast (ICN Biomedicals, Costa Mesa, California). Adult flies were raised on powdered milk + white granulated sugar (1:1 ratio by volume) and water *ad libitum*.

Bioassays. Three bioassay methods were evaluated in this study: topical application to the thoracic notum in 0.5 μ L of acetone (Shono & Scott 2003),

residual exposure in glass jars (Hamm et al. 2005), and via feeding. For feeding assays, spinosad (spinosyns A and D (88.5% purity) from Dow AgroSciences, Indianapolis, Indiana) was applied (0.25 mL in acetone solution) to individual cubes of sugar (Domino Dots, Tate and Lyle, London, United Kingdom). Treated cubes were allowed to dry for at least 3 h. One cube of sugar was placed into a 180-mL Sweetheart waxed paper cup with 20 flies and a 2.5-cm dental wick soaked in water. Cups were covered with nylon tulle and secured with rubber bands. All bioassays were conducted with 3- to 5-d-old female flies held at 25°C. Mortality was assessed after 48 h with flies that were ataxic being scored as dead. For determination of LD₅₀ or LC₅₀ values using the laboratory strains, a minimum of four doses (or concentrations), giving >0% and <100% mortality, were used for each replication and the entire bioassay was replicated a minimum of three times. Bioassay data were pooled and analyzed by standard probit analysis (Finney 1971), as adapted to personal computer use by Raymond (Raymond 1985) using Abbott's (Abbott 1925) correction for control mortality. Field collected flies were tested within four generations of being collected. Field collected flies were bioassayed by topical application at the LD₉₉, 3 × LD₉₉ and 10 × LD₉₉ of the susceptible strain (acetone only was used for the controls). CS flies were periodically tested side-by-side with the field collected flies. Percent mortality was arcsine transformed and differences were evaluated using Student's *t*-test.

Results and Discussion

Spinosad was toxic to house flies by feeding, exposure to a residue, and by topical application (Table 1). The NYSPINR strain was resistant to spinosad by all of these methods (Table 1), which is consistent with target site insensitivity being the mechanism of resistance (Scott 1990), as was previously suggested (Shono & Scott 2003). Considering the cost (supplies, etc.), efficiency (time per assay), and heterogeneity of response (i.e., slope) between the three bioassay methods we chose to use topical application (low cost, time efficient, and a low heterogeneity of response) at three diagnostic doses to monitor resistance in field populations of house flies.

To generate baseline data for the effectiveness of spinosad against field collected house flies, we determined the percent survival at three diagnostic doses (susceptible strain LD₉₉, 3 × LD₉₉ and 10 × LD₉₉) by topical application. Susceptibility of the different house fly strains to spinosad varied between collection sites (Fig. 1). At the susceptible strain LD₉₉, survival ranged from 1% (Alachua Co., Florida) to 61% (Wayne Co., New York), with all except one facility having <30% survival. At 3 × LD₉₉, survival ranged from 0% (three sites) to 2% (Wayne Co., New York). There were no survivors from any strain at 10 × LD₉₉. The higher percent survival at the Wayne Co. New York site (at the LD₉₉ and 3 × LD₉₉) suggests that there may be populations of house flies against which spinosad is less effective. Flies from the Florida dairy were highly susceptible, with few survivors at any dose. Although the highest percent survival was seen at two dairies, there was no correlation between type of facility and percent survival.

Evaluation of the New York dairies in 2005, before the use of spinosad, showed variability in response to spinosad similar to what was seen in 2004 with percent survival ranging from 16% to 21% at the susceptible strain LD₉₉ (Fig. 2). However, survival of flies from the four California dairies was lower, ranging from

Table 1. Comparison of spinosad toxicity to susceptible (CS) and resistant (NYSPINR) strains of house fly by three bioassay methods.

Method	CS		NYSPINR	
	LC ₅₀ or LD ₅₀ (CI)	Slope (SE)	LC ₅₀ or LD ₅₀	RR
Topical	0.054 ^a (0.049–0.058)	5.8 (0.9)	>10 ^b	>150
Feeding	2.85 ^c (2.53–3.30)	3.0 (0.3)	>1000 ^d	>300
Residue	0.064 ^e (0.038–0.108)	3.1 (1.6)	>60 ^f	>900

^aLD₅₀ in units of µg/fly at 48 h (Shono & Scott 2003).

^bLess than 50% mortality at 10 µg/fly.

^cLD₅₀ in units of µg/g at 48 h.

^dLess than 50% mortality at 1,000 µg/g.

^eLD₅₀ in units of µg/cm² at 48 h.

^fLess than 50% mortality at 60 µg/cm².

0.5% to 3.0%. After the use of spinosad for fly control during 2005, there was no indication that resistance was evolving (Fig. 2). To the contrary, the survival after a season of spinosad use was significantly lower at two of the collection sites. It is unclear why the flies collected in California were more sensitive (in most cases) than flies collected in New York.

To evaluate the sensitivity of our resistance monitoring bioassay for detection of homozygous- and heterozygous-resistant individuals, we evaluated survival of NYSPINR and F₁ (NYSPINR females × aabys males) house flies. The NYSPINR strain had nearly 100% survival at the LD₉₉ and 3 × LD₉₉ doses, whereas the F₁

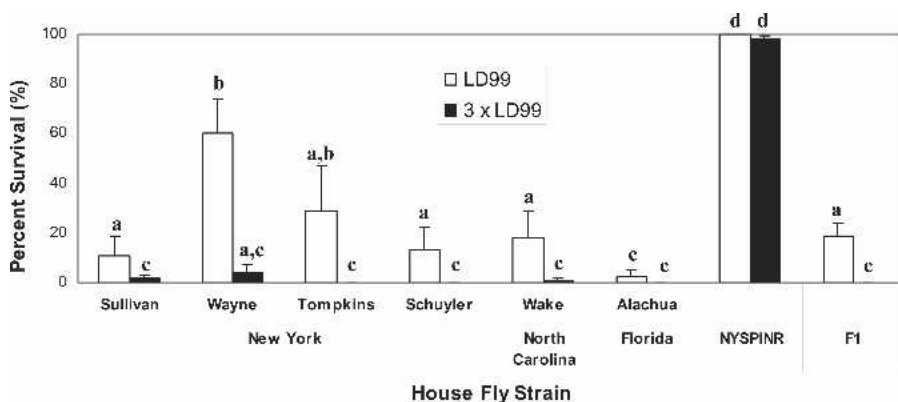


Fig. 1. Mean percent survival of house flies collected in 2004 from six sites in the Eastern United States at the susceptible strain LD₉₉ and 3 × LD₉₉. All flies were from dairies, except for Wake (hog), Sullivan, and Wayne (poultry) counties. The spinosad-resistant NYSPINR and F₁ (NYSPINR × susceptible aabys strain) are shown for comparison. Bars represent the standard deviation from the mean. CS house flies treated at these doses had 0% survival.

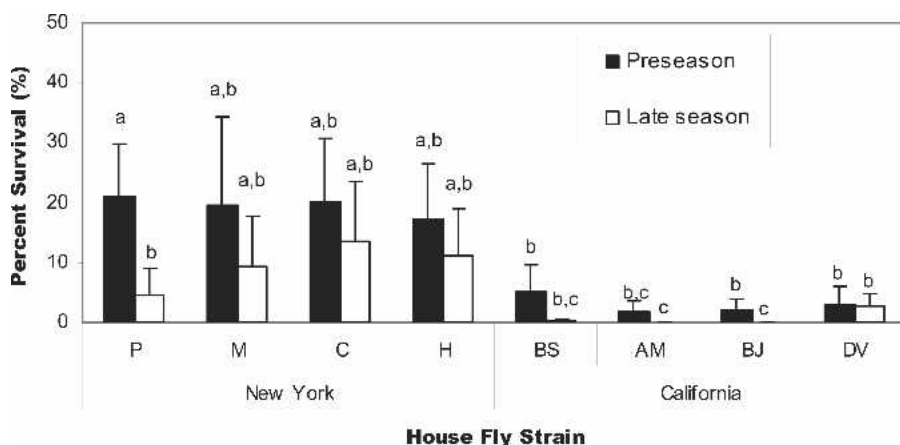


Fig. 2. Mean percent survival of house flies collected in 2005 before or after a season using spinosad for fly control, except for the H and DV dairies, which served as negative controls (no spinosad was used at these dairies). Flies were tested at the susceptible strain LD₉₉. Bars represent the standard deviation from the mean.

had 19% and 0% survival at these doses, respectively (Fig. 1). Thus, the homozygous-resistant house flies (NYSPINR) are readily detected, but the heterozygous-resistant house flies (NYSPINR $\times$ aabys F₁) are indistinguishable from field collected house flies that have never been exposed to spinosad (Fig. 1). The highly recessive nature of this resistance (combined with inherent variability in the bioassay with field collected flies) will make it very difficult to detect heterozygous resistant individuals (at least when they are present at low frequencies) in field populations.

Our results indicate that there is variation in susceptibility to spinosad in flies collected from different sites. Whereas selection of field collected house flies produced a highly resistant strain of house fly after eight generations of selection (Shono & Scott 2003), we did not detect a decrease in percent survival at the diagnostic dose at any site after one season of use. Spinosad works at a novel target site (Salgado & Sparks 2005) and resistance in the house fly is highly recessive (Shono & Scott 2003), which would be expected to slow the rate of evolution of resistance in field populations (Georghiou 1983). However, spinosad must be used judiciously and periodic monitoring of resistance should continue. Spinosad resistance is highly recessive and heterozygous individuals can not be readily detected (especially against the normal variation that exists in populations) using insecticide bioassays. Given this limitation it will be important to identify the gene (and allele) responsible for spinosad resistance, so that a more sensitive detection method can be developed.

Acknowledgments

We thank C. Reasor for assistance with the house fly collections. This work was supported by Elanco Animal Health and Hatch project 414.

References Cited

- Abbott, W. S. 1925.** A method of computing the effectiveness of an insecticide. *J. Econ. Entomol.* 18: 265–267.
- Bret, B. L., L. L. Larson, J. R. Schoonover, T. C. Sparks & G. D. Thompson. 1997.** Biological properties of spinosad. *Down to Earth* 52: 6–13.
- Darbro, J. M. & B. A. Mullens. 2004.** Assessing insecticide resistance and aversion to methomyl-treated toxic baits in *Musca domestica* L (Diptera: Muscidae) populations in southern California. *Pest Manage. Sci.* 60: 901–908.
- Finney, D. J. 1971.** Probit Analysis. Cambridge University Press, Cambridge, United Kingdom.
- Georgioui, G. P. 1983.** Management of resistance in arthropods, pp. 769–792. *In* G. P. Georgioui and T. Saito [Eds.], *Pest Resistance to Pesticides*. Plenum Press, New York.
- Hamm, R., T. Shono & J. G. Scott. 2005.** A cline in frequency of autosomal males is not associated with insecticide resistance in house fly (Diptera: Muscidae). *J. Econ. Entomol.* 98: 171–176.
- Kaufman, P. E. & D. A. Rutz. 2001.** Susceptibility of house flies (Diptera: Muscidae) exposed to commercial insecticides on painted and unpainted plywood panels. *Pest Manag. Sci.* 58: 174–178.
- Kaufman, P. E., J. G. Scott & D. A. Rutz. 2001.** Monitoring insecticide resistance in house flies (Diptera: Muscidae) from New York dairies. *Pest Manag. Sci.* 57: 514–521.
- Raymond, M. 1985.** Presentation d'un programme Basic d'analyse log-probit pour micro-ordinateur. *Cah. ORSTROM, Ser. Ent. Med. Parasitol.* 23: 117–121.
- Salgado, V. L. & T. C. Sparks. 2005.** The spinosyns: chemistry, biochemistry, mode of action, and resistance, pp. 137–173. *In* L. I. Gilbert, K. Iatrou and S. S. Gill [Eds.], *Comprehensive Molecular Insect Science*. Elsevier, Boston.
- Scott, J. G. 1990.** Investigating mechanisms of insecticide resistance: methods, strategies and pitfalls, pp. 39–57. *In* R. T. Roush and B. Tabashnik [Eds.], *Pesticide Resistance in Arthropods*. Chapman and Hall, New York.
- Scott, J. G. 1998.** Toxicity of spinosad to susceptible and resistant strains of house flies, *Musca domestica*. *Pestic. Sci.* 54: 131–133.
- Scott, J. G., R. T. Roush & D. A. Rutz. 1989.** Insecticide resistance of house flies from New York dairies (Diptera: Muscidae). *J. Agric. Entomol.* 6: 53–64.
- Scott, J. G., T. G. Alefantis, P. E. Kaufman & D. A. Rutz. 2000.** Insecticide resistance in house flies from caged-layer poultry facilities. *Pest Manage. Sci.* 56: 147–153.
- Shono, T. & J. G. Scott. 2003.** Spinosad resistance in the house fly, *Musca domestica*, is due to a recessive factor on autosome 1. *Pestic. Biochem. Physiol.* 75: 1–7.
-

Effects of Transgenic *Bacillus thuringiensis* Corn and Permethrin on Nontarget Arthropods¹

Denny J. Bruck, Miriam D. Lopez,² Leslie C. Lewis,² Jarrad R. Prasifka,²
Robert D. Gunnarson²

USDA-ARS, Horticultural Crops Research Laboratory, 3420 N.W. Orchard Avenue,
Corvallis, Oregon 97330 USA

J. Agric. Urban Entomol. 23(3): 111–124 (July 2006)

ABSTRACT Planting of insect-resistant transgenic corn has led to concerns about potential effects on nontarget arthropods. Various sampling methods (visual counts, yellow sticky cards, pitfall traps) were used at 7-to 10-day intervals to test for possible changes in abundance for common nontarget arthropods caused by transgenic or insecticide-based pest management. Large-scale field studies arranged in randomized complete blocks were conducted in 2001 and 2002 in central Iowa. Treatments included Bt corn (Pioneer 34M95, event MON810), a nonBt isolate (Pioneer 34M94), and the nonBt isolate treated with permethrin (Pounce 1.5G®). There was high variation in population estimates for the majority of the arthropods collected. Consequently, only data from taxa with coefficients of variation ≤ 75 using a particular sampling method were used in statistical analysis. Insecticide applications negatively impacted populations of arachnids (Araneae and Opiliones), Cantharidae, Elateridae and *Macrocentrus cingulum* (Hymenoptera: Braconidae). In the transgenic treatment, fewer *M. cingulum* and Nitidulidae were present, presumably because of a lack of *Ostrinia nubilalis* (Lepidoptera: Crambidae), which serve as hosts for *M. cingulum* and provide habitat for Nitidulidae, which are known to frequent *O. nubilalis* tunnels. However, application of a conventional insecticide for *O. nubilalis* control had a broader impact on populations of various nontarget arthropods. These results are consistent with other studies that suggest nontarget impacts resulting from insecticides may be greater than those caused by Bt corn.

KEY WORDS nontargets, coefficient of variation, arthropods, Bt corn

The European corn borer *Ostrinia nubilalis* (Hübner) (Lepidoptera: Crambidae) can cause severe economic injury to corn and has been historically controlled by chemicals, host-plant resistance, and biological control (Bergman et al. 1985). In recent years, *O. nubilalis* control has been achieved by planting genetically modified insecticidal corn. Several varieties of corn have been modified by insertion of specific genes to continuously express the crystalline (Cry) proteins of *Bacillus thuringiensis* (Berliner) (Bacillales: Bacillaceae) (Bt). These proteins dis-

¹Accepted for publication 12 June 2007

²USDA-ARS, Corn Insects and Crop Genetics Research Unit, Genetics Laboratory, c/o Insectary Building, Ames, Iowa 50011 USA

rupt the digestive systems of some insects, leading to gut paralysis and death (Tanada & Kaya 1993). Strains of Bt-producing different Cry proteins have selective toxicity because of the conditions necessary to solubilize, activate, and bind the proteins. The earliest varieties of Bt corn were modified by the insertion of lepidopteran-specific genes for *O. nubilalis* control. Because Bt must be ingested to have toxic effects, lepidopteran-specific Bt corn should kill only *O. nubilalis* or other susceptible Lepidoptera feeding on corn, leaving the remainder of the arthropod community unharmed.

However, because Bt corn has been genetically modified and has insecticidal properties, there are still questions concerning its safety regarding nontarget organisms. The potential for negative effects on nontarget arthropods has been an area of active research after a report that monarch larvae *Danaus plexippus* (L.) (Lepidoptera: Danaidae) experienced toxicity as a result of Bt pollen (Losey et al. 1999). These findings led to additional research into nontarget effects of Bt corn, including community-level responses (Dively 2005, Prasifka et al. 2005), effects of Bt pollen consumption (Hellmich et al. 2001, Dively et al. 2004), Bt plant material in the soil (Al-Deeb et al. 2003), and effects of Bt on predators and parasitoids (Dutton et al. 2003, Bhatti et al. 2005). Because of the complexity of these relationships and the possibility of additional unknown interactions, in-field studies are needed to assess what risks Bt corn may pose to nontarget arthropods.

Determining the effects of Bt crops on nontarget arthropods is complicated because clear protocols for selecting an experimental design, which taxa to sample, sampling methods, and analysis of results have not been established. The experimental design used must address plot size, replication, nontarget groups to be sampled, and a method for estimating nontarget group population size. Plot size and replication are important because inadequately sized plots can make it difficult to find statistically significant differences (Prasifka et al. 2005) and insufficient replication leads to decreased statistical power (Lopez et al. 2005). The choice of taxa can be narrow (e.g., sampling predators and parasitoids of the target pest [Pilcher et al. 2005], or all members of a group such as Collembola [Bitzer et al. 2005]), or broad (census-type sampling, i.e., sampling all arthropods present in the field [Dively 2005]). Methods for estimating population size should be suited to the arthropods being sampled; although both ground and ladybird beetles are coleopterans, general differences in locomotion (crawling versus flying) and microhabitat (ground versus foliage) suggest different sampling methods are appropriate for each type of beetle.

The objective of this study was to determine whether nontarget arthropods in Iowa corn fields were impacted by Bt corn and conventional insecticide applications using a standard randomized complete block experimental design and various sampling methods. The arthropods recorded were common in corn and could be identified with taxonomic certainty without requiring external professional expertise. In an attempt to include many potential avenues for nontarget effects, arthropods that occupied various ecological roles (predator, parasitoid, herbivore, and fungivore) were included. Rather than using a broad (community-wide) analysis or looking for effects on a narrow group of arthropods, we sampled many of the nontarget taxa using three common methods of population estimation, but restricted our data analysis to those groups that met a minimum sampling standard based on abundance and variability.

Materials and Methods

Field plots. Plots at the Sorenson farm, 13 km west of Ames, Iowa, were located within a single 198 × 190-m field that had been under corn-soybean rotation for 6 years. A randomized complete block experimental design with three replications was used. Replicates (61 × 198 m) were separated by 3-m alleyways, and plots were surrounded by five border rows (4 m) of nonBt isoline corn. Each replicate was divided into three treatments that were 80 rows (61 m) wide. Treatments included Bt corn (Pioneer 34M95, event MON810), an untreated nonBt isoline (Pioneer 34M94), and the nonBt isoline treated with permethrin (Pounce 1.5G®, permethrin, FMC Agricultural Products, Philadelphia, Pennsylvania), which is approved for use against first- and second-generation *O. nubilalis*. Corn was planted 16 May 2001 and 9 May 2002. Insecticide applications were made to coincide with the flight of *O. nubilalis* adults on 2 August (2001), 27 June, and 21 August (2002) at the rate of 11,340 g/ha. The experiment was designed to receive an insecticide application to coincide with both generations in 2001 and 2002, but weather conditions did not allow for first-generation applications in 2001. In 2002, treatments were planted in the same locations within the field as in 2001. Within each plot (80 rows), there were six rows designated for trapping, each containing five trapping stations (pitfall and yellow sticky cards) 4.5 m apart for a total of 10 traps per treatment and 30 per replicate. During each sampling period, two rows (11.4 m apart) of every treatment were sampled. Traps were operated for 7–10 days at 2-week intervals from June through October (2001) or September (2002).

Pitfall traps. Pitfall trapping was conducted as described by Lopez et al. (2005). After collection, pitfall traps were kept at 4°C until processed. To remove soil and antifreeze, the trap contents were first sifted through a 2.36-mm mesh screen. Large (more than 3 mm in length) arthropods were removed, rinsed with tap water, and placed in 70% ethanol. The remaining contents were then rinsed with tap water over a 45-μm mesh screen to remove fine soil particles, and the arthropods, along with the larger soil particles, were placed in a flat-topped modified separatory funnel filled with a saturated (45%) sugar solution. Arthropods floated in the sugar solution, whereas soil settled out and was removed. The contents of the separation funnel were poured through a 45-μm screen, rinsed with 70% ethanol, and placed into 70% ethanol along with the large arthropods removed previously. Processed samples were held at 4°C until arthropods could be identified.

Each sample was poured into a shallow 9-cm Petri plate. Relatively large arthropods were removed and identified and the remaining alcohol (containing smaller arthropods) was poured into a Büchner funnel lined with filter paper and vacuum-filtered. The filter paper was examined under a dissecting microscope and arthropods identified. Total numbers of the arthropods that were both small and numerous were estimated by counting the number that fell more than half-way into a marked area (10% of the total) of the filter paper and extrapolating. Numbers of aphids and collembola were commonly estimated in this way, and occasionally Staphylinidae and arachnids when they were numerous.

Sticky traps. Yellow 7.6 × 12.7 cm sticky traps (Olson Products, Medina, Ohio) with one sticky side exposed were placed within corn rows attached to 1.2-m × 1.0-cm unpainted bamboo stakes. They were attached to the stake with clothes

pins with cards perpendicular to the ground and at right angles to the corn row. Sticky traps were placed at canopy height until the corn reached 1 m, after which all sticky traps were placed at the top of each bamboo stake. Sticky traps collected from the field were placed into individual 15- × 20-cm clear plastic bags with the sticky side of the card adhered smoothly to one side of the bag. The bags were held at 4°C until the cards were examined (through the plastic bags) and the arthropods identified.

Visual plant examinations. Ten plants within each treatment were selected at random and visually inspected for arthropods. Plants were destructively sampled to account for arthropods in the leaf collars and ear tips. Starting at the base of the plant, the stalk, top and bottom of the leaves, leaf collars, ear tips, silks, and tassels were examined, and the arthropods recorded. At the beginning of the season, each plant was examined by a single person, but as the corn grew taller than 1 m, plants were sampled by a pair of individuals to reduce the number of arthropods missed as they flew away. Sampling occurred every 7–10 days from mid-June to mid-October 2001 and early June to late September 2002.

Data analysis. Individual taxa with coefficients of variation ($CV = 100 \times \text{standard deviation/mean}$) of 75 or less in the untreated control plots, for a given sample method (pitfall trap, sticky card, visual inspection) on the sample date immediately after insecticide application in 2001 or 2002 (either sample date) were selected for analysis. The CV is a scale-independent measure of variability among samples (Snedecor & Cochran 1989), allowing it to be used as a measure of variability for arthropods, which differ greatly in mean abundance. Coefficient of variation has been successfully used to evaluate the suitability of test systems for evaluating nontarget effects on plants (Reuter et al. 2002). Duan et al. (2006) noted that nontarget arthropods sampled with a $CV < 100$ tended to have greater statistical power. Insecticide treatment was included as a positive control for comparative risk assessment (Romeis et al. 2006). Therefore, we focused analysis on data collected immediately after insecticide applications. These dates were believed most likely to show impacts on nontarget arthropod populations. Taxa with a $CV > 75$ on the sample date immediately after insecticide applications were not analyzed because the likelihood of appropriately detecting treatment effects was considered to be similar to the chance of committing an error (particularly Type I errors; Snedecor & Cochran 1989). Although the use of a repeated measures analysis was considered, it was not used because these analyses would include data before and between insecticide applications, during which time arthropod populations could recover. Data were analyzed with analysis of variance using the General Linear Models Procedure and Tukey's multiple range test was used to separate means (SAS Institute 1999). Data were normalized by a $\log_{10}(x + 1)$ transformation where x is the mean number of individuals per plot for a taxon (using a given sampling method). Because of the relatively greater consequence of failing to detect changes in nontarget abundance, the significance level used for all statistical analysis was $P \leq 0.10$. A companion paper focuses on the impacts of Bt corn and insecticide treatment on the populations of individual ground beetle species (Lopez et al. 2005).

Results

Data on several taxa were collected from more than one sampling-method [e.g., Arachnida (Araneae and Opiliones), corn rootworm adults (*Diabrotica* spp.) (Co-

leoptera: Chrysomelidae), and leafhoppers (Cicadellidae)]. However, despite the large number of taxon $\times$ sampling method combinations used, the number of taxa with a CV ≤ 75 or less on a sample date immediately after insecticide application was modest. A complete list of the taxa sampled using each method, including the taxon $\times$ method combinations with a CV ≤ 75 is presented in Table 1.

Bt effects. There were few significant differences in nontarget arthropod populations resulting from Bt corn. In 2001, there were significantly fewer *Macrocentrus cingulum* Reinhard (Hymenoptera: Braconidae) captured on yellow sticky cards in the Bt corn plots than in the isoline control plots ($F = 18.93$; df = 2, 6; $P = 0.003$; Table 2). There were clear reductions in the mean numbers of *M. cingulum* captured on sticky cards in the Bt corn plots in 2001 and 2002 compared with the isoline (Fig. 1). However, in 2002 the CV for *M. cingulum* captures was too high to warrant analysis. Nitidulidae was the only taxon significantly impacted in the Bt corn plots in 2002 (Fig. 2). There were significantly fewer Nitidulidae in pitfall traps in the Bt corn than in the isoline control plots after the second insecticide application in 2002 ($F = 6.08$; df = 2, 6; $P = 0.04$; Table 2).

Insecticide effects. In 2001, visual inspections indicated that insecticide-treated plots contained fewer Arachnida ($F = 16.80$; df = 2, 6; $P = 0.004$; Table 2) but significantly more Aphidae ($F = 4.25$; df = 2, 6; $P = 0.07$; Table 2) compared with isoline control plots. Fewer *M. cingulum* were captured on yellow sticky cards ($F = 18.93$; df = 2, 6; $P = 0.003$; Table 2) and fewer Cantharidae larvae were found in pitfall traps ($F = 3.36$; df = 2, 6; $P = 0.10$; Table 2) in the insecticide treated plots compared with the isoline control plots.

After the first insecticide application in 2002, fewer Arachnida ($F = 4.95$; df = 2, 6; $P = 0.05$; Table 2) and Elateridae adults ($F = 11.90$; df = 2, 6; $P = 0.008$; Table 2) were collected from pitfall traps in insecticide-treated plots than the isoline control plots. Also, more adult Staphylinidae were found on yellow sticky cards in the insecticide-treated plots than in the untreated control ($F = 4.25$; df = 2, 6; $P = 0.07$; Table 2). After the second insecticide application in 2002, fewer Arachnida were found in visual inspections ($F = 3.75$; df = 2, 6; $P = 0.09$), and captured on yellow sticky cards ($F = 4.79$; df = 2, 6; $P = 0.06$; Table 2) in insecticide-treated plots than the control plots on the second sample date.

Discussion

Overall, there were few significant differences detected as the result of Bt corn or insecticide treatment. In 2001, there were two taxa from the visual plant inspections (Aphidae, Arachnida), one from yellow sticky card captures (*M. cingulum*), and one from pitfall traps (Cantharidae) that were significantly affected by Bt or insecticide treatment. In 2002, there was one taxon with a significant difference in the number of individuals observed from visual plant inspections (Arachnida), two from yellow sticky card captures (Arachnida, Staphylinidae) and three from pitfall traps (Arachnida, Elateridae, Nitidulidae). The two taxa that were significantly impacted as the result of Bt corn in this study are both known to be closely associated with *O. nubilalis*, and the observed decrease in their populations was likely caused by the absence of *O. nubilalis* in the Bt plots. These results suggest drastically decreasing the target pest population (as is the case in Bt-corn) can effect multitrophic level interactions in corn that occur with normal pest populations.

Table 1. List of nontarget arthropods sampled from near Ames, Iowa and analyzed in 2001–2002.

Nontarget group	Life stages	Pitfall	Sticky trap	Visual count
Aleyrodidae	Adults		A ^a	A
Alticinae	Adults			A
Aphidae	Nymphs, adults	B	B	B ^b
Arachnida (Araneae and Opiliones)	Nymphs, adults	B	B	B
Cantharidae	Adults	B	A	A
Cantharidae	Larvae	A		
Carabidae	Larvae	B		
Chilopoda	Larvae, adults	B		
Chrysopidae	Eggs			B
Chrysopidae	Larvae	B		A
Chrysopidae	Pupae			A
Chrysopidae	Adults		B	A
Cicadellidae	Nymphs, adults		B	B
Cicindellidae	Adults	A		
<i>Coccinella septempunctata</i> L.	Adults		A	A
Coccinellidae ^c	Eggs			A
Coccinellidae ^c	Larvae			A
Coccinellidae ^c	Pupae			B
<i>Coleomegilla maculata</i> DeGeer	Adults		A	A
Collembola	Nymphs, adults	B		
<i>Cycloneda munda</i> (Say)	Adults		B	A
<i>Diabrotica</i> spp.	Adults		A	B
Elateridae	Adults	B	A	A
Formicidae	Adults	A		A
Gryllidae	Nymphs, adults	B		
<i>Harmonia axyridis</i> (Pallas)	Adults		A	A
Hemerobiidae	Adults		A	A
<i>Hippodamia convergens</i> Guerin-Meneville	Adults		A	A
<i>Hippodamia tredecimpunctata</i> (L.)	Adults		A	A
Lampyridae	Adults	A	B	
<i>Macrocentrus cingulum</i> Reinhard	Adults		B	
Nitidulidae	Adults	B		B
<i>Orius insidiosus</i> (Say)	Adults		B	B
Pentatomidae	Eggs			A
Pentatomidae	Nymphs, adults	A	A	A
Scarabeidae	Adults	A		
Staphylinidae	Adults	B	B	A

^aA indicates the nontarget group was sampled using the method indicated by column heading.

^bB indicates the taxon was sampled using the method indicated by column heading and had a CV ≤ 75 on at least one selected sample date. These data were subsequently analyzed with analysis of variance, and those with a statistical difference between treatments on any sample date are presented (Table 2).

^cAll species combined for this life stage.

Table 2. Mean abundance ($\pm$ SD) of taxa for which Bt or permethrin treatments differed from the untreated isoline, collected near Ames, Iowa, 2001–2002.

Sampling method	Taxon	Treatment	2001	2002	
			August 2 ^a	June 27	August 21
Visual counts	Aphidae	Transgenic	0.46 (0.47)b	NA ^b	0.93 (0.90)a
		Insecticide	1.80 (1.28)a		1.50 (1.40)a
		Isoiline	0.30 (0.10)b		0.83 (0.40)a
	Arachnida	Transgenic	0.70 (0.20)a	0.23 (0.32)a	0.70 (0.17)ab
		Insecticide	0.10 (0.10)b	0.10 (0.10)a	0.63 (0.15)b
		Isoiline	0.63 (0.15)a	0.17 (0.06)a	0.93 (0.06)a
Sticky traps	Arachnida	Transgenic	NA	NA	1.05 (0.47)ab
		Insecticide			0.63 (0.15)b
		Isoiline			1.37 (0.06)a
	<i>Macrocentrus cingulum</i>	Transgenic	0.49 (0.16)b	NA	NA
		Insecticide	0.90 (0.20)b		
		Isoiline	1.96 (0.54)a		
	Staphylinidae	Transgenic	NA	0.11 (0.11)b	0.21 (0.18)a
		Insecticide		0.34 (0.13)a	0.20 (0.10)a
		Isoiline		0.17 (0.05)b	0.36 (0.06)a
	Arachnida	Transgenic	16.53 (9.01)a	10.42 (4.17)a	35.00 (15.00)a
		Insecticide	8.00 (4.95)a	4.24 (2.18)b	18.83 (5.69)a
		Isoiline	16.37 (8.12)a	14.78 (8.61)a	35.87 (16.49)a
Pitfall traps	Cantharidae	Transgenic	1.43 (1.07)a	NA	NA
		Insecticide	0.23 (0.06)b		
		Isoiline	3.27 (2.60)a		
	Elateridae	Transgenic	NA	2.57 (0.61)a	NA
		Insecticide		0.29 (0.51)b	
		Isoiline		4.52 (2.94)a	
	Nitidulidae	Transgenic	1.50 (1.13)a	3.13 (2.50)a	0.47 (0.06)b
		Insecticide	3.10 (1.75)a	2.02 (0.85)a	1.40 (0.75)ab
		Isoiline	1.90 (0.56)a	5.06 (2.81)a	1.80 (0.69)a

^aMeans ($n = 3$) from the same taxa and year followed by the same letter are not significantly different ($P < 0.10$).

^bThe coefficient of variation was too large to warrant analysis.

Macrocentrus cingulum is a specialist larval parasitoid of *O. nubilalis* imported from Europe and the Orient (Baker et al. 1949) that is well synchronized with *O. nubilalis* larval presence in the field (Bruck & Lewis 1998) and remains the most common larval parasitoid in Iowa (Lewis 1982, Bruck & Lewis 1998, Bruck & Lewis 1999). Bt corn provides 85–100% control of *O. nubilalis*, depending on the genetic event (Ostlie et al. 1997). A presumed lack of *O. nubilalis* larvae in the Bt corn plots is likely the cause for the reduction in *M. cingulum* captures. Because *M. cingulum* actively searches for and finds hosts based on olfactory cues (Ding et al. 1989), with no *O. nubilalis* larvae present, *M. cingulum* would not be likely to remain in fields of Bt corn. Although we did not directly sample *O. nubilalis* in the experimental field, we know from capture data from two light traps in central Iowa that in 2001 and 2002, *O. nubilalis* populations were below

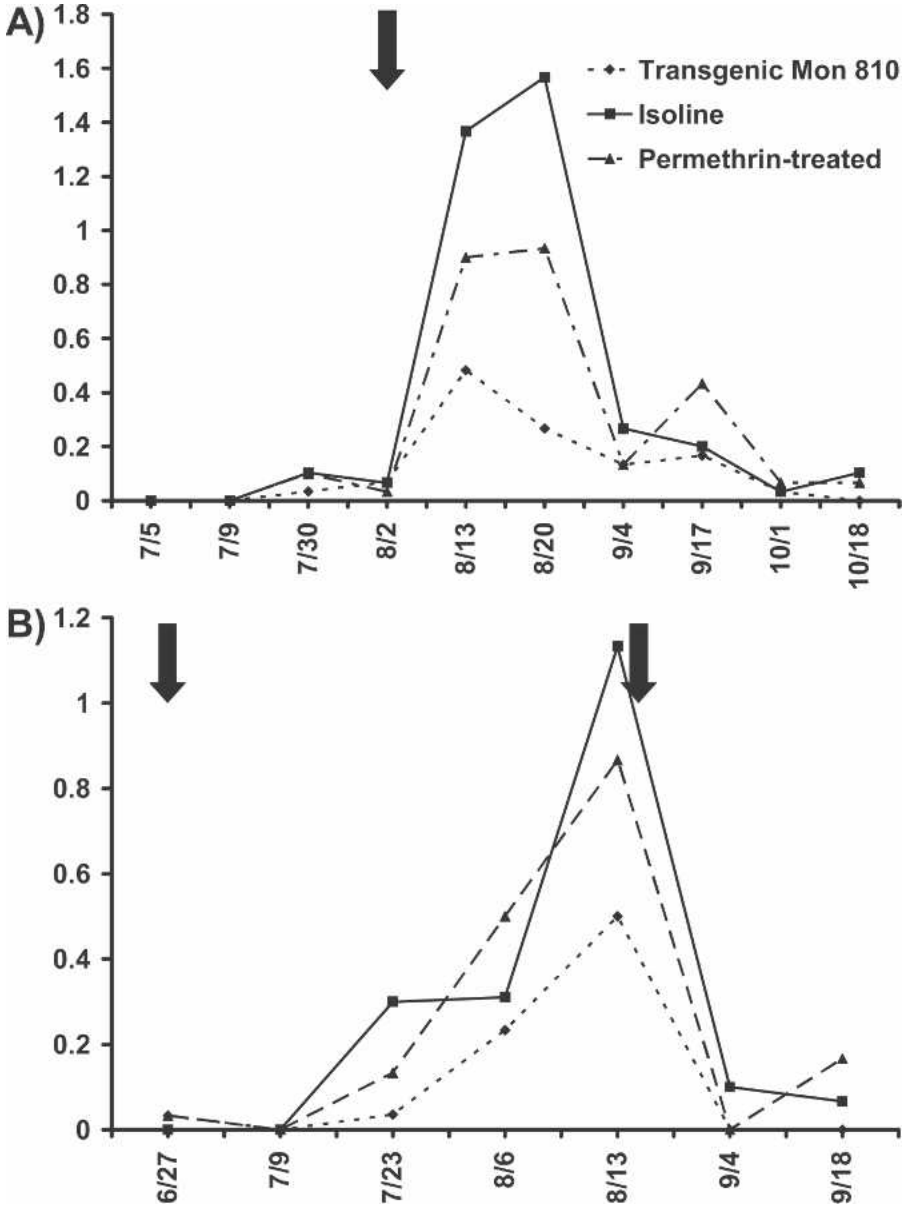


Fig. 1. Mean number of *Macrocentrus cingulum* (Hymenoptera: Braconidae) captured on sticky cards in 2001 (A) and 2002 (B). Timing of insecticide applications noted by downward arrow. Traps were placed into the field 24 h after insecticide application, and removed from the field 7–10 days later.

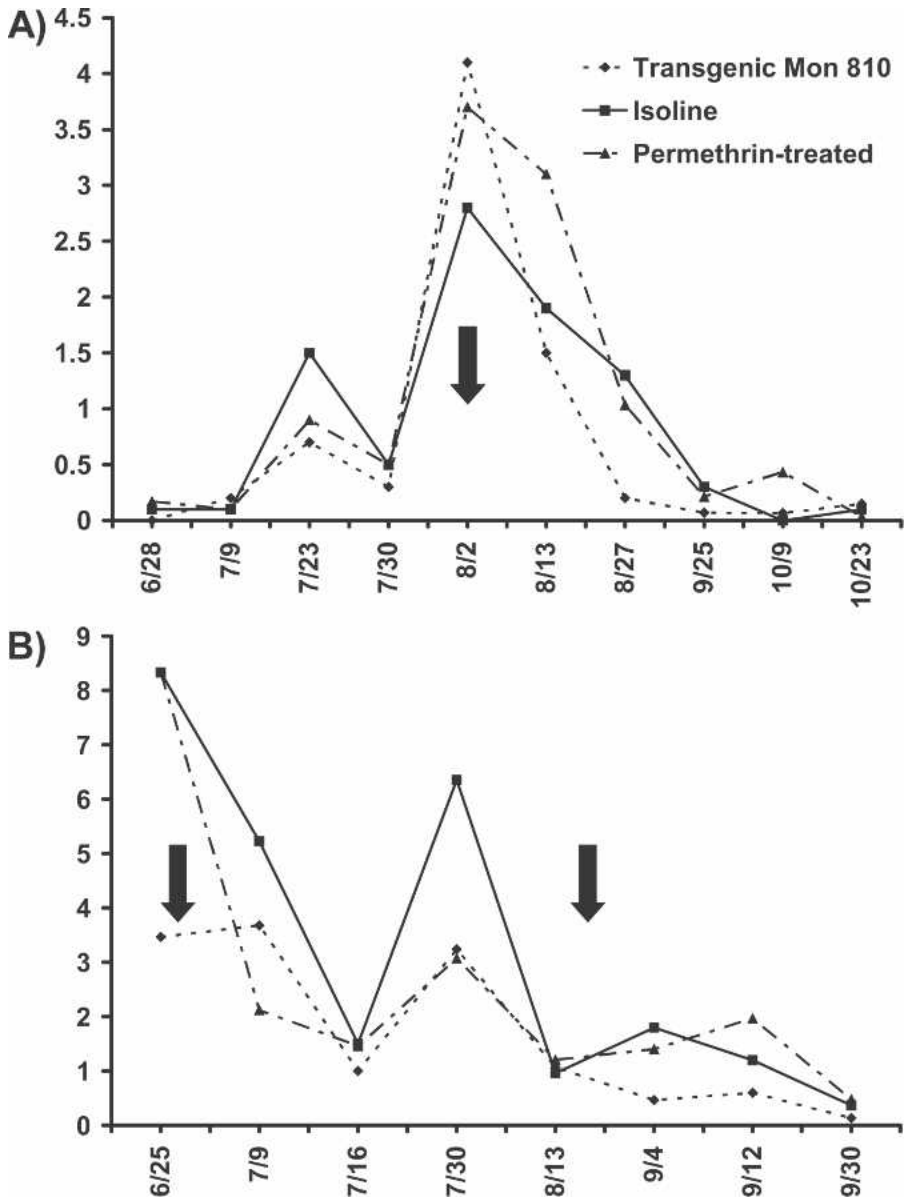


Fig. 2. Mean number of Nitidulidae captured in pitfall traps in 2001 (A) and 2002 (B). Timing of insecticide applications noted by downward arrow. Traps were placed into the field 24 h after insecticide application, and removed from the field 7–10 days later.

average when compared with the previous ten years (Hellmich 2006), so any minor effect detected in these years would only be more pronounced in years of greater *O. nubilalis* populations. Pilcher et al. (2005) also found a 29–60% reduction in *M. cingulum* numbers in Bt corn compared with nonBt corn. *Macrocentrus cingulum* is preferentially recruited to and increases over time in nonBt corn plots (Pilcher et al. 2005). We also observed an increase in *M. cingulum* numbers over time in the isoline control plots in 2001 and 2002 (Fig. 1).

Several species of Nitidulidae have long been associated with corn (Vinal & Caffery 1919). Many Nitidulidae are fungivores and have frequently been found in tunnels in corn made by *O. nubilalis* larvae (McCoy & Brindley 1961). The beetles likely feed on the fungi growing on the plant exudates in these tunnels. Nitidulidae adults are also attracted to corn ears that have been damaged by other insects (Dowd 2000). Although we did not quantify the *O. nubilalis* infestation in these plots, a lack of *O. nubilalis* tunneling in the Bt corn would make those plots less attractive to Nitidulidae. Our data, along with reports from Daly & Buntin (2005), support the hypothesis that Bt corn is less attractive to Nitidulidae. The reduction in attractiveness of Bt corn to Nitidulidae may also be attributable to the reduction of exposed kernels (Dowd 2000).

Many studies have shown that crop treatment with a synthetic insecticide against a target pest population has broad toxicity against many groups of nontarget arthropods (Cherry & Pless 1971, Croft 1994, Boyd & Boethel 1998, Amalin et al. 2001, Duan et al. 2004) as well as resulting in pest resurgence (Pedigo 2002). *Ostrinia nubilalis* is a chronic bivoltine pest of corn in central Iowa (Mason et al. 1996), and several different insecticides are labeled for *O. nubilalis* control (Mason et al. 1996, Catangui & Berg 2002), including Pounce 1.5G. The use of insecticides may cause greater harm to nontarget organisms than Bt corn, which is why we believe that the safety of Bt corn to nontarget arthropods must be made in comparison with that of insecticide application. Bhatti et al. (2005) found that foliar permethrin application significantly reduced arachnid populations compared with isoline control plots. Dively (2005) also found that foliar insecticide application had a more consistent and sustained impact on nontarget arthropod populations than Bt corn.

In this study, we attempted to sample and test a wide variety of taxa for nontarget effects using a standard experimental design. An insecticide treatment was included to provide a positive control to verify that differences between treatments were detectable; however, only a few significant effects of insecticide treatment were detected. Although the sample dates in these studies were focused on those following insecticide applications, it may be that enough time elapsed for most arthropods to repopulate insecticide treated plots from surrounding untreated control and Bt plots. The daily dispersal rate of some of the most commonly encountered nontarget arthropods in our study is 25–32 m/day (Prasifka et al. 1999). In addition, because pitfall and sticky traps were active in the field for 7–10 days at a time, immediate effects of the insecticide applications may have been diluted between collections. This may have been a limitation in the insecticide treated plots, but because event MON810 continuously expresses the Cry1Ab protein, we believe that the low number of differences observed between Bt corn and the isoline control plots is realistic.

Our observations of few detrimental impacts to nontarget arthropods caused by Bt corn are similar to those seen by others (Daly & Buntin 2005, Dively 2005,

Lopez et al. 2005, Romeis et al. 2006). Al Deeb et al. (2001) reported no significant effects of Bt corn on *Orius insidiosus* (Say) (Hemiptera: Anthocoridae). In addition, Al-Deeb et al. (2003) reported no deleterious effects of Bt corn on beneficial and other nontarget arthropods from Bt corn expressing Coleopteran-specific (Cry3Bb1) protein. Orr & Landis (1997) also saw no significant differences in the numbers of three predators and an egg parasite of *O. nubilalis* in Bt corn. Ingesting prey fed Coleopteran-specific Bt proteins did not adversely affect *C. maculata* (Riddick & Barbosa 1998). Al-Deeb & Wilde (2003) also found no significant differences in *C. maculata* populations between Bt corn plots and plots of its nonBt isoline. Dively (2005) found the community disturbances caused by Bt corn were significantly less profound than those caused by insecticide application. In laboratory tests, the fitness of *C. maculata* feeding on transgenic corn pollen (Cry3Bb) was not significantly different than of those fed nontransformed pollen (Lundgren & Wiedenmann 2002). Losey et al. (1999) did find reduced *D. plexippus* leaf consumption, larval survival, and larval weight when fed milkweed leaves dusted with Bt pollen. However, the pollen dosage used in this study was not controlled and may not have accurately mimicked the pollen distribution rates in the field (Hellmich et al. 2001). For some nontarget arthropods (particularly generalist predators), corn planting date and subsequent crop phenology can have a stronger influence on their population size than Bt-corn (Pilcher et al. 2005). Shelton et al. (2002) proposed that regardless of the technique used to reduce a pest population (Bt plants, insecticides, biological control, etc.) there will be an impact on the overall biological community.

In summary, only two of the taxa sampled experienced a significant decrease in Bt corn. It is likely that a lack of *O. nubilalis* in the Bt plots directly and indirectly impacted *M. cingulum* and Nitidulidae via host elimination and removal of preferred habitat, respectively. In contrast, five taxa experienced significant declines in the insecticide treated plots. These results are consistent with several other studies that suggest nontarget impacts because insecticides may be greater than those caused by Bt corn.

Acknowledgments

We thank Maya Hayslett, Vanessa Ware, Suzie Ahn, Christa Hoffman, and Shauna McCord for field and laboratory support. We would also like to thank Galen Dively, University of Maryland, Department of Entomology, for discussions assisting with the initial design of these experiments as well as Thomas Sappington and Edwin Berry for helpful comments on earlier drafts of the manuscript. Mention of trade names of commercial products in this publication is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the US Department of Agriculture.

References Cited

- Al-Deeb, M. A., G. E. Wilde & R. A. Higgins. 2001. No effect of *Bacillus thuringiensis* corn and *Bacillus thuringiensis* on the predator *Orius insidiosus* (Hemiptera: Anthocoridae). Environ. Entomol. 30: 625–629.
- Al-Deeb, M. A. & G. E. Wilde. 2003. Effect of Bt corn expressing the Cry3Bb toxin for corn rootworm control on aboveground nontarget arthropods. Environ. Entomol. 32: 1164–1170.
- Al-Deeb, M. A., G. E. Wilde, J. M. Blair & T. C. Todd. 2003. Effect of Bt corn for corn

- rootworm control on nontarget soil microarthropods and nematodes. *Environ. Entomol.* 32: 859–865.
- Amalin, D. M., J. E. Pena, R. McSorley, H. W. Browning & J. H. Crane. 2001.** Comparison of different sampling methods and effect of pesticide application on spider populations in lime orchards in south Florida. *Environ. Entomol.* 30: 1021–1027.
- Baker, W. A., W. G. Bradley & C. A. Clark. 1949.** Biological control of the European corn borer in the United States. U.S. Dept. Agric. Tech. Bull. 983.
- Bergman, P. W., R. A. Davis, S. N. Ferting, F. Kuchlerf, R. McDowell, C. Osteen, A. L. Padula, K. L. Smith & R. F. Torla. 1985.** Pesticide assessment of field corn and soybeans: Corn belt states. National Agricultural Pesticide Impact Assessment Program (NAPIAP) Ext. Serv., U.S. Dept. Agric., Washington, DC.
- Bhatti, M. A., J. Duan, G. Head, C. Jiang, M. J. McKee, T. E. Nickson, C. L. Pilcher & C. D. Pilcher. 2005.** Field evaluation on the impact of corn rootworm (Coleoptera: Chrysomelidae)-protected *Bt* corn on ground-dwelling invertebrates. *Environ. Entomol.* 34: 1325–1335.
- Bitzer, R. J., M. E. Rice, C. D. Pilcher, C. L. Pilcher & W. K. F. Lam. 2005.** Biodiversity and community structure of epedaphic and euedaphic springtails (Collembola) in transgenic rootworm *Bt* corn. *Environ. Entomol.* 34: 1346–1376.
- Boyd, M. L. & D. J. Boethel. 1998.** Susceptibility of predaceous hemipteran species to selected insecticides on soybean in Louisiana. *J. Econ. Entomol.* 91: 401–409.
- Bruck, D. J. & L. C. Lewis. 1998.** Influence of adjacent cornfield habitat, trap location, and trap height on capture numbers of predators and a parasitoid of the European corn borer (Lepidoptera: Pyralidae) in central Iowa. *Environ. Entomol.* 27: 1557–1562.
- Bruck, D. J. & L. C. Lewis. 1999.** *Ostrinia nubilalis* (Lepidoptera: Pyralidae) larval parasitism and infection with entomopathogens in cornfields with different border vegetation. *J. Agric. Urban Entomol.* 16: 255–272.
- Catangui, M. A. & R. K. Berg. 2002.** Comparison of *Bacillus thuringiensis* corn hybrids and insecticide-treated isolines exposed to bivoltine European corn borer (Lepidoptera: Crambidae) in South Dakota. *J. Econ. Entomol.* 95: 155–166.
- Cherry, E. T. & C. D. Pless. 1971.** Effects of carbofuran and disulfoton on parasitism of tobacco budworms and hornworms of burley tobacco. *J. Econ. Entomol.* 64: 187–190.
- Croft, B. A. 1994.** Arthropod biological control agents and pesticides. Wiley, New York.
- Daly, T. & D. Buntin. 2005.** Effect of *Bacillus thuringiensis* transgenic corn for lepidopteran control on nontarget arthropods. *Environ. Entomol.* 34: 1292–1301.
- Ding, D., P. D. Swedenborg & R. L. Jones. 1989.** Chemical stimuli in host-seeking behavior of *Macrocentrus grandii* (Hymenoptera: Braconidae). *Ann. Entomol. Soc. Am.* 82: 232–236.
- Dively, G. P. 2005.** Impact of transgenic VIP3A × Cry1Ab lepidopteran-resistant field corn on the nontarget arthropod community. *Environ. Entomol.* 34: 1267–1291.
- Dively, G. P., R. Rose, M. K. Sears, R. L. Hellmich, D. E. Stanley-Horn, D. D. Calvin, J. M. Russo & P. L. Anderson. 2004.** Effects on monarch butterfly larvae (Lepidoptera: Danaidae) after continuous exposure to Cry1Ab-expressing corn during anthesis. *Environ. Entomol.* 33: 1116–1125.
- Dowd, P. F. 2000.** Dusky sap beetles (Coleoptera: Nitidulidae) and other kernel damaging insects in *Bt* and non-*Bt* sweet corn in Illinois. *J. Econ. Entomol.* 93: 1714–1720.
- Duan, J. J., G. Head, A. Jensen & G. Reed. 2004.** Effects of transgenic *Bacillus thuringiensis* potato and conventional insecticides for Colorado potato beetle (Coleoptera: Chrysomelidae) management on the abundance of ground-dwelling arthropods in Oregon potato ecosystems. *Environ. Entomol.* 33: 275–281.
- Duan, J. J., M. S. Paradise, J. G. Lundgren, J. T. Bookout, C. Jiang & R. N. Wiedemann. 2006.** Assessing nontarget impacts of *Bt* corn resistant to corn rootworms: tier-1 testing with larvae of *Poecilus chalcites* (Coleoptera: Carabidae). *Environ. Entomol.* 35: 135–142.

- Dutton, A., J. Romeis & F. Bigler. 2003.** Assessing the risks of insect resistant transgenic plants on entomophagous arthropods: Bt-maize expressing Cry1Ab as a case study. *BioControl* 48: 611–636.
- Hellmich, R. L. 2006.** European corn borers: a diminished insect threat from now on? Illinois Crop Protection Technology Conference, 2006 Proceedings, University of Illinois, Champaign, Illinois 62–64.
- Hellmich, R. L., B. D. Siegfried, M. K. Sears, D. E. Stanley-Horn, M. J. Daniels, H. R. Mattila, T. Spencer, K. G. Bidne & L. C. Lewis. 2001.** Monarch larvae sensitivity to *Bacillus thuringiensis*-purified proteins and pollen. *Proc. Natl. Acad. Sci. USA* 98: 11925–11930.
- Lewis, L. C. 1982.** Present status of introduced parasitoids of the European corn borer, *Ostrinia nubilalis* (Hübner), in Iowa. *Iowa State J. Res.* 56: 429–436.
- Lopez, M. D., J. R. Prasifka, D. J. Bruck & L. C. Lewis. 2005.** Utility of ground beetle species in field tests of potential nontarget effects of Bt crops. *Environ. Entomol.* 34: 1317–1324.
- Losey, J. E., L. S. Rayor & M. E. Carter. 1999.** Transgenic pollen harms monarch larvae. *Nature* 399: 214.
- Lundgren, J. G. & R. N. Wiedenmann. 2002.** Coleopteran-specific Cry3Bb toxin from transgenic corn pollen does not affect the fitness of a nontarget species, *Coleomegilla maculata* DeGeer (Coleoptera: Coccinellidae). *Environ. Entomol.* 31: 1213–1218.
- Mason, C. E., M. E. Rice, D. D. Calvin, J. W. Van Duyn, W. B. Showers, W. D. Hutchison, J. F. Witkowski, R. A. Higgins, D. W. Onstad & G. P. Dively. 1996.** European corn borer ecology and management. Iowa State University, Ames, Iowa.
- McCoy, C. E. & T. A. Brindley. 1961.** Biology of the four-spotted fungus beetle, *Glischrochilus q. quadrisignatus* and its effect on European corn borer populations. *J. Econ. Entomol.* 54: 713–717.
- Orr, D. B. & D. A. Landis. 1997.** Oviposition of the European corn borer (Lepidoptera: Pyralidae) and impact of natural enemy population in transgenic versus isogenic corn. *J. Econ. Entomol.* 90: 905–909.
- Ostlie, K. R., W. D. Hutchison & R. L. Hellmich. 1997.** Bt corn and the European corn borer. University of Minnesota. St. Paul, Minnesota.
- Pedigo, L. P. 2002.** Entomology and pest management, 4th ed. Prentice Hall, Upper Saddle River, New Jersey.
- Pilcher, C. D., M. E. Rice & J. J. Obrycki. 2005.** Impact of transgenic *Bacillus thuringiensis* corn and crop phenology on five nontarget arthropods. *Environ. Entomol.* 34: 1302–1316.
- Prasifka, J. R., P. C. Krauter, K. M. Heinz, C. G. Sansone & R. R. Minzenmayer. 1999.** Predator conservation in cotton: using grain sorghum as a source for insect predators. *Biol. Contr.* 16: 223–229.
- Prasifka, J. R., R. L. Hellmich, G. P. Dively & L. C. Lewis. 2005.** Assessing the effects of pest management on nontarget arthropods: the influence of plot size and isolation. *Environ. Entomol.* 34: 1181–1192.
- Reuter, P., P. Frank & R. Kubiak. 2002.** Validity of non-target plant tests under greenhouse conditions. *J. Plant Dis. Prot.* 18: 1005–1012.
- Riddick, E. W. & P. Barbosa. 1998.** Impact of Cry3A-intoxicated *Leptinotarsa decemlineata* (Coleoptera: Chrysomelidae) and pollen on consumption, development, and fecundity of *Coleomegilla maculata* (Coleoptera: Coccinellidae). *Ann. Entomol. Soc. Am.* 91: 303–307.
- Romeis, J., M. Meissle & F. Bigler. 2006.** Transgenic crops expressing *Bacillus thuringiensis* toxins and biological control. *Nat. Biotechnol.* 24: 63–71.
- SAS Institute. 1999.** SAS Software: Version 8e. SAS Institute, Cary, North Carolina.
- Shelton, A. M., J.-Z. Zhao & R. T. Roush. 2002.** Economic, ecological, food safety, and

social consequences of the deployment of *Bt* transgenic plants. *Annu. Rev. Entomol.* 47: 845–881.

Snedecor, G. W. & W. G. Cochran. 1989. Statistical methods. 8th edition. Iowa State University Press, Ames Iowa.

Tanada, Y. & H. K. Kaya. 1993. Insect pathology. Academic Press Inc., San Diego California.

Vinal, A. C. & D. J. Caffery. 1919. The European corn borer and its control. *Mass. Expt. Stat. Bull.* 189.

**South Carolina
Entomological Society, Inc.**

**3517 Flowering Oak Way
Mt. Pleasant, SC 29466**

**MEMBERSHIP
APPLICATION
FORM**

NAME: _____ DATE: _____

MAILING ADDRESS: street _____

city: _____ state: _____ zip code: _____

HOME ADDRESS: street _____

city: _____ state: _____ zip code: _____

WORK PHONE: _____ HOME PHONE: _____

FAX # _____ E-MAIL _____

EDUCATION:

	DEGREE	INSTITUTION	YEAR
1.	_____	_____	_____
2.	_____	_____	_____
3.	_____	_____	_____
4.	_____	_____	_____

EMPLOYER: _____ SINCE: _____

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS:

1. _____
2. _____
3. _____
4. _____
5. _____

I hereby make application for membership in the South Carolina Entomological Society, Inc. with the status of active (\$30), student (\$15), emeritus (\$10) membership.

Signature: _____ Date: _____

Return form with membership dues to address above.

Journal of Agricultural and Urban Entomology

Contents

 Volume 23, No. 3

 July 2006

- DENNY J. BRUCK, MIRIAM D. LOPEZ, LESLIE C. LEWIS, JARRAD R. PRASIFKA
AND ROBERT D. GUNNARSON – Effects of Transgenic *Bacillus thuringiensis* Corn
and Permethrin on Nontarget Arthropods 111
- WEE L. YEE – GF-120, Nu-Lure, and Mazoferm Effects on Feeding Responses and
Infestations of Western Cherry Fruit Fly (Diptera: Tephritidae) 125
- D. MICHAEL JACKSON AND JANICE R. BOHAC – Evaluation of Pheromone Traps for
Monitoring Sweetpotato Weevils. 141
- RAMAZAN CANHILAL AND GERALD R. CARNER – Natural Occurrence of
Entomopathogenic Nematodes (Rhabditida: Steinernematidae and
Heterorhabditidae) in South Carolina 159
- WM. MICHAEL HOOD – The First Fifty Years of the South Carolina
Entomological Society 167
- ALLEN L. SZALANSKI, JAMES W. AUSTIN, JACKIE A. MCKERN, C. DAYTON
STEELMAN, DINI M. MILLER AND ROGER E. GOLD – Isolation and
Characterization of Human DNA From Bed Bug, *Cimex lectularius* L., (Hemiptera:
Cimicidae) Blood Meals. 189

GF-120, Nu-Lure, and Mazoferm Effects on Feeding Responses and Infestations of Western Cherry Fruit Fly (Diptera: Tephritidae)¹

Wee L. Yee

USDA-ARS, Yakima Agricultural Research Laboratory, 5230 Konnowac Pass Road, Wapato, Washington 98951 USA

J. Agric. Urban Entomol. 23(3): 125–140 (July 2006)

ABSTRACT A study was conducted to determine feeding responses of western cherry fruit fly, *Rhagoletis indifferens* Curran (Diptera: Tephritidae), to 40% (vol/vol) GF-120, Nu-Lure, and Mazoferm bait and to the insecticide spinosad in 2005 and 2006 in Washington state. When water or baits were sprayed on branches or were placed as single drops on leaves 1–2 cm from flies, the numbers of flies subsequently seen near or feeding on the water and baits were similar. Of flies that contacted water or baits, 0–23.1% did not respond by feeding or maintaining contact with drops. Females and males responded similarly. Later in the season (31 May to 9 June), responses to all baits were greater than early in the season (19–30 May). When single trees (with higher fly pressure than found in commercial orchards) were sprayed with treatments, Nu-Lure and Mazoferm significantly reduced larval infestations in fruit compared with the control in one test, but they, GF-120, and spinosad alone did not significantly reduce larval infestations compared with controls in two other tests. Results suggest the baits tested were not attractive to *R. indifferens* compared with water and that most flies find them through chance encounters, which may partially explain the lack of consistent larval control using baits. Based on the results of the current study and others, the choice of baits may not be vital for controlling larval infestations of *R. indifferens*, and even spinosad alone may have the same effects, at least when fly populations are lower than those encountered in this study.

KEY WORDS *Rhagoletis indifferens*, spinosad, protein baits

The use of protein bait sprays for control of tephritid fruit flies dates back many years (Roessler 1989). However, with the greater concern about potential long-term harmful effects of organophosphate insecticides and the development of newer protein baits, the feeding responses of tephritids to these baits have recently received much attention. This is especially true with subtropical tephritids such as *Ceratitis capitata* (Wiedemann), *Bactrocera* spp., and *Anastrepha ludens* (Loew) (Diptera: Tephritidae) (Barry et al. 2003, Prokopy et al. 2003, Vargas et al. 2003, Revis et al. 2004, Barry et al. 2006, Mangan et al. 2006, Vargas & Prokopy 2006). For these fly species, protein baits (those with hydrolyzed proteins, which

¹Accepted for publication 19 June 2007

consist of amino acids or peptides) have some attractiveness, but the attractiveness of and feeding propensity on various baits differ (Moreno and Mangan 1995, Fabre et al. 2003, Vargas et al. 2003, Vargas & Prokopy 2006), and feeding responses to baits among species vary (Barry et al. 2006, Vargas & Prokopy 2006).

Bait sprays have also been tested against temperate *Rhagoletis* (Diptera: Tephritidae) flies. However, they have been less well studied than for subtropical flies. For several *Rhagoletis* species, there appears to be little or no attraction to GF-120 Fruit Fly Bait (Dow AgroSciences, Indianapolis, Indiana) (Pelz et al. 2005, Yee & Chapman 2005, Pelz-Stelinski et al. 2006), but it is likely attraction and feeding responses of *Rhagoletis* to various baits differ among species.

The western cherry fruit fly, *Rhagoletis indifferens* Curran, is the major insect pest of sweet cherry, *Prunus avium* (L.) L., in the Pacific Northwest of the United States. There is a zero tolerance for its larvae in commercial fruit (Anonymous 1968). The fly is managed in many areas of Washington state using GF-120. Although commercial cherry orchards generally have few or no flies, they need to be treated with GF-120 because of the zero tolerance. Also, GF-120 can be used on backyard and abandoned trees, which usually are (and often heavily) infested and are potential sources of flies in managed orchards. GF-120 suppressed larval populations when sprayed on single trees, but it did not eliminate them after one season (Yee & Chapman 2005), possibly because some flies were not attracted to and did not feed on GF-120 (Yee & Chapman 2005, Yee 2006), or some flies oviposited before ingesting the bait and dying. The attraction and feeding responses of *R. indifferens* to baits in nature need to be studied to provide support for these hypotheses.

GF-120 is the only bait that has been tested for its effects on both attraction responses and control of *R. indifferens* (Yee & Chapman 2005). Nu-Lure (Scentry Biologicals, Billings, Montana) and Mazoferm (Corn Products, Bedford Park, Illinois) are amino-acid (autolyzed protein) baits that have been tested extensively or to a fair degree against several tephritids (Peck & McQuate 2000, Vargas et al. 2002, Fabre et al. 2003, Vargas & Prokopy 2006) and need to be tested against *R. indifferens*. Aside from possible differences in fly attraction to them and their acceptability, another reason GF-120, Nu-Lure, and Mazoferm need to be compared is that Nu-Lure and Mazoferm are about 6 to 13 times less expensive than GF-120. Bait spray costs could be reduced by using Nu-Lure or Mazoferm if they are as or more effective than GF-120 against *R. indifferens*.

In this study, the main objective was to determine the effects of GF-120, Nu-Lure, and Mazoferm on feeding responses of *R. indifferens* in the field. I tested the hypotheses that flies are not attracted to the baits from mid-range distances, but that they are attracted to them at close-range distances and that feeding responses to baits differ. I also determined seasonal effects on feeding responses. Finally, I determined if feeding responses are related to effects of the three baits on control of larval infestations in cherries.

Materials and Methods

Study sites and trees. Study sites were located in Yakima County, Washington. In 2005, three observational tests were conducted at site 1 in Zillah (46.40°N, 120.26°W), and a spray test was conducted using 22 residential sweet cherry trees in Yakima (46.56°N, 120.53°W). In 2006, one observational test was

conducted at site 2 in Zillah, and two spray tests were conducted using 43 residential sweet cherry trees in Zillah, Toppenish (46.38°N, 120.31°W), and Yakima. Site 1 was a yard that had 15 cherry trees (14 sweet [13 cv. 'Bing' and one cv. 'Rainer'] and one tart cherry). Each of the used trees was 3.5–4.5 m tall and 4.5–8 m in diameter. Trees were spaced approximately 3–5 m apart. Site 2 had one sweet cherry (cv. 'Bing') that was 5.5 m tall and 7 m in diameter. Trees used in the 2005 spray test were 4–5 m tall and 5–7 m in diameter. Trees used in 2006 sprays tests were 4–5.5 m tall and 4–7 m in diameter.

Baits. Undiluted GF-120, Nu-Lure, and Mazoferm had densities of 1.16–1.23 g/mL. Major components of the three baits are described elsewhere and include amino acids, polypeptides (from corn), and simple and complex carbohydrates (Buttery et al. 1983, Hull et al. 1996, Lee et al. 1997, Mangan et al. 2006). GF-120 has 1% ammonium acetate (Thomas & Mangan 2005), but the amount in Nu-Lure is proprietary (Barry & Polavarapu 2004) and there apparently is none detectable in Mazoferm (Lee et al. 1997). All attraction/feeding and spray tests in 2005 and 2006 were conducted using baits diluted to 40% concentration (vol/vol). Even though Nu-Lure is used at 9% concentration for subtropical and tropical tephritid flies (Barry et al. 2006), there is no recommendation for *R. indifferens*. The 40% GF-120, Nu-Lure, and Mazoferm contained 26.4, 24.0, and 24.6% solids (wt/wt), and at 23.5°C had viscosities of 76.7, 2.4, and 3.9 centipoises, respectively (water, 1.0 cP) (results using no. 2 and 3 falling ball viscometers, Gilmont Instruments, Barrington, Illinois). In 2005 attraction/feeding tests, baits did not contain spinosad, but in the spray test, baits were mixed with spinosad (Entrust®, Dow AgroSciences, Indianapolis, Indiana). In 2006, baits mixed with spinosad and spinosad alone were used in both attraction/feeding and spray tests. The spinosad concentration used in both years was 0.0096% (wt/vol).

Mid-range attraction and feeding. Two tests were conducted in 2005 at site 1 by using each sample of bait on each of three or five trees. To confirm fly presence in the trees, numbers of flies seen in each tree over 6 min were counted before observations of fly responses. Test 1 was conducted using a total of 500 µL of water or baits, with 100 µL of each (no spinosad) applied on each of five randomly selected leaves. Volumes were distributed as six to 16 drops per leaf using a micropipette with a 250 µL LST tip (LTS 200, Rainin Instrument LLC, Oakland, California), depending on the thickness of the solution. The leaf areas that had bait drops were approximately 700–1,400 cm². Test 2 was conducted using 10 mL of the water or baits applied to 45- to 60-cm lengths of randomly selected branches with 20–40 leaves, using a 32-oz volume spray bottle (Consolidated Plastics Co., Twinsburg, Ohio). For both tests, the control and three baits were applied on south sides of each tree. Treatments were approximately 1–1.5 m apart (tree diameters were 4.5–8 m). Some of the trees had branches that were approximately 1 m above ground, whereas others had them at approximately 2.5 m above ground, so treatments were applied at these heights. Within each tree, all treatments were applied at approximately the same height. On each date, data from one tree constituted one replicate. Test 1 was conducted on 24 and 27 (five trees each day) and 31 May (three trees). Test 2 was conducted on 7, 10, and 14 June (three trees each day).

Numbers of female and male flies $\leq$ 15 cm from drops or that drank or fed on drops were recorded every 1.5–2 min for 30 min between 0930–1500 h (Pacific Standard Time) (which include times of highest activity [Yee 2002]). Flies that

were in the same position at successive intervals were counted as one. Flies were not removed. All treated leaves were immediately removed after observations. One observer recorded all the data; when treatments were 2.5 m above ground, the observer stood on a ladder. Temperatures in the shade during observation periods were 21.0–31.6°C.

Close-range attraction and feeding. One test was conducted in 2005 at site 1 and one in 2006 at site 2 to simulate a situation where flies forage near bait drops incidentally or drops fall close to flies as they rest on leaves. In 2005, one 25 μ l drop of water or bait (no spinosad) was slowly placed 1–2 cm from a randomly selected fly on the top or bottom of a leaf using a micropipette. This was done instead of spraying because spraying a leaf could result in coverage of flies with bait. Flies did not fly off when an observer approached them slowly and deliberately, and thus did not appear to be disturbed as long as movements were slow. Drops were generally placed in front of still or walking flies. However, because some flies turned at some point after application, drops sometimes ended up behind or on the side of flies. Flies were observed on east and south sides of trees. A fly was watched for a maximum of 5 min between 0745 and 1130 h on sunny days, and numbers and durations (measured with a stopwatch) of all feeding bouts were recorded. Flies that flew off before 5 min were also used in analyses as long as they stayed a minimum of 15 sec. After observations, flies were removed whenever possible using glass vials. Treated leaves were also removed. Observations were made approximately 1–3 m above ground with the observer standing on a ladder if necessary (for 3 m: approximately 0.5 m above the lowest branches and leaves). Flies on one to three trees were observed on 11 dates: 25, 26, 27, and 31 May and 2, 3, 6, 7, 9, 10, and 13 June. On each day, treatments were alternated so that water and each of the baits was tested before another complete set was tested again. There were one to three flies of each sex tested per water or bait treatment per day (20–27 total females or males). Solutions were made fresh daily. Temperatures in the shade at the end of observations were 19.8–25.8°C.

In 2006, methods at site 2 were similar to those at site 1 in 2005, except for five differences: (1) spinosad was added to all treatments; (2) spinosad alone was an additional treatment; (3) drops were applied only to the tops of leaves; (4) more details of fly movements were recorded: (a) still, defined as no movement except for grooming or movement of $\leq$ one body length, or (b) active, which was walking while grazing on leaves or walking toward water or bait to drink or feed or walking but not drinking or feeding; (5) there were detectable differences in weather during the first and second halves of the season (four weeks when fly populations were highest), so data from the two periods were compared. Mean temperature and RH ranges from 19–30 May were 18.8–19.1°C and 49–51%, respectively; except on 19 May, temperatures at 1100 h were all < 21.1°C (70°F). Most of the days were cloudy or partly cloudy. Mean temperature and RH ranges from 31 May to 9 June were 20.8–22.7°C and 45–39%, respectively. Temperatures at 1100 h were > 21.1°C. Most of the days were sunny. The difference in high temperatures between first and second periods was significant (one-way analysis of variance [ANOVA]: $F = 7.27$; $df = 1, 10$; $P = 0.022$), justifying a separation of the two as discrete factors. During the first period, most fruit were green to yellow; during the second period, most were yellow or orange to red. Observations were made on one tree on 12 dates: 19, 22, 23, 24, 25, 26, 30, and 31 May and 5,

6, 8, and 9 June between 0800 and 1100 h. For periods 1 and 2, one to seven and five to eight flies were in the control or each bait treatment per day, respectively.

Effects of bait sprays on fly infestations. In 2005, GF-120, Mazoferm, and Nu-Lure with spinosad were applied to residential sweet cherry trees from May to June. Treatments were replicated eight (control) or four or five (baits) times. Trees were isolated or occurred in groups of two to five. A 14- × 23-cm sticky yellow panel trap (Scenturion, Clinton, Washington) was placed on each tree on 9 May and checked every one or 2 d. Traps were baited with 10 g ammonium carbonate (Keystone Universal Corp., Melvindale, Michigan) contained in a plastic vial with two 1-mm holes. Three days after the first fly capture (on 13 May), the first sprays were made, using 1.18 L RL Flo-Master® pressurized sprayers (Root-Lowell Manufacturing Co., Lowell, Michigan). All baits were applied at a total volume of 225 ml per tree, which was the recommended high label rate for “spot spray of individual plants” for GF-120 (Dow AgroSciences 2006). Sprays were applied as approximately eight streaks using an upward motion around the periphery of each tree every 7 d, except once when it was reapplied 3 d after it rained (also 3 d after the previous spray). Applications were made on 16, 19, and 26 May, and 2, 9, and 16 June. Numbers of flies on traps were counted on spray dates. Only trees that had at least one fly on a trap were included in analyses. Fruit from all trees were picked by 24 June. Fruit loads on most trees were low because of frost during fruit set in April and May and all fruit were removed from most trees. Fruit were laid on hardware cloth on tubs and held outdoors, and numbers of pupae in the tubs after ≥ 30 d were recorded. Precipitation data during the test were based on Yakima records and were obtained from The Weather Underground, Inc (2006).

In 2006, methods were similar to those in 2005. However, in addition to 2005 treatments, spinosad alone was included and two tests using different spray volumes were conducted. Traps were hung 11 and 12 May. In test 1, site 1 (same one as for 2005 observational tests) was used and only 75 ml of spray was applied per tree because trees at this site had been pruned back about 50% from 2005. There were three replicate blocks (locations within the yard) of trees, each occupying approximately 800 m² and each with a control and four treatments. In test 2, isolated residential trees in Zillah, Toppenish, and Yakima were used. The spray volume was 150 mL per tree. There were seven control and three or six treatment trees scattered throughout these sites. In both tests, there were five to seven spray applications, the variability caused by some varieties of cherries ripening over longer periods than others. Fruit from all trees were picked by 24 June. Unlike in 2005, trees bore heavy fruit loads. As in 2005, precipitation data were obtained from The Weather Underground, Inc (2006).

Statistics. Data from mid-range attraction/feeding tests in 2005 were analyzed with two-way ANOVA, using date and treatment as factors. Fisher's exact test for R × C tables (SAS Institute 2003) (independent of ANOVAs) was also used to analyze (1) 2005 mid-range attraction/feeding data and (2) 2005 and 2006 close-range attraction/feeding data (SAS Institute 2003), specifically comparing numbers of flies that responded to treatments versus those that did not respond. Percentage responses presented are not means, but are based on total numbers of flies of each sex, of sexes combined over the season (2005 and 2006), or on total flies of each sex within periods (2006). The null hypotheses tested were numbers of flies responding (drinking or feeding or near water or baits) were independent

of (1) treatments, (2) sex, and (3) periods. For close-range data (numbers of feeding bouts and feeding duration) from 2005 and 2006, two-ANOVA (treatment and sex as factors) or three-way ANOVA (treatment, sex, and periods as factors) was also conducted. Data from all flies and from responders only were analyzed. To examine responses in greater detail, three other measures were analyzed using Fisher's exact test: (1) numbers of flies that fed on water or bait drops ≤ 5 sec after drops were applied, (2) numbers of flies that touched the water or baits but did not feed, and (3) numbers of flies that remained still near drops and did not feed. For effects of bait spray on fly infestation data, one-way or randomized block design (RBD) ANOVA was conducted. Counts + 1 were square-root transformed before analyses. Fisher's least significant difference (LSD) test was used for mean separation (SAS Institute 2003).

Results

Mid-range attraction and feeding. Flies were active in trees as determined by 6-minute observations: the mean number of flies observed over three days ranged from 6.4–13.6 and from 14.3–13.3 for tests 1 and 2, respectively. In presentation of results, "feeding" instead of "drinking" water will be used throughout for simplification. Also, water, baits, and spinosad alone will be referred to as "treatments" throughout the results. In test 1 (500 μ L volume) and test 2 (10 ml volume), few flies were seen near or fed on treatments during 30-min observations. No differences in numbers of flies attracted to different treatments were detected in either test (two-way ANOVA, test 1: treatment: $F = 1.57$; $df = 3, 40$; $P = 0.212$; date: $F = 0.14$; $df = 2, 40$; $P = 0.872$; treatment $\times$ date: $F = 1.96$; $df = 6, 40$; $P = 0.095$; test 2: treatment: $F = 2.12$; $df = 3, 24$; $P = 0.124$; date: $F = 0.97$; $df = 2, 24$; $P = 0.392$; treatment $\times$ date: $F = 1.30$; $df = 6, 24$; $P = 0.296$). In test 1, mean numbers of flies $\pm$ SE/30 min were: water, 0.0 ± 0.0 ; GF-120, 1.3 ± 0.6 ; Nu-Lure, 0.5 ± 0.3 ; Mazoferm, 0.6 ± 0.5 . In test 2, they were: water, 1.6 ± 0.6 ; GF-120, 1.9 ± 0.5 ; Nu-Lure, 0.7 ± 0.4 ; Mazoferm, 0.8 ± 0.3 . Fisher's exact test indicated that total flies feeding in the 3 d of test 1 (390 min of observations, combined from all trees) were independent of treatments ($P > 0.05$, water not included because of zero responses to it). Only two flies fed, both upon GF-120, whereas 30 flies were near treatments (combined) but did not feed. During the 3 d of test 2 (270 min of observations), total numbers seen feeding were also independent of treatments ($P > 0.05$). Only four, one, and three flies fed upon GF-120, Nu-Lure, and Mazoferm, respectively, whereas 38 flies were near treatments but did not feed.

Close-range attraction and feeding. In 2005, when drops of water or baits were placed 1–2 cm from flies, responses of females were independent of treatments ($P > 0.05$), as was the case with males ($P > 0.05$). Within each treatment, responses were independent of sex ($P > 0.05$). The overall response was 42.0% (sexes and treatments combined). Numbers of feeding bouts were not significantly affected by treatment (two-way ANOVA, $F = 0.93$; $df = 3, 168$; $P = 0.425$) or sex ($F = 1.57$; $df = 1, 168$; $P = 0.212$), and the interaction between these factors was not significant ($F = 0.59$; $df = 3, 168$; $P = 0.624$) (Table 1). When numbers of feeding bouts from only responders were analyzed, results were similar ($P > 0.05$). Fly feeding durations (only responders) were affected by treatments, with longest duration on GF-120 (two-way ANOVA, $F = 8.49$; $df = 3, 68$; $P < 0.0001$). There

Table 1. Effects of placing water and bait drops 1–2 cm from *Rhagoletis indifferens* on leaves on mean feeding responses $\pm$ SE, May to June 2005, Zillah, Washington.

Treatment	Sex	N	No. feeding bouts ^a	Feeding duration (min) (n) ^b
Water	F	25	0.72 $\pm$ 0.19	0.15 $\pm$ 0.03 (12)b
	M	20	0.80 $\pm$ 0.28	0.26 $\pm$ 0.09 (8)b
40% GF-120	F	27	1.15 $\pm$ 0.31	1.29 $\pm$ 0.29 (13)a
	M	20	1.05 $\pm$ 0.30	0.99 $\pm$ 0.32 (10)a
40% Nu-Lure	F	20	1.20 $\pm$ 0.44	0.55 $\pm$ 0.24 (7)b
	M	21	0.33 $\pm$ 0.24	0.21 $\pm$ 0.04 (9)b
40% Mazoform	F	23	1.04 $\pm$ 0.47	0.45 $\pm$ 0.21 (12)b
	M	20	0.30 $\pm$ 0.13	0.11 $\pm$ 0.03 (5)b

Baits did not contain spinosad.

^aAll flies (N).

^bSample sizes of only flies that fed; feeding durations based on n.

Means within columns followed by the same letter are not significantly different (ANOVA, LSD test, $P > 0.05$).

was no sex effect ($F = 1.78$; $df = 1, 68$; $P = 0.186$) and no treatment $\times$ sex interaction ($F = 0.50$; $df = 3, 68$; $P = 0.686$) (Table 1). Feeding responses ≤ 5 sec after application were independent of treatments (sexes combined, $P > 0.05$), and overall percentages of responders ranged from 25.0–62.5%. Numbers of flies that touched the treatments but did not feed were independent of treatment ($P > 0.05$) and percentages of these flies ranged from 4.3–18.8%. Two females (on 9 and 10 June) flicked their wings around Mazoform and Nu-Lure drops.

In 2006, within periods 1 and 2 (19–30 May and 31 May to 19 June, respectively), female responses were independent of treatments ($P > 0.05$), as they were in males ($P > 0.05$). They were also independent of sex ($P > 0.05$), so sexes were combined (Fig. 1). However, fly responses (all treatments combined) were dependent on period, with greater responses in period 2 than period 1 ($P < 0.0001$) (Fig. 1). The overall response was 44.4% (sexes, treatments, and periods combined).

In 2006, three-way ANOVA indicated there were no significant differences in numbers of feeding bouts among different treatments ($F = 1.30$; $df = 4, 343$; $P = 0.271$) and between sexes ($F = 1.72$; $df = 1, 343$; $P = 0.191$), but there were more feeding bouts in period 2 than period 1 ($F = 39.59$; $df = 1, 343$; $P < 0.0001$) (Table 2). There were no treatment $\times$ sex or period, sex $\times$ period, and treatment $\times$ sex $\times$ period interactions ($P > 0.05$). In 2006, there was only one female that fed on spinosad alone in period 1, so this affected the analyses for responders only and for feeding duration. Thus for analyzing numbers of feeding bouts for responders only and for feeding duration, a (1) three-way ANOVA was conducted leaving the spinosad alone treatment out and (2) two-way ANOVA was conducted on male data only for all treatments. Three-way ANOVA indicated that the numbers of feeding bouts of responders only on GF-120 and Nu-Lure were greater than on water and Mazoform ($F = 2.83$; $df = 3, 118$; $P = 0.042$). Mean numbers of feeding bouts $\pm$ SE (sexes combined) for GF-120, Nu-Lure, water, and Mazoform were

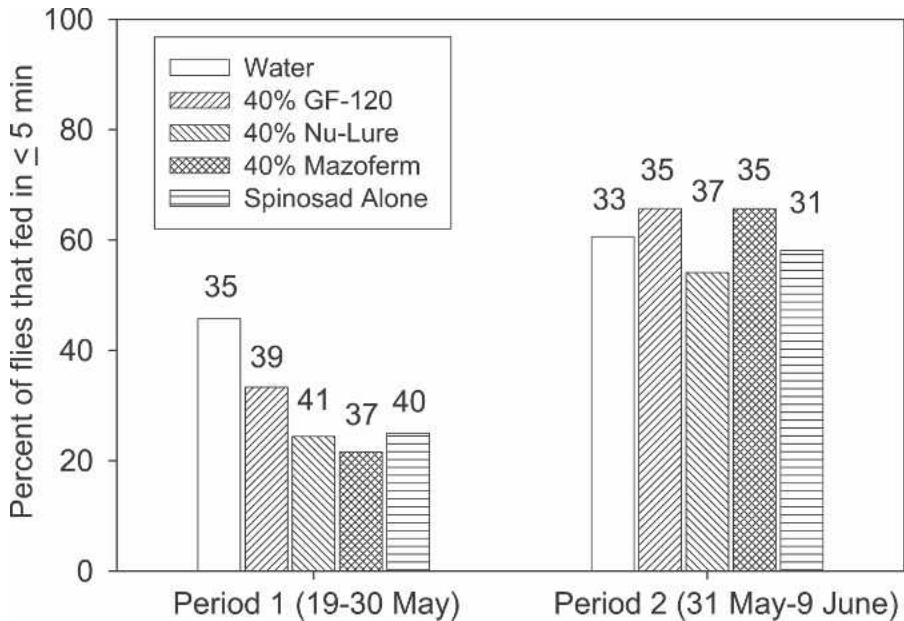


Fig. 1. Female and male *Rhagoletis indifferens* (combined) feeding responses to water or bait drops placed 1–2 cm away from them on tops of leaves, May to June 2006, Zillah, Washington. Figures above bars are total numbers of flies.

2.31 ± 0.26 , 2.26 ± 0.25 , 1.58 ± 0.15 , and 1.55 ± 0.18 , respectively. There were no interactions among factors ($P > 0.05$). Two-way ANOVA of feeding bouts of male responders only showed that there were no differences among treatments ($F = 1.36$; $df = 4, 70$; $P = 0.257$) and no treatment $\times$ period interaction ($P > 0.05$) (Table 2).

Three-way ANOVA indicated that feeding duration was longer on GF-120 than on water, Nu-Lure, and Mazoferm ($F = 10.18$; $df = 3, 118$; $P < 0.0001$), but that there were no sex ($F = 1.26$; $df = 1, 118$; $P = 0.263$) and period effects ($F = 0.10$; $df = 1, 118$; $P = 0.756$) on feeding duration (Table 2). There were no treatment $\times$ period, treatment $\times$ sex, sex $\times$ period, and treatment $\times$ sex $\times$ period interactions ($P > 0.05$). Two-way ANOVA indicated that for males, feeding duration on GF-120 was longer than on all other treatments ($F = 7.16$; $df = 4, 70$; $P < 0.0001$), but there was no period effect ($F = 0.16$; $df = 1, 70$; $P = 0.6871$). There was no treatment $\times$ period interaction ($P > 0.05$) (Table 2).

In all the following other measures of responsiveness, analyses were conducted for each sex and for sexes combined; percentages are for sexes combined, unless noted. In periods 1 and 2, numbers of flies that fed ≤ 5 sec after application were independent of treatments ($P > 0.05$). In period 1, percentages that fed ≤ 5 sec after application on water, GF-120, Nu-Lure, Mazoferm, and spinosad alone were 43.8, 23.1, 0, 50.0, and 40.0%, respectively ($n = 8$ –16 flies in which this behavior was recorded). In period 2, respective percentages were 35.0, 34.8, 50.0,

Table 2. Effects of placing water and bait drops 1-2 cm from *Rhagoletis indifferens* on leaves on mean feeding responses $\pm$ SE, May to June 2006, Zillah, Washington.

Treatment	Sex	N		No. feeding bouts ^a		Feeding duration (min) (n) ^b	
		Period 1	Period 2	Period 1	Period 2	Period 1	Period 2
Water	F	18	9	0.56 $\pm$ 0.18	1.11 $\pm$ 0.42	0.22 $\pm$ 0.06 (8)	0.21 $\pm$ 0.06 (5)
	M	17	24	0.71 $\pm$ 0.21	1.04 $\pm$ 0.24	0.27 $\pm$ 0.16 (8)	0.31 $\pm$ 0.09 (15)
40% GF-120	F	27	16	0.74 $\pm$ 0.22	2.19 $\pm$ 0.52	1.53 $\pm$ 0.43 (11)	1.04 $\pm$ 0.23 (12)
	M	12	19	0.33 $\pm$ 0.26	1.26 $\pm$ 0.37	0.93 $\pm$ 0.40 (2)	1.09 $\pm$ 0.18 (11)
40% Nu-Lure	F	18	20	0.78 $\pm$ 0.33	1.25 $\pm$ 0.41	0.57 $\pm$ 0.27 (6)	0.40 $\pm$ 0.08 (10)
	M	23	17	0.43 $\pm$ 0.20	1.24 $\pm$ 0.32	0.38 $\pm$ 0.02 (5)	0.43 $\pm$ 0.15 (10)
40% Mazoferm	F	21	20	0.38 $\pm$ 0.18	1.10 $\pm$ 0.28	0.41 $\pm$ 0.13 (5)	0.57 $\pm$ 0.16 (13)
	M	16	15	0.19 $\pm$ 0.10	1.00 $\pm$ 0.28	0.26 $\pm$ 0.05 (3)	0.20 $\pm$ 0.05 (10)
Spinosad Alone	F	14	16	0.07 $\pm$ 0.07	1.38 $\pm$ 0.33	0.17 (1)	0.44 $\pm$ 0.14 (11)
	M	26	15	0.54 $\pm$ 0.16	0.87 $\pm$ 0.27	0.23 $\pm$ 0.06 (9)	0.21 $\pm$ 0.05 (7)

Each bait and spinosad alone contained 0.0096% spinosad (wt/vol).
Period 1, 19-30 May (18.8-19.1°C and 49-51% RH); Period 2, 31 May to 9 June (20.8-22.7°C and 45-39% RH).

^aAll flies (N).

^bSample sizes of only flies that fed; feeding durations based on n.

See text for results of three- and two-way ANOVAs.

47.8, and 61.1% ($n = 18-23$). In period 1, numbers of females that touched treatments but did not feed on them were dependent of treatments ($P = 0.048$), with percentages for GF-120, Nu-Lure, Mazoferm, and spinosad alone being 5.9, 0, 0, 16.7, and 66.7%, respectively ($n = 3-11$). However, numbers of males that displayed this behavior were independent of treatments ($P > 0.05$). This was also the case when sexes were combined ($P > 0.05$), with percentages for respective treatments being 5.9, 0, 23.1, 20.0, and 16.7% ($n = 10-17$). In period 2, numbers of flies that touched treatments but did not feed were independent of treatments ($P > 0.05$). Percentages for water, GF-120, Nu-Lure, Mazoferm, and spinosad alone were 4.8, 8.0, 16.7, 0, and 5.3%, respectively ($n = 19-25$). In periods 1 and 2, the numbers of flies that remained still near treatments and did not feed were independent of treatments ($P > 0.05$). In period 1, percentages for water, GF-120, Nu-Lure, Mazoferm, and spinosad alone were 40.0, 26.5, 33.3, 42.4, and 48.6%, respectively ($n = 34-39$). In period 2, the respective percentages were 12.5, 5.9, 8.8, 9.1, and 20.7%, respectively ($n = 29-34$). One female flicked her wings around a Nu-Lure drop on 9 June.

Effects of bait sprays on fly infestations. In 2005, there were no significant differences in numbers of flies trapped in treatments when 225 ml of bait was applied per tree, although numerically there were fewer flies in the bait treatments than in the control (Table 3). However, the numbers of larvae per fruit in Nu-Lure and Mazoferm treatments were significantly lower than in the control, but the number in the GF-120 treatment was not significantly lower than in the control (Table 3). During the entire 40-d test (first spray to last fruit picking), there were 7 d of rain, for 1.25 cm total precipitation.

In 2006, there were no significant differences in numbers of flies trapped and in larval infestations among treatments when 75 (test 1) or 150 ml (test 2) of bait was applied per tree, although numerically there were always fewer larvae per fruit in bait treatments than in controls (Table 3). Also, only one of six trees sprayed with GF-120 (17%) had a detectable infestation. During the entire 36 d of test 1, there were 13 d of rain, for 3.17 cm total precipitation. During the entire 50 d of test 2, there were 15 d of rain, for 3.32 cm total precipitation.

Discussion

The hypothesis that *R. indifferens* is not attracted to protein baits from mid-range distances was supported. It is known that *R. indifferens* is attracted to ammonia from lures (Frick et al. 1954), so it appears even 10 mL of the three baits did not release sufficient ammonia to elicit responses. In one study, no ammonia was detected from Mazoferm (Lee et al. 1997). Work with other tephritids testing more or less mid- or long-range attraction shows effects differ among fly species. Similar to *R. indifferens*, in the blueberry maggot, *Rhagoletis mendax* Curran, and the eastern cherry fruit fly, *Rhagoletis cingulata* (Loew), there appeared to be no attraction to GF-120, although there appeared to be arrestment (Pelz et al. 2005, Pelz-Stelinski et al. 2006). There also appeared to be arrestment in *R. mendax* exposed to high concentrations of ammonium acetate (Barry & Polavarapu 2004). In contrast to *R. indifferens*, for *Anastrepha ludens*, Mazoferm is more attractive than Nu-Lure (Moreno & Mangan 1995), for *Bactrocera cucurbitae* (Coquillett), Solbait (precursor of GF-120) was more attractive than Nu-Lure and Mazoferm (Fabre et al. 2003), and for *Bactrocera dorsalis* (Hendel),

Table 3. Effects of bait sprays on mean numbers of flies trapped and numbers of larvae of *Rhagoletis indifferens* $\pm$ SE per cherry using single tree replicates in Yakima County, Washington, 2005 and 2006.

Yakima (16 May to 24 June 2005), 225 mL spray/tree						
Treatment	<i>N</i>	Flies/trap ^a	Larvae/fruit	% Fewer	% Pos. ^b	Fruit picked/tree
Control	8	257.2 $\pm$ 133.1	0.907 $\pm$ 0.137a	—	100	214.6 $\pm$ 21.9
40% GF-120	4	16.0 $\pm$ 6.5	0.501 $\pm$ 0.218ab	45	100	73.8 $\pm$ 29.0
40% Nu-Lure	5	49.0 $\pm$ 15.2	0.412 $\pm$ 0.103b	55	100	233.4 $\pm$ 45.4
40% Mazoferm	5	53.0 $\pm$ 29.1	0.125 $\pm$ 0.062b	86	80	273.2 $\pm$ 80.0
One-way ANOVA		<i>F</i> = 2.31	<i>F</i> = 6.90	—	—	
df = 3, 18		<i>P</i> = 0.111	<i>P</i> = 0.003	—	—	
2006 Test 1: Zillah (22 May to 26 June), 75 mL spray/tree						
Treatment	<i>N</i>	Flies/trap ^c	Larvae/fruit	% Fewer	% Pos. ^b	Fruit picked/tree
Control	3	40.0 $\pm$ 9.7	0.086 $\pm$ 0.067	—	100	194.3 $\pm$ 67.8
40% GF-120	3	71.7 $\pm$ 16.5	0.050 $\pm$ 0.028	42	67	401.0 $\pm$ 175.9
40% Nu-Lure	3	115.7 $\pm$ 54.7	0.004 $\pm$ 0.001	95	100	496.7 $\pm$ 132.4
40% Mazoferm	3	57.7 $\pm$ 15.3	0.034 $\pm$ 0.015	60	100	236.3 $\pm$ 118.6
Spinosad Alone	3	48.3 $\pm$ 7.7	0.007 $\pm$ 0.004	92	67	284.7 $\pm$ 66.7
RBD ANOVA	—	<i>F</i> = 1.22	<i>F</i> = 0.86	—	—	
df = 4, 8	—	<i>P</i> = 0.376	<i>P</i> = 0.527	—	—	
2006 Test 2: Zillah, Toppenish, Yakima (22 May to 10 July), 150 mL spray/tree						
Treatment	<i>N</i>	Flies/trap ^d	Larvae/fruit	% Fewer	% Pos. ^b	Fruit picked/tree
Control	7	14.9 $\pm$ 8.9	0.031 $\pm$ 0.013	—	100	368.0 $\pm$ 76.0
40% GF-120	6	7.0 $\pm$ 2.9	0.001 $\pm$ 0.001	97	17	590.7 $\pm$ 85.4
40% Nu-Lure	6	6.0 $\pm$ 1.8	0.004 $\pm$ 0.001	87	83	576.0 $\pm$ 68.6
40% Mazoferm	3	16.0 $\pm$ 2.6	0.004 $\pm$ 0.002	87	67	419.0 $\pm$ 204.2
Spinosad Alone	6	20.0 $\pm$ 11.3	0.027 $\pm$ 0.015	13	83	454.0 $\pm$ 54.5
One-way ANOVA		<i>F</i> = 0.70	<i>F</i> = 2.04	—	—	
df = 4, 23		<i>P</i> = 0.601	<i>P</i> = 0.122	—	—	

Each bait and spinosad only contained 0.0096% spinosad (wt/vol).

Dates inside parentheses are first spray to last fruit picking.

Fly catches: ^aover 42 d; ^bpercent of trees infested.

Fly catches: ^cover 46 d; ^dover 57 d.

Means within columns followed by the same letter are not significantly different (LSD test, *P* > 0.05).

Mazoferm and Provesta® 621 were more attractive than Nu-Lure (Vargas & Prokopy 2006). In the current study, only fresh baits were tested, because there is no evidence aged baits are more attractive (Yee 2006). Also, for *B. cucurbitae* under greenhouse conditions, GF-120 was highly attractive within 1 h of bait spray application, but it lost all attractiveness within 24 h (Prokopy et al. 2003),

and GF-120 aged for 2 h was 11 times less attractive than fresh bait (Revis et al. 2004).

The hypotheses that *R. indifferens* is attracted to baits at close range and that responses to baits differ were not supported in 2005 and 2006 tests, using baits without or with spinosad. The kinds or amounts of volatiles from among baits may have differed, but even at close range, they apparently were not attractive to flies. This may explain why 57.2% of flies (2005 and 2006 combined) remained motionless, ignored the bait drops, or grazed around the bait drops only 1–2 cm away. If testing had been done for longer periods, overall responses may have been higher, but it is unclear if this would have caused differential responses to the three baits, unless aging baits increase their attractiveness, and evidence suggest this does not occur (references above).

Based on mean numbers of feeding bouts per fly in 2006, there appeared to be differences in the palatability or acceptability among baits. When nonresponders were excluded from analyses, GF-120 and Nu-Lure caused more feeding bouts than water and Mazoferm. Apparently Mazoferm was not as phago-stimulatory as the other baits, perhaps because of its taste or lower sugar content (see next paragraph).

A significant finding was that feeding duration on GF-120 was greater than on other baits, indicating greatest acceptability for this bait. The percent total solids in 40% GF-120 was similar to that in 40% Nu-Lure and 40% Mazoferm, but its viscosity was 32 and 20 times higher, respectively, caused by thickeners (vegetable gums), and this may have contributed to longer feeding. In addition, 40% GF-120 has a higher sugar concentration than 40% Nu-Lure and 40% Mazoferm (11.9%, 2–5%, and < 1%, respectively, based on sugar amounts in undiluted baits [Buttery et al. 1983, Hull et al. 1996, Yee & Chapman 2005]). Nu-Lure also has ca. 6–13% salt content (Buttery et al. 1983), which is considered high, and which may have reduced its stimulatory power. *Anastrepha ludens* possibly consumed low quantities of Nu-Lure because of the salt (Lee et al. 1997). The rate of 9% Nu-Lure recommended for tropical and subtropical fruit flies would have less salt than 40% Nu-Lure, but in *R. mendax*, feeding duration even on 9% Nu-Lure was approximately 5.5 times shorter than on 50% Solbait (Barry & Polavarapu 2004). Increasing feeding duration by adding sugar in theory is important because baits that cause longer feeding might need only low insecticide concentrations to kill flies.

Despite no bait differences, 42.8% of the *R. indifferens* fed on baits, similar to the 48.7% that drank water and 39.4% that fed on spinosad alone (2005 and 2006 pooled). Thus most flies likely found the baits and water through chance while foraging on leaves or by sight—whether the drops were clear (water) or dark (Nu-Lure)—or by detecting moisture emanating from the baits and water. A few females flicked their wings around bait drops, suggesting they saw and perceived the drops as flies. Wing flicking is an aggressive behavior commonly observed when one fly sees another fly land near by (AliNiazee 1974).

Other measures of responsiveness to treatments were not significant, except in one case, again suggesting there were no differences among water, baits, and spinosad alone in attractiveness. The one case of significance could be a statistical artifact. Most flies that responded did so quickly, perhaps because these flies were hungrier and their stimulation threshold was lower than in nonresponders. Most flies (sexes combined) that contacted water, baits, and spinosad alone drank or fed

($\leq 23.1\%$ rejected them), indicating that baits were not repellent. Based on the percentages of flies that remained still near the water and baits without responding, there also was no evidence any of the baits arrested the flies. Because none of the baits was fed upon 100% of the time after flies contacted it and some flies simply did not contact the baits, the chances the baits do not kill at least a small percentage of flies before oviposition occurs could be high.

The similarity in female and male fly responses to baits was somewhat surprising. Female flies presumably need to feed on more protein (or amino acids) than males (as in *R. pomonella*, Webster et al. 1979), but results here indicate males respond to protein baits as much as females. Thus, the baits probably remove both sexes from the population at equal rates. Perhaps the females were mature and already consumed enough protein.

Another interesting and important finding was that feeding responses of flies in 2006 were greater later than earlier in the season. Because fly activity is relatively low when temperatures are $<20^{\circ}\text{C}$ (Yee 2002), cooler temperatures in early season rather than fly age may have caused this. In *R. pomonella*, protein and sucrose intake in the laboratory at fairly constant temperatures ($25 \pm 2^{\circ}\text{C}$) was higher when flies were <2 weeks old than when they were 2–3 weeks old (Webster et al. 1979). From a management standpoint, *R. indifferens* may be more difficult to control during early than late season if significant temperature differences between periods exist, if females at cooler temperatures can mature eggs by 7–10 d post emergence, and if suitable fruit are available for oviposition shortly after that.

In the 2005 spray test, none of the baits reduced adult fly captures, indicating flies were likely trapped before they fed on them or that the flies were newly emerged, but Mazoferm and Nu-Lure significantly reduced larval infestations compared with the control, suggesting that although they were not attractive, enough flies were able to find them to result in significant reductions. GF-120, however, did not significantly reduce larval infestations, in contrast with previous findings using single trees (Yee & Chapman 2005) and orchard trees, where GF-120 reduced infestations in late-season tart cherry collections by 92.3% (Alston & Lindstrom 2006). The present results may have been caused by flies not responding to GF-120 quickly enough to prevent oviposition and also by high variability in fly numbers among trees and low replication size. Mature flies from nontest trees may have caused infestations in some of the test trees, because GF-120 sprayed on fruit is unable to prevent oviposition by older females (Yee & Alston 2006), although outside sources of infestations seem unlikely in isolated trees. Rainfall may have reduced amounts of bait on trees. Under ideal conditions of little or no rain, GF-120 is effective in killing tephritid flies for 14 d (Mangan et al. 2006, Yee & Alston 2006).

In the 2006 spray tests, no significant differences among baits were seen, but unlike in 2005, where Nu-Lure and Mazoferm had significant effects on larval infestations, none of the baits was different from the controls, even though larval infestations in treatments numerically were lower than in the controls. The lack of any statistical differences between baits and controls again suggests flies did not find any of the baits quickly enough to prevent oviposition. The large variability in fly populations among trees also may have played a role. Spray volumes used in 2006 were lower than in 2005, so even fewer flies may have found the Nu-Lure and Mazoferm simply through chance encounters. Rainfall also may

have washed away or diluted the bait. Even though significant differences were not detected, results likely underestimated the ability of the baits to control flies under orchard conditions, where fly densities are lower and all trees would have been treated (Alston & Lindstrom 2006).

Despite the focus on baits in this study, it was not clear if baits were needed for control of flies, based on 2006 results. The lack of significant differences made the results difficult to interpret, but evidence from elsewhere suggests spinosad alone can be quite effective in controlling infestations. In tart cherry, spinosad alone (Success®, Dow AgroSciences, Indianapolis, Indiana) sprayed at 654 l/ha in an orchard eliminated infestations from late season tart cherry collections (Alston & Lindstrom 2006).

In summary, results suggest the baits tested were not attractive to *R. indifferens* compared with water and that most flies find them through chance encounters, especially when it is warm. Chance encounters may partially explain the lack of consistent larval control using baits. GF-120 did not retain flies longer once they began to feed. Addition of compounds such as ammonium carbonate in baits may increase their attractiveness (e.g., in GF-120 for *R. cingulata* [Pelz-Stelinski et al. 2006]) and the percentages of flies that respond. However, adding more ammonia to GF-120 did not reduce infestations by larval *R. indifferens* compared with using GF-120 alone (Alston & Lindstrom 2006). Based on the results of the current study and others, the choice of baits may not be vital for controlling larval infestations of *R. indifferens*, and even spinosad alone may have the same effects, at least when fly populations are lower than those encountered in this study.

Acknowledgments

The author thanks Peter Chapman and Katie Mackie for field assistance; Gene Miliczky, Carol Gefre and other homeowners for use of trees; and Robert Mangan (USDA-ARS, Weslaco, Texas), Diane Alston (Utah State University), and Peter Landolt (USDA-ARS) for helpful comments on the manuscript. This research was partially funded by the Washington Tree Fruit Research and Oregon Sweet Cherry Commissions. This article reports the results of research only. Mention of a proprietary product does not constitute an endorsement or recommendation for its use by USDA.

References Cited

- AliNiazee, M. T. 1974.** The western cherry fruit fly, *Rhagoletis indifferens* (Diptera: Tephritidae) 2. Aggressive behavior. Can. Entomol. 106: 1201–1204.
- Alston, D. & T. Lindstrom. 2006.** Western cherry fruit fly control: can addition of adult attractants enhance the efficacy of GF-120? Research reports – integrated pest management - <http://utahpests.usu.edu/ipm/htm/publications/research>, accessed 19 January 2007
- Anonymous. 1968.** Permanent order no. 1099 prohibiting the sale and/or movement of infested cherries. Washington Administrative Code 16-463-001, 16-463-010. Effective 30 September 1968. <http://www.leg.wa.gov/>, accessed 1 December 2006.
- Barry, J. D. & S. Polavarapu. 2004.** Feeding activity and attraction of blueberry maggot (Diptera: Tephritidae) to protein baits, ammonium acetate, and sucrose. J. Econ. Entomol. 97: 1269–1277.
- Barry, J. D., R. I. Vargas, N. W. Miller & J. G. Morse. 2003.** Feeding and foraging of wild

- and sterile Mediterranean fruit flies (Diptera: Tephritidae) in the presence of spinosad bait. *J. Econ. Entomol.* 96: 1405–1411.
- Barry, J. D., N. W. Miller, J. Piñero, A. Tuttle, R. F. L. Mau & R. I. Vargas. 2006.** Effectiveness of protein baits on melon fly and oriental fruit fly (Diptera: Tephritidae) attraction and feeding. *J. Econ. Entomol.* 99: 1161–1167.
- Buttery, R. G., L. C. Ling, R. Teranishi & T. R. Mon. 1983.** Insect attractants: volatiles of hydrolyzed protein insect baits. *J. Agric. Food Chem.* 31: 689–692.
- Dow AgroSciences. 2006.** Specimen Label, GF-120 NF Naturalyte Fruit Fly Bait, Dow AgroSciences, revised 04/11/06. Indianapolis, Indiana.
- Fabre, F., P. Ryckewaert, P. F. Duyck, F. Chiroleu & S. Quilici. 2003.** Comparison of the efficacy of food attractants and their concentration for melon fly (Diptera: Tephritidae). *J. Econ. Entomol.* 96: 231–238.
- Frick, K. E., H. G. Simkover & H. S. Telford. 1954.** Bionomics of the cherry fruit flies in eastern Washington. *Wash. Agric. Expt. Stat. Tech. Bull.* 13, 66 pp.
- Hull, S. R., B. Y. Yang, D. Venzke, K. Kulhavy & R. Montgomery. 1996.** Composition of corn steepwater during steeping. *J. Agric. Food Chem.* 44: 1857–1863.
- Lee, C.-J., A. B. DeMilo, D. S. Moreno & R. L. Mangan. 1997.** Identification of the volatile components of E802 Mazoferm Steepwater, a condensed fermented corn extractive highly attractive to the Mexican fruit fly (Diptera: Tephritidae). *J. Agric. Food Chem.* 45: 2327–2331.
- Mangan, R., D. Moreno & G. Thompson. 2006.** Bait dilution, spinosad concentration, and efficacy of GF-120 based fruit fly sprays. *Crop Protection J.* 25: 125–133.
- Moreno, D. S. & R. L. Mangan. 1995.** Responses of the Mexican fruit fly (Diptera: Tephritidae) to two hydrolyzed proteins and incorporation of Phloxine B to kill adults. *ACS Symp. Ser.* 616: 257–279.
- Peck, S. L. & G. T. McQuate. 2000.** Field tests of environmentally friendly malathion replacements to suppress wild Mediterranean fruit fly (Diptera: Tephritidae) populations. *J. Econ. Entomol.* 93: 280–289.
- Pelz, K. S., R. Isaacs, J. C. Wise & L. J. Gut. 2005.** Protection of fruit against infestation by apple maggot and blueberry maggot (Diptera: Tephritidae) using compounds containing spinosad. *J. Econ. Entomol.* 98: 432–437.
- Pelz-Stelinski, K. S., L. J. Gut & R. Isaacs. 2006.** Behavioral responses of *Rhagoletis cingulata* (Diptera: Tephritidae) to GF-120 insecticidal bait enhanced with ammonium acetate. *J. Econ. Entomol.* 99: 1316–1320.
- Prokopy, R. J., N. W. Miller, J. C. Piñero, J. D. Barry, L. C. Tran, L. Orde & R. I. Vargas. 2003.** Effectiveness of GF-120 Fruit Fly Bait spray applied to border area plants for control of melon flies (Diptera: Tephritidae). *J. Econ. Entomol.* 96: 1485–1493.
- Revis, H. C., N. W. Miller & R. I. Vargas. 2004.** Effects of aging and dilution on attraction and toxicity of GF-120 Fruit Fly Bait spray for melon fly control in Hawaii. *J. Econ. Entomol.* 97: 1659–1665.
- Roessler, Y. 1989.** Insecticidal bait and cover sprays, pp. 329–335. *In* A. S. Robinson and G. Hooper [Ed.], *Fruit Flies: Their Biology, Natural Enemies and Control*. World Crop Pests. Vol. 3B. Elsevier Science Publishers, Amsterdam, The Netherlands.
- SAS Institute. 2003.** SAS 9.1. SAS for windows. Cary, North Carolina.
- The Weather Underground, Inc. 2006.** Ann Arbor, Michigan. <http://www.wunderground.com>, accessed 1 December 2006.
- Thomas, D. B. & R. L. Mangan. 2005.** Nontarget impact of spinosad GF-120 Bait sprays for control of the Mexican fruit fly (Diptera: Tephritidae) in Texas citrus. *J. Econ. Entomol.* 98: 1950–1956.
- Vargas, R. I., N. W. Miller & R. J. Prokopy. 2002.** Attraction and feeding responses of Mediterranean fruit fly and a natural enemy to protein baits laced with two novel toxins, phloxine B and spinosad. *Entomol. Expt. et Appl.* 102: 273–282.
- Vargas, R. I. & R. Prokopy. 2006.** Attraction and feeding responses of melon flies and

- oriental fruit flies (Diptera: Tephritidae) to various protein baits with and without toxicants. *Proc. Hawaiian Entomol. Soc.* 38: 49–60.
- Vargas, R. I., E. B. Jang & L. M. Klungness. 2003.** Area-wide pest management of fruit flies in Hawaiian fruits and vegetables. pp. 37–46. *In* Recent Trends on Sterile Insect Technique and Area-wide Integrated Pest Management. Research Institute for Subtropics.
- Webster, R. P., J. G. Stoffolano & R. J. Prokopy. 1979.** Long-term intake of protein and sucrose in relation to reproductive behavior of wild and laboratory cultured *Rhagoletis pomonella*. *Ann. Entomol. Soc. Am.* 72: 41–46.
- Yee, W. L. 2002.** Daily and seasonal activity patterns of *Rhagoletis indifferens* (Diptera: Tephritidae) in Washington state. *Ann. Entomol. Soc. Am.* 95: 701–710.
- Yee, W. L. 2006.** Feeding history effects on feeding responses of *Rhagoletis indifferens* (Dipt., Tephritidae) to GF-120 and Nulure. *J. Appl. Entomol.* 130: 538–550.
- Yee, W. L. & P. S. Chapman. 2005.** Effects of GF-120 Fruit Fly Bait concentrations on attraction, feeding, mortality, and control of *Rhagoletis indifferens* (Diptera: Tephritidae). *J. Econ. Entomol.* 98: 1654–1663.
- Yee, W. L. & D. G. Alston. 2006.** Effects of spinosad, spinosad bait, and chloronicotinyl insecticides on mortality and control of adult and larval western cherry fruit fly (Diptera: Tephritidae). *J. Econ. Entomol.* 99: 1722–1732.
-

Evaluation of Pheromone Traps for Monitoring Sweetpotato Weevils^{1,2}

D. Michael Jackson and Janice R. Bohac

USDA, ARS, U.S. Vegetable Laboratory, Charleston, South Carolina USA

J. Agric. Urban Entomol. 23(3): 141–158 (July 2006)

ABSTRACT Ten types of pheromone traps for monitoring sweetpotato weevils, *Cylas formicarius* (F.) (Coleoptera: Brentidae), were evaluated in sweetpotato fields at the U.S. Vegetable Laboratory, Charleston, South Carolina, during 2001 and 2002. A funnel trap, a modification of a water-pan trap, and the Pherocon sticky trap were the most effective for capturing male sweetpotato weevil adults. A universal trap, Multipher trap, boll weevil trap, a trap constructed from a five-gallon pail, and a trap made from a milk jug also captured significant numbers of weevils. A Japanese beetle trap and the plastic Pherocon CRW kairomone trap were not effective for trapping sweetpotato weevils. In separate experiments in 2005 and 2006, a trap made from a recycled plastic (PET) soft drink bottle was significantly less effective than the universal trap. Universal traps and PET bottle traps with soapy water were as effective as traps with a dichlorvos insecticide strip for a killing agent. Captures of male sweetpotato weevils in universal traps ranged 0.0–6.0 adults per trap per day during a 6-year period, 2001–2006 at the U.S. Vegetable Laboratory. Except for 2005, when population levels were unusually low, weevil captures increased rapidly during August and continued at high levels until freezing weather in November or December each year. Peak captures were generally from mid-September to mid-October.

KEY WORDS *Cylas formicarius*, Brentidae, pheromones, sweetpotato, *Ipomoea*, population dynamics, traps

The sweetpotato, *Ipomoea batatas* (L.) Lam. (Convolvulaceae), is one of the world's most important food crops, especially in developing countries, where it is a major source of sustenance and food security (Woolfe 1992, CIP 2004). The sweetpotato is also an important commercial crop in the United States (LaBonte & Cannon 1998). Throughout the world, production of the sweetpotato is severely limited by several insect pests, and improved pest management approaches for this crop are needed (Cuthbert 1967, Chalfant et al. 1990). The sweetpotato weevil, *Cylas formicarius* (F.) (= *Cylas formicarius elegantulus* [Summers]) (Coleoptera: Brentidae) (Anderson & Kissinger 2002), is the most important worldwide insect pest of the sweetpotato (Sutherland 1986a). This species is

¹Accepted for publication 19 June 2007

²Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture.

widely distributed throughout much of the tropical and subtropical regions of the world, including coastal areas of the southeastern United States (Wolfe 1991). Losses inflicted by this pest can be severe both in storage facilities (Ray & Ravi 2005) and in the field, where yield losses can be as great as 60–97% (Mullen 1984, Sutherland 1986a, 1986b, Jansson et al. 1987).

Sweetpotato weevils damage sweetpotatoes by feeding on storage roots or in the crown of the plant (Sutherland 1986a, Jansson et al. 1990a). Adults are capable of flying at least 1 mile, they are nocturnal, and they are seldom seen on the plants during the daytime (Sakuratani et al. 1994, Shimizu & Moriya 1996). Jansson et al. (1990a) reported that approximately 80–90% of the weevil population is below the soil surface. Thus, it is difficult to determine infestation levels or root damage. The severity of infestations can only be accurately determined by digging up and dissecting the storage roots (Sutherland 1986b). This is not only a difficult process, but it also destroys a portion of the crop. Therefore, monitoring adult populations with pheromone traps has become an important component of integrated pest management (IPM) programs in many countries in Asia (Talekar 1991, Hwang & Hung 1991, Palaniswamy et al. 1992, Chiranjeevi & Reddy 2003), Africa (Smit et al. 1997, 2001), and the Caribbean (Alvarez et al., 1996, Alcázar et al. 1997, Jackson et al. 2002).

Coffelt et al. (1978) isolated and bioassayed the female-produced sex pheromone of *C. formicarius* in the laboratory. Heath et al. (1986) later purified, identified, and synthesized the active component as (*Z*)-3-dodecen-1-ol (*E*)-2-butenolate. Subsequent studies have reported on new synthesis procedures or formulations that improve safety, reduce costs, or enhance efficacy of this material (Yen & Hwang 1990, Lo et al. 1992, Pawar et al. 1993, Mithran & Subbaraman 1999, Sureda & Quero 2006). A synthetic sex pheromone has been evaluated in field experiments in the United States (Jansson et al. 1989, Heath et al. 1991), St. Croix, U. S. Virgin Islands (Proshold et al. 1986), Taiwan (Talekar & Lee 1989), Japan (Yasuda et al. 1992), India (Teli & Salunkhe 1993), Venezuela (Marcano et al. 1994), Indonesia (Braun & van de Fliert 1999), and Vietnam (Hao et al. 1996). Several studies have shown a positive relationship between trap captures and pheromone dosage between 10 ng and 1 mg (Proshold et al. 1986, Mason et al. 1990, Jansson et al. 1990b, 1992, Hao et al. 1996), but trap counts were not consistently correlated with pheromone dosage at low population densities of weevils (Jansson et al. 1991, 1992). Sugimoto et al. (1994a) reported that the average attractive distance for a lure with 100 or 400 µg of pheromone was 55 and 64 m, respectively. Jansson et al. (1991) suggested that a dosage of 10 µg was adequate for monitoring traps, but that higher dosages might be required for mass trapping or mating disruption applications. Pheromone lures for sweetpotato weevils have been used up to 3 months in the field (Pillai et al. 1993) and, in Florida, 2-month-old rubber septa with 10 µg of pheromone performed as well as new lures (Mason et al. 1990). Jansson et al. (1991) reported that the sweetpotato weevil sex pheromone is relatively stable on rubber septa and that lures remain active for at least 30–64 days. However, Hwang (2000) showed that the effectiveness of 2-month-old lures was reduced. Thus, it is often recommended that lures be changed monthly (Hwang 2000).

The sweetpotato weevil sex pheromone has several IPM applications, including quarantine pest detection (Hammond et al. 1989, Anonymous 2006a), population monitoring (Jansson et al. 1990a, Teli & Salunkhe 1993, Sugimoto et al.

1994b), mass trapping (male annihilation) (Setokuchi et al. 1991, Yasuda 1995, Smit et al. 2001), mating disruption (Mason & Jansson 1991), and eradication efforts (Komi 2000). This pheromone can also be used in attracticide formulations (Yasuda et al. 2004) or as the attractant for an auto-infection (autodissemination) device for distributing entomopathogens, such as the fungus *Beauveria bassiana* (Balsamo) Vuillemin (Deuteromycotina) (Yasuda 1999). Continuous trapping can significantly reduce male weevil populations and potentially protect storage roots (Talekar & Lee 1989, Jansson et al. 1991). Trapping has been shown to reduce the proportion of roots damaged by sweetpotato weevils in the Caribbean (Alvarez et al. 1996, Alcázar et al. 1997). Hwang (2000) estimated that use of pheromone-baited traps as part of an IPM program for sweetpotato weevils could eliminate one to three insecticide applications.

Several states in the southern United States have quarantine programs for limiting the spread of sweetpotato weevils (Nilakhe 1991, Anonymous 2006a, 2006b). Through quarantine detection and suppression efforts, sweetpotato weevils have been eradicated several times from the weevil-free zones of states with sweetpotato production (Jackai et al. 2006). The North Carolina Department of Agriculture (NCDA) uses screen-cone boll weevil traps with the sex pheromone at a density of one trap per 4.1 ha (10 acres), with a minimum of two traps per field (Anonymous 2006a). A commercial boll weevil trap is also listed for monitoring sweetpotato weevils by the National Agricultural Pest Information System (NAPIS) (Anonymous 2002). Hammond et al. (1989) showed that a screen-cone boll weevil trap was more effective than one with a solid cone at relatively low sweetpotato weevil populations. In a study from Louisiana and North Carolina, where there were low population densities of sweetpotato weevils, trap captures for funnel traps, universal traps, and boll weevil traps could not be differentiated (Jansson et al. 1992). However, Jansson et al. (1989, 1992) reported that the boll weevil trap was much less effective than universal traps or funnel traps at higher population levels of sweetpotato weevils.

Proshold et al. (1986) tested several trap designs in St. Croix and found that aluminum and plastic funnel traps were the most effective for capturing adult males. Jansson et al. (1989) reported that a modification of this trap with a plastic funnel was 85% efficient, compared with 49% efficiency for a plastic boll weevil trap in south Florida. Subsequently, hand-made variations of the funnel trap design have been evaluated in several countries (Talekar & Lee 1989, Yasuda et al. 1992, Sugiyama et al. 1996, Smit et al. 1997). Under high population densities in southern Florida, Jansson et al. (1992) found that the funnel trap worked better than the universal traps, partly because the lip of the universal trap impeded movement of weevils into this trap. Most weevils are captured within one meter of the ground (Sugimoto et al. 1994a), and it has been reported that the optimum height of the openings into the funnel trap is just above the height of the plant canopy (Proshold et al. 1986, Jansson et al. 1992). Because the nocturnal male weevils often crawl to the uppermost leaves of sweetpotato plants (Proshold et al. 1986), pheromone traps surrounded by foliage capture more insects than those not surrounded by plants (Jansson et al. 1992, Marcano et al. 1994). It also has been suggested that trap efficiency is increased if a wire-mesh "ladder" is provided for weevils to crawl up into the funnel trap (Proshold et al. 1986, Jansson et al. 1989, Heath et al. 1991).

Setokuchi et al. (1991) developed a square funnel-plastic box trap that was a

modification of the original funnel trap of Proshold et al. (1986). This design was used by Sugimoto et al. (1994a, 1994b) to estimate the attractive area of the pheromone and to estimate weevil population sizes in Japan. Sugiyama et al. (1996) developed a cylindrical trap from polyvinyl chloride (PVC) pipe, which was as effective as the funnel trap, captured fewer nontarget organisms, and cost less to construct.

Simple water-pan traps are often used in developing countries (Talekar & Lee 1989, Islam et al. 1992, Pillai et al. 1993, Braun & van de Fliert 1999). Talekar (1988) illustrated a simple, inexpensive trap with a water-pan supported in the middle of a truncated woven-wire cone 25 cm from the soil surface, and he recommended that the pheromone lure be suspended from the trap cover above the water pan so that it is 10–20 cm above the crop canopy. Other researchers have used a trap made from a pan or other open container filled with soapy water and with the pheromone lure suspended above it (Marcano et al. 1994, Alvarez et al. 1996, Braun & van de Fliert 1999). Although soapy water has been used most often as the killing agent for trapping sweetpotato weevils (Pillai et al. 1993, Smit et al. 1997), dichlorvos strips have also been used in traps with closed collection chambers (Proshold et al. 1986, Sugiyama et al. 1996, Hwang 2000). Pillai et al. (1993) illustrated a metal water-pan trap similar to the one Talekar (1988) used, and they reported that traps made from empty oil tins were used by farmers in Kerala, India. A similar trap was illustrated by Ray & Ravi (2005). Islam et al. (1992) described a new water pan trap for use in mass trapping sweetpotato weevils in Bangladesh.

Sticky traps of various designs have been tested for monitoring sweetpotato weevils (Proshold et al. 1986, Talekar & Lee 1989, Setokuchi et al. 1991, Yasuda et al. 1992, Smit et al. 1997, Sureda & Quero 2006). The pheromone-baited sticky trap tested by Yasuda et al. (1992) caught as many weevils as water-pan traps. When set on the ground, these sticky traps caught more weevils than traps at heights up to 300 cm (Yasuda et al. 1992). Disk-shaped sticky traps worked better than homemade traps constructed from 1-L soft drink bottles, 5-L jerry cans, or 4-L oil cans in Africa (Smit et al. 1997). However, because they were inexpensive, easy to construct, robust, and effective, 5-L jerry can traps were the best design for trapping African *Cylas* species (Smit et al. 1997). The 5-L jerry can traps were later used in mass-trapping studies (Smit et al. 2001). These traps had the added advantage of being larger than other designs so the trapped weevils did not have to be removed and the soapy water did not have to be replenished as often as in smaller traps.

Several variations of weevil traps made from recycled plastic containers have been tested (Talekar & Lee 1989, Hwang & Hung 1991, Alvarez et al. 1996, Smit et al. 1997, Downham et al. 1999, Lawrence & Meyers 1999). Hwang et al. (1989) developed a double funnel trap from discarded 1-L polyethylene terephthalate (PET) soft drink bottles. Clear bottles worked better than green ones, and there was no significant difference in trapping efficiency for traps with funnel diameters between 8.8 and 12 cm (Hwang 2000). Lo et al. (1992) illustrated a double-funnel version of this PET-bottle trap that they used for field evaluations of the formulation of *C. formicarius* sex pheromone they had synthesized. A commercial variation of this trap is now produced in China and Taiwan at a low cost (Hwang 2000). A slight modification of this trap was a good design for capturing *C. puncticolis* in Africa, but it was difficult to construct and not sturdy (Smit et al. 1997).

PET-bottle traps of similar designs have also been used to trap cutworms, *Spodoptera litura* (F.) (Lepidoptera: Noctuidae), in Korea (Anonymous 2001), *Agriotes brevis* Candeze (Coleoptera: Elateridae) in Europe (Tóth et al. 2002), and wasps and bees (Baxendale et al. 2002). Many other examples of home-made bottle traps of similar design can be found on the Internet.

For their IPM project in the Dominican Republic, Alvarez et al. (1996) used pheromone traps made from recycled plastic gallon containers with windows cut into them. This trap is similar to the one illustrated by Lawrence & Meyers (1999) for use in IPM of sweetpotato in the Caribbean. While visiting a grower's field in Antigua in November 2000, we were shown a new sweetpotato weevil trap that had been constructed by a grower from a five-gallon pail. This design has several advantages over the milk-jug trap (Lawrence & Meyers 1999), which has been recommended for use in Jamaica.

Although some current trap designs are effective, they may be cumbersome, difficult to maintain, or expensive. Thus, simple, cost-effective, and efficient traps are needed for monitoring sweetpotato weevils in the United States and in developing countries. The purpose of the research described herein was to evaluate commercially available and home-made pheromone traps for monitoring sweetpotato weevils.

Materials and Methods

U.S. Vegetable Laboratory (USVL) survey traps, 2001–2006. As part of the sweetpotato weevil reduction protocols at the USVL, Charleston, South Carolina, universal traps (Great Lakes IPM, Vestaburg, Michigan) were monitored continuously from April 2001 to December 2006. Each year, a single universal trap was placed on the ground near the edge of each of three to six fields (0.7–1.1 ha.) at the USVL. Traps were baited with rubber septa lures containing 12 µg of synthetically produced sex pheromone of the sweetpotato weevil (Product no. AGS-SPW, Great Lakes IPM, Vestaburg, Michigan). Traps were checked and emptied at least weekly during the sweetpotato growing season, but were checked less frequently (every 2–4 weeks) during the winter months. During this 6-year period (2050 days), traps were monitored 362 times, and pheromone lures were changed monthly.

Evaluation of different trap types, 2001–2002. Ten types of pheromone-baited traps were evaluated for their effectiveness in trapping male sweetpotato weevils at the USVL in 2001–2002. Each year, two fields (0.7–1.1 ha.) that had been planted to sweetpotato genotypes from the USDA-ARS sweetpotato breeding program (Jackson & Bohac 2006) were used. Local production practices were followed, except that no insecticides were used. When rainfall was not adequate during the growing season, supplemental irrigation was applied.

Seven types of traps were evaluated in the two field replications in 2001: 1) modified water-pan trap (Talekar 1988) constructed by us; 2) modified funnel trap (Proshold et al. 1986) constructed by us; 3) universal trap with yellow funnel, white bottom, and green cover (Great Lakes IPM, Vestaburg, Michigan); 4) prototype 5-gallon (18.9-L) plastic-pail trap constructed by us; 5) modified 1-gallon (3.8-L) plastic milk-jug trap (Lawrence & Meyers 1999) constructed by us; 6) Japanese beetle trap (Great Lakes IPM, Vestaburg, Michigan); and 7) Pherocon® CRW (corn rootworm) kairomone trap (Trécé, Inc., Salinas, California) (Lingren

2000, 2002; Trécé Inc. 2006). Eight types of traps were evaluated in the two field replications in 2002: 1) modified water-pan trap, 2) modified funnel trap, 3) universal trap, 4) plastic-pail trap, 5) milk-jug trap, 6) Pherocon 1C sticky trap (Trécé, Inc., Salinas, California), 7) Multiplier 3 trap (Les Services Bio-Contrôle, Inc., Quebec, Canada), and 8) screen-cone boll weevil trap (Great Lakes IPM, Vestaburg, Michigan) (Dickerson et al. 1981). Photographs of these traps in the field were shown by Jackson (2003).

Within a field, traps were randomly assigned to flagged locations in two rows that were at least 20 m apart and at least 20 m from the edge of the field. Within rows, traps were evenly spaced at least 20 m apart. In 2001, baited traps were put into sweetpotato fields on 29 June, and they were monitored twice a week (each Tuesday and Friday) until 29 November. There were 43 sampling dates over the course of 22 weeks in 2001. In 2002, baited traps were put into sweetpotato fields on 7 June, and they were monitored as in 2001 until 13 November. There were 44 sampling dates over the course of 23 weeks in 2002.

All traps were baited with rubber septa lures containing 12 µg of pheromone. Lures were changed monthly, and traps were rotated to a new field position each week (Tuesday). A pheromone lure was hung by a thin wire in the center of each funnel trap so that the septum was centered in the holes in the funnel. For the water-pan trap, a pheromone lure was hung from the cover by a thin wire so that the septum was suspended just above the water reservoir. A pheromone lure was hung inside the Japanese beetle trap near the bottom of the funnel. For the milk jug trap and plastic-pail trap, pheromone lures were hung inside the trap by a thin wire so that the septum was centered with the openings into the trap. A pheromone lure was placed inside the collection chamber of the CRW kairomone trap above the soapy water. The universal and Multiplier 3 traps have a small plastic holder with slits to hold a pheromone lure. A pheromone lure was placed in the center of the bottom panel of the sticky trap.

Modifications to the funnel trap (Proshold et al. 1986) included a plastic funnel (17 cm diameter × 17 cm long), a plywood cover (40 cm square), and a cylindrical support (35 cm diameter × 50 cm tall) made from galvanized hardware cloth (1.27 cm × 1.27 cm mesh) (The Home Depot, Atlanta, Georgia). This trap design is similar to the one illustrated by Heath et al. (1991) and Jansson et al. (1991). We modified the water-pan trap after the design of Talekar (1988). We used the bottom portion of a universal trap (15 cm diam. × 12 cm deep), to which four wooden dowels (6-mm diameter × 15 cm long) were attached vertically with clear hot-melt glue (Glue Stix®, Model AP10-4, Arrow Fastener Co., Inc., Saddle Brook, New Jersey). These dowels were then pushed into the existing sockets on the green cover (16-cm diameter) of the universal trap (the yellow funnel portion of the universal trap was not used). In the field, this trap was seated tightly into the top end of a truncated cone (60 cm tall × 60-cm diameter) constructed with galvanized hardware cloth (0.32 cm × 0.32 cm mesh; The Home Depot, Atlanta, Georgia). The plastic-pail trap was constructed from a round 3.8-L plastic pail (37 cm tall × 30 cm diam., 0.070 mil high-density polyethylene) with a lid (Leaktite, Leominster, Massachusetts). Two rectangular holes (5 cm × 29 cm) were cut into the side of the pail 15 cm from the bottom. The milk-jug trap was made from recycled white plastic (high-density polyethylene) milk containers that had been purchased from a local supermarket. Large rectangular holes were cut into the front and back of the milk jug to allow entrance of weevils.

Soapy water was placed in the collection reservoirs of the water-pan trap, funnel trap, milk-jug trap, plastic pail-trap, universal trap, CRW kairomone trap, and Multiplier trap. The soapy water contained 5 ml of liquid Ultra Palmolive® Antibacterial Hand Soap (Colgate-Palmolive Company, New York, New York) per liter of tap water. Small holes in the water reservoirs of the universal trap and the modified Talekar trap were plugged with clear hot-melt glue. The boll weevil trap, Japanese beetle trap, and sticky trap do not have reservoirs for water.

The universal trap, milk-jug trap, Japanese beetle trap, Multiplier, and sticky trap were suspended by a wire attached to a conduit-pipe support that was bent at a right angle. The boll weevil trap was supported by a 50 cm-long wooden dowel that was attached to the bottom of the trap at one end and pushed into the soil at the other end. The CRW kairomone trap was supported by a 45-cm long piece of 1/2-inch (1.27-cm) PVC pipe (Schedule 40 rigid conduit, Cantex, Mineral Wells, Texas). One end of the PVC pipe was inserted into a socket on the bottom of the collection chamber and the other end was pushed into the soil. The plastic-pail trap was set directly on the soil within the plant canopy. A wire-mesh cylinder supported the collection funnel of the modified funnel trap and was part of its design. The water-pan trap was supported by a hardware-cloth cone as discussed above. The height of each trap was positioned so that the pheromone lure was located about the same height as the top of the plant canopy (approximately 45–60 cm above the soil).

Current prices for the commercially available traps from Great Lakes IPM (Vestaburg, Michigan) are 1) \$2.21 for the Pherocon 1C sticky trap, 2) \$3.25 for the boll weevil trap, 3) \$3.68 for the CRW trap, 4) \$8.95 for the universal trap, and 5) \$10.50 for the Japanese beetle trap. The primary cost of the plastic-pail trap is for the pail, which can be purchased for \$4.00–\$7.00 (various internet sources) depending on quality of the plastic and the quantity purchased. There was essentially no cost for the recycled materials and minimal costs for labor to construct the milk-jug traps (Lawrence & Meyers 1999). On the other hand, construction times and material costs are substantial for the water-pan trap (\$12.00 plus labor costs) and funnel trap (\$6.50 plus labor costs). The water-pan trap would be much less expensive if a cheaper pan were used instead of the bottom portion of a universal trap, which contributed \$8.95 of the cost.

Evaluation of PET-bottle trap, 2005–2006. In 2005 and 2006, a separate experiment was set up to evaluate a trap made from recycled 2-L PET soft drink bottles, which were compared with the universal trap. The PET-bottle traps were similar in design to those illustrated by Hwang (2000). Each type of trap was filled about half full with soapy water. In 2005, two replications were evaluated in two sweetpotato fields (0.7 ha. each) at the USVL. Traps were monitored approximately weekly for the entire year (6 January to 23 December 2005), which resulted in 45 trap collections. In 2006, there were six replications in six different sweetpotato fields at the USVL. Traps were monitored weekly for 20 weeks (13 July to 22 November 2006). Pheromone lures were changed monthly during each year.

Evaluation of killing agents, 2006. In 2006, a separate experiment was set up to determine the effectiveness of the killing agent in pheromone-baited universal traps. For this experiment, there were three treatments: 1) universal trap with 3–5 cm of soapy water, 2) universal trap with a 2.5-cm × 5-cm strip impregnated with 10% dichlorvos (2,2-dichlorovinyl dimethyl phosphate) (Hercon

Vaportape® II, Hercon Environmental Co., Emigsville, Pennsylvania), and 3) universal trap with both soapy water and dichlorvos strip. Vaportape II is labeled for use as a toxicant strip in insect traps for sweetpotato weevil (Anonymous 2006c). The dichlorvos strips were taped to the inside of the universal trap above the soapy water. Three replications were set up about 25 m apart within a 0.7 ha sweetpotato field at the USVL. Each replication had three trap locations that were 8 m apart. Three times per week, traps were monitored, emptied of trapped weevils, and rotated to a new within-replication trap location. This experiment was conducted for 7 weeks (28 July to 18 September 2006), during which time traps were monitored for sweetpotato weevils 21 times. New pheromone lures and dichlorvos strips were put in the traps on 28 July, 9 August, and 21 August.

In another experiment, the PET-bottle trap was tested with and without a dichlorvos strip, but soapy water was used in both treatments. The dichlorvos strips were taped to the inside of the PET-bottle trap above the soapy water. This experiment was conducted in four sweetpotato fields (four replications) for three months (28 July to 26 October 2006). Traps were monitored for sweetpotato weevils 11 times at weekly intervals. New pheromone lures and dichlorvos strips were put in the traps on 28 July, 19 August, and 21 September.

Data analyses. Data sets were analyzed by Bartlett's test for homogeneity of variance (Steel & Torrie 1960) using HOVTEST in PROC GLM (SAS Institute 1999). Where appropriate, data were transformed by $\log_{10}(x + 1.0)$ to correct for nonnormal distributions. Transformed data were subjected to repeated measures analysis using PROC MIXED with an autoregressive covariance structure, which effectively handles missing data points (SAS Institute 1999, Littell et al. 1996, 1998). The repeated measures were dates that traps were checked. Treatment means were separated by Tukey's ω -procedure ($P < 0.05$) (Steel & Torrie 1960). Untransformed data are presented in the tables.

Results

USVL survey traps 2001–2006. Sweetpotato weevil trap captures varied widely among collection dates (Fig. 1). Few individuals were captured until about June of each year. Trap captures typically peaked in mid-September through mid-October each year, and populations remained high until sustained cold winter weather. Because of continued warm weather, weevils were collected well into the new year of 2002, 2003, and 2005.

Evaluation of different trap types, 2001–2002. The 2001 experiment showed highly significant effects for trap type ($F = 15.88$, $df = 6, 301$, $P < 0.0001$); thus, treatment means were separated by Tukey's ω -procedure ($P < 0.05$) (Table 1). Analysis also revealed that there were significant effects of sampling date ($F = 1.51$, $df = 42, 301$, $P = 0.029$), but trap type by sampling date interactions were not significant ($F = 0.39$, $df = 252, 301$, $P = 1.000$). In 2001, the water-pan trap, funnel trap, and plastic-pail trap worked significantly better than the other trap designs for collecting sweetpotato weevils (Table 1). The CRW kairomone trap and the Japanese beetle trap performed poorly in 2001, and they were dropped from the study for 2002.

The data showed highly significant effects for trap type in 2002 ($F = 29.37$, $df = 7, 352$, $P < 0.0001$). Because of significant treatment effects, means were separated by Tukey's ω -procedure ($P < 0.05$) (Table 1). Analysis also revealed that

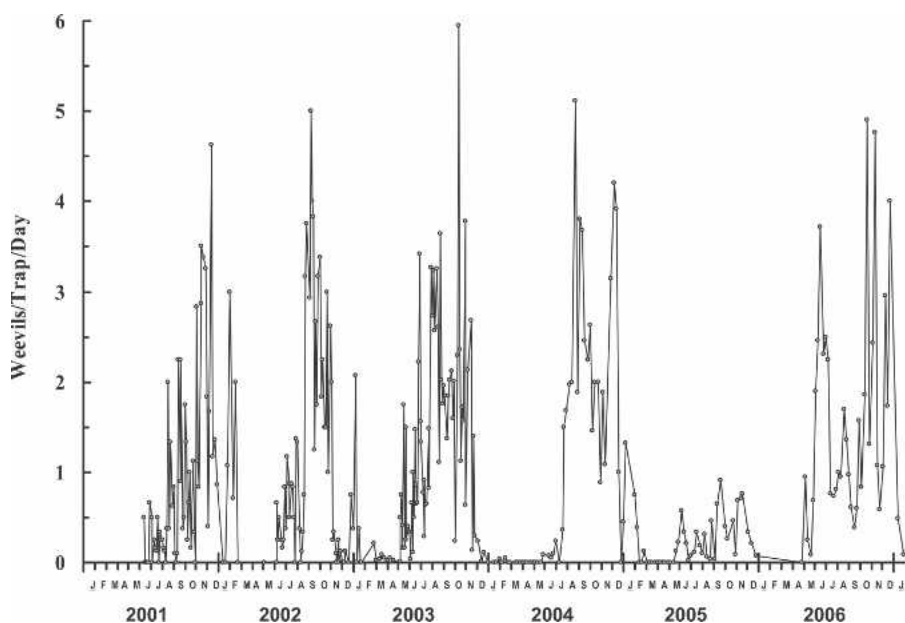


Fig. 1. Average number of adult male sweetpotato weevils captured per day in pheromone-baited universal traps over six seasons at Charleston, South Carolina, 2001–2006.

there were significant effects of sampling date ($F = 9.15$, $df = 43$, 352 , $P < 0.0001$), but trap type by sampling date interactions were not significant ($F = 1.11$, $df = 301$, 352 , $P = 0.181$).

In 2002, the water-pan trap, funnel trap, sticky trap, and universal trap worked significantly better than the other trap designs (Table 1). The Multipher trap was more effective than the milk-jug trap, plastic-pail trap, and boll weevil trap, which were the least effective traps in 2002. Interestingly, the plastic-pail trap did not work as well in 2002 as it did in 2001.

Evaluation of PET-bottle trap, 2005–2006. For the evaluation of the PET-bottle trap, repeated measures analysis showed highly significant effects for trap type in 2005 ($F = 15.95$, $df = 1$, 101 , $P < 0.0001$) and 2006 ($F = 10.11$, $df = 1$, 192 , $P = 0.0017$). Each year, the universal trap captured significantly more male sweetpotato weevils than did the PET-bottle trap (Table 2). This analysis also revealed that there were significant effects of sampling date (2005, $F = 9.15$, $df = 44$, 101 , $P < 0.0001$; 2006, $F = 2.98$, $df = 19$, 193 , $P < 0.0001$), but trap type by sampling date interactions were not significant (2005, $F = 1.18$, $df = 44$, 101 , $P = 0.243$; 2006, $F = 0.45$, $df = 19$, 192 , $P = 0.9783$).

Evaluation of killing agents, 2006. There were no significant differences in the average numbers of male sweetpotato weevils captured in universal traps containing different killing agents ($F = 2.49$, $df = 2$, 4 , $P = 0.198$). Although the traps with a dichlorvos strip alone tended to capture higher numbers of weevils (3.07 ± 0.37 [$\pm$ SE] weevils/trap/day) than the traps with soapy water (2.16 ± 0.26

Table 1. Mean number ($\pm$ SE) of adult male sweetpotato weevils captured in 10 types of pheromone-baited traps at Charleston, SC, 2001 and 2002.^a

Trap type	Mean number of weevils per trap per day ($\pm$ SE)		Total number of weevils captured	
	2001 ^b	2002 ^b	2001	2002
Water-pan trap	3.52 $\pm$ 0.70 a	2.28 $\pm$ 0.34 a	1045	716
Funnel trap	3.21 $\pm$ 0.53 a	2.09 $\pm$ 0.34 ab	963	658
Sticky trap	— ^c	1.86 $\pm$ 0.22 ab	—	614
Plastic-pail trap	2.29 $\pm$ 0.42 a	0.56 $\pm$ 0.11 d	712	183
Universal trap	0.95 $\pm$ 0.18 b	1.45 $\pm$ 0.17 b	288	474
Multipher 3 trap	— ^c	.89 $\pm$ 0.11 c	—	288
Milk-jug trap	0.66 $\pm$ 0.16 bc	0.54 $\pm$ 0.13 d	213	175
Boll weevil trap	— ^c	49 $\pm$ 0.08 d	—	155
CRW kairomone trap	0.32 $\pm$ 0.11 bc	— ^c	89	—
Japanese beetle trap	0.17 $\pm$ 0.04 c	— ^c	50	—

^aFor each year, means followed by a common letter are not significantly different as determined by Tukey's ω -procedure ($P < 0.05$) (Steel & Torrie 1960).

^bIn 2001, traps were monitored 43 times over 22 weeks (29 June–29 November), and in 2002, traps were monitored 44 times over 23 weeks (7 June–13 November); thus each mean was calculated from 86 data points in 2001 and 88 data points in 2002.

^cTrap not evaluated this year.

weevils/trap/day) or the soapy water-dichlorvos combination (2.48 $\pm$ 0.33 weevils/trap/day), these results were not significantly different.

The addition of a dichlorvos strip also did not affect captures of male sweetpotato weevils in the PET-bottle traps ($F = 0.24$, $df = 1, 63$, $P = 0.625$). A total of 328 male weevils were trapped in the PET-bottle traps without insecticide (1.06

Table 2. Mean number ($\pm$ SE) of adult male sweetpotato weevils captured in two types of pheromone-baited traps at Charleston, South Carolina, 2005 and 2006.

Trap type	Mean number of weevils per trap per day ($\pm$ SE)		Total number of weevils captured	
	2005 ^a	2006 ^a	2005	2006
Universal moth trap	0.51 $\pm$ 0.05** ^b	1.47 $\pm$ 0.25** ^b	293	1225
PET-bottle trap	0.15 $\pm$ 0.12	0.86 $\pm$ 0.12	84	716

^aIn 2005, 2 replications of traps were monitored 45 times over the entire year (6 January–23 December), and in 2006, 6 replications of traps were monitored weekly for 20 weeks (13 July–22 November).

^bSignificant treatment differences ($P < 0.01$) as determined by analysis of variance with repeated measures.

± 0.24 [$\pm$ SE] weevils/trap/day) and 265 male weevils were trapped in PET-bottles with the dichlorvos strips (0.85 ± 0.20 weevils/trap/day).

Discussion

Several of the trap designs evaluated in this study were effective and could be used in the management of sweetpotato weevils. However, when choosing a pheromone trap, its primary purpose must be considered first. Traps for mass trapping, eradication, or autodissemination are likely to have different designs than those for population monitoring (Sugiyama et al. 1996, Yasuda 1999, Komi 2000, Smit et al. 2001). Also, a trap for quarantine detection (low population levels) might have a different design than a trap used to determine spray thresholds in production fields. The trap user must consider effectiveness, convenience, and cost. Certain traps, such as the Japanese beetle trap, CRW kairomone trap, and Multiplier 3 trap, simply were not effective enough for trapping sweetpotato weevils to merit use. Other traps were effective, but inconvenient. For example, sticky traps are messy and the sticky panel must be replaced often as it quickly becomes covered with weevils, nontarget organisms, or soil and other debris (Talekar & Lee 1989, Smit et al. 1997). Some traps may be effective but relatively expensive to purchase (universal trap) or construct (funnel trap), especially for sweetpotato growers in developing countries (Jansson et al. 1991, Smit et al. 1997, Hwang 2000). Even in the United States, the cost of traps is not insignificant, as some states use large numbers of them in their monitoring programs. For example, in 2004 NCDA used 10,438 boll weevil traps to monitor 17,983 ha of sweetpotatoes in 4,954 fields (Anonymous 2006a).

No insect trap is perfect; each has its deficiencies. For example, the southern black widow spider, *Latrodectus mactans* (F.) (Araneae: Theridiidae) (Garb et al. 2004), was a frequent problem in funnel traps as the spiders often placed webbing over the entrance of the plastic funnel, thus preventing weevils from entering the collection bottle. Sugiyama et al. (1996) reported other disadvantages with the funnel trap were that it is expensive to construct, it catches many nontarget arthropods, it is bulky and hard to use, and it is sometimes dislodged by the wind.

The water-pan trap had the disadvantage of overflowing during heavy rain events, as the small cover allowed enough rain to enter and wash out trapped weevils. Conversely, during sunny periods, evaporation of water from the reservoir could be a problem; therefore, routine maintenance of water-pan traps is needed. These problems also affected the milk-jug trap, plastic-pail trap, and PET-bottle trap that had exposed water reservoirs. Another problem with these traps was the growth of mold and bacteria in the soapy water, which occasionally made accurate counting of weevils difficult. This was especially problematic when large numbers of nontarget organisms contaminated the water and when traps were monitored less frequently. The problem could be partially mitigated by the addition of a few drops of 6% sodium hypochlorite (Clorox Germicidal Bleach, Clorox Professional Products Company, Oakland, California). However, we observed that excessive sodium hypochlorite negatively affects trap captures, so researchers should use this product with caution.

Although the PET-bottle trap caught fewer total sweetpotato weevils than the universal trap over the course of the experiments in 2005 and 2006, there were several dates in which the PET-bottle trap caught more insects. A closer inspec-

tion of the data indicates that on certain dates the PET-bottle trap worked poorly compared with the universal trap. These poor trapping dates correspond to periods of high rainfall when weevils were apparently washed out of the uncovered PET-bottle traps. Interestingly, the PET double-funnel bottle-trap illustrated by Hwang et al. (1989) also was uncovered, but the authors did not report problems caused by excessive rain. We now construct our PET-bottle traps with small drain holes on the side to allow rain to drain without washing out the insects. Unfortunately, heavy downpours still can overflow the traps, so we are exploring ways of providing inexpensive covers for the traps.

Interestingly, some of the sweetpotato weevil traps commonly used are less effective than the water-pan trap and funnel trap. For example, the milk jug trap that is recommended for use in the Caribbean (Lawrence 1999) was relatively ineffective in our study. The larger, plastic-pail trap, first observed in Antigua, worked somewhat better in our study in 2001, but it was no better than the milk-jug trap in 2002. Because the milk-jug trap and other traps made from recycled containers are economical and easy to produce, their use in developing countries is understandable.

As previously reported (Jansson et al. 1989, 1992) and as shown in the present study (Table 1), the boll weevil trap is less effective than other trap designs for capturing adult male sweetpotato weevils. Nevertheless, commercial boll weevil traps are used in sweetpotato weevil quarantine programs in several states (Anonymous 2006b, Hammond et al. 1989). Their general availability, low cost, and ease of maintenance make them attractive for large-scale employment (Anonymous 2006a). Because there is no soapy water to replenish, traps can be left for long periods of time between monitoring without loss of data. The boll weevil trap has worked sufficiently well to detect sweetpotato weevil infestations into weevil-free areas of several states, and regulators are familiar with how to effectively use them in their monitoring programs (Anonymous 2006a, 2006b).

After being attracted by the pheromone lure, but before being captured, a sweetpotato weevil may leave the area and not be trapped. Also, weevils may escape from a trap, especially if no killing agent is used (Proshold et al. 1986, Jansson et al. 1992). This problem was not specifically addressed in this study, but it could potentially be a problem for certain trap designs like the boll weevil and Japanese beetle traps that did not have killing agents. Our study showed that the type of killing agent was not important, as soapy water worked as well as the insecticide pest strip in the universal and PET-bottle traps. Also, a combination of both soapy water and insecticide pest strip did not increase trap captures in universal traps. Hwang (2000) also reported that dichlorvos-impregnated strips did not affect capture of weevils in double-funnel PET-bottle traps with soapy water. These data indicate that once a weevil enters a trap, it is equally likely to be killed by falling into the soapy water as it is from dying from the insecticide. Researchers, extension personnel, and growers can use either technique, depending on which method best suits their circumstances.

The population levels of sweetpotato weevils at the USVL (maximum of 6 insects per trap day, Fig. 1) were much lower than reported for some tropical locations, where trap counts in the thousands of weevils per trap per day have been reported (Jansson et al. 1989, Talekar & Lee 1989). Jansson et al. (1991) reported that 20 traps in Florida averaged more than 2500 weevils per plastic-funnel trap per night during a 4-night period, and as many as 10,277 weevils have

been caught in one trap in one night. Even greater numbers have been reported outside the United States (Jansson et al. 1991). We suspect that pheromone traps perform differently at high population densities than they do at low population densities. We observed highly fluctuating trap counts from one sample period to the next (Fig. 1), which has also been reported in other studies (Jansson et al. 1989). It is not uncommon to have large fluctuations in trap counts from one sample (weekly or biweekly) to another (Proshold et al. 1986, Jansson et al. 1989). Jansson et al. (1989) attributed these fluctuations to environmental conditions, such as temperature, wind speed and direction, and rainfall. However, no significant correlations could be found between local temperature data and trap counts in our study.

Although pheromone traps may be effective in capturing male sweetpotato weevils, traps alone are usually not effective at controlling weevil populations (Talekar 1988, 1991). Current comprehensive IPM programs for the sweetpotato weevil recommend a combination of several cultural, biological, and chemical control practices, including the use of pheromone traps (Talekar 1988, 1991, Talekar & Lee 1989, Chalfant et al. 1990, Palaniswamy et al. 1992, Alvarez et al. 1996, Smit et al. 1997, Alcázar et al. 1997, Hwang 2000, Lagnaoui et al. 2000, Jackson et al. 2002). The best trap to integrate into a particular IPM program is a decision that must be based on a balance of trap efficacy, cost, availability, ease of use, and the economic situation of the country where it will be used. Thus, it is likely that some of the traps tested in this study may find use in sweetpotato IPM programs worldwide.

A sweetpotato weevil trap kit that is marketed commercially (Great Lakes IPM, Vestaburg, Michigan) uses the yellow and white universal trap. At the USVL, we also have chosen the universal trap as our standard for surveying sweetpotato weevils. Although this trap caught fewer sweetpotato weevils than some other trap designs in this study, it has several advantages that make it appropriate for our application. For instance, the universal trap can be used in buildings, greenhouses, and other locations not suitable for the sticky, funnel, or water-pan traps. Also, trap catches were less variable than for other traps as indicated by low standard error of the mean (Table 1) than some of the other designs. Data from this study are useful for establishing the efficacy of various trap designs for capturing male sweetpotato weevils, however, choice of a trap for a particular application must also take into account other considerations such as maintenance, cost, availability, and utility.

Acknowledgments

This article reports the results of research only. Mention of a proprietary product does not constitute endorsement or recommendation for its use by USDA ARS. Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture. The technical assistance of Jennifer Cook, Louise Cauthen, John Fender, Horton Howard, and Jeremiah Berry are greatly appreciated.

References Cited

- Alcázar, J., F. Cisneros & A. Morales. 1997. Large-scale implementation of IPM for sweetpotato weevil in Cuba: A collaborative effort, pp. 185–190 *In* Program Report 1995–1996, International Potato Center, Lima, Peru, 323 pp.

- Alvarez, P., P. Escarraman, E. Gomez, A. Villar, R. Jimenez, O. Ortiz, J. Alcázar & M. Palacios. 1996.** Economic impact of managing sweetpotato weevil (*Cylas formicarius*) with sex pheromones in the Dominican Republic, pp. 83-94. In T. Walker & C. Crissman [Eds.], Case Studies of the Economic Impact of CIP Related Technology. International Potato Center, Lima, Peru. Available at <http://www.cipotato.org/Market/ImpactCS/ipmdr.htm>, accessed 21 November 2006.
- Anderson, R. S. & D. G. Kissinger. 2002.** Family 129, Brentidae Billberg 1820, pp. 711-719. In Arnett, R. H., Jr., M. C. Thomas, P. E. Skelley, & J. H. Frank [Eds.], American Beetles, Volume 2. CRC Press, Boca Raton, Florida.
- Anonymous. 2001.** Use of plastic bottles for pheromone traps. Food & Fertilizer Technology Center, Taipei, Taiwan. Available at <http://www.agnet.org/library/article/pt2001033.html>, accessed 21 November 2006.
- Anonymous. 2002.** Survey method listing: Survey method 00038 'Trap; Hardee for weevils'. Nat. Agric. Pest Info. System (NAPIS), U. S. Dept. Agric., Anim. Plant Health Inspect. Serv., Plant Pest Quarant. Div. Available at <http://ddis.ifas.ufl.edu/napis/survey-method-ref.htm>, accessed 27 September 2006.
- Anonymous. 2006a.** Sweetpotato weevil program. N. C. Dept. Agric. Consumer Serv., Plant Industry Div., Plant Protect. Section, Entomological Serv. Available at <http://www.ncagr.com/plantind/plant/entomol/SPW.htm>, accessed November 7, 2006.
- Anonymous. 2006b.** *Cylas formicarius elegantulus*, sweetpotato weevil. Pest Tracker; Nat. Agric. Pest Info. System (NAPIS), U. S. Depart. Agric., Animal Plant Health Inspect. Serv., Plant Protect. Quarant. Div., Coop. Agric. Pest Surv. Available at <http://ceris.purdue.edu/napis/pests/spw/index.html>, accessed 11 October 2006.
- Anonymous. 2006c.** Hercon® Vaportape™ Insecticidal strips, Product No. 100311. Available at http://www.herconenviro.com/pdf/hercon_vaportape.pdf, accessed 21 November 2006.
- Baxendale, F. P., S. T. Kamble & M. D. Ellis. 2002.** Stinging wasps and bees. NebGuide G1447, Univ. Nebraska, Lincoln Exten. Inst. Agric. Natural Resources. Available at <http://www.ianrpubs.unl.edu/epublic/live/g1447/build/g1447.pdf>, accessed 21 November 2006.
- Braun, A. R. & E. van de Fliert. 1999.** Evaluation of the impact of sweetpotato weevil (*Cylas formicarius*) and of the effectiveness of *Cylas* sex pheromone traps at the farm level in Indonesia. Int. J. Pest Manage. 45: 101-110.
- Chalfant, R. B., R. K. Jansson, D. R. Seal & J. M. Schalk. 1990.** Ecology and management of sweetpotato insects. Annu. Rev. Entomol. 35: 157-180.
- Chiranjeevi, C. & D. D. R. Reddy. 2003.** Evaluation of an integrated pest management practice for sweetpotato weevil *Cylas formicarius* (F) in Andhra Pradesh. J. Res. ANGRAU 31: 22-28.
- CIP (Centro Internacional de la Papa). 2004.** The CIP Vision: Preserving the core, stimulating progress. International Potato Center (CIP), Lima, Peru. Available at <http://www.cipotato.org/projects/Documents/CIPVISION04.pdf>, accessed 21 November 2006.
- Coffelt, J. A., K. W. Vick, L. L. Sower & W. T. McClellan. 1978.** Sex pheromone of the sweetpotato weevil, *Cylas formicarius elegantulus*: Laboratory bioassay and evidence for a multiple component system. Environ. Entomol. 7: 756-758.
- Cuthbert Jr., F. P. 1967.** Insects affecting sweetpotatoes. USDA Agric. Handb. No. 329, 28 pp.
- Dickerson, W. A., G. H. McKibben, E. P. Lloyd, J. F. Kearney, J. J. Lam Jr. & W. H. Cross. 1981.** Field evaluation of a modified in-field boll weevil trap. J. Econ. Entomol. 74: 280-282.
- Downham, M. C. A., N. E. J. M. Smit, P. O. Laboke, D. R. Hall, D. I. Farman, A. Braun & B. Odongo. 1999.** Specificity of response to sex pheromones among the sweetpotato weevils, *Cylas puncticollis* Boheman and *C. brunneus* F. (Coleoptera: Apionidae). J. Chem. Ecol. 25: 591-609.

- Garb, J. E., A. González & R. G. Gillespie. 2004.** The black widow spider genus *Latrodectus* (Araneae: Theridiidae): Phylogeny, biogeography, and invasion history. *Mol. Phylogenet. Evol.* 31: 1127–1142.
- Hammond, A. M., T. N. Hardy, S. J. Toth, Jr., L. H. Rolston, E. L. Freeman, L. A. Hampton, W. A. Hogan, Jr. & J. L. Bagent. 1989.** Monitoring sweetpotato weevils with sex pheromone-baited traps. *La. Agric. Exp. Stn., Baton Rouge, La. The Station* 32: 12–13, 18.
- Hao, N. C., N. C. T. H. Giang, N. C. Khoa & N. Thanh Son. 1996.** Synthesis and application of insect attractants in Vietnam. *Resourc. Conserv. Recycling* 18: 59–68.
- Heath, R. R., J. A. Coffelt, P. E. Sonnet, F. I. Proshold, D. Dueben & J. H. Tumlinson. 1986.** Identification of sex pheromone produced by female sweetpotato weevil, *Cylas formicarius elegantulus* (Summers). *J. Chem. Ecol.* 12: 1489–1503.
- Heath, R. R., J. A. Coffelt, F. I. Proshold, R. K. Jansson & P. E. Sonnet. 1991.** Sex pheromone of *Cylas formicarius*: History and implications of chemistry in weevil management, pp. 79–96. *In* R. K. Jansson & K. V. Raman [Eds.], *Sweet Potato Pest Management: A Global Perspective*. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Hwang, J. S. 2000.** Integrated control of sweetpotato weevil, *Cylas formicarius* Fabricius, with sex pheromone and insecticide, pp. 25–43. *In* C. Chien-The [Ed.], *Control of Weevils in Sweet Potato Production*. Proceed. 12th Symp. Internat. Soc. Trop. Root Crops, 11–15 September 2000, Tsukuba, Ibaraki, Japan.
- Hwang, J. S. & C. C. Hung. 1991.** Evaluation of the effect of integrated control of sweetpotato weevil, *Cylas formicarius* Fabricius, with sex pheromone and insecticide. *Chinese J. Entomol.* 11: 140–146.
- Hwang, J. S., C. C. Hung & Y. P. Yen. 1989.** Evaluation of formulations of synthetic sex pheromone and trap designs for trapping sweetpotato weevil, *Cylas formicarius*. *Chinese J. Entomol.* 9: 37–43.
- Islam, M. N., M. A. Karim & Z. Neesa. 1992.** Field response of the male sweetpotato weevil, *Cylas formicarius* F. (Coleoptera: Curculionidae) to the synthetic pheromone in a newly developed water pan trap. *Bangladesh J. Zool.* 20: 247–256.
- Jackai, L. E. N., B. Sosinski, D. M. Jackson, K. A. Sorenson, C. K. Bonsi, A. Addo-Bediako, R. Ali, B. Tameru, F. Quarcoo & M. N. Alvarez. 2006.** Occurrence and intra-specific variation of sweetpotato weevil (Brentidae: Coleoptera) in relation to its potential spread in southern United States of America and the Caribbean, pp. 197–203. *In* S. L. Tan [Ed.], *Proceed. Second Internat. Symp. Sweetpotato & Cassava: Innovative Technologies for Commercialization*, 14–17 June 2005, Kuala Lumpur, Malaysia. *Internat. Soc. Hortic. Sci., Acta Horticulturae* No. 703, 267 pp.
- Jackson, D. M. 2003.** Trap designs for monitoring sweetpotato weevils. Poster presented at IPM-CRSP Caribbean Site, On-Site Planning Meeting, 24–28 February 2003, Kingston, Jamaica. Available at <http://www.ag.vt.edu/mrich/ipmcrsp/presentations/WeevilTrapping2002.pdf>, accessed 26 September 2006.
- Jackson, D. M. & J. R. Bohac. 2006.** Improved dry-fleshed sweetpotato genotypes resistant to insect pests. *J. Econ. Entomol.* 99: 1877–1883.
- Jackson, D. M., J. R. Bohac, K. M. Dalip, J. Lawrence, D. Clarke Harris, L. McComie, J. Gore, D. McGlashan, P. Chung, S. Edwards, S. Tolin & C. Edwards. 2002.** Integrated pest management of sweetpotato in the Caribbean, pp. 143–154. *In* T. Ames [Ed.], *Proceed. First Internat. Conf. Sweetpotato: Food and Health for the Future*, 26–30 November 2001, Lima, Peru. *Internat. Soc. Hortic. Sci., Acta Horticulturae* No. 583, 244 pp.
- Jansson, R. K., H. Bryan & K. A. Sorensen. 1987.** Within vine distribution and damage of sweetpotato weevil, *Cylas formicarius elegantulus* (Coleoptera: Curculionidae), on four cultivars of sweetpotato in southern Florida. *Fla. Entomol.* 70: 523–526.
- Jansson, R. K., R. R. Heath & J. A. Coffelt. 1989.** Temporal and spatial patterns of

- sweetpotato weevil (Coleoptera: Curculionidae) counts in pheromone-baited traps in white-fleshed sweetpotato fields in southern Florida. *Environ. Entomol.* 18: 691–697.
- Jansson, R. K., A. G. B. Hunsberger, S. H. Lecrone & S. K. O'Hair. 1990a.** Seasonal abundance, population growth, and within plant distribution of sweetpotato weevil (Coleoptera: Curculionidae) on sweetpotato in southern Florida. *Environ. Entomol.* 19: 313–321.
- Jansson, R. K., F. I. Proshold, L. J. Mason, R. R. Heath & S. H. Lecrone. 1990b.** Monitoring sweetpotato weevil (Coleoptera: Curculionidae) with sex pheromone: Effects of dosage and age of septa. *Trop. Pest Manage.* 36: 263–269.
- Jansson, R. K., L. J. Maon & R. R. Heath. 1991.** Use of sex pheromone for monitoring and managing *Cylas formicarius*, pp. 97–138. *In* R. K. Jansson & K. V. Raman [Eds.], *Sweet Potato Pest Management: A Global Perspective*. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Jansson, R. K., L. J. Mason, R. R. Heath, K. A. Sorensen, A. M. Hammond & J. V. Robinson. 1992.** Pheromone-trap monitoring system for sweetpotato weevil (Coleoptera: Apionidae) in the southern United States: Effects of trap type and pheromone dose. *J. Econ. Entomol.* 85: 416–423.
- Komi, K. 2000.** Eradication of sweetpotato weevil, *Cylas formicarius* Fabricius from Muroto City, Kochi, Japan, pp. 14–24. *In* C. Chien-The [Ed.], *Control of Weevils in Sweet Potato Production*. Proceed. 12th Symp. Internat. Soc. Trop. Root Crops, 11–15 September 2000, Tsukuba, Ibaraki, Japan.
- LaBonte, D. R. & M. J. Cannon. 1998.** Production and utilization of sweetpotato in the United States, pp. 29–32. *In* D. R. LaBonte, M. Yamashita, & H. Mochida [Eds.], *Sweetpotato Production Systems Toward the 21st Century*. Proceed. Internat. Workshop, 9–10 December 1997, Miyakonojo, Miyazaki, Japan. Kyushu National Agric. Exp. Stn., 357 pp.
- Lawrence, J. 1999.** Integrated pest management of the sweetpotato weevil. Factsheet, Caribbean Agric. Info. Serv., 4 pp.
- Lawrence, J. & P. Meyers. 1999.** Construction and use of pheromone traps in sweetpotato production. Factsheet, Caribbean Agric. Info. Serv., 4 pp.
- Lingren, B. W. 2000.** Kairomone insect trap for capture of corn rootworm. U. S. Patent No. 6,018,905, 1 February 2000.
- Lingren, B. W. 2002.** Kairomone and kill bait containing insect trap. U. S. Patent No. 6,393,760, 28 May 2002.
- Littell, R. C., G. A. Milliken, W. W. Stroup & R. D. Wolfinger. 1996.** SAS system for mixed models. SAS Institute, Cary, North Carolina.
- Littell, R. C., P. R. Henry & C. B. Ammerman. 1998.** Statistical analysis of repeated measures data using SAS procedures. *J. Anim. Sci.* 76: 1216–1231.
- Lo, C.-C., M.-D. Hung & C.-J. Liu. 1992.** Replacement of carcinogenic alkylating agent ethylene oxide in the synthesis of (Z)-3-dodecen-1-yl (E)-2-butenate, sex pheromone of sweetpotato weevil *Cylas formicarius elegantulus* (Summers) and *C. formicarius formicarius* (F.). *J. Chem. Ecol.* 18: 95–103.
- Marcano, A. J. J., P. N. Hernandez & E. D. Medina. 1994.** Evaluacion de una trampa para captura de adultos del picudo de la batata *Cylas formicarius elegantulus* Summers utilizando feromonas y la planta hospedera como atrayentes. *Agron. Trop.* 44: 217–232.
- Mason, L. J. & R. K. Jansson. 1991.** Disruption of sex pheromone communication in *Cylas formicarius* (Coleoptera: Apionidae) as a potential means of control. *Fla. Entomol.* 74: 469–472.
- Mason, L. J., R. K. Jansson & R. R. Heath. 1990.** Sampling range of male sweetpotato weevils (*Cylas formicarius elegantulus*) (Summers) (Coleoptera: Curculionidae) to pheromone traps: Influence of pheromone dosage and lure age. *J. Chem. Ecol.* 16: 2493–2502.
- Mithran, S. & A. S. Subbaraman. 1999.** Synthesis of (3Z)-dodecenyl-(E)-2-butenate, the pheromone of sweetpotato weevil. *Molecules* 4: 159–164.

- Mullen, M. A. 1984.** Influence of sweetpotato weevil infestation on the yields of twelve sweetpotato lines. *J. Agric. Entomol.* 1: 227–230.
- Nilakhe, S. S. 1991.** Quarantine programs for *Cylas formicarius* in the United States, pp. 247–262. *In* R. K. Jansson & K. V. Raman [Eds.], *Sweet Potato Pest Management: A Global Perspective*. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Palaniswamy, M. S., N. Mohandas & A. Visalakshi. 1992.** An integrated package for sweetpotato weevil (*Cylas formicarius* F.) management. *J. Root Crops* 18: 113–119.
- Pawar, A. S., S. Chattopadhyay & V. R. Mamadapur. 1993.** Two approaches for the synthesis of sweetpotato weevil and sugar beet moth pheromones. *Ind. J. Chem.* 32B: 463–464.
- Pillai, K. S., P. Rajamma & M. S. Palaniswami. 1993.** New technique in the control of sweetpotato weevil using synthetic sex pheromone in India. *Int. J. Pest Manage.* 39: 84–89.
- Proshold, F. I. 1983.** Mating activity and movement of *Cylas formicarius elegantulus* (Coleoptera: Curculionidae) on sweetpotato. *Proc. Trop. Reg., Am. Soc. Hort. Sci., Annu. Meet.* 27: 81–92.
- Proshold, F. I., J. L. Gonzalez, C. Asencio & R. R. Heath. 1986.** A trap for monitoring the sweetpotato weevil (Coleoptera: Curculionidae) using pheromone or live females as bait. *J. Econ. Entomol.* 79: 641–647.
- Ray, R. & V. Ravi. 2005.** Post harvest spoilage of sweetpotato in tropics and control measures. *Crit. Rev. Food Sci. Nutr.* 45: 623–644.
- Sakuratani, Y., T. Sugimoto, O. Setokuchi, T. Kamikado, K. Kiritani & T. Okada. 1994.** Diurnal changes in microhabitat usage and behavior of *Cylas formicarius* (Fab.) adults. *Appl. Entomol. Zool. (Jpn.)* 29: 307–315.
- SAS Institute, Inc. 1999.** SAS/STAT User's Guide, Version 8.02. SAS Institute, Inc., Cary, North Carolina.
- Setokuchi, O., Y. Nakamura & Y. Kubo. 1991.** A fiberboard formulation of the sweetpotato weevil, *Cylas formicarius* (Fabricius), sex pheromone and an insecticide for mass-trapping. *Japan J. Appl. Entomol. Zool.* 35: 251–253.
- Shimizu, T. & S. Moriya. 1996.** Flight time and flight age in the sweetpotato weevil, *Cylas formicarius* (Fabricius) (Coleoptera: Brentidae). *Appl. Entomol. Zool. (Jpn.)* 31: 575–580.
- Smit, N. E. J. M., M. C. A. Downham, B. Odongo, D. R. Hall & P. O. Laboke. 1997.** Development of pheromone traps for control and monitoring sweetpotato weevils, *Cylas puncticollis* (Bohe.) and *C. brunneus* (F.) in Uganda. *Entomol. Exp. Appl.* 85: 95–104.
- Smit, N. E. J. M., M. C. A. Downham, P. O. Laboke, D. R. Hall & B. Odongo. 2001.** Mass-trapping male *Cylas* spp. with sex pheromone: A potential IPM component in sweetpotato production in Uganda. *Crop Prot.* 20: 643–651.
- Steel, R. G. D. & J. H. Torrie. 1960.** Principles and procedures of statistics: With special reference to the biological sciences. McGraw-Hill Book Co., New York, New York, 481 pp.
- Sugimoto, T., Y. Sakuratani, O. Setokuchi, T. Kamikado, K. Kiritani & T. Okada. 1994a.** Estimations of the attractive area of pheromone traps and dispersal distance, of male adults of sweetpotato weevil, *Cylas formicarius* (Fabricius) (Coleoptera: Curculionidae). *Appl. Entomol. Zool. (Jpn.)* 29: 349–358.
- Sugimoto, T., Y. Sakuratani, O. Setokuchi, K. Kawazoe, K. Kiritani & T. Okada. 1994b.** Estimation of male adult population of sweetpotato weevils, *Cylas formicarius* (Fabricius) (Coleoptera, Curculionidae) on Kikai Island in Japan. *Appl. Entomol. Zool. (Jpn.)* 29: 359–367.
- Sugiyama, M., Y. Shimoji & T. Kohama. 1996.** Effectiveness of a newly designed sex pheromone trap for the sweetpotato weevil, *Cylas formicarius* (Fabricius) (Coleoptera: Brentidae). *Appl. Entomol. Zool. (Jpn.)* 31: 547–550.
- Sureda, T. & C. Quero. Ma. Pilar Bosch, R. Avilés, F. Coll, M. Renou & Á. Guerrero. 2006.** Electrophysiological and behavioral responses of a Cuban population of the sweetpotato weevil to its sex pheromone. *J. Chem. Ecol. (On-Line First)*. Available at <http://>

- www.springerlink.com/content/f5257v6642772747/fulltext.pdf, accessed 21 November 2006.
- Sutherland, J. A. 1986a.** A review of the biology and control of the sweetpotato weevil *Cylas formicarius* (Fab). Trop. Pest Manage. 32: 304–315, 360–361, 364.
- Sutherland, J. A. 1986b.** Damage by *Cylas formicarius* Fab. to sweetpotato vines and tubers, and the effect of infestations on total yield in Papua, New Guinea. Trop. Pest Manage. 32: 316–323.
- Talekar, N. S. 1988.** How to control sweetpotato weevil: A practical IPM approach. AVRDC Publ. 88–292, Taiwan, R.O.C., 6 pp.
- Talekar, N. S. 1991.** Integrated control of *Cylas formicarius*, pp. 139–156. In R. K. Jansson & K. V. Raman [eds.], Sweet Potato Pest Management: A Global Perspective. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Talekar, N. S. & S. T. Lee. 1989.** Studies on the utilization of a female sex pheromone for the management of sweetpotato weevil, *Cylas formicarius formicarius* (F.) (Coleoptera: Curculionidae). Bull. Inst. Zool. Acad. Sinica 28: 281–288.
- Teli, V. S. & G. N. Salunkhe. 1993.** Monitoring adults of sweetpotato weevil, *Cylas formicarius* Fab. with sex pheromone. J. Insect Sci. 6: 283–284.
- Tóth, M., L. Furlan, V. G. Yatsynin, I. Ujváry, I. Szarukán, Z. Imrei, M. Subchev, T. Tolasch & W. Francke. 2002.** Identification of sex pheromone composition of click beetle *Agriotes brevis* Candeze. J. Chem. Ecol. 28: 1641–1652.
- Trécé, Inc. 2006.** Cidetrak® Catalog; Trécé Inc., Adair, Oklahoma. Available at <http://www.trece.com/cidetrak.html>, accessed 21 November 2006.
- Wolfe, G. W. 1991.** The origin and dispersal of the pest species of *Cylas* with a key to the pest species groups of the world, pp. 13–43. In R. K. Jansson & K. V. Raman [Eds.], Sweet Potato Pest Management: A Global Perspective. Westview Press, Inc., Boulder, Colorado, 458 pp.
- Woolfe, J. A. 1992.** Sweetpotato: An untapped food resource. Cambridge University Press, Cambridge, England.
- Yasuda, K. 1995.** Mass trapping of the sweetpotato weevil *Cylas formicarius* (Fabricius) (Coleoptera: Brentidae) with a synthetic sex pheromone. Appl. Entomol. Zool. (Jpn.) 30: 31–36.
- Yasuda, K. 1999.** Auto-infection system for the sweetpotato weevil, *Cylas formicarius* (Fabricius) (Coleoptera: Curculionidae) with entomopathogenic fungi, *Beauveria bassiana*, using a modified sex pheromone trap in the field. Appl. Entomol. Zool. (Jpn.) 34: 501–506.
- Yasuda, K., T. Hongo & N. Etoo. 2004.** A new type of pheromone formulation with visual stimulation for control of sweetpotato weevil, *Cylas formicarius* (Fabricius). Japan. J. Appl. Entomol. Zool. 48: 87–94.
- Yasuda, K., H. Sugie & R. R. Heath. 1992.** Field evaluation of synthetic sex-attractant pheromone of the sweet-potato weevil, *Cylas formicarius* Fabricius (Coleoptera: Brentidae). Japan J. Appl. Entomol. Zool. 36: 81–87.
- Yen, Y. P. & J. S. Hwang. 1990.** Improved synthesis method and bioactivity of sex pheromone of sweetpotato weevil, *Cylas formicarius*. Plant Protect. Bull. (Taiwan) 32: 239–241.
-

Natural Occurrence of Entomopathogenic Nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) in South Carolina¹

Ramazan Canhilal and Gerald R. Carner²

Plant Protection Department, Seyrani Faculty of Agriculture, Erciyes University 38039 Melikgazi, Kayseri, Turkey

J. Agric. Urban Entomol. 23(3): 159–166 (July 2006)

ABSTRACT One-hundred thirty soil samples from five ecologically diverse habitats in 10 locations in South Carolina were assessed to detect natural populations of entomopathogenic nematodes using the *Galleria* baiting technique. Twenty-two soil samples (16.9%) in eight sites contained entomopathogenic nematodes. Sixteen samples (12.3%) were positive for *Heterorhabditis*. *Heterorhabditids* in 13 samples were identified as *H. bacteriophora*. Two sites produced *H. zealandica*, and one isolate was *H. megidis*. Six samples (4.6%) yielded *Steinernema*. Four of these samples were identified as *S. glaseri*, and the other two samples were *Steinernema* sp. Only one site yielded both *Heterorhabditis* and *Steinernema*. The soil texture of positive soils were sand, loamy sand, and sandy loam with strongly acidic (pH 4.3) to neutral (pH 7.0) and average of 3% organic content for *Heterorhabditis*-positive soil samples, and loamy sand and sandy loam with strongly (pH 4.3) to weak (pH 6.0) acidic and average of 4% organic content for *Steinernema*-positive soil samples

KEY WORDS Entomopathogenic nematodes, *Heterorhabditis*, *Steinernema*, natural occurrence, South Carolina

Entomopathogenic nematodes have been used in controlling insects since the 1930s (Smart 1995) in various climatic regions of the world. They are important biological control agents of soil-inhabiting insects (Gaugler 1981, Georgis and Poinar 1984, Klein 1990) such as Japanese beetles, mole crickets, and root weevils. They have also been used successfully against above-ground insects in cryptic habitats (Bedding & Miller 1981, Ralph 1981, Kaya 1988, Begley 1990, Kaya 1990, Vreditelyami et al. 1992), for example, navel orange worm, the codling moth and the artichoke plume moth, carpenter worms, and clearwing moths. These nematodes have many ideal attributes as biological control agents, such as a wide host range, rapid host mortality, ease of commercial production in vivo or in vitro, active host-seeking ability, long-term efficacy, easy application, compatibility with many chemical pesticides, environmental safety, and no registration requirement with the Environmental Protection Agency. Nematode species and

¹Accepted for publication 21 August 2007

²Department of Entomology, Clemson University, 113 Long Hall Box 340365 Clemson, South Carolina 29634-0365

strains are variable in pathogenicity, host-searching behavior, and survivability, making them suitable in biological control programs.

After Bedding & Akhurst (1975) developed the *Galleria*-bait method, a number of surveys have been conducted around the world (Hominick 2002). The habitat and soil characteristic where these nematodes have been isolated vary: pastures, forests, field crops, orchards, beaches, and in calcareous, sandy loam, loam, humus and sandy, and humus and organo-mineral soils (Akhurst & Brooks 1984, Hara et al. 1991, Hominick 2002)

There has been no published information on the natural distribution of entomopathogenic nematodes in South Carolina. Our objective was to conduct a survey for these nematodes to document their occurrence in different habitats and soil types in different locations and to obtain local isolates for possible use in biological control programs.

Materials and Methods

A total of 130 soil samples from five different habitats at 10 sites in South Carolina were collected at various times in 1999 and 2000 (Table 1 and Fig. 1). The habitats were forest, pasture, cropland (soybean/corn/cotton), vegetable, and orchard/vineyard. A field where *Heterorhabditis* sp. and *Steinernema* spp. had been applied against the squash vine borer in 1997 and 1998 (Canhilar & Carner 2006) also was sampled. At each site, five to eight random samples were taken in a 50-m² area at 1- to 3-m intervals to a depth of 10–15 cm with a hand shovel/shovel or soil corer with a volume of 1500–3000 mL. The shovel was surface sterilized between sampling sites. These samples were put in a plastic bag (43 cm × 46 cm × 20 µm) and mixed well. Approximately 500 mL of soil was taken and placed in a 591-mL plastic container with a cover containing nine 1-mm diameter holes. Associated vegetation, date, and site location were recorded on the container. The samples were kept in a cooler at 12–15°C during transportation.

Late-stage larvae of *Galleria mellonella* (L.), the greater wax moth, (New York Worms, New York, New York) were used to bioassay the soil samples within 24 h after sampling (Bedding & Akhurst 1975). Five to seven larvae were buried in each sample, and the containers were held at 25 ± 2°C. Checks for mortality were made 6–8 d after larvae were buried by removing the larvae from soil. Dead larvae from each soil sample, after rinsing in sterile distilled water, were placed on a modified White trap individually where infective juveniles (IJs) were collected (White 1929). The trap consisted of a folded 11-cm filter paper (3 mm in depth after folding) in a Petri dish (100 × 15 mm). The Petri dish was filled with 15–20 mL of distilled water and the dead larva was kept on the filter paper in the Petri dish 10–12 d to collect IJs. The IJs that migrated into water were exposed to 5–8 *G. mellonella* larvae at the rate of 20–25 IJs per larva on a filter paper in a Petri dish (100 × 15 mm) to verify pathogenicity and complete Koch's postulates. The pathogenic isolates were cultured in the laboratory (Kaya & Stock 1997).

The nematodes were identified morphologically by examining morphometrics for IJs and first-generation males. The key for the genera *Steinernema*, *Neosteinernema*, and *Heterorhabditis* by Nguyen & Smart (1996) was used for identification. Dr. Khuong Nguyen in the Entomology and Nematology Department, the University of Florida, verified the identifications.

Table 1. Location, habitat types, and percentage of sites in which nematodes were detected in 1999 and 2000, South Carolina.

Location ^a	Habitats (No. positive samples/no. samples tested)						Nematode test field	% of H ^b	% of S ^b	Total %
	Forest	Pasture	Orchard/ vineyard	Cropland	Vegetable					
C.Farm	1(S)/6	1(H)/5	1(H)/8	0/3	0/3	0/4	6.9	3.4	10.3	
Mus.E.St.	1(S)/3	1(H)/4	0/5	—	0/1		7.7	7.7	15.4	
WP.Stv.	0/2	1(H), 1(S)/2	—	—	—		25	25	50	
B.Garden	0/2	0/2	—	—	0/2		0	0	0	
Smp.E.St.	0/3	0/3	0/4	1(H)/3	0/3		6.3	0	6.3	
Eds.REC	1(H)/2	0/3	1(H)/3	0/3	0/1		16.7	0	16.7	
Cstl.REC	1(S)/5	2(S)/5	3(H)/5	0/3	0/4		13.6	13.6	27.2	
Sndh.REC	0/1	—	0/1	—	0/1		0	0	0	
Lexing.	0/3	0/3	1(H)/2	0/2	0/2		8.3	0	8.3	
P.D.REC	0/4	2(H)/3	1(H)/2	1(H)/2	1(H)/2		38.5	0	38.5	

^aC.Farm = Cherry Farm, Mus.E.St. = Musser Experimental Station, WP.Stv. = WestPoint Stevens Clenson Complex, B.Garden = Botanical Garden, Smp.E.St. = Simpson Experimental Station, Eds.REC = Edisto Research Education Center, Cstl.REC = Coastal Research Education Center, sndh.REC = Sandhill Research Education Center, Lexing. = Lexington, and P.D.REC = Pee Dee Research Education Center.

^bH = *Heterorhabditis* and S = *Steinernema*.

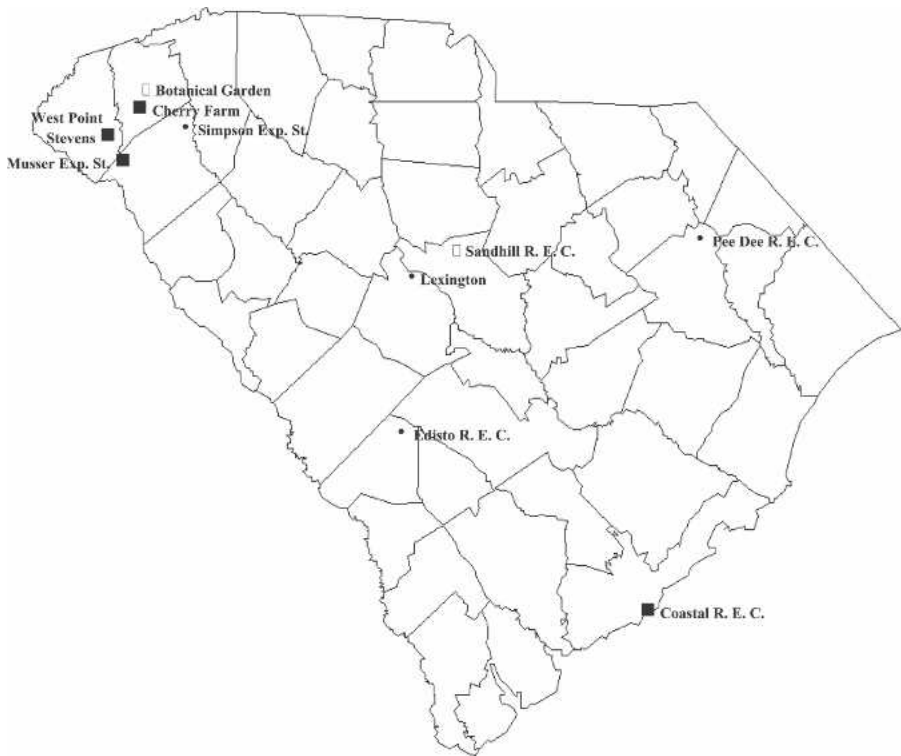


Fig. 1. Locations in South Carolina where soil samples were taken and distribution of naturally occurring nematodes. ■, *Heterorhabditis* and *Steinernema*; ●, *Heterorhabditis*; □, no entomopathogenic nematodes.

Differences in the prevalence of entomopathogenic nematodes in different habitats and locations were analyzed using the analysis of variance procedure (SAS Institute 2004). Positive soil samples for nematodes were analyzed for pH, electrical conductivity, and organic matter in the Agricultural Service Laboratory, and for soil texture in the Department of Soil Science, Clemson University.

Results and Discussion

Twenty-two samples of 130 (16.9%) from eight locations were positive for entomopathogenic nematodes (Fig. 1 and Table 1). Six isolates (4.6%) of *Steinernema* were obtained from pasture (two isolates) and forest at the Coastal Research Education Center (REC), from forest at the Musser Experiment Station (Exp. St.), Clemson University, from forest at the Cherry Farm, and from pasture at the WestPoint Stevens Clemson Complex (Table 1). Two isolates from pasture and one from forest in Coastal REC, and one from pasture in WestPoint Stevens Complex were identified as *Steinernema glaseri* (Stecken). The other two isolates of *Steinernema* were not identified to species level because the isolates failed rearing in *G. mellonella*.

Prevalence of *Heterorhabditis* was 12.3%. Thirteen isolates of *Heterorhabditis* were identified as *Heterorhabditis bacteriophora* Poinar from pasture and a red maple orchard at the Cherry Farm, pasture at the Musser Exp. St., forest at the Edisto REC, soybean field at the Simpson Exp. St., vegetable, pasture (two isolates), pecan orchard and corn field at the Pee Dee REC, Clemson University, pasture at the WestPoint Stevens Clemson Complex and two isolates from a vineyard at the Coastal REC, Clemson University. Two samples; from blueberry in Coastal REC and from a pecan orchard in Edisto REC, produced *H. zealandica* Wouts. One isolate from a peach orchard in Lexington was identified as *H. megidis* Poinar, Jackson & Klein. Four soil samples taken from the field that *Steinernema* spp. and *Heterorhabditis* sp. had been applied in previous years were negative for these nematodes. Only one sample (site) from WestPoint Stevens Clemson Complex yielded both *Heterorhabditis* and *Steinernema* (Table 1). Differences in the prevalence of nematodes in different locations were not statistically significant ($F = 1.24$; $DF = 9$; $P = 0.282$).

Nematodes were obtained in all habitats. Heterorhabditid nematodes were isolated more in the orchard/vineyard, with 23.3%, followed by pasture, cropland, vegetable, and forest with 16.7, 12.5, 5.3, and 3.2%, respectively. *Steinernema* was recovered from only pasture and forest, in 10 and 9.7% of the samples, respectively (Table 2). Differences in the prevalence of nematodes in different habitats were not statistically significant ($F = 1.48$; $DF = 5$; $P = 0.204$).

All *Steinernema* and *Heterorhabditis*-positive soil samples were sandy. Soil texture for *Heterorhabditis*-positive soil samples was sand, loamy sand, and sandy loam, and soil texture for *Steinernema*-positive soil samples was loamy sand and sandy loam (Table 3). Electrical conductivity averaged 0.19 dS/m for all *Heterorhabditis*-positive soil and 0.07 dS/m for *Steinernema*-positive soil samples (Table 3). Apparently, entomopathogenic nematodes seemed well adapted to different soil salinity as indicated in Thurston et al. (1994).

Soil pH ranged from 4.3 (strongly acidic) to 7.0 (neutral) for *Heterorhabditis*-positive soils and 4.3 (strongly acidic) to 6.0 (weakly acidic) for *Steinernema*-positive soil samples (Table 3). Kung et al. (1990) found that survival and pathogenicity of steinernematids was reduced only slightly as the tested soil pH decreased from pH 8 to pH 4, but survival and pathogenicity drastically decreased at pH 10. Organic matter of *Heterorhabditis*-positive soils averaged 3% (0.7–7.8), and it was 4% (2–7.2) for *Steinernema*-positive soils (Table 3).

Frequencies of entomopathogenic nematodes varied. We found heterorhabditids more than steinernematids in our survey (12.3% versus 4.6%). Hara et al. (1991) obtained 6.3% *Heterorhabditis* and 0.06% *Steinernema*. Soil samples positive for heterorhabditid species tended to be sandy, which is similar to the positive soil samples in our study.

Akhurst & Brooks (1984) isolated heterorhabditids (*H. bacteriophora*), in 16.9% of samples and steinernematids (*S. glaseri*, *S. feltiae* (Filipjev), and *S. sp.*), in 2.2% of samples in North Carolina. Our findings are similar to these results. However, in a survey conducted in Tennessee nursery soils, half of positive soil samples contained heterorhabditid nematodes. *H. bacteriophora* was the only heterorhabditid species occurring more in sandy loam soils. Only *S. carpocapsae* was obtained more in silt loam soil (Rueda et al. 1993). Hominick & Briscoe (1990) recovered *Steinernema* from 48.6% of 403 sites but only one site yielded *Heterorhabditis* in British soils. Whereas steinernematids occurred in 33 of 35 sites,

Table 2. Habitats, no. samples taken from each habitat, no. positive samples, % of habitats where *Heterorhabditis* and *Steinernema* detected, and total % in 1999 and 2000, South Carolina.

Habitats	No. positive sites/ no. sampling sites	No. and % of heterorhabditids	No. and % of steinernematids	Total %
Pasture	8/30	5–16.7	3–10.0	26.7
Orchard/ vineyard	7/30	7–23.3	0–0	23.3
Forest	4/31	1–3.2	3–9.7	12.9
Cropland	2/16	2–12.5	0–0	12.5
Vegetable	1/19	1–5.3	0–0	5.3
Nematode test field	0/4	0–0	0–0	0

heterorhabditid nematodes were isolated from only two sites in Spanish soils (Gracia & Palomo 1996). Although Glazer et al. (1991) stated that heterorhabditid nematodes were more common in warmer regions such as North Carolina, the Hawaiian Islands, and most parts of Australia (Akhurst & Brooks 1984, Akhurst & Bedding 1986, Hara et al. 1991), only steinernematids were recovered in cooler regions like Czechoslovakia (Mracek 1980), Sweden (Burman et al. 1986), and Northern Ireland (Blackshaw 1988); however, steinernematid occurrence in Tennessee soils (Rueda et al. 1993) and Spanish soils (Gracia & Palomo 1996) does not support that statement. These differences in the distribution of entomopathogenic nematodes are probably the result of the availability of susceptible hosts

Table 3. Mean ($\pm$ SE) of characterizations of soil containing *Heterorhabditis* spp. and *Steinernema* spp. and % of positive samples in 1999 and 2000, South Carolina.

Soil texture	% Positive samples	% Organic matter content	Electrical conductivity (dS/m)	pH
<i>Heterorhabditis</i> spp.				
Sand	23.8	1.1 $\pm$ 0.2	0.18 $\pm$ 0.04	5.5 $\pm$ 0.5
Loamy sand	23.8	3.0 $\pm$ 0.2	0.24 $\pm$ 0.04	4.9 $\pm$ 0.3
Sandy loam	23.8	4.8 $\pm$ 0.8	0.14 $\pm$ 0.10	5.5 $\pm$ 0.3
Overall mean $\pm$ SE		3.0 $\pm$ 0.5	0.19 $\pm$ 0.04	5.3 $\pm$ 0.2
<i>Steinernema</i> spp.				
Sand	—	—	—	—
Loamy sand	9.5	2.0 $\pm$ 0.0	0.09 $\pm$ 0.00	5.1 $\pm$ 0.0
Sandy loam	19.1	1.6 $\pm$ 0.8	0.05 $\pm$ 0.02	5.1 $\pm$ 0.4
Overall mean $\pm$ SE		4.0 $\pm$ 0.8	0.07 $\pm$ 0.02	5.1 $\pm$ 0.2

and environmental factors such as soil texture, soil moisture, and cultural practices.

The prevalence of nematodes in different habitats varies in different regions of the world. Mracek (1980) isolated nematodes more from forest than cropland in Czechoslovakia, whereas Gracia & Palomo (1996) found no difference for the prevalence of nematodes among cultivated fields, woodlands, or pastures in Spain. Akhurst & Brooks (1984) recovered more nematodes in cropland than forest, orchard/vineyard, or pasture. In contrast, we obtained more nematodes in pasture than in cropland (26.7% versus 12.5%, respectively).

Acknowledgment

We thank the Research and Education Centers of Clemson University for providing guidance of the centers during soil sampling and Dr. Khuong Nguyen of the University of Florida for verification of our identification.

References Cited

- Akhurst, R. J. & W. M. Brooks. 1984.** The distribution of entomophilic nematodes (Steinernematidae and Heterorhabditidae) in North Carolina. *J. Invertebr. Pathol.* 44: 140–145.
- Akhurst, R. & R. A. Bedding. 1986.** Natural occurrence of insect pathogenic nematodes (Steinernematidae and Heterorhabditidae) in soil in Australia. *J. Austr. Entomol. Soc.* 25: 241–244.
- Bedding, R. A. & R. J. Akhurst. 1975.** A simple technique for the detection of insect parasitic rhabditid nematodes in soil. *Nematologica* 21: 109–116.
- Bedding, R. A. & L. A. Miller. 1981.** Disinfesting black currant cuttings of *Synanthedon tipuliformis*, using the insect parasitic nematode, *Neoaplectana bibionis*. *J. Entomol. Sci.* 10: 449–453.
- Begley, J. W. 1990.** Efficacy against insects in habitats other than soil, pp. 215–231. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic nematodes in biological control*. CRC Press, Boca Raton, FL.
- Blackshaw, R. P. 1988.** A survey of insect parasitic nematodes in Northern Ireland. *Ann. Appl. Biol.* 113: 561–565.
- Burman, M., K. Abrahamsom, J. Ascard, A. Sjoeborg & B. Erikson. 1986.** Distribution of insect parasitic nematodes in Sweden. pp. 300–303. *In* R. A. Samson, J. M. Vlak, and D. Peters [Eds.], *Fundamentals and applied aspects of invertebrate pathology*, Wageningen, Netherlands.
- Canhilal, R. & G. R. Carner. 2006.** Efficacy of Entomopathogenic Nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) against the Squash Vine Borer, *Melittia cucurbitae* (Lepidoptera: Sesiidae) in South Carolina. *J. Agric. Urban Entomol.* 23: 1–6.
- Gaugler, R. 1981.** Biological control potential of neoaplectanid nematodes. *J. Nematol.* 13: 241–249.
- Georgis, R. & G. O. Poinar. 1984.** Field control of the strawberry root weevil, *Nemocestes incomptus*, by neoaplectanid nematodes (Steinernematidae: Nematoda). *J. Invertebr. Pathol.* 43: 103–131.
- Glazer, I., N. Liran & Y. Steinberg. 1991.** A survey of entomopathogenic nematodes (Rhabditida) in the Negev desert. *Phytoparasitica* 19: 291–300.
- Gracia, F. D. P. & A. Palomo. 1996.** Natural occurrence of entomopathogenic nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) in Spanish soils. *Invertebr. Pathol.* 68: 84–90.
- Hara, A. H., R. Gaugler, H. K. Kaya & L. M. Lebeck. 1991.** Natural populations of

- entomopathogenic nematodes (Rhabditida: Heterorhabditidae, Steinernematidae) from Hawaiian Islands. *Environ. Entomol.* 20: 211–216.
- Hominick, W. M. 2002.** Biogeography, pp. 100–115. *In* R. Gaugler [Ed.], *Entomopathogenic Nematology*. CABI Publications, New York, NY.
- Hominick, W. M. & B. R. Briscoe. 1990.** Occurrence of entomopathogenic nematodes (Rhabditida: Steinernematidae and Heterorhabditidae) in British soils. *Parasitology* 100: 295–302.
- Kaya, H. K. & S. P. Stock. 1997.** Techniques in insect nematology, pp. 282–324. *In* L. A. Lacey (Ed.), *Manual of Techniques in Insect Pathology*. Academic Press, London.
- Kaya, H. K. 1988.** Princes from todes. *Am. Nurseryman* 168: 63, 65–69.
- Kaya, H. K. 1990.** Soil ecology, pp. 93–115. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic Nematodes in Biological Control*. CRC, Boca Raton, FL.
- Klein, M. G. 1990.** Efficacy against soil-inhabiting insect pests, pp. 195–214. *In* R. Gaugler and H. K. Kaya [Eds.], *Entomopathogenic nematodes in biological control*. CRC, Boca Raton, FL.
- Kung, S. P., R. Gaugler & H. K. Kaya. 1990.** Influence of soil pH and oxygen on persistence of *Steinernema* spp. *J. Nematol.* 22: 440–445.
- Mracek, Z. 1980.** The use of “*Galleria* traps” for obtaining nematode parasites of insects in Czechoslovakia (Lepidoptera: Nematoda, Steinernematidae). *Acta. Entomol. Bohemos.* 77: 378–382.
- Nguyen, K. B. & G. C. Smart Jr. 1996.** Identification of entomopathogenic nematodes in the Steinernematidae and Heterorhabditidae (Nemata: Rhaditida). *J. Nematol.* 28: 286–300.
- Ralph, W. 1981.** Nematodes for insect control. *Rural Research* (No. 113): 11–14.
- Rueda, L. M., S. O. Osawaru, L. L. Georgi & R. E. Harrison. 1993.** Natural occurrence of entomopathogenic nematodes in Tennessee Nursery soils. *J. Nematol.* 25: 181–188.
- SAS Institute. 2004.** SAS/STAT software, release 9.0. SAS Institute, Cary, NC.
- Smart Jr., G. C. 1995.** Entomopathogenic nematodes for the biological control of insects. Supplement to *J. Nematol.* 27: 529–534.
- Thurston, G. S., Y. Ni & H. K. Kaya. 1994.** Influence of salinity on survival and infectivity of entomopathogenic nematodes. *J. Nematol.* 26: 345–351.
- Vreditelyami, S. V., V. P. C. Smorodiny & A. S. Zainalov. 1992.** Attempts to increase efficiency of control measures against the principle pests in black currant nurseries, pp. 17–20. Thesis Russian Academy of Agricultural Sciences, Moscow, Russia.
- White, G. F. 1929.** A method for obtaining infective nematode larvae from cultures. *Science* 66: 302–303.
-

The First Fifty Years of the South Carolina Entomological Society¹

Wm. Michael Hood

Department of Entomology, Soils, and Plant Sciences, 305 Long Hall, Clemson University,
Clemson, South Carolina 29634 USA

J. Agric. Urban Entomol. 23(3): 167–188 (July 2006)

ABSTRACT Historical information covering the first 50 years of the South Carolina Entomological Society is summarized in this article. The information was gathered from minutes of annual meetings, board of directors meetings, and from other sources in the historical files of the Society. The major event for the Society has been the annual meeting, which has included a varied program through the years. The Society has published annual meeting proceedings and newsletters infrequently through its 50-year history, and the desire to publish a professional research journal culminated in 1984 with the launching of the *Journal of Agricultural Entomology*, becoming the *Journal of Agricultural and Urban Entomology* in 1999. The Society has met jointly several times with other nearby state entomological societies, fostering scholarship and fellowship with other regional entomologists. Several awards have been presented at annual meetings to recognize students and professionals who have made exemplary contributions to the field of entomology. Other awards have been presented to individuals who delivered superior presentations at annual meetings. The Society has offered its members many opportunities to promote entomology in South Carolina.

KEY WORDS South Carolina Entomological Society, organization, entomology, history

The first organized effort to form an entomological society in South Carolina occurred at a meeting on 17 December 1954 at the Jefferson Hotel in Columbia, South Carolina. The primary purpose of this meeting was to celebrate the Centennial of Professional Entomology. However, the idea of forming a society may be traced to an informal meeting of entomologists, meeting at the Jefferson Hotel in Columbia on 26 January 1951. Attendees at this meeting expressed a strong desire to meet on an annual basis (Farrar 1951). The meeting, held on 17 December 1954, resulted in the selection of James H. Cochran of Clemson and Frank T. Arnold, Jr. of Columbia to appoint a committee to write a constitution and bylaws for the proposed organization. They appointed Max M. Askey, Jr. (Chairman), W. J. Goodwin, Julian E. Keil, Vernon M. Kirk, and William C. Nettles to serve on the founding committee. After two meetings, the committee had formulated an original draft of a constitution and bylaws. An organizational meeting was called by committee chair Askey to be held at the Jefferson Hotel in Columbia

¹Accepted for publication 4 September 2007



Fig. 1. Attendees of South Carolina Entomological Society Organizational Meeting, Jefferson Hotel, Columbia, South Carolina, August 26, 1955. Key to group picture: Front Row: L. H. Moore, G. C. Fenklea, J. E. Payne, F. Reeves Gressette, T. E. Skelton and G. M. Anderson; Second Row: L. H. Senn, J. K. Reed, N. Allen, D. Duncan, C. S. Creighton, J. A. Berly; Third Row: W. H. Tiller, R. E. Howell, W. C. Johnson, R. L. Walker, A. R. Hopkins, J. L. McKee; Fourth Row: J. P. Fulmer, W. A. Tarplay, J. H. Cochran, F. T. Arnold, W. J. Reid, Jr., L. M. Sparks; Standing: W. C. Nettles, J. E. Keil, M. M. Askey, Jr., W. J. Goodwin, and Frances McAlister.

on 26 August 1955; about 30 South Carolina entomologists attended (Fig. 1). The proposed constitution and bylaws were discussed in detail, revised, and approved at the meeting. The first slate of officers elected at the meeting included James H. Cochran, President; Leon H. Moore, Vice President; Frank T. Arnold, Jr., Secretary/Treasurer; and David Dunavan, Historian. The new officers were instructed to proceed in having the Society duly incorporated in the State of South Carolina. On 25 November 1955, officers Cochran, Moore, and Arnold prepared and submitted a Declaration and Petition for Incorporation of the Society to the Secretary of State of South Carolina. On 9 December 1955, the Entomological Society of South Carolina was duly incorporated by the Secretary of State for South Carolina under the laws of the State and issued a Certificate of Incorporation No. 848. The Charter was presented to the Society at the next meeting (Reid 1958). Program committee chair Norman Allen announced the first meeting of the Society, to be held at the Hotel Columbia, Columbia, on 28–29 March 1956 (Dunavan 1956). The first formal meeting was duly held and the Society continued to meet annually for the next 50 years.

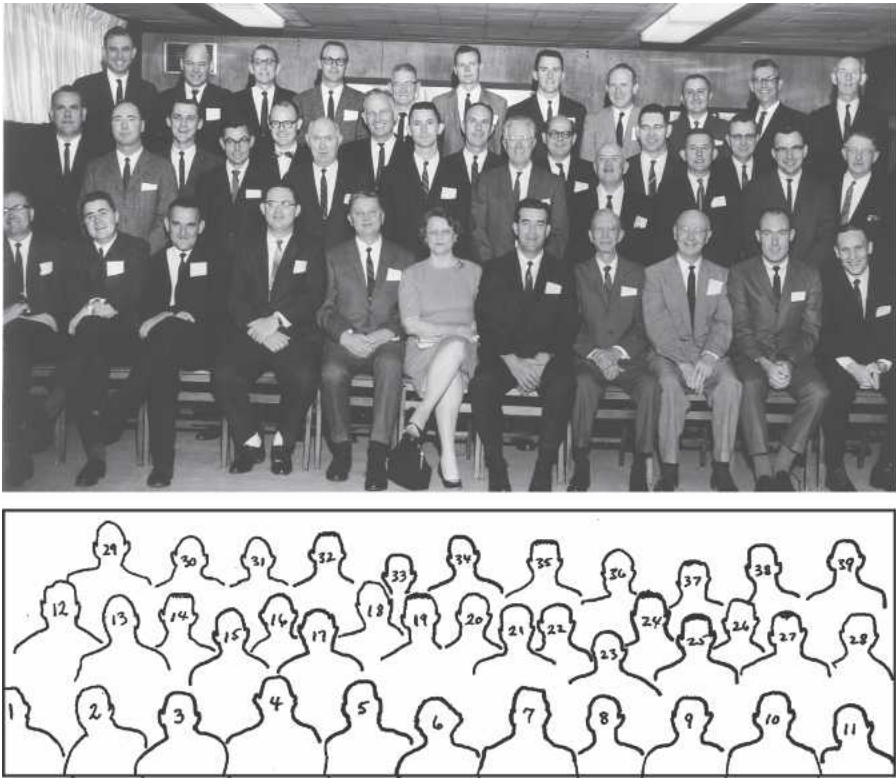


Fig. 2. Attendees of South Carolina Entomological Society's 10th Annual Meeting, Francis Marion Hotel, Charleston, South Carolina, 1965. Key to group picture: 1. L. H. Senn, 2. W. Kinard, 3. C. Creighton, 4. T. R. Adkins, 5. L. R. Morgan, 6. F. McAlister, 7. H. Taft, 8. W. H. Tiller, 9. W. Burnside, 10. W. Johnson, 11. M. Blum, 12. M. Askey, 13. Q. Chapman, 14. R. C. Fox, 15. F. P. Cuthbert, 16. H. Agee, 17. J. H. Cochran, 18. A. Day, 19. S. Turnipseed, 20. J. Mahaney, 21. M. D. Farrar, 22. W. Wymer, 23. W. C. Nettles, 24. E. W. King, 25. V. M. Kirk, 26. W. Westmoreland, 27. D. Broersma, 28. C. S. Harris, 29. J. H. Pruitt, 30. M. Love, 31. J. Keil, 32. S. B. Hays, 33. N. Allen, 34. D. Bowen, 35. A. Ward, 36. J. K. Reed, 37. C. Allee, 38. S. Benedict, and 39. W. J. Reid, Jr.

Annual Meeting Programs

The annual meeting programs varied over the years from professional presentations to professional and student presentations to symposia with invited speakers (Figs. 2 and 3). The first annual meeting (1956) of the South Carolina Entomological Society, Inc., followed several organizational meetings and was the first meeting of entomologists in the state to have a fully planned program. President James H. Cochran, Head of the Department of Entomology and Zoology, Clemson College, called the meeting to order in the State Room of Hotel Columbia at 9:30

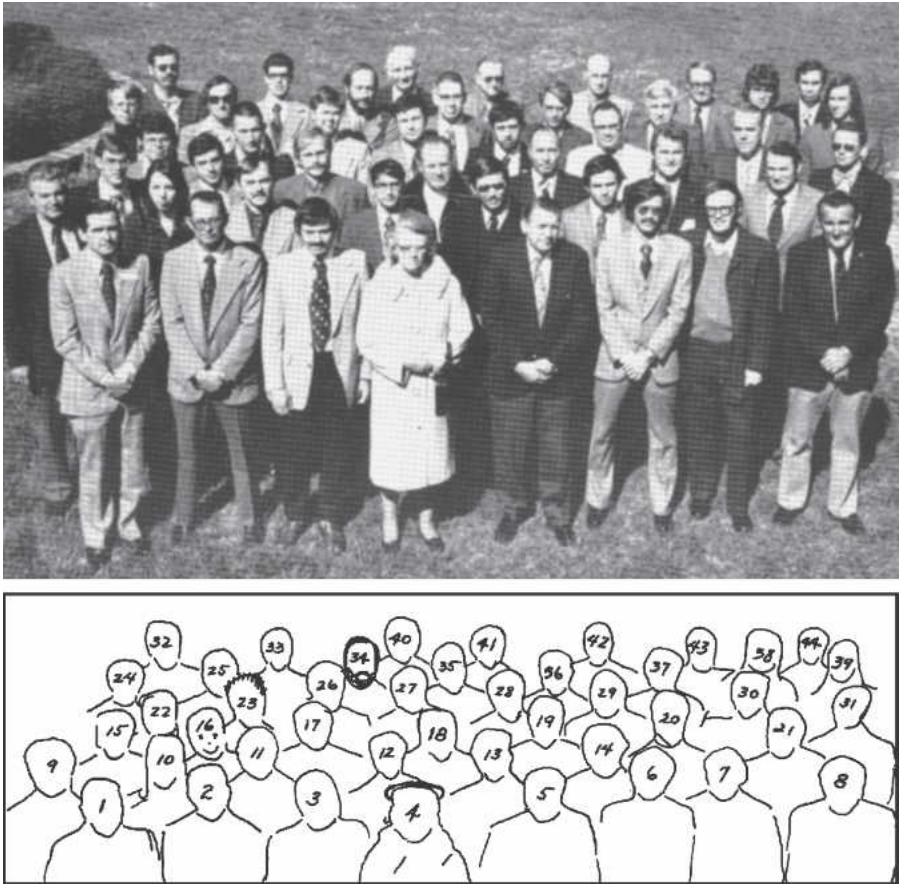


Fig. 3. Attendees of South Carolina Entomological Society's 20th Annual Meeting, Holiday Inn, Clemson, South Carolina, 1975. Key to group picture: 1. Lance Wallace, 2. C. A. Thomas, 3. John DuRant, 4. Frances McAlister, 5. Leo Ehnis, 6. Bill Kea, 7. Ed Quattlebaum, 8. Chas. Creighton, 9. Ben Kissam, 10. Gerry Wicker, 11. Jerry Burkett, 12. Albert Johnson, 13. Paul Horton, 14. Robert Moore, 15. Lawrence Hughes, 16. Gerry Carner, 17. David Montross, 18. "Lefty" Moore, 19. Quentin Chapman, 20. R. C. Fox, 21. Frank Eason, 22. Rodney Holloway, 23. S. G. Turnipseed, 24. Dan Brooks, 25. R. F. Nash, 26. Tommy Walker, 27. Gary Newman, 28. Toney Boozer, 29. T. R. Adkins, 30. Max Askey, 31. Bruce Ezell, 32. Mike Sullivan, 33. John Morse, 34. Von McCaskill, 35. E. W. King, 36. Alton Walker, 37. Julian Mikell, 38. Robert Herlong, 39. Frank Davis, 40. Augustine Day, 41. Bill Westmoreland, 42. Bill Lott, 43. L. H. Senn, and 44. Ray Noblet.

AM (Arnold 1956). President Cochran gave a presidential address based on the third objective of the Society's Constitution, which stated "[T]o bring about through concerted effort further improvements in the field of entomology throughout the State of South Carolina." He presented four main points: 1) entomology needs more workers and more money, 2) entomology students and workers need to be given better training in basic entomology and biology, 3) there is a great need for better and safer insecticides, and 4) entomologists need to sell their profession to the public. On the first day of the meeting, members toured the Public Health Laboratory at the State Hospital in Columbia, the Sandhill Experiment Station in Pontiac, and the State-Federal Livestock Sanitary Laboratories at the Sandhill Station. Three presentations were given on the second day of the meeting. John K. Reed, Associate Entomologist, South Carolina State Crop Pest Commission, Clemson, gave a talk on "South Carolina Pesticidal Testing Program." Reed outlined the steps the South Carolina Crop Commission follows in assuring the public that pesticides purchased in the state are legitimate. E. H. Smith, Associate Professor of Entomology, New York Agricultural Experiment Station, Geneva, New York State, gave a presentation titled "Promotion of Entomology" and C. R. Jordan, Extension Entomologist, University of Georgia, Athens, Georgia, spoke next on "Shall We Wait Another 100 Years?" During the business session, program Chair Norman Allen solicited from attendees their opinions on what type of program was appropriate for annual meetings. Members suggested a wide range of programs, which varied from informal programs having no paper presentations or demonstrations to formal programs having invited speakers and paper presentations. After much discussion, the members decided to delegate this responsibility to the program committee. A varied program continued for the next few years with professionals giving presentations, conducting workshops, or leading field and laboratory oratory tours.

The fifth annual meeting program in 1960 included a presentation by Hannah Holleman, a student in the Department of Zoology, Clemson College. She was the first graduate student to give a presentation at a Society meeting. The twenty-eighth meeting program in 1982 featured a themed symposium titled "Perspectives on Insect Control." This was the first Society meeting to include a symposium, which included three professional presentations. Waldemar Klassen, National Program Research Leader for IPM, USDA-ARS, Stoneville, Mississippi, presented a talk "Development of Coordinated Insect Control from the Single Farm to Regional Programs." Edward King, Jr., Director, Biological Control Laboratory, USDA-ARS, Stoneville, Mississippi, gave a talk on "Biological Methods for Managing Insect and Mite Pests," and Thomas M. Brown, Insecticide Toxicologist, Department of Entomology, Fisheries and Wildlife, Clemson University, presented a talk on "Benefits, Constraints and Prospects of Insecticide Use."

The thirtieth meeting program in 1984 included two symposia entitled "Naturalist's Symposium" and "Future Direction for *Heliothis* Research in South Carolina." The program of the thirty-third meeting in 1987 was the first to have sessions formally listed by specialty area. Two general sessions featured talks on crop protection, one general session featured talks on behavior and bionomics, and the other general session featured talks on medical, veterinary, and forensic entomology. A Lyme disease symposium was held the first afternoon of the thirty-sixth meeting on 8–9 November 1990. A naturalist symposium was held during the thirty-seventh annual meeting on 7–8 November 1991. Invited speaker Rudy

Mancke, Director of Science and Nature Programming, South Carolina, ETV, gave a talk on "The Dragonflies of South Carolina" to begin the Naturalist Symposium held during the second session. He also gave a presentation titled "To See a World" after the social hour and dinner.

M. B. Willey and Peter H. Adler gave a poster presentation at the thirty-eighth annual meeting of the Society held on 5–6 November 1992. The poster, "Systematic Position of the Basilica Spider, *Mecynogea lemniscata* (Araneae: Araneidae)," marked the first poster presentation at a Society meeting.

A photo session was included in the thirty-ninth annual meeting program (1994) of the Society, which was held jointly with the Georgia Entomological Society. This marked the first photo session held at a Society meeting.

The fortieth meeting program included the first high school student presentations that included student, Karen E. Mabry, from Union High School and three students, Meredith Mitchell, Alicia Syrett, and Marcus Washington, from the South Carolina Governors School for Science and Mathematics, Hartsville. Several later meeting programs included presentations by students from high schools and the South Carolina Governors School.

During the final business session of the forty-sixth annual meeting, held on 3 November 2000, the question of providing computer projection equipment for Power Point® presentations was discussed extensively. The call for papers for that meeting specified only 2 × 2 inch slide format. A motion was made which was approved to officially sanction both types of presentations at the next annual meeting and to revise the call-for-papers form accordingly.

A symposium titled "Natural History of the Sandhills" was held during the third general session of the forty-eighth annual meeting (2002) of the Society. Invited speaker Rudy Mancke, now of the School of the Environment, University of South Carolina, Columbia, gave a talk on "The Sandhills Then and Now." Other invited speakers and their respective topics included W. Grooms, The Nature Conservancy, South Carolina Chapter, Columbia, "A Layman's Overview of the Cultural and Agricultural History of the Sandhills;" Patrick D. McMillan, Department of Biological Sciences, Clemson University, Clemson, "A Brief Synopsis of the Vegetation and Ecology of the Fall-line Sandhills;" J. Glover, South Carolina Department of Health and Environmental Control, Columbia, "The Aquatic Macroinvertebrates of Streams and Rivers in the Sandhills of South Carolina;" Donald G. Manley, Department of Entomology, Clemson University, Clemson, "Velvet Ants of the Sandhills;" J. Ciegler, West Columbia, "Coleoptera from Ultraviolet Light Traps in the Sandhills of South Carolina;" and Alfred G. Wheeler, Department of Entomology, Clemson University, Clemson, "An African Grass in the Carolina Sandhills: Colonization of an Exotic Plant by Little-known, Rarely-Collected Herbivores."

On 7–8 October 2004, the Society met at the Sandhill Research and Education Center, Columbia, to celebrate the fiftieth meeting of the Society. Fifty-two entomologists, guests or invited speakers attended the 2004 meeting (Fig. 4), which was presided over by President Paul M. Horton. A special anniversary session entitled "A Look Back at 50 Years" was held at the beginning of the meeting to commemorate the history of the Society. Three charter members, Louie H. Senn, Thomas E. Skelton, and Mac Sparks, were present at the meeting. Skelton, retired entomologist and 1983 president of the Society, began the session with a talk on "The Early Years of the Society." J. Benjamin Kissam, retired extension en-

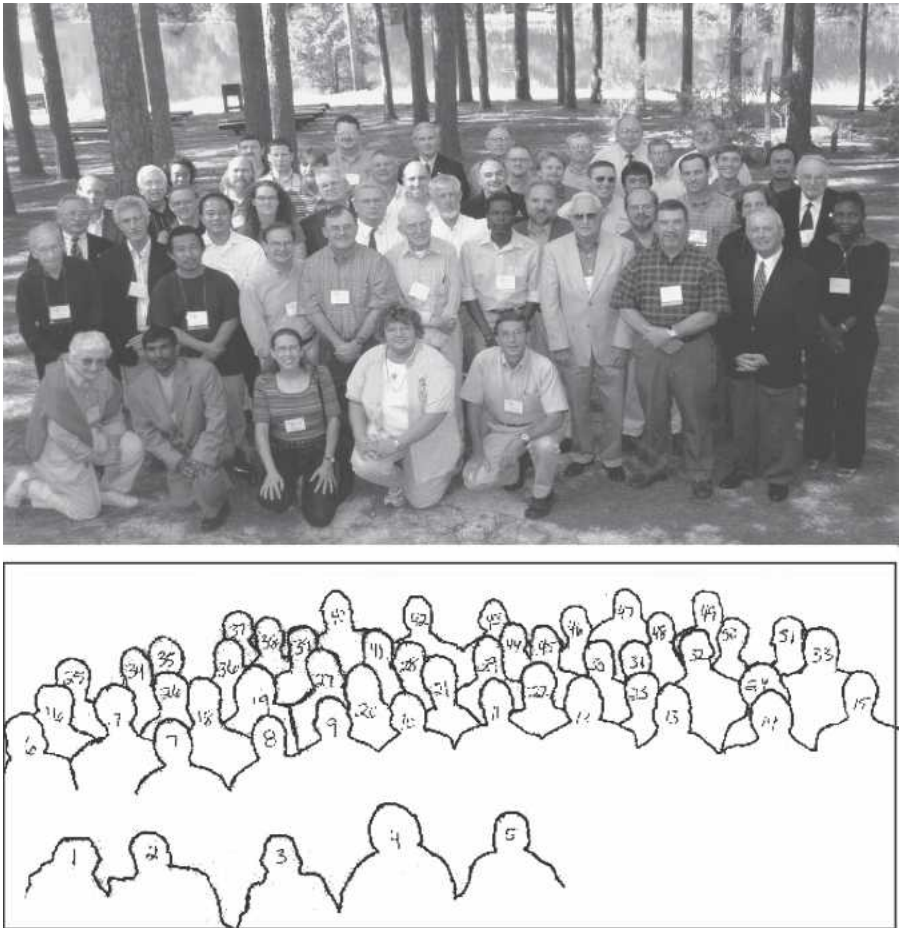


Fig. 4. Attendees of South Carolina Entomological Society's 50th Annual Meeting, Sandhill Research and Education Center, Columbia, South Carolina, 7–8 October 2004. Key to group picture: 1. Lynn Smith, 2. Mohamed F. R. Khan, 3. Paula Mitchell, 4. Barbara Padgett, 5. Gerry Carner, 6. Mac Sparks, 7. Idham Harahap, 8. Merle Shepard, 9. Mike Jackson, 10. John Reed, 11. Roger Francis, 12. Dick Baughman, 13. Randy Griffin, 14. Tom Skelton, 15. Margaret Nakakande, 16. Mac Horton, 17. Peter Adler, 18. Sam-Kyu Kim, 19. Layla Burgess, 20. Bruce Ezell, 21. Richard Porcher, 22. Morgan McClure, 23. Rolando Lopez-Guitierrez, 24. Nancy Osler, 25. Al Wheeler, 26. Joe Culin, 27. Ben Kissam, 28. Don Manley, 29. Randy McWhorter, 30. Tim Davis, 31. Yayi Kusumah, 32. Eric Benson, 33. Louie Senn, 34. Bill Nettles, 35. Gloria McCutcheon, 36. Mark Schaffer, 37. Hugh Conway, 38. Chad Smith, 39. James Korecki, 40. Phil Harpootlian, 41. Powell Smith, 42. Mike Hood, 43. Sam Turnipseed, 44. Mark Nelder, 45. Patrick McMillan, 46. Jay Chapin, 47. L. A. Williams, 48. Albert Johnson, 49. Dwight Williams, 50. Ben Powell, and 51. Ruly Anwar.

tomologist and 1991 president of the Society, gave a talk on "Extension Entomology thru the Years." Kissam also read a presentation entitled "Teaching Entomology in the State," which was submitted by Sidney B. Hays, 1972 president of the Society, who was unable to attend the meeting. Senn, retired entomologist and 1977 president of the Society, gave a presentation on "Regulatory Entomology in the State." Vice President and 2004 program committee Chair W. Michael Hood ended the session with a slide presentation on "The Fifty Year History of the Society."

Publications

The Society had various publications during the first 50-year history, including newsletters, brochures, proceedings of annual meetings, and a professional journal. At the fourth meeting of the Society held 11–12 March 1959, publications committee Chair Edward W. King reported his committee's discussion on establishment of a journal for the Society, saying "a journal should be scientifically respectable and reliable." Discussion followed with "prominent entomologists giving varying opinions." There were some favorable opinions given but the general opinion was against the establishment of a journal published by a single state society. A combination state-based journal was suggested, possibly with Georgia. However, Theodore R. Adkins, Jr. reported that in discussions during his visit to a recent Georgia Entomological Society meeting there was opposition to such a combination journal. Someone noted that a South Carolina Entomological Society questionnaire indicated little support or publishable material for a journal. Leon H. Moore noted that he would support a publication similar to the Georgia Entomologist. James H. Cochran stated that the journal should be a solid publication that will provide good publicity for the Society (Thornburg 1960). No further discussion was recorded on this topic in the minutes of the first day of this meeting. During the final business meeting of the second day of the 1959 meeting, publications committee Chair King solicited direction from the group for his committee concerning the status of a journal and newsletter. He noted four possibilities: 1) continue journal plans, 2) abandon journal plans and start a newsletter, 3) attempt both, or 4) abandon both. After much discussion, Vernon M. Kirk moved that plans for a journal be abandoned, which was seconded by W. H. Burnside. The members present voted and the motion failed to pass. By an informal show of hands, a newsletter was favored over the journal.

During the business session of the 1960 meeting, publications committee Chair King gave his committee's report, which was followed by a motion made by Richard C. Fox that the Society should publish an annual proceedings of the meeting, including presentation abstracts and other pertinent information. The motion passed to take this route (McAlister 1960). King edited the proceedings of the 1960 meeting, which included the text of remarks by speakers. The Proceedings were the result of several years' effort by the publications committee to provide the members some form of publication (King 1960).

At a preliminary business session held during the sixth annual meeting in 1961, publications committee Chair King distributed mimeographed copies of Volume I, No. 1 of Proceedings of the South Carolina Entomological Society, which was the Society's first publication. Chair King reported during the 1963 final business meeting that he had prepared a brochure designed to introduce the

field of entomology to South Carolina high school students and their advisors. The brochure was approved and the members voted to proceed with printing copies at a cost not to exceed \$250. The brochure was included on pages 7–9 of the Proceedings of the Society, 1963, Vol. III, No. 1. Apparently meeting proceedings were published through the 1969 meeting, after which the executive committee met on 10 December 1969 and voted to discontinue the publication of the proceedings of the Society's annual meeting.

During the final business session of the 1970 meeting, publications committee Chair Thomas E. Skelton noted that his committee had reconsidered the possibility and feasibility of publishing an annual meeting proceedings. He added that during the December 1969 board of directors meeting, President Samuel G. Turnipseed directed his committee to have published in a recognized journal the talk given by Wesley W. Gregory titled "History of Entomology in South Carolina" at the 1969 annual meeting. Chair Skelton reported that correspondence had been received recently from the Executive Secretary of the Entomological Society of America, stating that he had agreed to publish Gregory's talk (Gregory 1970) in the June 1970 issue of the Bulletin of the Entomological Society of America (Skelton & Kissam 1970).

During the final business session of the 1971 meeting, publications committee Chair Julian E. Keil recommended that the Society should have a publication and further noted that abstracts of papers presented at annual meetings should be published. Chair Keil made a motion that was approved that the Society have a publication. Chair Keil recommended that the executive committee should appraise the situation and make the final decision on the type of publication based on the Society's finances. The executive committee met on 6 December 1971 and approved a motion by former publications committee Chair Skelton to enlarge the annual meeting program to include abstracts of each presentation. During the final business session of the 1972 meeting, publications committee Chair Adkins reported that the board of directors had requested that abstracts of each paper be printed in the meeting program, which would result in much less expense to the Society than printing a separate meeting proceedings. Each presentation title and author was followed by a brief abstract in the 1972 meeting program.

During the final business session of the 1973 annual meeting held on 14 February, publications committee Chair Wm. Bruce Ezell, Jr., reported that his committee recommended that the Society should contact members of the North Carolina Entomological Society to investigate the possibility of a joint publication. He also reported that his committee recommended that the Society publish a newsletter. J. Benjamin Kissam made a motion that was approved that the Society president be responsible for preparing a quarterly newsletter. The first Society newsletter was published in July 1973 by President Theodore R. Adkins. The purpose of the newsletter was to inform the membership of Society activities and other information related to the field of Entomology. The first newsletter featured an article on Rocky Mountain spotted fever because an increasing number of cases had been reported recently in South Carolina and surrounding states. Several agriculture-related business updates were listed in the newsletter, giving new office openings, personnel changes, and new product status. Personal news items were included that gave updates on current and past members of the Department of Entomology and Economic Zoology, Clemson University. A brief summary of the Society's last annual meeting was given because many members were

unable to attend the meeting, which had to be rescheduled because of inclement weather. Society newsletters were published on a regular basis until 1983 (Kissam 1994).

In the final business session of the 1975 annual meeting, publications committee Chair Raymond Noblet reported that his committee had questioned many Society members and the general opinion was that because the newsletter had recently been established, this was not a good time to start a journal. He noted that the membership supported the idea of beginning a journal in the future and that consideration be given to a possible joint publication with the North Carolina Entomological Society. An undated memorandum addressed to the Society was located in the historical records. The memo "proposed that the South and North Carolina Entomological Societies jointly publish a refereed journal of general entomology" and the memo included some suggested journal names including *The Carolina Entomologist*, *MidAtlantic Journal of Entomology*, *Journal of Carolina Entomology*, and *Carolina Journal of Entomology*.

On 6 August 1975, Adkins and Noblet sent to all Society members a memorandum stating that they had assumed the responsibility of compiling the quarterly newsletter. The *Society Newsletter*, Vol. III, No. 1, published in November 1975, featured an article that announced the initiation of the Pesticide Applicator Certification Program in South Carolina. The program was designed to ensure that users have the competence to handle restricted-use pesticides without causing danger to themselves, the public, or the environment. The Clemson University Cooperative Extension Service was charged with responsibility for the certification training program and the Plant Pest Regulatory service was responsible for examinations and licensing.

A South Carolina Entomological Society Newsletter, Vol. X, No.1, was published in February 1982. The newsletter announced the upcoming Society meeting to be held 25–26 February 1982 in Columbia. An article titled "A Journal Sponsored by the South Carolina Entomological Society?" encouraged members to attend the meeting and be prepared to discuss the proposed journal during a block of time set aside on the first day of the meeting. As of 16 February, 47 members had returned a questionnaire that solicited their opinion of the proposed journal. Of the questionnaires returned, only three members were not in favor of the Society publishing a journal. Twenty-eight respondents indicated they would consider publishing in the proposed journal and 37 said they would be willing to serve on the editorial board. The article listed several items for members to give serious thought before the upcoming meeting. Adkins moderated a special session entitled "Discussion on the Proposal to Publish a Scientific Journal" during the first day of the twenty-seventh meeting of the Society held on 25–26 February 1982. No detailed minutes were found in the historian's files for this meeting.

A Society Newsletter was published in September 1982, Vol. X, No. 2. The newsletter began with an announcement of the next Society meeting to be held on 11–12 November 1982, which would be the second meeting in 1982. The board of directors decided to change the meeting time from spring to fall in an attempt to avoid the last big early spring snow storms.

The ad-hoc committee assigned to study the feasibility of a journal and other related issues met on 25 March 1982. Committee member David R. Alverson prepared a summary of the recommended proposals resulting from the meeting. The title approved by the committee was the "Journal of Agricultural Entomol-

ogy." The editorial policy recommended the journal should be published quarterly by the Society and that subscription should be included with membership. A policy statement recommended, "the Journal of Agricultural Entomology publishes contributions on all aspects of original research concerning insects and other arthropods of agricultural importance. The journal is particularly dedicated to the timely publication of articles and notes pertaining to applied entomology, although it will accept suitable contributions of a fundamental nature related to agricultural entomology." The committee recommended the editorial board shall consist of eight nationally recognized scientists representing major disciplines associated with agricultural entomology. And further, the board shall represent research efforts by university, USDA, and industry scientists from all regions of the United States. Each member shall serve a 3-year term and serve in an advisory capacity to the journal through the editor-in chief. The editor-in-chief shall be compensated by a suitable honorarium and shall be expected to manage the flow of manuscripts, ensure adequate peer review, and chair the editorial board. The committee recommended the journal be distributed to members of the Society, entomology departments at all land grant institutions, USDA, and their respective libraries, and indexing services. The committee recommended that the original issue should consist of approximately 25 invited papers from nationally recognized scientists. These initial papers would establish the quality of manuscripts accepted for publication and introduce the journal as national in scope. Journal costs would be recovered through page charges, subscription rates, and advertising. Five pages of the Journal would be dedicated to advertising at scale rates and at reduced rates for recurrent ads. The committee planned to meet again prior to the next Society meeting in November 1982.

The ad hoc committee had a second meeting on 9 November 1982 and a summary of the meeting was published in a report dated 11 November 1982 and signed by Chair Adkins. The report contained much of the same information resulting from the earlier meeting that was published in the summary by Alverson. Additional recommendations resulting from this second meeting included the objectives of the publication: a) national recognition in agricultural circles, b) highest-quality possible, and c) speedy publication of worthy manuscripts. The committee recommended that the Society commit funds to underwrite the cost of half the first issue not to exceed \$2,000. After the first issue, each issue of the journal should be self-sustaining through page charges and advertising. The committee recommended that Adkins be appointed Journal Editor, John Reed as Associate Editor, William P. DuBose as Business Manager, and Thomas E. Skelton as Associate Business Manager. All committee recommendations were approved and the first issue of the *Journal of Agricultural Entomology* was published in 1984. Adkins served as Journal editor until July 1989.

The final presentation of the thirtieth meeting, held on 8–9 November 1984, was presented by Adkins entitled "The Trials and Tribulations of Producing a High Quality Scientific Journal." During a board of directors meeting held on 18 July 1988, Adkins reported that the January issue of the journal was printed and ready to be mailed. He noted that the rate of rejection of manuscripts was extremely high and although there were enough papers to fill future issues, the papers were in the hands of reviewers, which continued to be a problem (Bellinger 1988a).

The Executive Committee of the Society met at 10 AM, before the thirty-fourth meeting held 10–11 November 1988. The status of the Journal of Agricultural Entomology (JAE) was discussed by the committee. Barry Dover, JAE Business Editor, reported on the financial status of the journal. He reported the JAE was presently in arrears resulting from several outstanding debits for which payments would be forthcoming. He noted that the Journal was presently losing approximately \$700 per issue, giving a projected JAE loss of \$2600 for 1988. Publications Committee Chair Adkins reported that the JAE manuscript review process was taking too long and that there was a backlog of about 10 papers. Much discussion followed on how to best address JAE's problems. President Dubose recommended the Society raise dues, tighten the review process, and seek to make the JAE more widely known (Bellinger 1988b).

A newsletter was published in October 1991 by Society President J. Benjamin Kissam. The newsletter contained proposed revisions to the Society's Constitution and Bylaws, which were mainly a result of necessary changes to meet Internal Revenue requirements under Section 501 [c] Internal Revenue Code of 1954. John C. Morse chaired the Ad Hoc Constitution Revision Committee. The proposed revisions were to be presented at the next Society meeting to be held in November 1991 (Kissam 1991).

A note was posted on the thirty-ninth meeting program stating that the South Carolina Entomological Society wished to express appreciation to Catherine Walgenbach from Hendersonville, North Carolina, for serving as editor of the South Carolina JAE from September 1989 to January 1994. Bob Cartwright, Oklahoma State University, was appointed as the new editor, effective 1 January 1994 and served in that capacity till August 1995. Howard W. Fascemyer, Clemson University, was appointed editor to succeed Cartwright and served till November 1996. Julie L. Todd, Iowa State University, followed Fascemyer as editor and served till October 1998.

During the business session of the forty-fourth annual meeting, held 12–13 October 1998, Paula L. Mitchell asked for discussion about changing the name of the JAE to the *Journal of Agricultural and Urban Entomology* (JAUE). The issue was discussed and a motion was made and approved that the journal would be renamed beginning with the first issue of 1999. Journal topics would continue to include the types of papers that had always been published therein as well as urban entomology to include nursery, lawn, and structural entomology. Other business included the announcement that David K. Weaver, Montana State University, would replace Julie Todd as JAUE editor after publication of the October 1998 issue (Ellis 1998).

During the business session of the forty-sixth annual meeting, held at the Sands Ocean Club Resort in Myrtle Beach on 2–3 November 2000, new business included a discussion on the JAUE. The slow rate of manuscript processing and the late publication of journal issues were voiced as concerns. Donald Manley and Paula Mitchell noted that Editor Weaver had replaced two subject editors recently and that the new subject editors were now receiving and processing manuscripts in a timely manner. Journal production delays were caused, at least partially, by a change in printer.

During the preliminary business session of the forty-seventh annual meeting in 2001, JAUE Editor Weaver announced that his first term as editor would end in November and that he would like to be relieved of this position. President

Gloria S. McCutcheon announced that Thomas Shanower, USDA-ARS Northern Plains Agricultural Research Laboratory, Sidney, Montana, was the new editor of JAUE. Shanower continued to serve as editor through August 2007. A goal was set to have Journal Volume 19 published by the end of 2002, which would put the JAUE back on schedule.

In the business session of the forty-ninth annual meeting in 2003, the Society's long range planning committee recommended the establishment of a special committee to review the needs of the JAUE in relation to the needs of the Society and to explain options for the Journal that would be less costly to the Society. Shanower noted in his editors report at the 2004 annual meeting that the journal still suffers from a low submission rate of quality articles. Executive Director Randy E. McWhorter reported that a society committee had been established to review the status of the JAUE, to look at the survivability of the JAUE, and to report back to the members at the 2005 annual meeting.

Joint Meetings

President Julian Mikell presided over a board of directors meeting on 7 November 1974 at the J. Marion Sims Building, Columbia. President Mikell read a letter that he had received from Harry Yates, President of the Georgia Entomological Society, proposing a joint meeting of the Georgia and South Carolina Entomological Societies on 17–19 March 1976, at the DeSota Hilton Hotel in Savannah, GA. A motion was made and approved that the Society accept the invitation. President Mikell appointed Wm. Bruce Ezell, Jr., to be chair of a special committee to work out the preliminary details for this joint meeting. On 6 August 1975, the executive committees of the South Carolina Entomological Society and the Georgia Entomological Society met at the DeSota Hilton Hotel in Savannah to plan the upcoming joint meeting (Adkins & Noblet 1975).

The twenty-first meeting of the Society was held jointly with the Georgia Entomological Society on 17–19 March 1976 at the DeSota Hilton, Savannah, Georgia. This was the first meeting held jointly with another state society. R. Chalfant reported that there were 178 paid registrants, with 53 from South Carolina and 104 from Georgia, and 21 from both societies in attendance (Noblet 1976). The meeting marked the fortieth meeting of the Georgia Entomological Society.

On 21–23 March 1979, the twenty-fourth meeting of the Society was held jointly with the Georgia Entomological Society at the Quality Inn, Hilton Head Island, South Carolina. One of the most interesting events of the meeting was the first annual 5,000-m “Bug Run” where 25 entomologists or their guests ran alongside the nature trails of Sea Pines Forest Preserve in team and individual competition. Sixty-five 10-min presentations were given in ten general sessions during the last 2 days of the meeting.

The twenty-fifth meeting of the Society was held jointly with the North Carolina Entomological Society at the Holiday Inn, Wilmington, North Carolina, on 12–14 March 1980, marking the first joint meeting of these two societies. A golf tournament and “3-Mile Bug Run” at Greenfield Lake were held on the first day of the meeting. Forty-eight 10-min presentations were given during four general sessions.

The thirty-first meeting of the Society was held jointly with the Georgia Entomological Society at the Savannah Sheraton Resort and Country Club, Savan-

nah, Georgia, on 24–26 March 1986. This marked the 50th meeting of the Georgia Entomological Society and was the third joint meeting of the two societies. An annual run and golf tournament were held prior to the beginning of the formal program. Sixty-five 12-min presentations were given during three concurrent sessions. A banquet luncheon and photo salon were held on the second day of the meeting.

The thirty-fourth meeting of the Society was held jointly with the North Carolina Entomological Society at the Landmark Resort Hotel, Myrtle Beach, on 10–11 November 1988. This marked the second joint meeting of the two societies. A special event, a “5K Beach Fly Run,” was held before the opening session of the meeting. Twenty-eight 12-min presentations were given during three general sessions (Bellinger 1988b).

The thirty-ninth meeting of the Society was held in conjunction with the Georgia Entomological Society at the Sheraton Charleston Hotel, Charleston, on 23–25 March 1994. This marked the fourth joint meeting of the two societies. A golf tournament and a 5K run were held prior to the beginning of the formal program. Fifty-two 12-min presentations were given during five general sessions, and 12 poster presentations were displayed in one poster session.

The forty-seventh meeting of the Society was held jointly with the Georgia Entomological Society at the Best Western Lanier Center Hotel and the Georgia Mountains Center, Gainesville, Georgia on 4–6 April 2001. This was the sixty-fifth annual meeting of the Georgia Entomological Society and the fifth joint meeting with the South Carolina Entomological Society. A golf tournament and a “Fun Bug Run/Walk” were special events held the first day of the meeting. The first Georgia Entomological Society and South Carolina Entomological Society Linnaean games were held during the meeting where three teams competed, the University of Georgia bulldogs, the Georgia Southern University eagles, and the Clemson University tigers. Thirty-seven 10-min presentations were given during four general sessions and thirteen poster presentations were displayed during the meeting.

Awards

Scholastic awards. During a business session of the 1960 annual meeting, James H. Cochran made a motion, which was approved, that the incoming president appoint a committee to investigate the idea of presenting an award annually to an outstanding student in entomology (McAlister 1960). At the next annual meeting (1961), awards committee Chair Cochran made a motion during the business session that “1) the Society approve an award to be presented to a second semester junior or senior Entomology major in South Carolina; the award is to be based on scholastic attainment, leadership ability and interest in the profession of Entomology; the award is to be called the South Carolina Entomological Society Scholastic Award and consist of \$25 cash and an inscribed scroll; and 2) a committee be appointed and authorized to select the candidate for the South Carolina Entomological Society Scholastic Award.” The motion was seconded and passed (Kirk 1961). According to the minutes of the 1962 and 1963 meetings, awards committee Chair Cochran noted that no student qualified to receive the “Scholastic Award.” Chair Cochran made a motion that was approved at the 1963 meeting that the incoming president should appoint a committee to consider changing the award to a savings bond to be presented to a deserving high school

4-H or FFA student who showed significant interest in entomology. No record exists in the minutes of the selection of a high school student to receive this award. During the final business meeting of the 1969 meeting, Awards Committee Chair Cochran made a motion, which was approved, that Clemson University senior student Thomas M. Hill be awarded the \$50 savings bond as the biological sciences major with the best academic record.

Cochran awards. After the untimely death of James H. Cochran on 9 May 1969, the Society members voted by a mail-in ballot to establish the J. H. Cochran Memorial Scholarship Fund. At a board of directors meeting on 10 December 1969, Secretary/Treasurer Louie H. Senn noted that the award would be established as a cash award and that recipients of the award would be selected by a three-member committee consisting of the Head of the Department of Entomology & Economic Zoology at Clemson University as permanent chair, the Vice President of the South Carolina Entomological Society, and a third member to be appointed by the president of the Society. It was decided that the amount of the annual award would have to coincide with income. A motion was made and approved to discontinue the presentation of the \$50 bond after 1970. At the final business session of the 1970 annual meeting, awards committee Chair Edward W. King made a motion that was approved that the first J. H. Cochran Memorial Scholarship Award in the amount of \$200 be presented to David C. Arnold, an entomology student. Table 1 lists other annual J. H. Cochran Memorial Scholarship Award recipients, with the exception of a few years when records were not available in the Society's historical files. In the minutes of the 1973 annual meeting, awards committee Chair King reported that Norman A. Williams had received the J. H. Cochran Award, which is presented annually by the South Carolina Pest Control Association.

Presentation awards. During the first joint meeting with the Georgia Entomological Society on 17–19 March 1979, Edward W. King received the best professional presentation award for a talk on "Mathematical Morphology." At the same meeting, John C. French announced the best student paper was by R. L. Osburn and J. H. Oliver. At the second joint meeting of the two societies held on 21–23 March 1979, the O. I. Snapp Best NonStudent Paper Award was presented to Maxcy P. Nolan, Jr., who gave three talks, and the C. M. Beckham Best Student Paper Award was presented to J. M. Pound for his paper entitled, "Evidence of a Role for Juvenile Hormone in the Reproduction of Ticks."

An article in the Society Newsletter dated September 1982 announced the presentation of the 4H Entomology Award to high school student Randy Cummings from Dorchester County. The presentation was made by Society President Thomas E. Skelton at the annual 4H Banquet, which was held in Clemson on 29 July 1982. The awardee received a collection of moths prepared by Society member Donald G. Manley, a \$75 savings bond given by Mobay, Inc., and a check for \$75 given by the Society.

During a board of directors meeting on 18 January 1988, awards committee Chair Skelton noted that there would be separate Society awards for best student paper and best nonstudent paper for the upcoming meeting (Bellinger 1988a).

Excellence in Entomology Award. The Executive Committee of the Society met at Clemson University on 21 August 1990. A lengthy discussion was held on the proposal that the Society establish a professional entomologist award. Many questions were asked concerning such an award, such as how would the

Table 1. Recipients of the J. H. Cochran Memorial Scholarship Award.

1970	David Arnold	1988	Kevin Hoffman
1971	H. S. Moore	1989	Charles N. Watson
1972	Keith Griffith	1990	Marianne B. Willey
1973	Roy Sutton	1991	Stuart R. Reitz
1974	Frank Davis	1992	
1975	Frank Davis	1993	Michael A. Floyd
1976	Wayne A. Gardner	1994	
1977	Gayle F. Holman	1995	Jeffery K. Tomberlin
1978	Jay W. Chapin	1996	Jennifer A. Zettler
1979	Ed Quattlebaum	1997	Kristen van den Meiracker
1980	Glenn Lee	1998	Mohammed Kahn
1981		1999	Will K. Reeves
1982	Robert Kelley	2000	Michelle Colacicco
1983	Jerome Grant	2001	J. Powell Smith
1984	Steven W. Hamilton	2002	Janet L. Kintz-Early
1985		2003	Ozlem Kalkar
1986	Eric Benson	2004	Eric Paysen
1987	Mike Chambers	2005	Donald A. Oswalt

award be funded, and what would be the Society's tax status relative to such an award? Society President John C. Morse announced that he would appoint an ad hoc committee made up of Thomas E. Skelton, J. Benjamin Kissam, and Robert G. Bellinger to investigate the various financial issues identified by the executive committee. A memorandum was addressed by Skelton, Chair of the Ad Hoc Committee on Strategic Financial Planning, to Society President Morse dated 24 September 1990. The Committee recommended: 1) that funds from the general fund be placed in short term certificates of deposit or be transferred to the Cochran Fund to enable the Society to give a Professional Entomologist Award as discussed below, 2) that the President appoint a committee to develop procedures for recognizing members who provide exemplary contribution or service to the profession of entomology, 3) that funds from the general fund be transferred to the Cochran Fund to help support this award and, 4) that the name of the Cochran Fund be changed from the J. H. Cochran Memorial Scholarship Fund to the J. H. Cochran Memorial Fund. An additional unsolicited recommendation was given in the memorandum that the Society limit the amount of funds given each year either to a percent of the earnings or to a percent of the total fund amount (Skelton 1990). During the business session of the thirty-sixth annual meeting on 8–9 November 1990, the Society voted to establish an Award for Excellence in Entomology to recognize deserving individuals in our discipline. Kissam provided much of the concept and momentum for establishment of the Award for Excellence in Entomology, and led the Society in a major fund raising effort to generate funds to support the award.

The first J. H. Cochran Award for Excellence in Entomology was given to Sidney B. Hays at the thirty-ninth annual meeting of the Society, held at the Sheraton Charleston Hotel, Charleston, on 25 March 1993. Thomas E. Skelton

received the second triannual J. H. Cochran Award for Excellence in Entomology at the forty-second annual meeting of the Society held at the Madren Center, Clemson University, on 13 November 1996. During the forty-fifth annual meeting of the Society, held at the Pee Dee Research and Education Center, Florence, on 16–17 November 1999, Peter H. Adler received the third triannual J. H. Cochran Award for Excellence in Entomology. J. A. Reinert and Alfred G. Wheeler were selected to receive the triannual J. H. Cochran Award for Excellence in Entomology at the forty-eighth annual Society meeting on 11 October 2002 at the Holiday Inn, Columbia, and John C. Morse was selected to receive the award at the fifty-first annual meeting.

Special Projects

Selection of insignia. During the final business session of the 1959 meeting, a motion was made and approved that the incoming executive committee should investigate the creation of an insignia for the Society. Three possible Society insignias were submitted by Vernon M. Kirk who presented them to the members present at the final business session of the 1960 meeting. Discussion followed, with members endorsing their preference for each insignia. A vote was held and a large majority of members voted in favor of a single insignia. President William J. Reid suggested that Kirk and Theodore R. Adkins further investigate the lettering on the selected insignia design and report back to the board of directors for motion at next year's meeting (Fig. 5). During the 1961 annual meeting, Secretary Kirk reported on the status of the Society's insignia and noted that 20 of the 25 tie tacks bearing the insignia had been sold.

Affiliation with ESA. James H. Cochran made a motion during a business session at the 1961 meeting, which was approved, that the incoming president petition for affiliation of the Society with the Entomological Society of America (ESA). On 28 February 1962, a letter was addressed to Cochran from R. H. Nelson, Executive Secretary ESA, stating that the Governing Board of ESA had approved the affiliation of the South Carolina Entomological Society with ESA at the 1961 annual meeting in Miami, Florida.

Registry of professional entomologists. During the final business session of the 1969 meeting, legislative committee Chair Frank T. Arnold, Jr. gave his committee's report in the form of a "pep talk" saying that entomology had passed into a new phase or crossroads and we should, "upgrade and look up to entomology as a profession." He made a motion, which was approved, that the Society should decide whether it wants to certify entomological professionals, and that the president be given the necessary authority to proceed with the proposal. After the meeting, a memorandum was mailed to all Society members, addressing the issue of the Society in seeking professional status as entomologists. The mail ballots were returned, which indicated a favorable vote count for the Society moving forward in establishing professional status as entomologists. During a business session of the 1971 annual meeting, legislative committee chair Theodore R. Adkins suggested that the society proceed slowly on the decision of establishing a state registry of professional entomologists to appraise how the registry works at the national level. At the 1972 annual meeting, Adkins presented a paper on "The American Registry of Certified Entomologists—A Progress Report." He reported that ESA maintains the American Registry of



Fig. 5. South Carolina Entomological Society Insignia.

Certified Entomologists. He noted that the ESA Professional Training, Standards, and Status Committee is studying the procedures for having the Registry recognized by law in all states.

Selection of state insect. In a letter dated 25 September 1975, Society President Wm. Bruce Ezell, Jr. wrote a letter to Edward B. King, Department of Entomology and Economic Zoology, Clemson University, asking for his assistance in a project of the Society, the adoption of a South Carolina state insect. Ezell requested that King provide an illustration of the ladybeetle *Adalia bipunctata* (L.) (Coleoptera: Coccinellidae) for state publicity, because the insect appears on the Society Seal and the Society would like to promote the beetle as the state insect. In a letter dated 19 April 1976, Ezell received a letter from Senator T. D. Wise, Charleston and Georgetown Counties, Senatorial District No. 16, informing him that a House Bill existed that provides that the honey bee shall be the official insect of the state. He noted that the bill had been favorably reported from the House Agriculture and Natural Resources Committee and that there had been only one objection to the bill by Representative W. McLeod. He advised Ezell to contact members of the Charleston House Delegation to have them amend the bill to the Society's desires. In a letter dated 28 June 1975 to Louie H. Senn, Ezell forwarded copies of requested materials regarding the Society's previous effort to establish *Adalia bipunctata* (L.) as the state insect. In the materials was a letter dated 2 July 1975, addressed to Ezell from Vernon M. Kirk, who had apparently been asked to give some background information on the selection of *Adalia bipunctata* (L.) for the society's emblem (Kirk 1975). Kirk responded:

"As for the emblem of the SCES, Inc., I chose the species *Adalia bipunctata* (L.) for several reasons. It is a beneficial insect (lady beetle) that feeds on many kinds of plant lice and other pest insects, it is widespread both in South Carolina and in the United States, it is

probably known and recognized by more people than most other insects, and it has a design that is easily adapted to reproduction on an emblem. Because I am primarily interested in Coleoptera, I suppose it is logical that I chose a beetle, even after considering species from other orders. The species is distinctive, therefore not to be confused with other species."

During the 1976 joint meeting with the Georgia Entomological Society, President Ezell discussed the importance and advantages of having a state insect and the type of campaign needed to get one adopted. King unveiled a painting of a Two-Spotted Lady Beetle to be used in the campaign for the state insect. Gerald R. Carner recommended that the Society promote the Two-Spotted Lady Beetle as the state insect and solicit needed support for passage through the state legislature. A suggestion was made that the Legislative Committee prepare a short synopsis of the two-spotted lady beetle to be used with the painting to gain adoption of a state insect. At the same meeting, Adkins reported for publications committee Chair John C. Morse that a membership certificate had been designed that was 5 × 7 inches, with the lettering in black, the beetle in orange, and the outline of the state shaded green. A motion was made, which was approved that the Society accept the certificate as presented.

According to a letter dated 19 May 1976, addressed to President Ezell from Allen R. Carter, State Senator of Charleston and Georgetown Counties, the "Ladybug Bill" had been introduced to the South Carolina Senate.

During the awards banquet of the thirty-fourth meeting held on 10–11 November 1988, President William P DuBose reported on the Society's efforts in support of the state insect. The Society's choice for state insect ran contrary to that of the Dacusville Elementary School student body. The elementary school students successfully persisted in their lobbying efforts of the South Carolina Legislature and won, selecting the Carolina mantid as the South Carolina State Insect, over the Society's choice, the lady beetle. However, the Society provided guidance in properly identifying the Carolina mantid, and provided wording in the state law to properly note the species and its scientific name, *Stagmomantis carolina* (Johannson) (Mantodea: Mantidae). President Dubose provided attendees copies of the bill.

Executive Director. During the preliminary business meeting of the forty-seventh annual meeting, held 4–6 April 2001, President Paula L. Mitchell announced that progress had been made in discussions with Randy E. McWhorter to serve as the Society's first Executive Director. A nomination was made and unanimously approved that McWhorter be appointed as the Executive Director to be effective upon completion of his current term as Secretary, which expired in April 2001.

Summary

The Society has been operational now for more than 50 years, and the founding organizers of the Society would be pleased with the accomplishments of the organization over this period of time. The Society has been influenced heavily by Clemson University entomologists. The first President of the Society, James H. Cochran, was Head of the Department of Entomology at Clemson and at least 30 other presidents were Clemson University entomologists (Table 2). The annual

Table 2. Presidents of the South Carolina Entomological Society.

1956	James H. Cochran	1981	Raymond Noblet
1957	Leon (Lefty) H. Moore	1982	John A. DuRant
1958	Frank T. Arnold, Jr.	1983	Thomas E. Skelton
1959	John K. Reed	1984	Alfred R. Hopkins
1960	William J. Reid, Jr.	1985	Kenneth A. Peeples
1961	Norman Allen	1986	Clyde S. Gorsuch
1962	James N. Smith	1987	Jay W. Chapin
1963	William Carl Nettles	1988	William P. DuBose
1964	Wendell H. Tiller	1989	David R. Alverson
1965	Vernon M. Kirk	1990	John C. Morse
1966	Edward W. King	1991	J. Benjamin Kissam
1967	Julian E. Keil	1992	Steven H. Roach
1968	Charles S. Creighton	1993	John D. Hopkins
1969	Richard C. Fox	1994	B. Merle Shepard
1970	Samuel G. Turnipseed	1995	Randall P. Griffin
1971	Quinton L. Chapman	1996	Michael J. Sullivan
1972	Sidney B. Hays	1997	Mitchell E. Roof
1973	Max M. Askey, Jr.	1998	Peter H. Adler
1974	Theodore R. Adkins, Jr.	1999	Ron Burnette
1975	Julian Mikell	2000	Donald G. Manley
1976	William Bruce Ezell, Jr.	2001	Paula Levin Mitchell
1977	Louie H. Senn	2002	Gloria S. McCutcheon
1978	Augustine Day	2003	Alfred G. Wheeler
1979	Gerald R. Carner	2004	Paul M. (Mac) Horton
1980	Charles A. Thomas	2005	Wm. Michael Hood

meetings have been held at various meeting sites throughout South Carolina, but three sites have been favored over the 50-year history: Charleston (9), Clemson (11), and Florence (11) (Hood 2005).

According to annual meeting minutes, membership has varied over the 50-year history of the Society. A membership of 56 was reported for the second annual meeting held on 4 April 1957 (McAlister 1957). Other annual meeting minutes reported membership of 91 in 1963 (McAlister 1963), 168 in 1969 (Nash 1969), 144 in 1975 (Skelton 1975), 279 in 1988 (Zungoli 1988), 273 in 1995 (Mitchell 1995), 143 in 2001 (Shepard 2001), and 210 in 2004 (McWhorter 2004).

The Society has contributed in many ways to the advancement of Entomology in South Carolina. Many of the activities of the Society have been centered on the annual meetings. The program chairs have developed interesting and informative annual meetings that have resulted in professional development for Society members. The meeting program content has evolved over the years from mostly professional entomologist oral presentations during the early years to a combination of professional entomologist and student oral and poster presentations. The annual meetings have provided students, faculty, and others an opportunity to present oral and poster presentations on their research. Impressive symposia have been offered at many meetings on entomological and related scientific topics.

The J. H. Cochran Memorial Scholarship has been presented to a deserving student annually for the past 34 years. Students and nonstudents have been recognized for their outstanding oral and poster presentations for many years. The J. H. Cochran Award for Excellence in Entomology has recognized a nonstudent member triannually since 1993 for demonstrated excellence or outstanding contributions to entomology. The Society's journal, *Journal of Agricultural and Urban Entomology*, has been published for the past 23 years, offering entomologists an opportunity to publish their research in a refereed professional journal devoted to agricultural and urban entomological issues. Joint meetings with other nearby state societies, the Georgia Entomological Society (1976, 1979, 1986, 1994, 2001) and the North Carolina Entomological Society (1980, 1988), have offered the Society members an opportunity to share knowledge and gain professionally by meeting with other regional entomologists. Without a doubt, the Society has accomplished the founding members' intent of providing many opportunities during the past 50 years to promote the welfare of persons engaged in the profession of entomology in South Carolina.

Acknowledgments

It has been my pleasure to write this manuscript of the first 50 years history of the South Carolina Entomological Society, Inc. The author thanks all former historians, secretaries, and others who recorded and preserved informative and detailed minutes of Society meetings and other activities. Thanks to them, a legacy of information on prominent South Carolina entomologists was recorded and passed to me that made this manuscript possible. I would like to give a special thanks to my former mentor, good friend and fellow extension entomologist J. Benjamin Kissam who encouraged and advised me in the production of this manuscript. Kissam is one of several prominent South Carolina entomologists who have made great contributions to our Society. This is technical contribution No. 5320 of the Clemson University Experiment Station.

References Cited

- Adkins, T. R. Jr. 1968.** Meeting minutes of the Board of Directors of the South Carolina Entomological Society, Inc. held on 11 December 1968. pp. 2 (unpublished).
- Adkins Jr., T. R. & R. Noblet. 1975.** South Carolina Entomological Society Newsletter. III: 12.
- Arnold, F. T. 1956.** Notes and minutes of the first annual meeting held 28–29 March 1956 at the Hotel Columbia, Columbia, South Carolina. Prepared by Frances McAlister. pp. 3 (unpublished).
- Bellinger, R. G. 1988a.** Meeting minutes of the Board of Directors of the South Carolina Entomological Society, Clemson University, Clemson, South Carolina, 18 July 1988, pp. 2 (unpublished).
- Bellinger, R. G. 1988b.** Minutes and general report of the meeting, The second joint meeting of the North Carolina and South Carolina Entomological Societies, Inc., Landmark Resort Hotel, Myrtle Beach, S.C., 10–11 November 1988. pp. 6 (unpublished).
- Dunavan, D. 1956.** Report of the Historian of the South Carolina Entomological Society, Inc. at the meeting of the Society at the Columbia Hotel, Columbia, South Carolina 29: 2. (unpublished)
- Ellis, J. P. 1998.** Minutes of the Forty-Fourth annual meeting of the South Carolina Entomological Society, Inc. held 11–12 October 1998. pp. 3 (unpublished).
- Farrar, M. D. 1951.** Chairman, Department of Entomology and Zoology, Clemson Agricultural College, Clemson, South Carolina. Report of Entomologist meeting held at the Jefferson Hotel, Columbia, South Carolina, 26: 1. (unpublished)

- Gregory, W. W. 1970.** South Carolina entomology—historical glimpse. *Bull. Entomol. Soc. Am.* 16: 3, 166–167.
- Hood, W. M. 2005.** A chronological history of the South Carolina Entomological Society—the first fifty years. pp. 47. (unpublished).
- King, E. W. 1960.** Proceedings of the South Carolina Entomology Society. Fifth annual meeting held on 24–25 March 1960 at the Clemson House, Clemson, South Carolina. Vol. I, No. 1. pp. 16.
- Kirk, V. M. 1961.** Minutes of the sixth annual meeting of the South Carolina Entomological Society, Inc., 23–24 March 1961, Florence, SC. pp. 3. (unpublished).
- Kirk, V. M. 1975.** Letter addressed to Wm. Bruce Ezell, Jr., Department of Biology, The Citadel, Charleston, South Carolina, from Vernon M. Kirk, Entomologist, USDA/ARS, Northwestern Grain Insects Research Laboratory, Brookings, South Dakota. p. 1. (unpublished).
- Kissam, J. B. 1991.** South Carolina Entomological Society Newsletter #3. October 1991, Editor J. B. Kissam, pp. 12.
- Kissam, J. B. 1994.** South Carolina Entomological Society Archives Review. Published by J. B. Kissam. *Soc. Hist.* (November): 1994.
- McAlister, F. 1957.** Minutes of the second annual meeting of the South Carolina Entomological Society. Inc. 4: 3. (unpublished)
- McAlister, F. 1960.** Minutes of the fifth annual meeting of the South Carolina Entomological Society. Meeting held on 24–25 March 1960 at the Clemson House, Clemson, South Carolina. pp. 7. (unpublished).
- McAlister, F. 1963.** Minutes of the eighth annual meeting of the South Carolina Entomological Society. Inc. 12–13: 2. (unpublished)
- McWhorter, R. E. 2004.** Minutes of the fiftieth annual meeting of the South Carolina Entomological Society. Inc. 7: 2. (unpublished)
- Mitchell, P. L. 1995.** Membership committee chair report recorded in the minutes of the annual meeting of the South Carolina Entomological Society. Inc. 13: 1. (unpublished)
- Nash, R. F. 1969.** Membership committee chair report recorded in the minutes of the annual meeting of the South Carolina Entomological Society. Inc. 5: 1. (unpublished)
- Noblet, R. 1976.** Minutes of the twenty-first annual meeting of the South Carolina Entomological Society. DeSoto Hilton Hotel, Savannah, Georgia 17–19: 6. (unpublished)
- Reid Jr., W. J. 1958.** Historian's report of the South Carolina Entomological Society. Inc. 21: 4. (unpublished)
- Shepard, B. M. 2001.** Membership committee chair report recorded in the minutes of the annual meeting of the South Carolina Entomological Society. Inc. 6: 1. (SCESH)
- Skelton, T. E. 1975.** Membership committee chair report recorded in the minutes of the twentieth annual meeting of the South Carolina Entomological Society, Inc. p. 1. (SCESH)
- Skelton, T. E. 1990.** A memorandum from the ad hoc committee on strategic financial planning addressed to the South Carolina Entomological Society President J. C. Morse. Committee report and recommendations, 24 September 1990, pp. 4. (unpublished).
- Skelton, T. E. & J. B. Kissam. 1970.** Report of the South Carolina Entomological Society's publication committee at the Society's fifteenth annual meeting, 25–26 February 1970, Florence-Darlington Technical Education Center, Florence, SC, p. 1. (unpublished).
- Thornburg, M. A. 1960.** Minutes of the fourth annual meeting of the South Carolina Entomological Society, Florence, South Carolina, 11–12 March 1959, submitted to the Society at its fifth annual meeting on 24 March 1960, pp. 6. (unpublished).
- Zungoli, P. A. 1988.** Membership committee chair report recorded in the minutes of the annual meeting of the South Carolina Entomological Society. Inc. 11: 1. (unpublished)
-

Isolation and Characterization of Human DNA From Bed Bug, *Cimex lectularius* L., (Hemiptera: Cimicidae) Blood Meals¹

Allen L. Szalanski, James W. Austin,² Jackie A. McKern, C. Dayton Steelman, Dini M. Miller,³ and Roger E. Gold²

Department of Entomology, University of Arkansas, Fayetteville, Arkansas USA

J. Agric. Urban Entomol. 23(3): 189–194 (July 2006)

ABSTRACT The ability to identify individual human hosts based on analyses of blood recovered from hematophagous insects is beneficial for both medical and forensic entomology. Bed bugs, *Cimex lectularius* L. (Heteroptera: Cimicidae), may have several advantages over other blood-feeding arthropods for forensics because they do not remain on the host after their blood-feeding activity and remain in close proximity to a crime scene. Successful isolation, amplification, and sequencing of human DNA obtained from adult bed bugs is reported for the first time from this study. Engorged bed bugs were recovered from a human volunteer and from field collected samples from New York, New York, and Brazos County, Texas. Samples were preserved by drying, stored in 70% ethanol, or freezing at –20°C. DNA was extracted from individual insects, and polymerase chain reaction was conducted using short tandem repeat (STR), human mitochondrial DNA (mtDNA) hypervariable region (HVR1), and insect mtDNA 16S markers. Amplification of a STR marker used in forensic investigations, D18S51, a HVR1 marker, and an insect mtDNA 16S marker was successful. These results demonstrate that DNA isolated from bed bugs is qualitatively and quantitatively sufficient for DNA typing and could be helpful to identify individuals for forensic analysis.

KEY WORDS bed bug, *Cimex lectularius*, forensic entomology, STR typing, hypervariable region

Cimex lectularius L. (Heteroptera: Cimicidae) is a hematophagous insect of humans and has regained worldwide attention as the result of its recent resurgence. Because the species is easily translocated by passive dispersal and adapts to multiple hosts (Usinger 1966, Marshall 1981, Lehane 2005) when preferred food sources are unavailable, it is extremely difficult to isolate the origins of recent infestations, even with genetic tools. However, basic molecular biology tools can be used to elucidate the identity of hosts fed upon by bed bugs, hence providing the potential for forensic applications into host identity from undi-

¹Accepted for publication 7 September 2007

²Center for Urban & Structural Entomology, Department of Entomology, Texas A&M University, College Station, Texas, USA

³Department of Entomology, Virginia Tech, Blacksburg, Virginia, USA

gested blood obtained from recently-fed bed bugs. Bed bugs are obligate hematophages, with both sexes feeding on blood and blood meals are required before eclosion into subsequent molts (Usinger 1966). Thus, these insects present an excellent means for criminologists to match human hosts to specific locations in criminal investigations while excluding other hosts (e.g., birds or bats). Their basic biological necessities from hosts make them temporally and spatially predictable, forming gregarious assemblages in enclosed spaces (Reinhardt & Siva-Jothy 2007) providing ample opportunity to quickly locate recently blood-fed samples for examination.

The ability to isolate human DNA and possibly genotype blood from insects may prove invaluable to future forensic cases in which associating the presence of a person to a specific location may be needed. Human DNA has been successfully isolated and characterized from mosquitoes (Kreike & Kampfer 1999), human crab louse, *Phthirus pubis* (L.) (Anoplura: Phthiridae) (Lord et al. 1998), fly larvae (Wells et al. 2001, Zehner et al. 2004), and nitidulid beetles (DiZinno et al. 2002). To date no studies have been conducted on the isolation and characterization of human blood from bed bugs. The goal of this study was to determine if human blood can be isolated and characterized from bed bugs and establish the potential for its use in forensic criminal investigations.

Materials and Methods

In this study, a total of 35 blood-fed bed bugs were subjected to DNA analysis. Thirteen bed bugs were obtained from human dwellings in New York, New York and preserved in 90% ethanol. Seventeen bed bugs were from human dwellings in Brazos County, Texas, and preserved by freezing at -20°C . Five additional bed bugs were allowed to feed on a human volunteer for 15 min and then were maintained at 25°C for 24 h before preservation in 100% ethanol. The bed bugs that fed on the human volunteer are from a laboratory strain maintained at Virginia Tech, Blacksburg, Virginia, and blood feeding was conducted as approved by the Virginia Tech Institutional Review Board (IRB 06–165). A reference saliva sample was obtained from the human volunteer and air-dried.

Adult *C. lectularius* were morphologically identified using descriptions outlined by Usinger (1966), and voucher specimens are deposited in the Arthropod Museum, Department of Entomology, University of Arkansas, Fayetteville, Arkansas, USA.

DNA was extracted from individual insects using a Puregene DNA isolation kit D-5000A (Gentra, Minneapolis, Minnesota). Extracted DNA was resuspended in 50 μL of Tris:EDTA and stored at -20°C . polymerase chain reaction (PCR) was conducted with three primer sets. DNA from the reference saliva sample was extracted from FTA cards (Whatman International Ltd., Maidstone, United Kingdom). Single 2.0-mm punches were taken from the FTA card and prepared for PCR by washing three times with 200 μL of FTA Purification Reagent and twice with 200 μL of TE buffer, as recommended by the manufacturer.

PCR was conducted using two human and one insect marker. PCR primers L15997 and R202 amplify the human mtDNA first hypervariable part (HRV1) of the control region (Cavelier et al. 2000); primers 5'-CAAACCCGACTACCAG-CAAC-3' and 5'-GAGCCATGTTCATGCCACTG-3' amplify a 271–343 bp region of

the human D18S51 locus (Barber & Parkin 1996); and primers LR-J-13007 (5'-TTA CGC TGT TAT CCC TAA-3') (Kambhampati & Smith 1995) and LR-N-13398 (5'-CGC CTG TTT ATC AAA AAC AT-3') (Simon et al. 1994) amplify a 428-bp fragment of the insect 16S rRNA gene. The PCR reactions were conducted with 2 μ L of the extracted DNA following Szalanski et al. (2003). PCR conditions for the D18S51 marker consisted of 35 cycles of denaturing at 94°C for 30 sec, annealing at 55°C for 45 sec, extension at 72°C for 45 sec, and final extension at 72°C for 5 min (Egyed et al. 2006). The HRV1 thermocycler profile consisted of 28 cycles of denaturing at 94°C for 30 sec, annealing at 60°C for 30 sec, with extension at 72°C for 45 sec, and final extension at 60°C for 30 min. The profile for the 16S marker consisted of 35 cycles of 94°C for 45 sec, 46°C for 45 sec and 72°C for 60 sec. The 16S and HVR1 PCR products were resolved on 1% agarose gels, whereas the STR PCR products were separated by 10% PAGE per Taylor et al. (1996). Amplified DNA from the HRV1 marker was purified and concentrated using Microcon-PCR Filter Units (Millipore, Bedford, Massachusetts). Samples were sent to the University of Arkansas Medical School DNA Sequencing Facility (Little Rock, Arkansas) for direct sequencing in both directions using an ABI automated sequencer (Foster City, California). DNA sequences were aligned using BioEdit v5.89 (Hall 1999) and adjusted manually.

Results and Discussion

From the 35 bed bugs subjected to DNA analysis, DNA was successfully isolated from alcohol preserved, air dried, and frozen specimens. The mtDNA 16S marker amplified successfully for all of the bed bugs, whereas the STR D18S51 marker amplified for 32 bed bugs (Fig. 1), and the HVR1 marker amplified for 30 bed bugs. The mtDNA 16S marker can be subjected to DNA sequencing analysis, allowing population genetics studies of the bed bugs themselves (Szalanski et al. 2007). We have conducted this on bed bugs populations from Arkansas, Texas, South Carolina, Michigan, Virginia, and Illinois, and several haplotypes have been observed (unpublished data). From the five blood-fed bed bugs collected from the human volunteer, all of the STR allelic patterns and HVR1 sequences matched those obtained from the reference saliva sample. The STR marker set revealed five distinct allelic patterns from the bed bugs obtained from Brazos Co. Texas, which provides evidence that the bed bugs had fed on at least five individuals. On the basis of our findings, profiling with additional Combined DNA Index System (CODIS) STR primer sets should allow genetic characterization to individuals using bed bug blood meals. In addition, PCR product from the HRV1 amplicon from three field collected bed bugs samples was subjected to DNA sequencing and was confirmed as human DNA using a Basic Local Alignment Search Tool (BLAST) (Altschul et al. 1990) search found at the National Centers for Biotechnology Information (NCBI) website <http://www.ncbi.nlm.nih.gov/>.

Host location is of paramount importance to cimicids: Nymphs of *C. lectularius* die within a few days of emerging from the egg if they do not feed (Usinger 1966). Marx (1955) suggests that *C. lectularius* detects human hosts from as far away as 1.5 m through the use of heat cues, host kairomone(s) and/or CO₂. Laboratory colonies of bed bugs have been shown to feed every 7 d (Usinger 1966, Siva-Jothy 2006). This feeding rate was similar to a study that looked at naturally infested

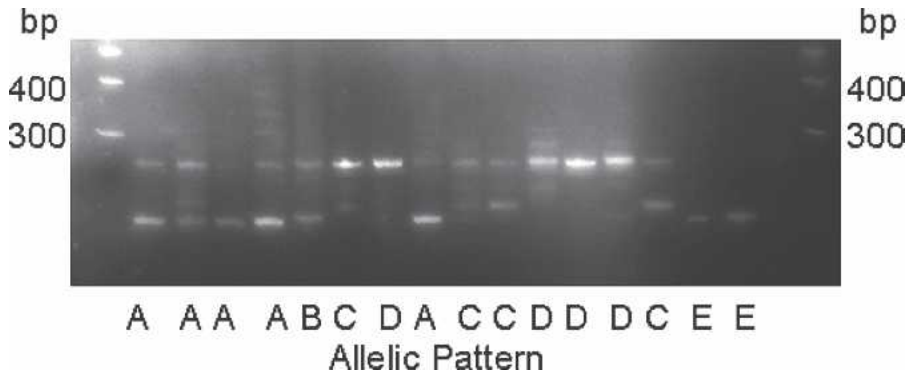


Fig. 1. STR D18S51 allelic patterns from 17 bed bugs resolved on a 5% polyacrylamide gel.

rat cages (Mellanby 1939), but considerably longer than studies of other cimicid species in hotter climates, which were observed feeding daily (Hase 1917). However, field surveys of other cimicid species suggest a feeding cycle that likely occurs between 3 and 7 d, suggesting that cimicids may not be as limited to blood meals in the field (Reinhardt & Siva-Jothy 2007). By evaluating local environmental conditions, the availability of alternative hosts, and the basic biological demands of bed bugs for any given location, the ability to use modern detection techniques such as molecular diagnostics provides yet another tool for forensic specialists to attempt to link hosts with a location. The next logical steps after identifying what hosts are being fed upon would be to attempt to identify the individual by genotyping, as would be common in any forensic investigation, and predict the duration from time of feeding to recovery of viable human DNA. This work is currently under investigation in our research laboratories. Future studies that are attempting to identify geographic relationships are also being evaluated and may yet provide additional important clues (unpublished data). Although our sample size here is small, this study demonstrates that human DNA can be isolated from bed bug blood meals and that the DNA can be characterized using both STR and DNA sequencing methods. Dried bed bugs, as well as samples preserved by freezing and in ethanol, all worked well for DNA profiling. Bed bugs may have several advantages over other blood feeding arthropods for human forensics. For example, bed bugs do not remain on the host after their blood feeding activity, and will remain in close proximity to a crime scene. These findings may have forensic applications in the use of bed bug derived blood meals for host identification and warrants future research into host-specific identification using molecular approaches.

Acknowledgment

Thanks are extended to T. McCoy, J. Meyers, T. Keefer, A. Thompson, and to local Pest Management Professionals from the College Station area for their assistance with recently

collected material. Research was supported in part by the University of Arkansas, Fayetteville, Arkansas, Agricultural Experiment Station, and the Center for Urban and Structural Entomology at Texas A&M University, College Station, Texas.

References Cited

- Altschul, S. F., W. Gish, W. Miller, E. W. Myers & D. J. Lipman. 1990. Basic local alignment search tool. *J. Mol. Biol.* 215: 403–410.
- Barber, M. D. & B. H. Parkin. 1996. Sequence analysis and allelic designation of the two short tandem repeat loci D18S51 and D8S1179. *Int. J. Legal Med.* 109: 62–65.
- Cavelier, L., E. Jazin, P. Jalonen & U. Gyllenstein. 2000. MtDNA substitution rate and segregation of heteroplasmy in coding and noncoding regions. *Hum. Genet.* 107: 45–50.
- DiZinno, J. A., W. D. Lord, M. B. Collins-Morton, M. R. Wilson & M. L. Goff. 2002. Mitochondrial DNA sequencing of beetle larvae (Nitidulidae: Omosita) recovered from human bone. *J. Foren. Sci.* 47: 1337–1339.
- Egyed, B., S. Furedi, M. Angyal, I. Balogh, L. Kalmar & Z. Padar. 2006. Analysis of the population heterogeneity in Hungary using fifteen forensically informative STR markers. *Forensic Sci. Int.* 158: 244–249.
- Hall, T. A. 1999. BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. *Nucleic Acids Symp. Ser.* 41: 95–98.
- Hase, A. 1917. Die Bettwanze *Cimex lectularius* L.: ihr Leben und ihre Bekämpfung. *Monogr. Angew. Entomol. Z. Angew. Entomol. Beiheft* 4: 1–144.
- Kambhampati, S. & P. T. Smith. 1995. PCR primers for the amplification of four insect mitochondrial gene fragments. *Insect. Mol. Biol.* 4: 233–236.
- Kreike, J. & S. Kampfer. 1999. Isolation and characterization of human DNA from mosquitoes (Culicidae). *Int. J. Legal Med.* 112: 380–382.
- Lehane, M. J. 2005. *Biology of blood-sucking Insects*. Cambridge, UK: Cambridge Univ. Press. 2nd Ed.
- Lord, W. D., J. A. DiZinno, M. R. Wilson, B. Budowle, D. Taplin & T. L. Meinking. 1998. Isolation, amplification, and sequencing of human mitochondrial DNA obtained from human crab louse, *Phthirus pubis* (L.) blood meals. *J. Forensic Sci.* 43: 1097–1100.
- Marshall, A. G. 1981. *The ecology of Ectoparasitic Insects*. London: Academic. 446 pp.
- Marx, R. 1955. Über die Wirtsfindung und die Bedeutung des artspezifischen Duftstoffes bei *Cimex lectularius* Linné. *Z. Parasitenkd.* 17: 41–73.
- Mellanby, K. 1939. The physiology and activity of the bed-bug (*Cimex letularius* L.) in a natural infestation. *Parasitol.* 31: 200–211.
- Reinhardt, K. & M. T. Siva-Jothy. 2007. Biology of the Bed Bugs (Cimicidae). *Annu. Rev. Entomol.* 52: 351–374.
- Simon, C., F. Frati, A. Beckenbach, B. Crespi, H. Liu & P. Flook. 1994. Evolution, weighting, and phylogenetic utility of mitochondrial gene sequences and a compilation of conserved polymerase chain reaction primers. *Ann. Entomol. Soc. Am.* 87: 651–701.
- Siva-Jothy, M. T. 2006. Trauma, disease and collateral damage: conflict in cimicids. *Phil. Trans. R. Soc. London B* 361: 269–275.
- Szalanski, A. L., J. W. Austin & C. B. Owens. 2003. Identification of *Reticulitermes* spp. (Isoptera: Rhinotermitidae) from South Central United States PCR-RFLP. *J. Econ. Entomol.* 96: 1514–1519.
- Szalanski, A. L., J. W. Austin, J. A. McKern, D. Miller, T. McCoy & R. E. Gold. 2007. Population genetics of the bed bug *Cimex lectularius* (Heteroptera: Cimicidae). *J. Med. Entomol.* in press.
- Taylor, D. B., A. L. Szalanski & R. D. Peterson II. 1996. Identification of screwworm

species (Diptera: Calliphoridae) by polymerase chain reaction–restriction fragment length polymorphism. *Med. Vet. Entomol.* 10: 63–70.

Usinger, R. 1966. Monograph of Cimidae (Hemiptera, Heteroptera). College Park, Maryland: Entomol. Soc. Am. 585 pp.

Wells, J. D., F. Introna Jr., G. Di Vella, C. P. Campobasso, J. Hayes & F. A. H. Sperling. 2001. Human and insect mitochondrial DNA analysis from maggots. *J. Forensic Sci.* 46: 685–687.

Zehner, R., J. Amendt & R. Krettek. 2004. STR typing of human DNA from fly larvae fed on decomposing bodies. *J. Forensic Sci.* 49: 1–4.

Journal of Agricultural and Urban Entomology

Contents

 Volume 23, No. 4

 October 2006

PHILLIP E. KAUFMAN, ALEC C. GERRY, DONALD A. RUTZ, AND JEFFREY G. SCOTT – Monitoring Susceptibility of House Flies (<i>Musca domestica</i> L.) in the United States to Imidacloprid.	195
MARGARET C. GENTZ AND J. KENNETH GRACE – A Review of Boron Toxicity in Insects With an Emphasis on Termites	201
GRETCHEN PALUCH, FREDRIC MILLER, JUNWEI ZHU, AND JOEL COATS – Influence of Elm Foliar Chemistry for the Host Suitability of the Japanese Beetle, <i>Popillia japonica</i> , and the Gypsy Moth, <i>Lymantria dispar</i>	209
WALTER E. RIEDELL AND MICHAEL A. CATANGUI – Greenhouse Studies of Soybean Aphid (Hemiptera: Aphididae) Effects on Plant Growth, Seed Yield and Composition	225
ALLEN L. SZALANSKI, JAMES W. AUSTIN, JACKIE A. MCKERN, TIM McCOY, C. DAYTON STEELMAN, AND DINI M. MILLER – Time Course Analysis of Bed Bug, <i>Cimex lectularius</i> L., (Hemiptera: Cimicidae) Blood Meals With the Use of Polymerase Chain Reaction	237
MARTIN SHAPIRO AND B. MERLE SHEPARD – The Gypsy Moth (Lepidoptera: Lymantriidae) Nucleopolyhedrovirus as a Synergist for Baculoviruses Against Beet Armyworm, Fall Armyworm and Corn Earworm (Lepidoptera: Noctuidae)	243
Author Index	253
Subject Index	254
Notice to Contributors	256
Membership Application	258

Monitoring Susceptibility of House Flies (*Musca domestica* L.) in the United States to Imidacloprid¹

Phillip E. Kaufman,² Alec C. Gerry,³ Donald A. Rutz, and Jeffrey G. Scott⁴

Department of Entomology, Comstock Hall, Cornell University, Ithaca, New York USA

J. Agric. Urban Entomol. 23(4): 195–200 (October 2006)

ABSTRACT Neonicotinoids are currently the fastest-growing class of insecticides. In 2004, imidacloprid (a neonicotinoid) was registered for house fly control in the United States. This study was undertaken to examine the relative susceptibility of laboratory strains of house flies to imidacloprid using a feeding assay, to determine the relative susceptibility of field collected house flies before imidacloprid use, and to determine the susceptibility 1 and 2 years after imidacloprid use. We found substantial variation in imidacloprid susceptibility in laboratory strains and in populations collected from the field before imidacloprid use. We observed an increase in susceptibility at one site from 2004 to 2005 and a decrease at one site from 2005 to 2006, although all populations in 2005 and 2006 were within the range of responses observed in 2004. The implications of these results to the future use of imidacloprid for house fly control are discussed.

KEY WORDS Neonicotinoid resistance, *Musca domestica*, Insecta, cross-resistance, imidacloprid

Neonicotinoids currently are the fastest-growing class of insecticides and exert their toxic effects via interactions with the nicotinic acetylcholine receptor (nAChR) (Jeschke & Nauen 2005). The most widely used neonicotinoid insecticide, imidacloprid, is highly effective against many important pests, including the house fly (Jeschke & Nauen 2005). Baits containing imidacloprid (QuickBayt®) were registered for house fly control in the United States in 2004. This represents the first granular insecticide bait for fly control that has been registered in the United States in nearly three decades. A potential limitation to the long-term usefulness of imidacloprid is the development of resistance, and early detection of resistance is considered critical for the implementation of successful resistance management strategies (Leeper et al. 1986). In addition to being an important pest, house flies are often a reliable indicator of the resistance mechanisms that develop in other pests.

Although cross-resistance to imidacloprid has been investigated in house flies

¹Accepted for publication 16 October 2007.

²Current address: Entomology and Nematology Department, P.O. Box 110620, University of Florida, Gainesville, Florida 32611-0620 USA

³Department of Entomology, University of California, Riverside, California 92521 USA

⁴Address Correspondence to: Dr. Jeffrey G. Scott, Department of Entomology, Comstock Hall, Cornell University, Ithaca, New York 14853-0901, USA. E-mail: JGS5@CORNELL.EDU

by topical application (Wen & Scott 1997), it is unknown whether cross-resistance would be conferred to imidacloprid in a feeding assay (representative of exposure to a bait). Despite the importance of imidacloprid for house fly control, it is unknown what level of variation in susceptibility exists in populations that have not been exposed to imidacloprid or whether there is any indication of resistance evolving in house fly populations since the introduction of imidacloprid.

We investigated baseline susceptibility to imidacloprid using laboratory strains of house flies that were susceptible or resistant to various other insecticides to evaluate the potential for cross-resistance to imidacloprid. We then monitored the susceptibility to imidacloprid in house fly populations from poultry, dairy, and hog facilities collected before the first use of imidacloprid in 2004, and after 1 (2005) or 2 (2005, 2006) years of use.

Materials and Methods

House flies. Nine strains of laboratory reared house flies were used. CS (Scott et al. 1996) is an insecticide-susceptible strain. SRS is a standard reference susceptible (WHO-SRS) strain created in 1961 (Keiding 1999), which was obtained from M. Kristensen (Lyngby, Denmark). OCR is a cyclodiene-resistant (*Rdl*, Kozaki et al. unpublished) strain that is susceptible to pyrethroids (Scott & Wen 1997), spinosad (Scott 1998), and indoxacarb (Shono et al. 2004). OCR was provided by Dr F.W. Plapp Jr. in 1996 and maintained under biannual selection with dieldrin. Cornell-R is an organophosphate-resistant strain with insensitive AChE. This strain was selected by tetrachlorvinphos (an organophosphate) from a colony collected in Tompkins Co., New York, in the 1970s (Tripathi & O'Brien 1973) and is maintained under biannual selection with tetrachlorvinphos. NYINDR is an indoxacarb-resistant strain derived from flies collected at seven different locations in New York in 1999 (Shono et al. 2004). LPR is a multiresistant strain with high levels of pyrethroid resistance as the result of increased oxidative metabolism by cytochrome P450 CYP6D1 (Scott & Georgiou 1986, Wheelock & Scott 1992), *kdr*, and decreased cuticular penetration (Scott & Georgiou 1986, Shono et al. 2002). Its parental colony was collected from the Learn Dairy (New York) in 1980 (Scott & Georgiou 1985). NG98 is a multiresistant strain with pyrethroid resistance resulting from monooxygenase-mediated detoxification (CYP6D1) and *kdr*. This strain was established from a colony collected in Georgia in 1998 (Kasai & Scott 2000). NYSPINR is a multi-resistant strain with high levels of resistance to spinosad (Shono & Scott 2003), established from collections made in New York in 1999 (Kaufman et al. 2001). YPER is a multiresistant strain established by the selection of permethrin from a Yumenoshima population from Japan in 1997 (pyrethroid resistance attributable to *super-kdr* and monooxygenase-mediated detoxification; Shono et al. 2002).

In 2004, house flies were collected from dairy (Alachua Co., Florida, Schuyler, Co, New York, and Tompkins Co., New York), poultry (Sullivan Co., New York, and Wayne Co., New York) and hog farms (Wake Co., North Carolina). Adult flies were captured with a sweep net from inside barns and around calf hutches, except for North Carolina where pupae were collected from around calf hutches. Field-collected animals were used to establish laboratory colonies. In 2005, house flies were collected from two dairies in Tompkins Co., New York, one poultry house in Sullivan Co., New York, and from two dairies (#1 and #2) in San Bernardino Co.,

California. The 2004 and 2005 collections were made before the start of insecticide applications for the season. In 2006, house flies were collected midseason from dairies in San Bernardino (#1) and Riverside Co., California. The three dairies in California all used QuickBayt intensively throughout the year starting in 2004. There was little or no use of QuickBayt at any of the other sites. House flies were reared using standard conditions as described previously (Scott et al. 2000). All bioassays on the field collected house flies were completed within two generations' rearing in the laboratory.

Bioassays. Susceptibility to imidacloprid was assessed using a feeding bioassay. Twenty female flies (3–5 d old) were placed in glass jars (230 mL, internal surface area is 180 cm²) and given two 2-cm pieces of cotton dental wick that were soaked in 20% sugar water containing different concentrations of imidacloprid (97.4%, Miles Inc. Kansas City, Missouri). Mortality (flies that were ataxic were considered dead) was recorded at 72 h (dental wicks were hydrated at 24 and 48 h). At least five concentrations were used for each bioassay. Bioassays were held at 25°C (approx 60% relative humidity). Bioassay data from a minimum of three replications were pooled and analyzed by standard probit analysis (Finney 1971), as adapted to personal computer use by Raymond (Raymond 1985) using Abbott's correction (Abbott 1925) for control mortality. LC₅₀ values were judged as significantly different if there was no overlap in 95% confidence intervals.

To evaluate the susceptibility of house fly populations after the introduction of imidacloprid, house flies collected in 2005 and 2006 were tested using a diagnostic concentration ($2 \times \text{LC}_{99}$ of the susceptible CS strain) of imidacloprid. At this concentration there was no survival in any of the four "imidacloprid susceptible" (CS, SRS, OCR and Cornell-R) strains. All other conditions were as described above. A minimum of 350 flies of each strain were tested. Data were arc sine transformed and means were compared by Tukey's *t*-test using inerSTAT-a software.

Results and Discussion

Susceptibility of the laboratory strains of house fly to imidacloprid varied considerably, with LC₅₀ values ranging from 9.0 µg/mL to 170 µg/mL (Table 1). There was a 2-fold difference in the imidacloprid LC₅₀ between the two susceptible strains (CS and SRS). The Cornell-R and OCR strains have been reared in the laboratory for decades, have never been selected with neonicotinoid insecticides, and have resistance mechanisms (altered acetylcholinesterase and *Rdl*, respectively) that can reasonably be expected to offer no protection to imidacloprid. Therefore, these strains could also be considered "imidacloprid susceptible." Comparison of SRS and CS with these "imidacloprid-susceptible" strains shows that the LC₅₀ values range from 9 µg/mL to 30 µg/mL, indicating more than a 3-fold variation in susceptibility to imidacloprid in "susceptible" strains. This is similar to the 2-fold difference in susceptibility noted between susceptible strains by topical application for imidacloprid (Wen & Scott 1997), fipronil (Scott & Wen 1997) and spinosad (Scott 1998). Variation in susceptibility to imidacloprid between susceptible strains makes the precise detection of resistance in field populations more difficult.

Two of the five insecticide-resistant strains (YPER and LPR) were significantly cross-resistant to imidacloprid. There was a 5.6-fold cross-resistance in the

Table 1. Toxicity of imidacloprid to nine laboratory and six field collected (2004) strains of house flies exposed using a no-choice feeding assay.

Strain	LC ₅₀ ^a (95% CI)	n	Slope (SE)	RR ^b
CS	30 (28–33)	1320	3.3 (0.2)	—
SRS	17 (14–20)	2760	3.4 (0.4)	0.6*
OCR	9.0 (7.0–11)	1160	3.1 (0.4)	0.3*
Cornell-R	11 (9.1–14)	600	2.2 (0.3)	0.4*
NYINDR	30 (26–34)	1420	1.7 (0.1)	1.0
NYSPINR	40 (33–47)	720	2.2 (0.2)	1.3
YPER	170 (130–240)	1240	2.9 (0.6)	5.6*
NG98	48 (31–53)	1120	3.5 (0.7)	1.6
LPR	58 (50–65)	560	2.9 (0.3)	1.9*
Schuyler, NY	210 (180–240)	560	2.9 (0.4)	7.0*
Sullivan, NY	240 (220–280)	1000	2.5 (0.3)	8.0*
Tompkins, NY	170 (130–230)	760	2.5 (0.5)	5.7*
Wayne, NY	92 (82–110)	2080	1.8 (0.1)	3.1*
Wake, NC	170 (150–190)	1100	2.7 (0.2)	5.7*
Alachua, FL	94 (78–110)	2060	2.1 (0.2)	3.1*

^aIn units of µg/ml.^bResistance ratio = LC₅₀ resistant strain / LC₅₀ susceptible (CS) strain.

*Significantly different from CS based on nonoverlap of 95% confidence intervals.

multiresistant YPER strain, which is highly resistant to numerous insecticides (Shono et al. 2002). The mechanism(s) responsible for this low level of cross-resistance in YPER is unknown. The multiresistant LPR strain was 1.9-fold cross-resistant to imidacloprid (relative to CS; Table 1), which is lower than the >4.2-fold cross-resistance to imidacloprid by topical application (Wen & Scott 1997). Spinosad resistance in NYSPINR is the result of an altered target site (Shono & Scott 2003), which may be a nicotinic acetylcholine receptor (Salgado & Sparks 2005). It is therefore interesting that this putative change in a nAChR did not confer cross-resistance to imidacloprid (Table 1).

All six colonies of the 2004 field collected house flies showed limited cross-resistance to imidacloprid, ranging from 3.1- to 8.0-fold (Table 1), even though neonicotinoid insecticides had not been used at these facilities. Results from the YPER and LPR strains (above), and from the Sullivan Co. site indicate that low levels (<10-fold) of cross-resistance exist in some house fly populations. Given that 64% of the house flies from the Sullivan Co., New York site survived the diagnostic concentration of imidacloprid (Fig. 1) suggests that there may be occasional facilities where imidacloprid is less effective than others.

House flies collected in 2005 showed similar susceptibility to imidacloprid, compared with 2004, with survival ranging from 12% to 35% at the diagnostic concentration (Fig. 1). One exception was the Sullivan Co., New York population, which had a significantly lower percent survival (35%) in 2005, relative to 2004 (64%).

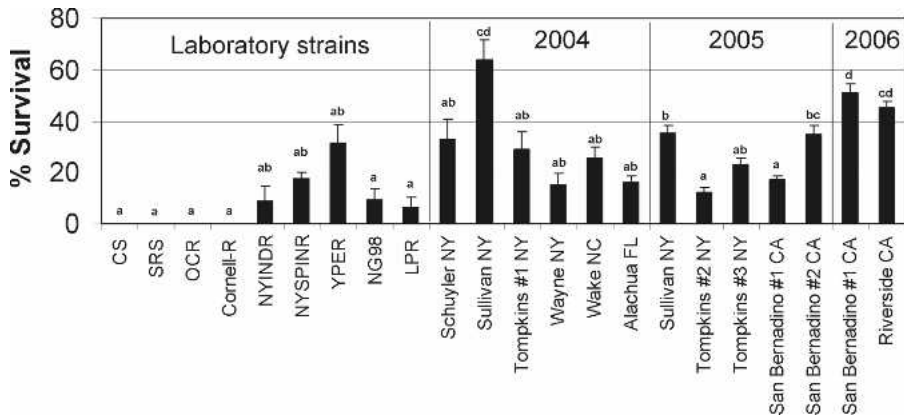


Fig. 1. Percent survival at a diagnostic concentration ($2 \times \text{LC}_{99}$ of susceptible CS strain) of imidacloprid (feeding assay) to laboratory and field collected house flies. Field collected flies came from dairies, except for Sullivan Co., New York, Wayne Co., New York, (poultry) and Wake Co. North Carolina (hog farm). Values represent the average of at least five replications. Error bars represent the standard deviation of the mean and values with different letters are significantly different ($P < 0.05$).

Overall, house flies collected in 2006 showed relatively greater percent survival, relative to 2004 and 2005 (Fig. 1). The San Bernardino Co., California #1 population had a significant increase in percent survival (51%) relative to 2005 (17%). This is consistent with the intensive use of QuickBayt at this dairy starting in 2004 and may be an indication that imidacloprid resistance is evolving at this dairy. Monitoring imidacloprid resistance at this site should be given a priority. Unfortunately, because of financial limitations, the number of sites we were able to monitor in 2006 was limited to two.

Results of this study provide important baseline data for susceptibility of field collected house flies to imidacloprid, as well as understanding the potential for cross-resistance to imidacloprid in house flies resistant to other insecticides. Further work is needed to determine why there is a 3-fold variation in toxicity between imidacloprid susceptible strains and to understand why some populations are cross-resistant to imidacloprid. In addition, it will be critically important to continue monitoring efforts, so that appropriate steps can be taken if resistance levels start to increase.

Acknowledgments

We thank D. W. Watson for the North Carolina fly collection, C. J. Geden for the Florida collection, and C. A. Leichter and A. Paul for technical assistance. This project was partially supported by Bayer Environmental Science, the Daljit S. and Elaine Sarkaria Professorship and Hatch project 414.

References Cited

- Abbott, W. S. 1925.** A method of computing the effectiveness of an insecticide. *J. Econ. Entomol.* 18: 265–267.

- Finney, D. J. 1971.** Probit Analysis. Cambridge University Press, Cambridge, UK.
- Jeschke, P. & R. Nauen. 2005.** Neonicotinoid Insecticides, pp. 53–105. *In* L. I. Gilbert, K. Iatrou and S. S. Gill [Eds.], Comprehensive Molecular Insect Science. Elsevier, New York.
- Kasai, S. & J. G. Scott. 2000.** Overexpression of cytochrome P450 CYP6D1 is associated with monooxygenase-mediated pyrethroid resistance in house flies from Georgia. *Pestic. Biochem. Physiol.* 68: 34–41.
- Kaufman, P. E., J. G. Scott & D. A. Rutz. 2001.** Monitoring insecticide resistance in house flies (Diptera: Muscidae) from New York dairies. *Pest Manag. Sci.* 57: 514–521.
- Keiding, J. 1999.** Review of the global status and recent development of insecticide resistance in field populations of the housefly, *Musca domestica* (Diptera: Muscidae). *Bull. Entomol. Res.* 89: S7–S67.
- Leeper, J. R., R. T. Roush & H. T. Reynolds. 1986.** Preventing or managing resistance in arthropods, *Pesticide Resistance: Strategies and Tactics for Management*, Committee on Strategies for the Management of Pesticide Resistant Pest Populations. National Academy Press, Washington, DC.
- Raymond, M. 1985.** Presentation d'un programme Basic d'analyse log-probit pour micro-ordinateur. *Cah. ORSTROM, Ser. Ent. Med. Parasitol.* 23: 117–121.
- Salgado, V. L. & T. C. Sparks. 2005.** The spinosyns: chemistry, biochemistry, mode of action, and resistance, pp. 137–173. *In* L. I. Gilbert, K. Iatrou and S. S. Gill [Eds.], Comprehensive Molecular Insect Science. Elsevier, Boston.
- Scott, J. G. 1998.** Toxicity of spinosad to susceptible and resistant strains of house flies, *Musca domestica*. *Pestic. Sci.* 54: 131–133.
- Scott, J. G. & G. P. Georghiou. 1985.** Rapid development of high-level permethrin resistance in a field-collected strain of house fly (Diptera: Muscidae) under laboratory selection. *J. Econ. Entomol.* 78: 316–319.
- Scott, J. G. & G. P. Georghiou. 1986.** The biochemical genetics of permethrin resistance in the Learn-PyR strain of house fly. *Biochem. Genet.* 24: 25–37.
- Scott, J. G. & Z. Wen. 1997.** Toxicity of fipronil to susceptible and resistant strains of German cockroaches (Dictyoptera: Blattellidae) and house flies (Diptera: Muscidae). *J. Econ. Entomol.* 90: 1152–1156.
- Scott, J. G., P. Sridhar & N. Liu. 1996.** Adult specific expression and induction of cytochrome P450_{pr} in house flies. *Arch. Insect Biochem. Physiol.* 31: 313–323.
- Scott, J. G., T. G. Alefantis, P. E. Kaufman & D. A. Rutz. 2000.** Insecticide resistance in house flies from caged-layer poultry facilities. *Pest Manag. Sci.* 56: 147–153.
- Shono, T. & J. G. Scott. 2003.** Spinosad resistance in the house fly, *Musca domestica*, is due to a recessive factor on autosome 1. *Pestic. Biochem. Physiol.* 75: 1–7.
- Shono, T., L. Zhang & J. G. Scott. 2004.** Indoxacarb resistance in the house fly, *Musca domestica*. *Pestic. Biochem. Physiol.* 80: 106–112.
- Shono, T., S. Kasai, E. Kamiya, Y. Kono & J. G. Scott. 2002.** Genetics and mechanisms of permethrin resistance in the YPER strain of house fly. *Pestic. Biochem. Physiol.* 73: 27–36.
- Tripathi, R. K. & R. D. O'Brien. 1973.** Insensitivity of acetylcholinesterase as a factor in resistance of houseflies to the organophosphate Rabon. *Pestic. Biochem. Physiol.* 3: 495–498.
- Wen, Z. & J. G. Scott. 1997.** Cross-resistance to imidacloprid in strains of German cockroach (*Blattella germanica*) and house fly (*Musca domestica*). *Pestic. Sci.* 49: 367–371.
- Wheelock, G. D. & J. G. Scott. 1992.** The role of cytochrome P450_{pr} in deltamethrin metabolism by pyrethroid resistant and susceptible strains of house flies. *Pestic. Biochem. Physiol.* 43: 67–77.
-

A Review of Boron Toxicity in Insects With an Emphasis on Termites¹

Margaret C. Gentz^{2,3} and J. Kenneth Grace⁴

J. Agric. Urban Entomol. 23(4): 201–207 (October 2006)

ABSTRACT Managing urban pests with “environmentally-friendly” compounds has gained importance in recent years with the global deregistration of persistent organochlorine insecticides. Boron compounds, especially boric acid, zinc borate, and disodium octaborate tetrahydrate, are considered excellent options for preserving lumber from insect attack as well as decay by fungi and bacteria. Remedial borate applications are used primarily to treat cockroach and flea infestations but have also achieved recent popularity as wood treatments to protect timber from colonization by subterranean termites. The goal of this review is to provide a synthesis of the research on the toxicity of boron in insects with a specific focus on termites and urban pests.

KEY WORDS Isoptera, Rhinotermitidae, boric acid, wood preservation

Boron, a ubiquitous element, is an essential nutrient but can be toxic at greater concentrations. Found naturally in rocks, soil, and water, boron is present in the soil at an average concentration of 10–20 ppm and always as a boron–oxygen complex. These compounds, which are mined in arid areas that previously experienced volcanic or hydrothermal processes, have been used by humans for a variety of purposes. Around 4000 years ago, the Babylonians imported tincal (mineral borax, $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) for goldsmithing. Boron has also been used in mummification rituals in ancient Egypt, in medicines, and for metalworking (Woods 1994). In modern history, borates have been useful as fire retardants in glass and ceramics, as a component of detergents, as a timber treatment to preserve lumber from wood decay, and as insecticides.

Boric acid is a stable and biologically benign compound at levels found in nature (Hall 2005). Boranes, which comprise chemical compounds of boron and hydrogen, are able to complex with a variety of substrates, especially those with adjacent hydroxyl groups, and are mild Lewis acids. Most small boronic acids are amphiphilic, containing both lipophilic and hydrophilic moieties.

Boron, which is present as a trace element in many foods, has been shown to be necessary for metabolism (Lloyd et al. 1990, Rainey et al. 1999). The most important documented role of boron is in stimulatory and inhibitory enzyme function. Without boron, plants and animals demonstrate decreased metabolic efficiency and sometimes exhibit severe symptoms of cellular level “starvation”

¹Accepted for publication 22 October 2007.

²Corresponding author (m.gentz@imb.uq.edu.au).

³Institute for Molecular Bioscience, University of Queensland, 306 Carmody Road, St. Lucia, Queensland, Australia 4072.

⁴University of Hawaii at Manoa, Department of Plant and Environmental Protection Sciences, 3050 Maile Way, Gilmore Hall 310, Honolulu, Hawaii 96822 USA.

(Lloyd et al. 1990, Woods 1994, Rainey et al. 1999). Lloyd et al. (1990) have suggested that borates in biological systems yield competitive inhibition as the result of a substrate-inhibitor complex, rather than the more commonly described enzyme-inhibitor complex. In addition to interfering with enzyme activity, research has suggested that boron deficiency in plants interferes with the integrity of root cells, in turn reducing the ability of cells to absorb ions and terminating root growth (Woods 1994).

Boron Compounds as Wood Preservatives

When used appropriately, borates are cost-effective biocides with low toxicity to vertebrates and to the environment. Toxicity is low in mammals and in vertebrates in general because of rapid excretion of excess boron by the kidneys (Lloyd et al. 1990). In rodents, the lethal dose of boron is comparable with that of table salt, or approximately 4000 mg/kg for sodium tetraborate decahydrate; during development in rats, the no-observed-adverse-effect level (NOAEL) was reported as 0.075% boric acid (55 mg/kg/day) during gestation and 0.1% boric acid (74 mg/kg/day) shortly after birth (Price et al. 1996). Boron treatments are effective at protecting wood from termite attack and colonization by harmful fungi, and also act as fire retardants at concentrations greater than 8% by weight (LeVan & Tran 1990, Barrett 1995). The diffusible nature of borates is an advantage in penetrating refractory timber species, such as spruce or Douglas fir. However, diffusion can also be a disadvantage: the main problem with the use of borate-treated lumber for construction is leaching. When placed directly in contact with damp soil, where a moisture gradient can be created, or if exposed to running water, the retention of boron in the treated wood is reduced over time (Barrett 1995).

Different boron formulations have been reported to produce similar behavioral and physiological responses in *Coptotermes formosanus* Shiraki (Isoptera: Rhinotermitidae), although anhydrous boric acid acted most rapidly (Gentz & Grace 2007). Although borates do not act as feeding repellents foraging termites have demonstrated a delayed avoidance of borate-treated wood after one week in field tests (Grace & Campora 2005, Campora & Grace 2007). In laboratory choice tests, termites fed less on borate-treated pieces than untreated samples (Ahmed et al. 2004, Kartal et al. 2004, Kartal & Ayilmis 2005). In field tests, similar results have been reported, with borate-treated lumber fed on less than untreated wood (Jones 1991, Tokoro & Su 1993a, Grace et al. 1995, Grace et al. 2006, Tsunoda et al. 2006).

The use of borates for termite prevention and control has been previously reviewed (Grace 1997). Disodium octaborate tetrahydrate (DOT), in particular, has proven to be a successful commercial treatment of structural lumber (Grace 2002). A variety of formulations for superficial application and lumber impregnation have been developed, and the solvent used may affect toxicity. Ethylene glycol, a solvent used with DOT, may act as a synergist and decrease the amount of boric acid needed for the delivery of a lethal dose (Tokoro & Su 1993b), although other work indicated that the greatest benefit of ethylene glycol was as an efficient solvent for DOT rather than as a synergist (Grace & Yamamoto 1994). The mechanism of this possible synergy is still unknown and is contrary to what might be expected, namely that the presence of an ethylene glycol/boric acid

complex would mean the boron would be bound and thus rendered less likely to bind in vivo and become biologically active.

Many efforts have been made to curtail leaching, the major setback to the widespread use of borates in unprotected exterior applications as wood treatments for the prevention of both insect and fungal damage (Manning et al. 1996). Kartal et al. (2004) successfully used chemical modifications of DOT with copolymerizations of allyl glycidyl ether and methyl methacrylate to limit boron leaching in laboratory experiments; even after severe leaching cycles, the wood specimens suffered reduced attack by *C. formosanus*. Working with blockboard veneer made of untreated fir (*Abies bornmulleriana* M.) strips between borate-treated or untreated veneers of Ekaba (*Tetraberlinia bifoliolata* Harms.), decreased damage was demonstrated by *C. formosanus* as well as by both brown-rot (*Fomitopsis palustris*) and white-rot fungus (*Trametes versicolor* (L. ex. Fr.) Quel. (Kartal & Ayirlmis 2005). In laboratory experiments with wood superficially treated with DOT in water alone or water plus ethylene glycol boron has been reported to accumulate in the wood immediately adjacent to termite galleries—perhaps as a result of termite activity (e.g., gallery construction) and/or moisture accumulation in these areas (Tokoro & Su 1993a).

Biological Role and Toxicity in Living Organisms

The common occurrence of boron in living organisms makes the mechanism of toxicity difficult to determine. While investigating whether boron facilitates wound healing in humans directly or indirectly, Nzietchueng et al. (2002) demonstrated that extracellular matrix regeneration and protein phosphorylation is increased in vivo in borate-treated human fibroblasts. In addition, they described how boron treatment, most likely through a cellular mediator, enhanced trypsin-like, collagenase, and cathepsin D activities. One of these cellular mediators was determined to be a tumor necrosis factor- α , a cytokine that is also present in the insect immune system (Franchini et al. 1996).

Although the mechanism of borate toxicity has not been fully elucidated, their toxic nature at high concentrations has been well-established. Previous research has shown that termite mortality is caused at both high and low borate concentrations more rapidly than defaunation and subsequent starvation (Khoo & Sherman 1979) would account for alone (Ahmed et al. 2004, Kartal & Ayirlmis 2005). The in vivo biochemistry of borates is complicated, particularly because the tetrahydroxyborate ion $[B(OH)_4]^-$ can complex with any molecule with two adjacent hydroxyl groups (Kim et al. 2004). In vitro, high concentrations of borates are toxic to all cells, and the toxicity appears to be a result of rapid esterification of borates with molecules of biological significance, although those molecules have not been clearly defined. The cells seem to be “starved,” pointing to a biostatic, rather than biocidal, toxicity (Lloyd et al. 1990). Borates have been reported to interact with molecules ranging from riboflavin to vitamin B6, coenzyme A, vitamin B-12, and nicotinamide adenine dinucleotide (NAD $^+$) (Lloyd et al. 1990, Williams et al. 1990, Woods 1994). Kim et al. (2004) have shown that boron-nucleotide complexes are affected by the pH of the solution and that at pH 7.4 only borate-NAD $^+$ complexes are detected. Additionally, borates can also act as purely ionic inhibitors; this may affect membrane stability and the polyols within them (Lloyd et al. 1990).

Recent research by Habes et al. (2006) suggests that boric acid induced glutathione S-transferase activity and inhibited acetylcholinesterase activity in the German cockroach, *Blattella germanica* (L.) (Blattaria: Blattellidae). Zurek et al. (2002), also working with *B. germanica*, found that the use of an entomopathogenic fungus, *Metarhizium anisopliae* (Metchnikoff) Sorokin (Deuteromycota: Hyphomycetes) in concert with topical applications of boric acid dust (12% w/w) or aqueous formulations of boric acid (0.1% w/v) as drinking water increased cockroach mortality in comparison with the use of either alone, suggesting a synergistic interaction between the two management strategies.

Organismal respiration, which can be an indicator of overall condition, has been shown to be affected in termites exposed to boron or boron-treated lumber. Nunes & Dickinson (1995) found an increase in respiration of *Reticulitermes lucifugus* (Isoptera: Rhinotermitidae) at certain boric acid concentrations (0.04, 0.64% boric acid equivalent [BAE] in paper). Toyoshima et al. (1997) reported an overall decrease in respiration in termites immediately after feeding on borate-treated timber across concentrations of boric acid (5, 10, and 20 kg/m³), although the 5 kg/m³ treatment caused respiration at 15 and 30 min to slightly increase over the initial rate. It is unlikely that a decrease in organismal respiration is the primary mechanism of action of boron compounds, because this would be expected to cause death more rapidly than has been reported from exposure to borates.

Toxicity parameters for boron have been well-established in the literature, and there is a clear dose-dependent mortality relationship. Because of the difficulty in measuring ingestion of boron by individual termites, the lethal dose of 50% of a test population (LD₅₀), as well as more general data on mortality, is commonly reported. Toyoshima et al. (1997) reported the LD₉₉ at 16 days as 930 µg/g in *C. formosanus* worker termites, and the lethal cumulative dose as 136 µg/g.

Other work with termites reports more general effects of boron on survival, for example Grace (1990) observed mortality >90% in *Reticulitermes flavipes* within 15 days when exposed to substrates containing concentrations of ≥30,000 ppm (≥30,000 µg/g) barium metaborate monohydrate, and mortality of >90% at ≥1000 ppm (≥1000 µg/g) within 30 days. Grace et al. (1992) found mortality of *C. formosanus* >90% after 16 days of exposure to DOT-impregnated filter paper at a solution concentration of 1.5 g/L (1500 µg/g), and 100% by day 12 with a 12.0 g/L (12,000 µg/g) DOT solution or 7 days with a 120.0 g/L (120,000 µg/g) solution. Su et al. (1994) determined the LD₅₀ of boric acid as 721.29 µg/g in *C. formosanus* workers and 264.02 µg/g in *R. flavipes* (Kollar). Mortality through the exchange of different boron dusts has also been reported, with boric acid powder-treated termites experiencing mortality more rapidly than DOT dusted ones in laboratory experiments when proportions as low as 10% of the experimental population were treated; the dust may be transferred via the close interactions between these social insects, e.g., mutual feeding or grooming (Grace et al. 1992). In laboratory tunneling assays, *R. flavipes* exposed to DOT (1.20 BAE) in sand showed greater mortality and less tunneling activity than those exposed to zinc borate (0.86 BAE), but the reverse pattern was found with *C. formosanus*, possibly because of differences in the tunneling behavior of these two species (Grace 1991).

Summary

Boron is an element crucial for survival, but toxic at concentrations above those necessary for physiological function. Boron, especially in its complexed

forms as DOT, zinc borate, and anhydrous boric acid, is used as a wood preservative to protect from attack by decay fungi and some insects, including termites and wood-boring beetles. Boron is also used in insecticide formulations against urban insects like cockroaches and fleas. Despite the benefits of boron as an insecticide and commercial use for over 200 years, the mechanism for toxicity caused by boron is not well understood. Recent research has focused on the effects of boron in termites and cockroaches, both physiological and behavioral, and has provided an excellent basis for future work on the mechanism of action. As a crucial component of biological function, boron serves a role in plants, insects, and animals, and work focused on understanding both beneficial and toxic mechanisms will surely be complex and exciting.

Acknowledgments

Research involved in preparing this review was supported by USDA-ARS Specific Cooperative Agreement 58-6615-4-237, and McIntire-Stennis funds for forestry research administered by the College of Tropical Agriculture and Human Resources, University of Hawaii at Manoa.

References Cited

- Ahmed, B. M., J. R. J. French & P. Vinden. 2004.** Evaluation of borate formulations as wood preservatives to control subterranean termites in Australia. *Holzforschung* 58: 446–454.
- Barrett, S. [Ed]. 1995.** Zinc borate in wood composites. *Addit Polymer*. 9:6.
- Campora, C. E. & J. K. Grace. 2007.** Foraging behavior of the Formosan subterranean termite (Isoptera: Rhinotermitidae) in response to borate treated wood. International Research Group on Wood Preservation. Stockholm, Sweden. IRG/WP 07-10605. 13 pp.
- Franchini, A., J. A. Miyan & E. Ottaviani. 1996.** Induction of ACTH- and TNF- α -like molecules in the hemocytes of *Calliphora vomitoria* (Insecta, Diptera). *Tissue Cell* 28: 587–592.
- Gentz, M. C. & J. K. Grace. 2007.** Different boron compounds elicit similar responses in *Coptotermes formosanus* (Isoptera: Rhinotermitidae). *Sociobiology* 50: 633–641.
- Grace, J. K. 1990.** Oral toxicity of barium metaborate to the Eastern subterranean termite (Isoptera: Rhinotermitidae). *J. Entomol. Sci.* 25: 112–116.
- Grace, J. K. 1991.** Response of Eastern and Formosan subterranean termites (Isoptera: Rhinotermitidae) to borate dust and soil treatments. *J. Econ. Entomol.* 84: 1753–1757.
- Grace, J. K. 1997.** Review of recent research on the use of borates for termite prevention. Proceedings of the 2nd International Conference on Wood Protection With Diffusible Preservatives and Pesticides. Forest Products Society, Madison, Wisconsin. pp. 85–92.
- Grace, J. K. 2002.** Hawaiian experience with treated building components, pp. 305–309. *In: Enhancing the Durability of Lumber & Engineered Wood Products*. Forest Products Society, Madison, Wisconsin.
- Grace, J. K. & R. T. Yamamoto. 1994.** Simulation of remedial borate treatments intended to reduce attack on Douglas-fir lumber by the Formosan subterranean termite (Isoptera: Rhinotermitidae). *J. Econ. Entomol.* 87: 1547–1554.
- Grace, J. K. & C. E. Campora. 2005.** Food location and discrimination by subterranean termites (Isoptera: Rhinotermitidae), pp. 437–441. *In* C.-Y. Lee and W. H. Robinson, eds. Proceedings of the 5th International Conference on Urban Pests, Executive Committee of the International Conference on Urban Pests, Singapore.
- Grace, J. K., R. T. Yamamoto & M. Tamashiro. 1992.** Resistance of borate-treated Douglas-fir to the Formosan subterranean termite. *Forest Prod. J.* 42: 61–65.

- Grace, J. K., K. Tsunoda, T. Byrne & P. I. Morris. 1995.** Field evaluation of borate-treated lumber under conditions of high termite hazard. *In* Wood Preservation in the '90s and Beyond. Forest Products Society, Madison, Wisconsin, Proceedings No. 7308: 240.
- Grace, J. K., A. Byrne, P. I. Morris & K. Tsunoda. 2006.** Performance of borate-treated lumber after 8 years in an above-ground termite field test in Hawaii. International Research Group on Wood Preservation. Stockholm, Sweden. IRG/WP 06-30390. 8 pp.
- Habes, D., S. Morakchi, N. Aribi, J.-P. Farine & N. Soltani. 2006.** Boric acid toxicity to the German cockroach, *Blattella germanica*: alterations in midgut structure, and acetylcholinesterase and glutathione S-transferase activity. *Pestic. Biochem. Physiol.* 84: 17–24.
- Hall, D. G., ed. 2005.** Boronic acids: preparation and applications in organic synthesis and medicine. Wiley-VCH: Weinheim, Germany. 549 pp.
- Jones, S. 1991.** Field evaluation of boron as a bait toxicant for control of *Heterotermes aureus* (Isoptera: Rhinotermitidae). *Sociobiology* 19: 187–207.
- Kartal, S. N. & N. Ayrlimis. 2005.** Blockboard with boron-treated veneers: laboratory decay and termite resistance tests. *Int. Biodeterior. Biodegradation* 55: 93–98.
- Kartal, S. N., T. Yoshimura & Y. Imamura. 2004.** Decay and termite resistance of boron-treated and chemically modified wood by in situ co-polymerization of allyl glycidyl ether (AGE) with methyl methacrylate (MMA). *Int. Biodeterior. Biodegradation* 53: 111–117.
- Khoo, B. K. & M. Sherman. 1979.** Toxicity of chlorpyrifos to normal and defaunated Formosan subterranean termites. *J. Econ. Entomol.* 72: 298–304.
- Kim, D. H., K. F. Faull, A. J. Norris & C. D. Eckhert. 2004.** Borate-nucleotide complex formation depends on charge and phosphorylation state. *J. Mass Spectrom.* 39: 743–751.
- LeVan, S. L. & H. C. Tran. 1990.** The role of boron in flame-retardant treatments, pp. 39–41. *In* M. Hamel [Ed.], 1st International Conference on wood protection with diffusible preservatives, 28–30 November 1990, Nashville, Tennessee. Forest Products Research Society, Madison, Wisconsin. Proceedings No. 47355.
- Lloyd, J. D., D. J. Dickinson & R. J. Murphy. 1990.** The probable mechanisms of action of boric acid and borates as wood preservatives. IRG/WP Document 90-1450. International Research Group on Wood Protection. Stockholm, Sweden. 21 pp.
- Manning, M. J., J. Lloyd & M. Schoeman. 1996.** The future of diffusible preservative and pesticide systems, pp. 157–168. *In* Proceedings of the 2nd International Conference on Wood Protection with Diffusible Preservatives and Pesticides. Forest Products Society, Madison, Wisconsin.
- Nunes, L. & D. J. Dickinson. 1995.** The influence of boric acid on respiratory quotients and methane production of subterranean termites. IRG/WP Document 95-10136. International Research Group on Wood Protection. Stockholm, Sweden. 9 pp.
- Nzietchueng, R. M., B. Dousset, P. Franck, M. Benderdour, P. Nabet & K. Hess. 2002.** Mechanisms implicated in the effects of boron on wound healing. *J. Trace Elem. Med. Biol.* 16: 239–244.
- Price, C. J., P. L. Strong, M. C. Marr, C. B. Myers & F. J. Murray. 1996.** Developmental toxicity NOAEL and postnatal recovery in rats fed boric acid during gestation. *Toxicol. Sci.* 32: 179–193.
- Rainey, C. J., L. A. Nyquist, R. E. Christensen, P. L. Strong, D. Culver & J. R. Coughlin. 1999.** Daily boron intake from the American diet. *J. Am. Diet. Assoc.* 99: 335–340.
- Su, N.-Y., M. Tokoro & R. H. Scheffrahn. 1994.** Estimating oral toxicity of slow-acting toxicants against subterranean termites (Isoptera: Rhinotermitidae). *J. Econ. Entomol.* 87: 398–401.
- Tokoro, M. & N.-Y. Su. 1993a.** Wood protection by surface treatment of two borate preservatives, Tim-Bor and Bora-Care, for the Formosan subterranean termite and eastern

- subterranean termite. IRG/WP 93-10044. International Research Group on Wood Protection. Stockholm, Sweden. 7 pp.
- Tokoro, M. & N.-Y. Su. 1993b.** Oral toxicity of Tim-Bor, Bora-Care, boric acid, and ethylene glycol against the Formosan subterranean termite and the Eastern subterranean termite. IRG/WP Document 93-10045. International Research Group on Wood Protection. Stockholm, Sweden. 6 pp.
- Toyoshima, I., T. Yoshimura, K. Tsunoda & M. Takahashi. 1997.** Comparative toxicity, residual nature and effect on respiration of boron compounds in a lower termite, *Coptotermes formosanus* Shiraki (Isoptera: Rhinotermitidae). Mater. Organ. 31: 217-226.
- Tsunoda, K., A. Byrne, P. I. Morris & J. K. Grace. 2006.** Performance of borate-treated lumber after 10 years in a protected, above-ground field test in Japan (final report). IRG/WP 06-30395. International Research Group on Wood Preservation. Stockholm, Sweden. 8 pp.
- Williams, L. H., S. I. Sallay & J. A. Breznak. 1990.** Borate-treated food affects survival, vitamin B-12 content, and digestive processes of subterranean termites. IRG/WP Document 90-1448. International Research Group on Wood Protection. Stockholm, Sweden. 16 pp.
- Woods, W. G. 1994.** An introduction to boron: history, sources, uses, and chemistry. Environ. Health Perspect. 102(Supplement 7): 5-11.
- Zurek, L., D. W. Watson & C. Schal. 2002.** Synergism between *Metarhizium anisopliae* (Deuteromycota: Hyphomycetes) and boric acid against the German cockroach (Dictyoptera: Blattellidae). Biol. Control. 23: 296-302.

Influence of Elm Foliar Chemistry for the Host Suitability of the Japanese Beetle, *Popillia japonica*, and the Gypsy Moth, *Lymantria dispar*¹

Gretchen Paluch, Fredric Miller,² Junwei Zhu, and Joel Coats

Department of Entomology, Iowa State University, Ames, Iowa, 50011 USA

J. Agric. Urban Entomol. 23(4): 209–223 (October 2006)

ABSTRACT The Asian elm trees that are closely associated with the David complex, *Ulmus davidana* (Planch), *U. propinqua* (Koidz), *U. japonica* (Rehd.), and *U. wilsoniana* (Schneid), show substantial levels of disease resistance and tolerance. Studies have shown that these species suffer moderate levels of injury from the adult Japanese beetle, *Popillia japonica* (Newman) (Coleoptera: Scarabaeidae). The research presented here reports the influence of elm leaf chemistry on the feeding activity of two generalist herbivores, the Japanese beetle and gypsy moth, *Lymantria dispar* (L.) (Lepidoptera: Lymantriidae). Elm leaf chemistry was surveyed for 19 species, 5 cultivars, and 2 varieties that are known to vary in susceptibility to Japanese beetle feeding damage. Extracts were analyzed for significant differences in lipid, phenolic, terpene, and alkaloid diversity; those data were then compared with previous findings in Japanese beetle laboratory and field studies, and a gypsy moth no-choice developmental study. Elms that belong to the *U. davidiana* complex were more suitable for gypsy moth development compared with other elms tested. These elms typically had greater levels of lipid and overall blend complexity in the leaf chemical extracts. Elm leaf lipid extractions were evaluated for deterrent effects on Japanese beetle feeding activity in a greenhouse study and only three (*U. propinqua* var *suberosa*, 'Commendation' and *U. wilsoniana*) deterred Japanese beetle feeding. Elm tree susceptibility to gypsy moth and Japanese beetle injury was correlated with the leaf chemistry surveyed in this study; more specifically elm leaf lipids influenced Japanese beetle and gypsy moth preference for the Asian elms tested.

KEY WORDS Elm, Japanese beetle, gypsy moth

Several studies have evaluated the effect of phytochemicals (phenolics, alkaloids and primary metabolites) on the feeding activities of the Japanese beetle and gypsy moth (Ladd 1986, Barbosa & Krischik 1987, Ladd 1988, Stamp & Harmon 1991, Fulcher et al. 1996, Keathley et al. 1999, Potter & Held 1999); however, there are many questions remaining about the direct and indirect (tritrophic) interactions between leaf chemistry and herbivore feeding behavior. The influence of other secondary chemicals (e.g., terpenes) on insect herbivores

¹Accepted for publication 9 January 2008.

²Morton Arboretum, Lisle, Illinois, 60532 USA

also requires further study. Some researchers have addressed the release of feeding-induced volatile terpenes and green-leaf compounds (C-6) and looked for effects on other trophic levels (Loughrin et al. 1998, Havill & Raffa 2000). Field studies measuring volatiles released from crabapple and maple trees varying in susceptibility to adult Japanese beetles concluded that individuals were attracted to a complex blend of terpenes, aliphatics, and aromatics (Spicer et al. 1995, Loughrin et al. 1996, 1997, Potter & Held 2002), many of which are known to be produced by a number of plant families (including both preferred and nonpreferred host plants). As the complexity and volume of volatiles emitted by a plant increase, so does the attractiveness of a plant to Japanese beetles (Loughrin et al. 1998). The specific chemical cues adults use to find food sources and how these messages are carried and conserved through the environment to signal receptive adults are still poorly understood. The direct effects of terpenes and other phytochemicals in determining host plant suitability of the gypsy moth and Japanese beetle are also uncertain.

With the development and introduction of new cultivars, it is becoming increasingly important to evaluate their suitability for generalist insect herbivores to predict their success in the urban environment. Several studies have evaluated different elms for injury by Japanese beetle (Miller et al. 1999, Miller 2000), although few data are available on gypsy moth. These studies reported that Asian elm species (*Ulmus glaucescens* (Franch), *U. lamellosa* (C. Wang et S.L. Chang, ex L. K. Fu et al.), and *U. macrocarpa* (Hance) were less preferred by adult beetles. *Ulmus davidiana* (Planch), *U. japonica* (Rehd.), *U. wilsoniana* (Schneid), and *U. propinqua* (Koidz), all elms belonging to the David complex, and the Szechuan elm, *U. szechuanica* (Fang), range from moderate to high levels of preference by the Japanese beetle.

In our current study, we surveyed leaf chemistry of 26 different elms (19 species, 5 cultivars, and 2 varieties) and made comparisons to past studies on feeding activity of Japanese beetles and more recent laboratory studies on the gypsy moth host plant suitability. Because there is little information in the literature on elm leaf chemistry, the first portion of this research reports qualitative data on leaf chemistry comparisons of several elm trees. Trends in Asian elm leaf chemistry were studied for relationships to Japanese beetle feeding activity. Recent assessment of gypsy moth growth and development on a selection of elm species and cultivars was completed, and then it was also analyzed for trends linking elm leaf chemistry to insect herbivore feeding activity. Because these analyses showed the strongest relationship between elm leaf lipid diversity and insect herbivore feeding activity, lipid extractions were performed and then used in a greenhouse bioassay to identify elms with deterrent surface lipids.

Materials and Methods

Evaluation of elm leaf chemistry and insect host suitability. *Chemical analysis.* Leaf material from 26 elms (19 species, 5 cultivars, and 2 varieties) was collected at the Morton Arboretum, Lisle, Illinois, on 17 June 2004 and sent to the Pesticide Toxicology Laboratory at Iowa State University for extraction. Leaf samples were weighed into 7-g quantities (total wet weight) and then placed into 500-mL French square bottles for solvent extraction. Hexane (300 mL) was used to extract nonpolar low-molecular-weight leaf chemicals for analysis. Ex-

tractions were conducted in a dark environment and maintained at room temperature ($22 \pm 1^\circ\text{C}$) for a 72-h period. Linalool and transcaryophyllene (Sigma-Aldrich, St. Louis, Missouri) were used as internal standards for mono and sesquiterpenes. Extracts were then filtered using Whatman no. 1 qualitative filter paper to remove leaf tissue. Samples were concentrated under a gentle stream of nitrogen and brought to a final 1-mL volume. Because of large amounts of waxes in the extracts, a final filtration step was performed with 0.45-micron MAGNA nylon filters (Osmonics, Inc., Minnetonka, Minnesota) before analysis with gas chromatography with flame-ionization detection (GC-FID) and GC with mass spectrometry (GC-MS). All sample extractions and GC preparation occurred over a 4-d period.

Elm profiles were determined first by analysis with a Hewlett Packard 5,890 Series II gas chromatograph with a 30-m $\times$ 0.25mm i.d. DB-5 column (J & W Scientific, Folsom, California) and flame ionization detection. Samples were injected at 250°C onto an initial column temperature of 75°C . This temperature was held for 3 min, then ramped at $10^\circ\text{C}/\text{min}$ to final temperature of 295°C , holding for 15 min.

Samples also were analyzed on a Hewlett Packard 5,890 Series II gas chromatograph interfaced to a Hewlett Packard 5,972 Mass Selective Detector. The gas chromatograph was equipped with either a DB-5 or a DB-wax column (30 m $\times$ 0.25 mm i.d., J & W Scientific). The injector temperature was 250°C , and the split valve was opened 1 min after injection. The oven initial temperature was set at 50°C for 3 min, and then increased to 250°C at a programmed rate of $15^\circ\text{C}/\text{min}$. Mass spectra were recorded from 30 to 550 a.m.u. with electronic impact ionization at 70 eV. The assignments of chemical identities to the chemical compounds detected were confirmed by comparison of the retention indices and mass spectra with those of authentic standards and reference spectra in a mass spectral library (Wiley 138K, John Wiley and Sons, Inc.). Some standards were available commercially and purchased including: eugenol (Sigma Aldrich, St. Louis, Missouri) and palmitic acid (Fisher Scientific, Hanover Park, Illinois).

The chemical components from the elm leaf chemistry analysis were classified into four chemical classes: phenolic, terpenoid, lipid, and alkaloid. The diversity of compounds in each class was analyzed in relation to the host plant suitability data from recently conducted gypsy moth developmental studies and Japanese beetle no-choice feeding trials and field defoliation observations (previously reported by Miller et al. 1999) using a principal component analysis with PROC PRINCOM (SAS Institute 2003). A correlation matrix was used to equalize the variances between different units of measurement. Variables that showed high levels of correlation and/or were large contributors to the first two eigenvectors were further analyzed for levels of significance in a multivariate analysis (as dependent variables) with PROC GLM (SAS Institute 2003), followed by univariate analyses containing only single variables. Elm tree parentage served as the independent variable, which included classifying elms according to whether they belonged to the David complex, including *U. davidiana*, *U. japonica*, *U. wilsoniana*, *U. propinqua*, and the cultivars 'Triumph' (*U. japonica-wilsoniana* $\times$ *U. japonica-pumilia*) and 'Accolade' (*U. japonica-wilsoniana*). *Ulmus szechuanica* was also grouped with the David complex in this analysis (Fu 1980).

Gypsy moth developmental study. Gypsy moth laboratory no-choice feeding studies were conducted in July of 2001 at the Morton Arboretum (Lisle,

Illinois) by placing one neonate larva into a 60-mL (2.5×6.5 cm) plastic tube with a snap cap lid and fresh leaf material provided every 2 days. Ten replicates of each elm (12 species and 4 cultivars) were run until adult emergence or until the larva or pupa died. Three trees for each of the 16 types of elm were used, each with 30 larvae total, to account for within-species/cultivar variability. All trees were growing on the grounds of the Morton Arboretum under natural conditions. The larvae were provided with fresh leaf material every 2 d. Vials were checked daily for larval mortality, pupation, and adult emergence. The vials containing the larvae were held in an incubator under a photoperiod of 16:8 (L:D)h at 25°C. Elm leaves for the no-choice laboratory bioassays were randomly collected from ground level from all four cardinal directions on a test tree and held in cold storage in plastic bags at 5°C for a maximum of 2 d. Measures of host suitability for gypsy larvae and adults were defined by larval longevity, percentage pupation, pupal fresh weight, and percentage of adult emergence. Student–Newman–Keuls multiple comparison tests were used to compare gypsy moth development between elms. Natural log transformations were performed on pupal fresh weight and percentage of adult emergence for principal component analysis with elm leaf chemistry data.

Japanese beetle laboratory and field studies. Laboratory no-choice bioassay and field defoliation studies were conducted on adult Japanese beetles (July–August) during the 1997, 1999, and 2000 growing seasons at the Morton Arboretum, Lisle, Illinois. The methodology and results have been previously reported (Miller et al. 1999). The measures of suitability for each elm (12 species and 2 cultivars) in the no-choice studies were mean percentage leaf tissue removed and mean dried fecal pellet weights. Trees evaluated in the field defoliation study were rated for evidence of feeding, using a scale of 0–4 as follows: 0, no feeding; 1, very light feeding (1–10% of canopy with damage); 2, light feeding (11–25%); 3, moderate feeding (26–50%); and 4, heavy feeding (>50%).

Japanese beetle feeding deterrent study. The elm leaf chemistry analysis detected the strongest relationship between insect feeding activity and lipid diversity. Japanese beetles were selected for further testing in a greenhouse bioassay to assess the deterrent effect of elm leaf lipids on the feeding behavior of the Japanese beetle. Adult beetles (mixed sexes) were collected in mid-July from wild populations in Ames, Iowa with three Japanese beetle traps, each containing the lure: 2-phenethyl propionate, eugenol, geraniol, and (R,Z)-5-(1-decenyl)-dihydro-2(3H)-furanone (Trécé, Inc., Adair, Oklahoma). Before testing, beetles were sustained in the greenhouse on a diet of ‘Tiffany’ and ‘Mirandy’ hybrid tea roses and peach fruit.

Extracts of elm leaf lipids from 23 different elms (16 species, 5 cultivars, and 2 varieties) were prepared from leaf material collected at the Morton Arboretum. Multiple trees (three) of each elm were sampled to account for variation within species, cultivar, or variety. Leaves were weighed into 2.5-g samples for lipid extraction, which comprised 6–7 large leaves of each. Leaf area was measured using an LI-3100 Area Meter (LI-COR, Inc., Lincoln, Nebraska). Lipid extractions were completed with a 10-s leaf dip in 40 mL of hexane. Extracts were then concentrated to a final 5-mL volume under nitrogen.

Whatman no.1 qualitative filter paper discs (radius = 0.95 cm) were soaked in a saturated sucrose solution and then allowed to air dry before application of the leaf lipid extract. The saturated sucrose solution was used to increase levels of

beetle feeding activity. The amount of extract to be applied to the filter papers (200 μ L) was calculated by using the average surface area from the leaf material extracted. The treated paper disc was suspended by a stainless-steel pin in the middle of the paper cup arena at a height of 3 cm. A moist cotton wick was placed in each feeding arena to serve as a water source. Beetles were maintained in the arena with a mesh screen covering the top.

Feeding arenas were placed in a greenhouse maintained at 27°C, and beetles were allowed to feed for a maximum of 5 d. Each arena was checked daily for evidence of beetle feeding and mortality. Frequency of beetle feeding activity was also recorded. Replicates containing beetles that did not survive the initial 48 h, of testing were not included in the final analysis. In most cases, feeding was observed within 24 h, and there was low beetle survival at the end of the 5-d test period. Lipid-treated discs were placed into envelopes, and the area consumed was measured by calculating the difference between complete circle and the part of the disc consumed. Adobe Photoshop CS2 (Version 9.0) and an HP Scan Jet 4,670 digital flatbed scanner with a linear sensor (Hewlett Packard Company, Palo Alto, California) were used to quantify surface area of the lipid extract-treated filter paper discs.

Data on surface-area consumption were analyzed with PROC MIX (SAS Institute 2003). A nested hierarchy was used to account for the variance component from within elm tree species. Pairwise comparisons using least squares means are reported along with the adult beetle feeding frequency noted by presence or absence of beetle feeding activity in individual arenas.

Results

Evaluation of elm leaf chemistry and insect host suitability. *Chemical analysis.* Results from the survey of leaf chemistry yielded more than 100 different compounds. Major components are presented for specific elm comparisons that are relevant in the gypsy moth developmental study (Table 1). More components from the elm leaf volatile extracts are presented for specific species, cultivars, and varieties, based on relevance to the Japanese beetle lipid-disc deterrent study (Table 2). The major components recovered included chemicals that are in the terpene, phenolic, lipid, and alkaloid groups.

Gypsy moth developmental study. Assessment of gypsy moth growth and development on selected elms is reported in Table 3. Gypsy moth survival on the elms included in this survey was low and shows that 13 of these are not suitable host plants. Significant levels of survival were observed on the Gaussen elm (*U. gaussenii*), 'Danada Charm' (*U. japonica* $\times$ *wilsoniana*), 'Commendation' (Accolade $\times$ *U. pumila* - *minor*), and a direct comparison of the elm leaf chemistry extracts of these elms is provided (Table 2).

A principal component analysis on gypsy moth larval longevity, pupal mass, and percentage adult emergence from 16 of the elms yielded two eigenvectors that accounted for 87% of the variability in the data. When the data were analyzed with a multivariate analysis with elm parentage, by assigning elms belonging to the David complex as the independent variable, there was a significant effect attributable to parentage ($F = 3.48$; $df = 3, 12$; $P = 0.050$).

Elm trees belonging to the David complex generally had a larger diversity of leaf terpenes, phenolics, alkaloids, and lipids. More specifically, the diversity

Table 1. Comparison of nonpolar compound mass spectra extracted during elm leaf soaks of suitable hosts for complete development of the gypsy moth in laboratory no-choice bioassays: Gaussen elm (*Ulmus gaussenii*), 'Danada Charm' (*U. japonica* x *wilsoniana*), 'Commendation' (Accolade x *U. pumila* - minor).

Chemical	Mass spectral	Percentage of fixed total		
	Match quality, %	<i>U.</i> <i>gaussenii</i>	'Danada charm' ^a	'Commendation' ^a
Pentacosane	93	—	—	0.6
Docosane	99	—	—	2.6
Octadecane	95	—	3.9	—
Eicosane	96	3.6	0.7	0.6
n-hexadecanoic acid	90	—	11.6	15.9
(z,z)-9,12-octadecadienoic acid	95	—	57.3	—
(z,z,z)-9,12,15- octadecatrienoic acid, me ester	95	21.0	—	1.5
2-me-,2,2-dimethyl-1- (2-hydroxy-1- methylethyl)propyl esterpropanoic acid	59	3.4	—	—
2-me-,2-ethyl-3- hydroxyhexyl ester propanoic acid	78	6.1	—	—
11,14,17-eicosatrienoic acid, me ester	86	—	1.1	—
2-me-,2-ethyl-1-propyl-1,3- propanediyl ester propanoic acid	43	—	—	0.6
butanoic acid, butyl ester	83	—	—	0.8
butanoic acid, butyl ester	78	—	0.1	—
1,2-epoxyundecane	50	—	1.2	—
1,3,5-tri(methylene)- cycloheptane	38	—	—	0.9
9,12,15-octadecatrienal	91	—	—	50.7
pentadecanal	91	—	—	0.9
(z)-9-octadecenamide	86	9.4	—	1.6
(e)-3-tetradecen-5-yne	78	—	—	3.8
2-me-3-buten-2-ol	32	—	0.9	—
2me-6-methylene-7-octen-2-ol	38	—	1.3	—
1,3-dimethyl-, (z)-cyclohexanol	43	—	1.5	—
Hexylpentyl ether	47	—	1.0	—
Germacrene d	95	—	0.8	—
(z)-linalool oxide	90	—	0.8	0.8
3,7-dimethyl-1,5,7-octatrien- 3-ol	47	—	0.8	0.8

Table 1. Continued.

Chemical	Mass spectral	Percentage of fixed total		
	Match quality, %	<i>U.</i> <i>gaussenii</i>	‘Danada charm’ ^a	‘Commendation’ ^a
5-ethenyltetrahydro- α,α,5-trimethyl-, (z)-2- furanmethanol	91	4.1	1.2	0.9
3,5,5-trimethyl-4-(3- oxobutyl)-2-cyclohexen- 1-one	97	—	0.7	—
2,6-dimethyl-3,7-octadien- 2,6-diol	78	5.0	2.4	0.9
α -farnesene	95	3.8	0.9	0.7
4-(2,6,6-trimethyl-1- cyclohexen-1-yl)-2-buten- 2-one	84	—	—	0.6
2H-pyran-3-ol,6- ethenyltetrahydro-2,2,6- trimethyl	86	—	—	0.8
3-buten-2-ol,2-me	43	—	—	0.6
1,7-octadiene-3,6-diol,2,6- dimethyl-	87	—	—	0.8
[(2-phenylethenyl)sulfonyl]- benzene	43	—	—	0.6
4-me-2,7-octadiene	49	—	—	0.8
2-oxacyclo[2.2.2]octan-6- ol,1,3,3-trimethyl-,acetate	32	—	—	0.9
5-methyl-2-(1-methylethyl)-, (1 α, 2β,5β) cyclohexanol	58	—	—	0.4
9-hydroxyfarnesol	56	—	—	0.8
Caryophyllene oxide	43	—	—	0.7
Methyl salicylate	95	4.0	—	—
α-me-, α,-[4-me-3- pentynal] oxiranemethanol	86	2.1	—	—
phenylethyl alcohol	93	5.6	3.5	2.1
3-allyl-6-methoxyphenol	97	5.8	—	—
eugenol	94	—	1.1	1.8
3-isopropoxy-5-me-phenol	72	—	—	0.4
2-amino-4-methylpyrimidine	50	1.8	1.6	1.3

Leaf samples were collected at the Morton Arboretum, Lisle, Illinois, on 17 June 2004.

^aElms belonging to, or closely associated with the David complex.

Table 2. Comparison of nonpolar compound mass spectra extracted during elm leaf soaks of nonpreferred food sources by the Japanese beetle in the lipid extract-treated disc greenhouse study: *Ulmus wilsoniana*, ‘Commendation’; (Accolade × *U. pumila* – *minor*), and *U. propinqua* var *suberosa*.

Chemical	Mass spectral	Percentage of fxd total		
	Match quality	<i>U.</i> <i>wilsoniana</i> ^a	‘Commendation’ ^a	<i>U. propinqua</i> var <i>suberosa</i> ^a
Octadecane	95	3.8	—	—
Eicosane	96	—	0.6	1.8
Heneicosane	95	—	—	13.9
Docosane	99	—	2.6	—
Pentacosane	93	5.3	0.6	—
Heptacosane	98	8.9	—	—
Hexatriacontane	89	—	—	1.3
n-hexadecanoic acid	90	0.9	15.9	8.1
(z,z) 9,12- octadecadienoic acid	90	24.5	—	—
2-me-,2-ethyl-3- hydroxyhexyl ester propanoic acid	74	—	—	2.7
2-me-,2,2-dimethyl-1-(2- hydroxy-1-methylethyl) propyl esterpropanoic acid	59	—	—	1.5
butyl ester butanoic acid	83	—	0.8	—
11,14,17-eicosatrienoic acid, me ester	87	—	—	1.4
2-me-2-ethyl-1-propyl-1,3- propanediyl esterpropanoic acid	43	—	0.6	—
9-octyl-heptadecane	95	19.0	—	—
3-ethyl-2,5-dimethyl-1,3- hexadiene	53	—	—	3.6
1,3,5-tri(methylene)- cycloheptane	38	—	0.9	—
9,12,15-octadecatrienal	91	—	50.7	26.3
4-me-2,7-octadiene	49	—	0.8	—
(e)3-tetradecen-5-yne- Pentadecanal	78	—	3.8	—
	91	—	0.9	—
3-ethyl-2,5-dimethyl-1,3- hexadiene	53	3.9	—	—
(z) 9-octadecenamide	68	10.4	1.6	3.2
2-me-3-buten-2-ol	43	—	0.6	—
2,6-dimethyl-1,7-octadiene- 3,6-diol	87	—	0.8	—

Table 2. Continued.

Chemical	Mass spectral	Percentage of fixed total		
	Match quality	<i>U.</i> <i>wilsoniana</i> ^a	'Commendation' ^a	<i>U. propinqua</i> <i>var suberosa</i> ^a
[(2-phenylethenyl) sulfonyl]-benzene	43	—	0.6	—
9-hydroxyfarnesol	56	—	0.8	—
5-methyl-2-(1- methylethyl)-(1 α ,2 β ,5 β) cyclohexanol	58	—	0.4	—
4-(2,6,6-trimethyl-1- cyclohexen-1-yl)-2- buten-2-one	84	—	0.6	—
3,7-dimethyl-1,5,7- octatrien-3-ol	47	—	0.8	—
(z)-linalool oxide	90	—	0.8	—
α -caryophyllene	97	—	—	2.4
Caryophyllene oxide	43	—	0.7	—
Alpha-farnesene	95	—	0.7	—
2,6-dimethyl-3,7-octadien- 2,6-diol	90	—	0.9	—
5-ethenyltetrahydro- α , α ,5- trimethyl-,cis-2- furanmethanol	74	—	0.9	1.8
Benzeneacetaldehyde	94	—	—	2.3
α -me-, α -, [4-me-3-pentynal] oriranemethanol	90	—	—	1.6
benzyl alcohol	96	—	—	3.0
2-methoxy-3-(2-propenyl)- phenol	98	—	—	2.2
4'-methylpropiophenone	70	—	—	1.7
(z,z)9,12-octadecadienoyl chloride	76	—	0.5	—
Eugenol	94	5.3	1.8	—
Phenylethyl alcohol	93	7.2	2.1	4.1
3-isopropoxy-5-me-phenol	72	—	0.4	—
6-ethenyltetrahydro-2,2,6- trimethyl-2h-pyran-3-ol	86	—	0.8	—

Leaf samples were collected at the Morton Arboretum, Lisle, Illinois on 17 June 2004.

^aElms belonging to, or closely associated with the David complex.

(total number) of lipids showed a significant effect relating to David complex parentage ($F = 7.18$; $df = 1, 15$; $P = 0.018$). An examination of the first and second eigenvectors from the data on leaf chemistry and gypsy moth larval development suggests that elms belonging to the David complex are better suited for gypsy moth larval development than other elms included in this survey. This

Table 3. Suitability of Asian elms for gypsy moth larval development, conducted in Lisle, Illinois, May-July 2001.

Elm	Larval longevity ^{b,c} (days)	Percentage pupation ^{b,c}	Pupal fresh Wt ^{b,c} (g)	Percentage adult emergence ^{b,c}
<i>U. castenifolia</i>	6a	0a	—	0a
<i>U. macroarpa</i>	4a	0a	—	0a
<i>U. chenmoui</i>	4a	0a	—	0a
<i>U. lamellosa</i>	4a	3a	0.3a	3a
<i>U. glaucesens</i>	4a	0	—	0a
<i>U. gausenii</i>	4a	17ab	0.2a	13ab
<i>U. parvifolia</i>	5a	0a	—	0a
<i>U. davidiana</i> ^a	5a	0a	—	0a
<i>U. szechuanica</i> ^a	10ab	3a	0.5ab	3a
<i>U. japonica</i> ^a	4a	0a	—	0a
<i>U. wilsoniana</i> ^a	5a	0a	—	0a
<i>U. propinqua</i> ^a	6a	0a	—	0a
‘Commendation’ ^a	8ab	60bc	0.6b	60bc
‘Accolade’ ^a	6a	3a	0.5ab	3a
‘Triumph’ ^a	7a	3a	0.4a	3a
‘Danada Charm’ ^a	10ab	53b	0.7b	53b
Reference				
<i>Malus spp.</i>	16b	93c	0.6b	93c
<i>Quercus palustris</i>	10ab	70bc	0.8b	70bc

^aElms belonging to, or closely associated with the David complex.
^bAverages from 10 replicates per elm, 3 trees per species/cultivar (30 larvae total).
^cValues within columns followed by the same letter are not significantly different. (*, $P < 0.05$; Student-Newman-Keuls multiple comparison test).

relationship was supported in a correlation between leaf chemical diversity and gypsy moth adult emergence ($r = 0.530$; $df = 1, 15$; $P = 0.003$; Fig. 1) and a weak (nonsignificant) correlation detected between adult emergence and a diverse foliar lipid content ($r = 0.429$; $df = 1, 15$; $P = 0.097$).

Japanese beetle no-choice and field defoliation study. Principal component analysis with results from Japanese beetle no-choice (percentage leaf tissue consumed and dry fecal pellet weight) and field defoliation studies (Table 4) on 14 different elms yielded two eigenvectors that accounted for 94% of the variability. Elm parentage had a significant effect on percentage leaf consumed ($F = 7.59$; $df = 1, 12$; $P = 0.017$) and fecal pellet masses ($F = 27.73$; $df = 1, 12$; $P \leq 0.001$) in the no-choice bioassay. Field defoliation rating was not significant when tested individually, but a multivariate test with all three Japanese beetle host-plant-suitability measurements was significant ($F = 7.75$; $df = 3, 10$; $P = 0.005$). Japanese beetle adults showed higher levels of feeding activity on the elm trees that belonged to the David complex; however, there was no significant correlation linking elm leaf chemical diversity (terpenes, phenolics, alkaloids, and

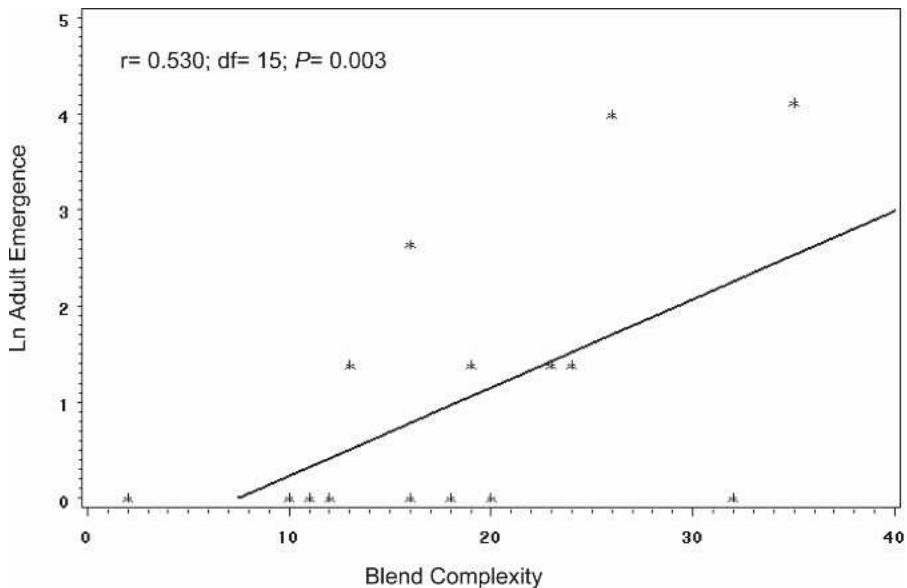


Fig. 1. Correlation between elm leaf chemical diversity (classified into four chemical classes: phenolics, terpenoids, lipids, and alkaloids) found in the David complex and gypsy moth, *Lymantria dispar*, adult emergence.

lipids). This could be attributable to the lower number of observations, which resulted in a limited number of degrees of freedom in the analysis.

Japanese beetle feeding deterrent study. Elm leaf lipid extracts were selected for further study as they were the chemical class found most significantly linked with insect herbivore host plant suitability. Adult beetles were observed consuming significant amounts of the sugar-only treated filter papers (positive control) within 24 h. There was no consumption of the untreated filter paper discs (negative control); however, there was evidence of tasting and regurgitating. Considerable variation was found in surface area consumption of the paper discs treated with different elm surface lipids (Table 5). Deterrent effects were observed in three of the treatments tested. There was only one elm that showed significantly lower rates of consumption than the positive control, *U. propinqua* var *suberosa*. Discs treated with lipid extracts from 'Commendation' and *U. wilsoniana* also received low averages of disc consumption and minimal feeding frequency, but were only significantly different from the control at $\alpha = 0.10$. Because none of the leaf discs showed significant levels of disc consumption higher than the positive control, no conclusions could be made related to the presence/absence of feeding stimulants.

Discussion

The present study surveyed leaf chemistry of selected Asian elm species, cultivars, and varieties. Results were then analyzed with data obtained in previous studies on Japanese beetle no-choice assays and a field defoliation survey and

Table 4. Summary of Japanese beetle no-choice feeding studies and the field defoliation survey conducted at the Morton Arboretum, Lisle, Illinois, 1997 (data from Miller et al. 1999).

Elm	1997 no-choice feeding study		1997 field study
	Leaf tissue consumed ^b	Dried fecal pellet mass ^b (mg)	Defoliation rating ^b
<i>U. pumila</i>	50%b	22.5a	4.0b
<i>U. macrocarpa</i>	14%a	13.7a	0.5ab
<i>U. chenmoui</i>	61%b	27.9a	2.0ab
<i>U. glaucescens</i>	25%a	23.1a	1.0a
<i>U. gausenii</i>	41%b	50.6b	1.0a
<i>U. parvifolia</i>	10%a	7.2a	0.0a
<i>U. lamellosa</i>	14%a	18.7a	0.0a
<i>U. japonica</i> ^a	69%c	94.0c	1.8ab
<i>U. szechuanica</i> ^a	50%b	57.7b	1.0ab
<i>U. davidiana</i> ^a	60%b	70.0c	2.4ab
<i>U. propinqua</i> ^a	49%b	72.3c	2.2ab
<i>U. wilsoniana</i> ^a	50%b	51.25b	0.20a
'Accolade' ^a	53%b	84.3c	2.0ab
'Triumph' ^a	43%b	45.5b	3.0b

^aElms belonging to, or closely associated with the David complex.

^bValues within columns followed by the same letter are not significantly different. (* $P < 0.05$; Student-Newman-Keuls multiple comparison test).

additionally, the reported suitability of elms for gypsy moth. Not only are these two generalist herbivores important pests in the urban forestry environment, but they also provide an interesting means for comparison of factors that potentially govern host plant suitability. Some of the strategies for reducing the detrimental effects of allelochemicals favor dietary mixing or the use of multiple food sources that could potentially benefit the herbivore by the dilution of toxic secondary plant compounds, by induction of detoxification enzyme systems, or through increased nutrient balance. Trends found in the principal component and multivariate analyses with both Japanese beetle feeding activity and gypsy moth suitability showed that elms with more diverse leaf chemistry (terpenes, phenolics, lipids, and blend complexity) were preferred. The strongest connection to any one chemical group was the adult emergence of gypsy moth, which showed a positive correlation to elm leaf lipid diversity. There was no significant correlation found between elm leaf lipid diversity and Japanese beetle preference, as measured in no-choice studies (percentage leaf tissue removed and dried fecal pellet mass). In a separate greenhouse study designed to measure deterrent effects of elm leaf lipids on Japanese beetle feeding activity, extracts from only 3 of the 23 elms surveyed were found to deter feeding. This result could be related to lipid diversity, but might also depend on the presence or absence of a specific chemical.

Of these three elms, *U. wilsoniana* was the only one that was included in the principal component analysis described above and also was the only species that

Table 5. Japanese beetle feeding frequency (noted by presence and absence of beetle feeding activity in individual arenas) and surface area consumption of elm leaf lipid extract-treated disc in the greenhouse feeding deterrent bioassay conducted at Iowa State University, Ames, Iowa, 2005.

Elm	Feeding frequency ^b	Disc consumption ^b (cm ²)	Standard deviation ^b (disc consumption)
<i>U. parvifolia</i>	0.63	0.12	0.11
<i>U. lamellosa</i>	0.44	0.06	0.09
<i>U. davidiana</i>	0.67	0.12	0.12
<i>U. crassifolia</i>	0.75	0.19	0.29
<i>U. gausenii</i>	0.44	0.07	0.10
<i>U. americana</i>	0.80	0.13	0.12
<i>U. thomassii</i>	0.83	0.13	0.13
<i>U. bergamanniana</i>	0.67	0.09	0.07
<i>U. pumilia</i>	0.67	0.10	0.09
<i>U. castenifolia</i>	0.56	0.10	0.10
<i>U. chenmoui</i>	0.63	0.13	0.13
<i>U. macrocarpa</i>	0.44	0.08	0.11
<i>U. japonica</i> ^a	0.78	0.16	0.11
<i>U. szechuanica</i> ^a	0.89	0.14	0.09
<i>U. propinqua</i> ^a	0.50	0.06	0.09
<i>U. wilsoniana</i> ^a	0.67	0.01*	0.14
<i>U. propinqua</i> var <i>suberosa</i> ^a	0.00	0**	0.00
<i>U. glaucesens</i> var <i>lasiophylla</i>	0.67	0.12	0.13
'Vanguard'	0.83	0.15	0.10
'Triumph' ^a	0.67	0.11	0.09
'Commendation' ^a	0.44	0.04*	0.06
'Accolade' ^a	0.89	0.17	0.07
'Danada Charm' ^a	0.44	0.07	0.01
Positive Control	1.00	0.16	0.08
Negative Control	0.00	0**	0

^aElms belonging to, or closely associated with the David complex.^bAverages from individual leaf lipid extracts were pooled (nine replications per elm).*Treatments that significantly differ from the positive control (sugar treated) by least-squares means analysis (* α = 0.10; ** α = 0.05).

had a lower lipid diversity compared with other elms in the David complex. Table 2 provides a qualitative report on the composition of elm leaf soaks from *U. propinqua* var *suberosa*, 'Commendation' and *U. wilsoniana*. Future work should include a more quantitative study of these elms; more detailed feeding behavior assays that limit specific components of the lipid extracts are needed also to sort out the differences observed. None of the lipid-treated discs prompted an increase in beetle-feeding activity relative to the positive control, limiting any conclusions about the lipid extracts containing feeding stimulants. A different study design,

with lower levels of sugars, would be valuable to test for stimulant effects of leaf-surface lipids. Recent advances in genomics continue to increase our understanding of the functionality of genes involved in plant lipid biosynthesis (enzymes like thioesterases, fatty acid synthases, etc.), and further research on identification and expression of these metabolic pathways in Ulmaceae would be beneficial (Mekhedov et al. 2000).

These results might share some relationship with previous studies in which other researchers speculated that the high density of trichomes present on the foliage of *U. glaucescens*, *U. lamellosa*, and *U. macrocarpa* could explain the low levels of feeding seen in no-choice and multiple-choice laboratory studies (Miller et al. 1999). It is possible that additional physical factors, including leaf surface color, size, or texture, possibly presence or absence of specific foliar trichomes and their densities may have an impact on insect herbivores like the Japanese beetle and gypsy moth. It is also important to consider relevant sources of variation for the chemical parameters analyzed in this study. Other research has shown that leaf chemical composition can fluctuate throughout the season (Fischbach et al. 2002). Future studies could also incorporate sampling from a larger number of trees to allow for any differences amongst individuals.

Part of the study design allowed for comparisons of selected leaf chemicals with elm parentage. Trends seen in the principal component and multivariate analysis with both Japanese beetle feeding activity and gypsy moth developmental suitability showed that elms belonging to the *U. davidiana* complex (*U. davidiana*, *U. japonica*, *U. wilsoniana*, 'Triumph', 'Accolade', and *U. propinqua*) and the closely related species *U. szechuanica*, is related to the diversity of leaf chemical composition (terpenes, phenolic, lipids, and overall blend complexity).

To date much of the research on elms shows that those closely associated with the David complex have good levels of resistance and tolerance to Dutch elm disease, as well as elm yellows resistance and tolerance to the urban environment. These trees are more tolerant of poor soil types, are less susceptible to elm leafminer injury (Miller 2000), and have excellent foliage and growth habits. Unfortunately, some of these trees may not be as useful in areas with high Japanese beetle and gypsy moth populations based on suitability and preferences testing (Miller et al. 1999 and Miller unpublished). This research provides evidence that gypsy moth host suitability and Japanese beetle preference rely on the complexity of leaf chemical composition. It also suggests that generalist insect herbivores might selectively use host plants containing a larger diversity of compounds.

Acknowledgments

Financial support for this research was provided by the Tree Research and Education Endowment Fund, Champaign, Illinois and the Morton Arboretum, Lisle, Illinois. We thank Bryan Clark for assistance in the laboratory, Dr. Michael Collyer for statistical guidance, and the Iowa Department of Agriculture and Land Stewardship for use of the Japanese beetle lure traps. This journal paper is a product of the Iowa Agriculture and Home Economics Experiment Station Project No. 5075.

References Cited

- Barbosa, P. & V. A. Krischik. 1987.** Influence of alkaloids on feeding preference of eastern deciduous forest trees by the gypsy moth (*Lymantria dispar*). Am. Nat. 130: 53–69.

- Fischbach, R. J., M. Staudt, I. Zimmer, S. Rambal & J.-P. Schnitzler. 2002.** Seasonal pattern of monoterpene synthase activities in leaves of the evergreen tree *Quercus ilex*. *Physiol. Plant.* 114: 354–360.
- Fu, L. 1980.** Studies in the genus *Ulmus* in China. Northeastern Forest Institute, Harbin, Heilongjiang.
- Fulcher, A. F., T. G. Ranney, J. D. Burton, J. F. Walgenbach & E. P. Maness. 1996.** Natural resistance to Japanese beetle among *Malus* taxa: role of endogenous foliar phenolics. *Proc. S. Nursery Res. Conf.* 41: 57–60.
- Havill, N. P. & K. F. Raffa. 2000.** Compound effects of induced plant responses on insect herbivores and parasitoids: implications for tritrophic interactions. *Ecol. Entomol.* 25: 171–179.
- Keathley, C. P., D. A. Potter & R. L. Houtz. 1999.** Freezing-altered palatability of Bradford pear to Japanese beetle: evidence for decompartmentalization and enzymatic degradation of feeding deterrents. *Entomol. Exp. Appl.* 90: 49–59.
- Ladd Jr., T. L. 1986.** Influence of sugars on the feeding response of Japanese beetles (Coleoptera: Scarabaeidae). *J. Econ. Entomol.* 79: 668–671.
- Ladd Jr., T. L. 1988.** Japanese beetle (Coleoptera: Scarabaeidae): influence of sugars on feeding response of larvae. *J. Econ. Entomol.* 81: 1390–1393.
- Loughrin, J. H., D. A. Potter & T. R. Hamilton-Kemp. 1998.** Attraction of Japanese beetles (Coleoptera: Scarabaeidae) to host plant volatiles in field trapping experiments. *Environ. Entomol.* 27: 395–400.
- Loughrin, J. H., D. A. Potter, T. R. Hamilton-Kemp & M. E. Byers. 1996.** Volatile compounds from crabapple cultivars (*Malus* spp.) differing in susceptibility to the Japanese beetle (*Popillia japonica* Newman). *J. Chem. Ecol.* 22: 1295–1305.
- Loughrin, J. H., D. A. Potter, T. R. Hamilton-Kemp & M. E. Byers. 1997.** Response of Japanese beetles (Coleoptera: Scarabaeidae) to leaf volatiles of susceptible and resistant maple species. *Environ. Entomol.* 26: 334–342.
- Mekhedov, S., O. M. Ilárduya & J. Ohlrogge. 2000.** Toward a functional catalog of the plant genome. A survey of genes for lipid biosynthesis. *Plant Physiol.* 122: 389–401.
- Miller, F. 2000.** Insect resistance of elm genotypes, pp. 137–154. *In* C. P. Dunn [Ed.], *The Elms: Breeding, Conservation, and Disease Management*. Kluwer Academic Publishers, Boston.
- Miller, F., S. Jerdan & G. Ware. 1999.** Feeding preference of adult Japanese beetles (Coleoptera: Scarabaeidae) for Asian elm species and their hybrids. *J. Econ. Entomol.* 92: 421–426.
- Potter, D. A. & D. W. Held. 1999.** Absence of food-aversion learning by a polyphagous scarab, *Popillia japonica*, following intoxication by geranium, *Pelargonium hortorum*. *Entomol. Exp. Appl.* 91: 83–88.
- Potter, D. A. & D. W. Held. 2002.** Biology and management of the Japanese beetle. *Annu. Rev. Entomol.* 47: 175–205.
- SAS Institute. 2003.** SAS system for Windows version 9.1. SAS Institute, Cary, NC.
- Spicer, P. G., D. A. Potter & R. G. McNiel. 1995.** Resistance of crabapple (*Malus* spp.) cultivars to defoliation by the Japanese beetle (Coleoptera: Scarabaeidae). *J. Econ. Entomol.* 88: 979–985.
- Stamp, N. E. & G. D. Harmon. 1991.** Effect of potassium and sodium on fecundity and survivorship of Japanese beetles. *Oikos* 62: 299–305.
-

Greenhouse Studies of Soybean Aphid (Hemiptera: Aphididae) Effects on Plant Growth, Seed Yield and Composition¹

Walter E. Riedell and Michael A. Catangui²

USDA-ARS, North Central Agricultural Research Laboratory, Brookings, South Dakota 57006

J. Agric. Urban Entomol. 23(4): 225–235 (October 2006)

ABSTRACT There is little published information available that describes the effects of soybean aphids (*Aphis glycines* Matsumura, Hemiptera: Aphididae) on soybean [*Glycine max* (L.) Merr.] growth, yield, and seed composition. The objective of this research was to measure how soybean growth, yield, and yield components are affected by soybean aphid infestations under controlled environments. Greenhouse-grown plants were infested with aphids at the first node stage (V1), the third node stage (V3), the beginning bloom stage (R1), the full pod stage (R4), and the beginning maturity stage (R7). Aphids remained on the plants for a total of 6,000 aphid-d (e.g. 600 aphids present for 10 days) after which time the insects were removed with insecticide. Increases in stem length, as compared with uninfested controls, were observed in plants infested at the V1 and V3 stages but not at the R7 development stage. Aphid infestation had no significant effect on seed yield, number of pods plant⁻¹, or number of seeds pod⁻¹. Aphid treatment reduced seed number plant⁻¹. Individual seed weights were greater in aphid infested plants than control when treated at the V1 and V3 development stage but were not different when aphids were applied to plants in the reproductive stages. Seed protein concentration was not affected by aphid infestation treatment. Infestation at the R4 and R7 stages resulted in dramatic reductions in seed oil concentrations compared with controls. Although aphid injury in this study did not lead to a significant loss in total seed yield, aphid-induced reductions in oil concentration suggest that seed composition responses to aphid injury should be an important consideration when conducting research to develop IPM systems for this soybean pest.

KEY WORDS Soybean aphid, *Aphis glycines* Matsumura, soybean, *Glycine max* L. Merrill, seed yield, seed composition, plant insect interactions

The soybean aphid (*Aphis glycines* Matsumura, Hemiptera: Aphididae), a common soybean insect pest in Asia (Wang et al. 1962), was first found in Wisconsin during July 2000 (Alleman et al. 2002). By 2003, the insect was found in 21 U.S. states and 3 Canadian provinces (Venette & Ragsdale 2004) and is now considered a major soybean insect pest in the north central United States. (Myers et al. 2005). Predictions are that this insect will expand its range to include all regions of the United States where soybeans are grown (Venette & Ragsdale

¹Accepted for publication 10 January 2008.

²Plant Science Department, South Dakota State University, Brookings, South Dakota 57007.

2004). Thus, the soybean aphid has the potential to continue to cause serious economic loss to soybean farmers and the value-added enterprises that depend on the production of large amounts of high-quality soybeans.

Because the soybean aphid is a new pest in the United States, a comprehensive integrated pest management (IPM) approach to manage aphid populations has yet to be developed (Hill et al. 2004). Because soybean is an important crop in the United States, development of an IPM system for soybean aphid is a high-priority research need. An integral component of IPM programs for pest population management is the knowledge of the effects of pest injury on plant growth, yield, and seed composition (Kogen 1998). The objectives of this study were to measure the effects of soybean aphid infestation, applied at different soybean development stages, on soybean plant growth, yield, and seed composition.

Materials and Methods

Aphids used in the study came from colonies maintained on soybeans (Pioneer brand '91B01') in a growth chamber. Colony aphids were initially collected from natural populations in a soybean field near Aurora, South Dakota, during the 2001 growing season. Colony aphids were not viruliferous (M. A. C. Langham, personal communication).

Greenhouse experiments were conducted at the North Central Agricultural Research Laboratory near Brookings, South Dakota. Soybean seeds (Pioneer brand '91B01'), treated with *Bradyrhizobium japonicum* ('S' culture; Nitragin Inc., Milwaukee, Wisconsin), were used. Plants were grown in a greenhouse under natural light supplemented with a daily 12h:12h light:dark photoperiod from metal halide and high pressure sodium lamps providing $250 \mu\text{mol s}^{-1} \text{m}^{-2}$ photosynthetically active radiation (PAR) at canopy height. Supplemental lighting was provided during the first and last 3 h of the daily photoperiod. Greenhouse temperatures were maintained at 25°C (days) and 14°C (nights) with 20–50% relative humidity. The Barnes sandy clay loam soil used in these experiments was obtained from continuous corn fields located at the Eastern South Dakota Soil and Water Research Farm near Brookings, South Dakota.

Soybean seeds were planted 2.5 cm deep in a 2:1:1 soil:peat:vermiculite mixture contained in 19-cm diam. fiber pots (3 seeds pot⁻¹). Pots were watered to field capacity and placed in the greenhouse. Eighteen days after planting, 1 seedling pot⁻¹ was selected for uniform growth and the other seedlings were removed. Pots were assigned to specific experimental treatments and then were arranged in a completely randomized design in the greenhouse. Plants were allowed to grow under these greenhouse conditions until the soybean plants reached the first node (V1; 27 DAP), third node (V3; 41 DAP), beginning bloom (R1; 52 DAP), full pod (R4; 59 DAP), or beginning maturity (R7; 67 DAP) development stages (Ritchie et al. 1997).

At these specific soybean development stages, 10 pots were removed from the greenhouse and 5 each were transferred to 2 separate smaller greenhouse cubicles for soybean aphid infestation treatments. These cubicles were maintained at 30°C (day) and 18°C (night), 20–50% relative humidity, and natural light supplemented with a 14-h photoperiod provided by a sodium vapor lamp ($100 \mu\text{mol s}^{-1} \text{m}^{-2}$ PAR). The plants contained in one of the cubicles were treated with

soybean aphids (initial infestation of approximately 750 aphids plant⁻¹), whereas the plants in the second cubicle received no aphids. Aphids were allowed to feed and reproduce freely on the plants and were counted every third day.

Once a total infestation of 6000 aphid-d (Kieckhefer et al. 1995) was attained, plants in both cubicles were sprayed with acephate systemic insecticide. This process was repeated on different sets of plants at the specific development stages of interest. The 6000 aphid-d mark across all development stage treatments was reached in 6–13 d after initial infestation. Plants were then returned to the main greenhouse. Plants were grown to maturity and harvested. Yield and yield components (total seed weight, number of pods, number of seeds, seeds per pod, and individual seed weight) on a plant⁻¹ basis were measured. Seed oil and protein concentrations were measured with a NIR spectrometer (Ward Laboratories Inc., Kearney, Nebraska).

Data were analyzed using procedures appropriate for a two-way analysis of variance (PROC GLM) in SAS (SAS Institute, Cary, North Carolina). With the occurrence of significant ($\alpha = 0.1$) treatment effects on dependent variables, means were separated using the LSD option of the GLM procedure. Highest-order significant interactions between independent variables were further investigated by plotting treatment means and standard errors on graphs.

Results

Plant growth responses to treatments. Plant stem length was not affected by plant development stage treatments ($P = 0.15$) but was significantly affected by aphid treatments (33.7 cm plant⁻¹ for control, 39.8 cm plant⁻¹ for infested; $F = 15.8$, $df = 1$, $P = 0.0003$). Simple interpretation of plant stem length data was confounded by a significant development stage $\times$ aphid treatment interaction ($F = 5.9$, $df = 4$, $P = 0.0008$). This interaction suggests that stem length responded differently to aphid treatments when plants were infested at different plant development stages. Marked increases in infested plant stem length as compared with uninfested controls, observed when plants were infested at early (V1 and V3) development stages, were not present when plants were treated with aphids at later (R7) development stages (Fig. 1).

To further illustrate this aphid infestation-induced stem elongation, specific internode length data were plotted for plants treated with aphids at the V1, V3, and R1 development stages (Fig. 2). Stem elongation in aphid infested plants took place in those internodes that were directly above the node that contained a fully expanded leaf when the aphids were applied. Thus, in a V1 plant, aphid infestation increased the length of internodes 2 through 5, whereas internodes 4 through 6 showed increased length in plants infested at the V3 development stage. Infestation at the R1 stage resulted in elongation of internodes 7 through 9. The magnitude of the elongation response of these specific internodes to aphid infestation appeared to decrease as the age of the plant at infestation increased (Fig. 2). There were no significant differences in the stem diameter of these internodes (data not shown).

Yield, yield components, and seed composition. Aphid infestation had no significant effect on seed yield ($P = 0.2$; 9.5 g plant⁻¹ for control, 8.8 for infested; Table 1), number of pods plant⁻¹ ($P = 0.14$; 21.2 pods plant⁻¹ in control,

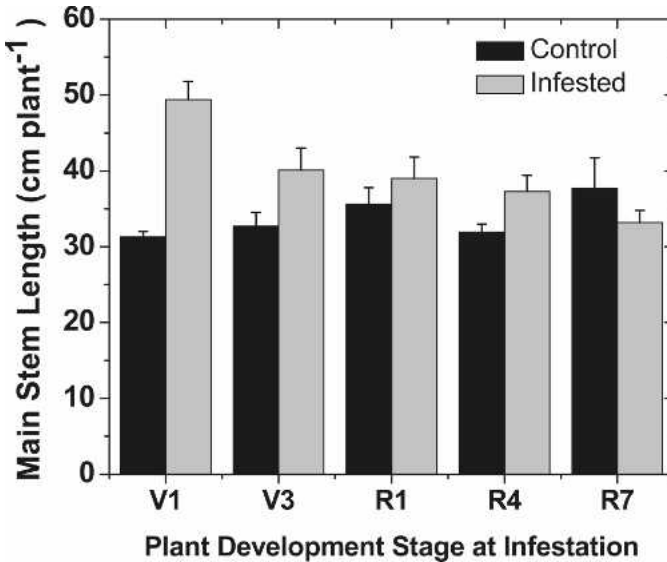


Fig. 1. Effect of soybean aphid infestation applied at different stages of plant development on main stem length. Values represent mean $\pm$ SD for 5 replicates treatment⁻¹.

19.4 in infested), or number of seeds pod⁻¹ ($P = 0.7$; 2.23 seeds pod⁻¹ in control, 2.25 in infested). However, aphid treatment significantly reduced seed number ($F = 2.7$, $df = 1$, $P = 0.08$; 47.4 seeds plant⁻¹ in control treatments, 43.2 in infested). There were no significant plant development stage $\times$ aphid treatment interactions for any of these dependent variables (Table 1).

Aphid treatments had significant ($F = 4.6$, $df = 1$, $P = 0.04$) effects on individual seed weight. Aphid infestation treatments increased the individual seed weight (199 mg seed⁻¹ in control; 208 for infested). A significant plant development stage $\times$ aphid infestation treatment interaction for individual seed weight ($P = 0.005$) suggests that this dependent variable responded differently to aphid infestation when it was applied at different plant development stages. Individual seed weights were greater in aphid-infested plants than control when treated at the V1 and V3 development stage but were not different when aphids were applied to plants in the reproductive stages (Fig. 3).

Seed protein concentration was not affected ($P = 0.5$; 351 g kg⁻¹ protein for control; 352 g kg⁻¹ for infested) by aphid infestation treatment. Oil concentration was significantly reduced by aphid treatments ($F = 6.2$, $df = 1$, $P = 0.02$; 161 g kg⁻¹ for control treatments; 149 g kg⁻¹ oil for infested). Significant plant development stage $\times$ aphid infestation treatment interactions for protein ($F = 3.2$, $df = 4$, $P = 0.02$) and oil ($F = 8.3$, $df = 4$, $P = 0.0001$) suggests that these dependent variables responded differently to aphid infestation when it was applied at different plant development stages. Protein concentrations were lower in infested plants when aphids were applied in the V1 and V3 stages but were greater when soybean aphids were applied at the R1, R3, and R7 stages (Fig. 4).

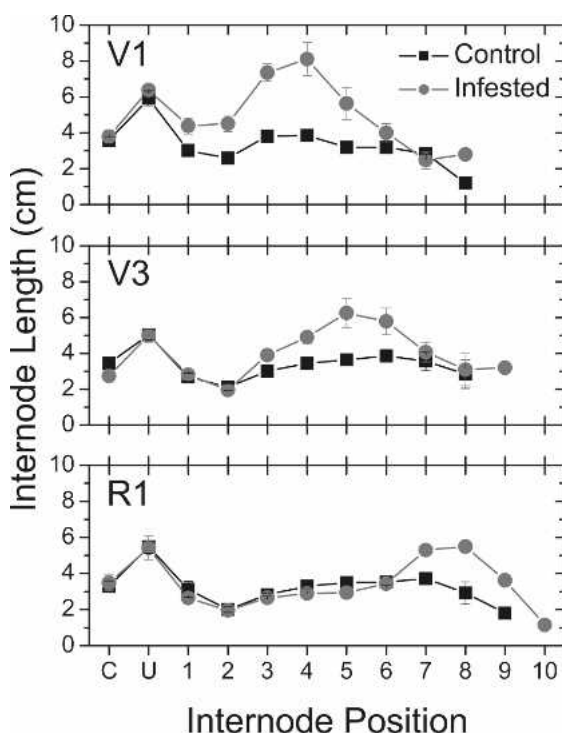


Fig. 2. Effect of soybean aphid infestation applied at the V1, V3, or R1 stage of plant development on length of specific stem internodes. The letter C in the x-axis represents the internode below the cotyledons, whereas the letter U represents the internode between the cotyledonary node and the unifoliolate leaf node. Internode C was measured from the soil surface to the cotyledonary node, internode U was from the cotyledonary node to the unifoliolate node, internode 1 was from the unifoliolate node to the first trifoliolate node, etc. Symbols and error bars represent mean $\pm$ SD for 5 replicates treatment⁻¹. In some cases, symbols were larger than error bars.

In contrast, oil concentration was equal or greater in infested plants when aphids were applied at the vegetative stages, whereas infestation at the R4 and R7 stages resulted in reduced oil concentrations in infested plants compared with controls (Fig. 4).

Discussion

Plant growth. In this experiment, plants at specific development stages were removed from the main greenhouse and placed into one of two smaller greenhouses for aphid infestation treatments. Supplemental light intensity ($250 \mu\text{mol s}^{-1} \text{m}^{-2}$ PAR in main house, $100 \mu\text{mol s}^{-1} \text{m}^{-2}$ PAR in smaller houses) and

Table 1. Analysis of variance mean squares and probability levels for soybean aphid treatments applied at different soybean crop developmental stages during the greenhouse experiment.

Source	df	Pods plant ⁻¹	Seeds plant ⁻¹	Seed pod ⁻¹	Seed weight	Yield	Protein	Oil
V-stage treatment	4	59**	293**	0.04	1400***	4.3	1076***	300
Aphid treatment	1	46	260*	0.004	953**	4.9	28	2035***
V-stage x aphid	4	10	25	0.03	910***	0.6	292**	2737***
Error	40	20	96	0.03	204	3.5	90	328

*, **, ***Significant at the 0.1, 0.05, and 0.01 probability levels, respectively.

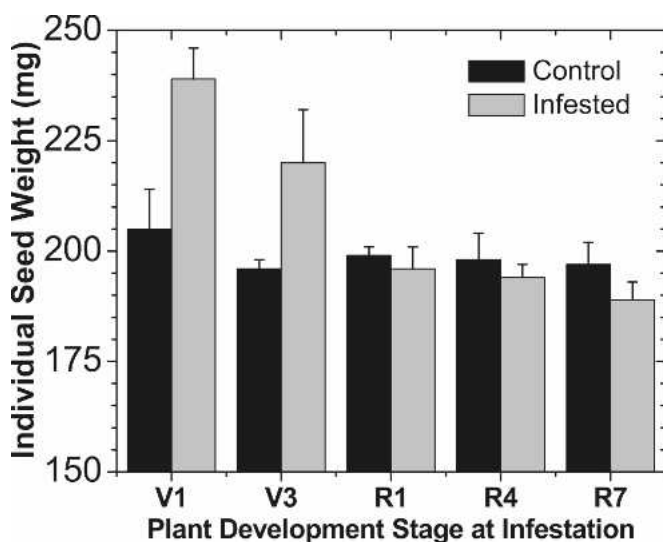


Fig. 3. Effect of soybean aphid infestation applied at different stages of plant development on individual seed weight. Values represent mean $\pm$ SD for 5 replicates treatment⁻¹.

light quality (6:4 ratio of metal halide:sodium vapor lamps in main house, sodium vapor lamps only in the small houses) were different in the main greenhouse compared with the small houses. Because soybean aphid injury causes plant stunting under field conditions (Macedo et al. 2003), the measured increase in total stem (Fig. 1) and specific internode lengths (Fig. 2) in soybean aphid-injured plants in the greenhouse was unexpected.

The unexpected stem elongation effects were observed on organs that were actively growing during the time the infested plants were in the small houses under relatively low intensity light provided by sodium vapor bulbs. Sodium vapor lamps emit little energy in the blue region of the spectrum, whereas metal halide lamps emit considerably greater blue wavelength energy (Dougher & Bugbee 2001). Low levels of blue light cause undesirable morphological responses (stem elongation) in soybean plants (Wheeler et al. 1991). As the percentage of blue light increased, soybean stem elongation was reduced regardless of the overall intensity ($200\text{--}500\ \mu\text{mol s}^{-1}\ \text{m}^{-2}$) of the photosynthetically active radiation (Dougher & Bugbee 2001).

During aphid infestation treatments, plants were subjected to a low-intensity lighting environment that contained only a small percentage of blue light. Thus, it is our contention that the low-intensity sodium vapor light treatment used in our experiment provided permissive environmental conditions for these stem elongation (Wheeler et al. 1991, Board 2001) symptoms to be expressed in aphid infested plants. The physiological mechanisms responsible for mediating these stem elongation responses to soybean aphid feeding in our experiment are not known. One possibility is that the plant wounding caused during soybean feeding activities stimulates the production of plant hormones, which in turn mediates

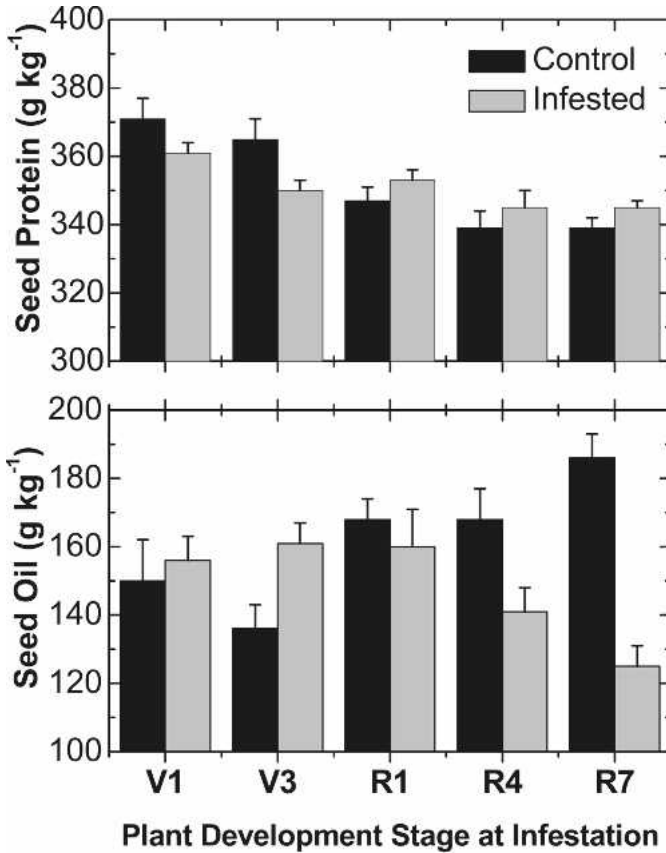


Fig. 4. Effect of soybean aphid infestation applied at different stages of plant development on seed oil and seed protein. Values represent mean $\pm$ SD for 5 replicates treatment⁻¹.

plant response to feeding injury. One potential candidate substance, jasmonate, a nontraditional plant growth regulator (Creelman & Mullet 1997), has been shown to rapidly and transiently increase in wounded soybean stem tissues (Creelman et al. 1992). A second candidate substance, ethylene, a gaseous hormone that increases in concentration in plants injured by insects (Yang & Hoffman 1984), has been shown to increase stem internode lengths when applied to soybeans under growth chamber conditions (Bugbee 2007). Further research would be needed to determine if plant hormones interacted with low light conditions to permit the expression of these aphid-induced stem elongation injury symptoms.

Yield and components. During the vegetative phase of soybean growth, root and shoot structures rapidly develop and increase in biomass. Soybeans accumulate about 25% of their total mature biomass before flowering. During the vegetative stage, soybean aphids tend to feed on newly expanded leaves, petioles, and stems (Ragsdale et al. 2004, Wu et al. 2004). In the present study, aphid

feeding injury reduced seeds plant⁻¹. When feeding injury took place during vegetative stages, individual seed weight increased (Fig. 3), seed protein concentration decreased, and seed oil concentration increased (Fig. 4). Soybean plants can compensate for stresses imposed during the vegetative phase if the stress conditions are removed in a timely manner (Ritchie et al. 1997). Because aphid infestation did not significantly reduce seed yield, and because there were no significant plant development stage $\times$ aphid treatment interactions for seed yield, we conclude that the measured increases in individual seed weight when aphid feeding injury was applied in the vegetative stage represent a successful compensatory response to the aphid injury-induced reduction in seed numbers plant⁻¹.

During the reproductive phase of the lifecycle, root and shoot biomass continues to rapidly accumulate, but biomass accumulation gradually shifts to the reproductive structures (seeds and pods). Soybeans set yield in a specific order: first, number of pods plant⁻¹; second, number of seeds plant⁻¹; and third, weight seed⁻¹ (Mathew et al. 2000). Thus, stresses imposed during early reproductive stages could reduce yield by reducing pods plant⁻¹ or seeds plant⁻¹ whereas stress imposed later may reduce weight seed⁻¹. If stress is minor in nature or if it is removed in a timely manner, reductions in one yield component may be compensated by another component so yields are not significantly changed (Ritchie et al. 1997).

When plants reach the reproductive stage, aphid tend to feed on the underside of fully expanded leaves, stems, petioles, and developing pods (Ragsdale et al. 2004). In our study, aphid feeding injury in the late reproductive stage (R7, Fig. 3) reduced weight seed⁻¹, but this reduction did not translate into significant reductions in yield. Thus, it appears that the soybean plants were able to compensate for feeding injury imposed during the reproductive phase so as to ameliorate yield loss.

Aphid infestation in the vegetative stages tended to reduce protein and increase oil concentrations, whereas aphid infestation in the reproductive stage increased protein and decreased oil concentrations (Fig. 4). These results support and extend the findings of Rose (1988) and Yazdi-Samadi et al. (1977), who found that stress (caused by drought) during early podfill stages reduced seed protein and increased seed oil whereas higher seed protein and lower seed oil concentrations were associated with stress occurring late in the soybean podfill stages. Because of the intimate association between lipid bodies, chloroplasts, and fatty acid synthesis in soybean (Adams et al. 1983, Stumpf 1980, Wilson 1987) and because of the reduction in photosynthesis caused by soybean aphid feeding (Macedo et al. 2003), it was not unexpected to see the dramatic reductions in seed oil concentration in soybean injured late in the podfill stage by soybean aphids.

Conclusions

One important requirement for insect pest IPM systems is the development of dynamic yield loss-to-injury relationships for each stage of crop development (Pedigo 1989). Because of limitations inherent to greenhouse experiments (e.g., reduced light intensity and limited soil volume for root development), greenhouse plant responses may not be directly representative of field grown plant responses. It is important to understand these limitations and to realize that plant responses to insects observed in greenhouse-grown plants may only be qualitatively representative of plant responses in field situations.

Greenhouse studies can play an important role in measuring yield-injury relationships at different crop stages, and may permit examining insect damage effects on yield in ways that are not possible using strictly field methods. For example, soybean aphid populations in field environment dramatically decrease once plants reach the early seed development stages, making it difficult to develop yield-injury relationships beyond R5 using field data (Ragsdale et al. 2007). In the present study, the effect of infestations at R7 on seed yield and composition were evaluated using greenhouse techniques. The greenhouse techniques used in the present study, therefore, allowed the collection of data that was unobtainable in field settings.

Although aphid injury in this study did not lead to a significant loss in total seed yield, measured differences in oil concentrations suggest that seed composition responses to aphid injury during late reproductive development stages should be an important consideration when conducting field research to develop IPM systems for this soybean pest.

Acknowledgments

We thank S. Osborne, K. Rosentrater, and W. French for review of an earlier version of this manuscript and D. Schneider and E. Beckendorf for excellent technical assistance. This research was supported in part by a grant from the South Dakota Soybean Research and Promotion Council. Mention of commercial or proprietary products does not constitute endorsement by the USDA. USDA offers its programs to all eligible persons regardless of race, color, age, sex, or national origin.

References Cited

- Adams, C. A., S. W. Norby & R. W. Rinne. 1983.** Ontogeny of lipid bodies in developing soybean seeds. *Crop Sci.* 23: 757–759.
- Alleman, R. J., C. R. Grau & D. B. Hogg. 2002.** Soybean aphid host range and virus transmission efficiency. *Proc. Wisc. Fertilizer Aglime Pest Manage. Conf.* (<http://www.soils.wisc.edu/extension/FAPM/fertaglime02.htm>). Accessed: 1 August 2007.
- Board, J. 2001.** Reduced lodging for soybean in low plant populations is related to light quality. *Crop Sci.* 41: 379–384.
- Bugbee, B. 2007.** Ethylene and yield physiology studies in controlled environments. p. 63. *In* Agronomy abstracts. American Society of Agronomy, Madison, WI.
- Creelman, R. A. & J. E. Mullet. 1997.** Biosynthesis and action of jasmonates in plants. *Annu. Rev. Plant Physiol. Mol. Biol.* 48: 355–381.
- Creelman, R. A., M. L. Tierney & J. E. Mullet. 1992.** Jasmonic acid/methyl jasmonate accumulate in wounded soybean hypocotyls and modulate wound gene expression. *Proc. Natl. Acad. Sci. USA* 89: 4938–4941.
- Dougher, T. A. O. & B. Bugbee. 2001.** Differences in the response of wheat, soybean, and lettuce to reduced radiation. *Photochem. Photobiol.* 73: 199–207.
- Hill, C. B., Y. Li & G. L. Hartman. 2004.** Resistance to the soybean aphid in soybean germplasm. *Crop Sci.* 44: 98–106.
- Kieckhefer, R. W., J. L. Gellner & W. E. Riedell. 1995.** Evaluation of the aphid-day standard as a predictor of yield loss caused by cereal aphids. *Agron. J.* 87: 785–788.
- Kogen, M. 1998.** Integrated pest management: Historical perspectives and contemporary developments. *Annu. Rev. Entomol.* 43: 243–270.
- Macedo, T. B., C. S. Bastos, L. G. Higley, K. R. Ostlie & S. Madhavan. 2003.** Photosynthetic responses of soybean to soybean aphid (Homoptera: Aphididae) injury. *J. Econ. Entomol.* 96: 188–193.

- Mathew, J. P., S. J. Herbert, S. Zhang, A. A. F. Rautenkranz & G. V. Litchfield. 2000.** Differential response of soybean yield components to the timing of light enrichment. *J. 92*: 1156–1161.
- Myers, S. W., C. Gratton, R. P. Wolkowski, D. B. Hogg & J. L. Wedberg. 2005.** Effect of soil potassium availability on soybean aphid (Hemiptera: Aphididae) population dynamics and soybean yield. *J. Econ. Entomol.* 98: 113–120.
- Pedigo, L. P. 1989.** Entomology and pest management. Macmillan Publishing Company, New York.
- Ragsdale, D. W., D. J. Voegtlin & R. J. O'Neil. 2004.** Soybean aphid biology in North America. *Ann. Entomol. Soc. Am.* 97: 204–208.
- Ragsdale, D. W., B. P. McCornack, R. C. Venette, B. D. Potter, L. V. Macrae, E. W. Hodgson, M. E. O'Neal, K. D. Johnson, R. J. O'Neil, C. D. Difonzo, T. E. Hunt, P. A. Glogoza & E. M. Cullen. 2007.** Economic threshold for soybean aphid (Hemiptera: Aphididae). *J. Econ. Entomol.* 100: 1258–1267.
- Ritchie, S. W., J. J. Hanway, H. E. Thompson & G. O. Benson. 1997.** How a soybean plant develops. Special Report 53. Iowa State University, Coop. Extension Ser., Ames, IA.
- Rose, I. A. 1988.** Effects of moisture stress on the oil and protein components of soybean seeds. *Aust. J. Agric. Res.* 39: 163–170.
- Stumpf, P. K. 1980.** Biosynthesis of saturated and unsaturated fatty acids, pp. 177–204. *In* P. K. Stumpf and E. E. Conn [Eds.], *The biochemistry of plants*, Vol. 4. Academic Press, New York.
- Venette, R. C. & D. W. Ragsdale. 2004.** Assessing the invasion by soybean aphid (Homoptera: Aphididae): where will it end? *Ann. Entomol. Soc. Am.* 97: 219–226.
- Wang, C. L., N. J. Siang, G. S. Chang & H. F. Chu. 1962.** Studies on the soybean aphid *Aphis glycines* Matsumura. *Acta Entomol. Sinica.* 11: 31–44.
- Wheeler, R. M., C. L. Mackowiak & J. C. Sager. 1991.** Soybean stem growth under high-pressure sodium with supplemental blue lighting. *Agron. J.* 83: 903–906.
- Wilson, R. F. 1987.** Seed metabolism, pp. 643–686. *In* J. R. Wilcox [Ed.], *Soybeans: improvement, production, and uses*, 2nd ed. American Society of Agronomy, Madison WI.
- Wu, Z., D. Schenk-Hamlin, W. Zhan, D. W. Ragsdale & G. E. Heimpel. 2004.** The soybean aphid in China: A historical view. *Ann. Entomol. Soc. Am.* 97: 209–218.
- Yang, S. F. & N. E. Hoffman. 1984.** Ethylene biosynthesis and its regulation in higher plants. *Ann. Rev. Plant Physiol.* 35: 155–189.
- Yazdi-Samadi, B., R. W. Rinne & R. D. Seif. 1977.** Components of developing soybean seeds: oil, protein, sugars, starch, organic acids, and amino acids. *Agron. J.* 69: 481–486.
-

Time Course Analysis of Bed Bug, *Cimex lectularius* L., (Hemiptera: Cimicidae) Blood Meals With the Use of Polymerase Chain Reaction¹

Allen L. Szalanski, James W. Austin,² Jackie A. McKern, Tim McCoy,³
C. Dayton Steelman, and Dini M. Miller³

Department of Entomology, University of Arkansas, Fayetteville, Arkansas USA

J. Agric. Urban Entomol. 23(4): 237–241 (October 2006)

ABSTRACT The ability to identify individual human hosts based on analyses of blood recovered from bed bugs, *Cimex lectularius* L. (Hemiptera: Cimicidae) may be useful for forensic and urban entomology. Time-course experiments can provide insight into how long bed bug bloodmeals may be detected with the use of polymerase chain reaction (PCR). Bed bugs were allowed to feed on a human volunteer and were then frozen at intervals of 1, 3, 7, 14, 30, 60, and 90 d after feeding. DNA was extracted from the individual insects and PCR was conducted with the use of short tandem repeat (STR) and human mitochondrial DNA (mtDNA) hypervariable region (HVR1) markers. Human DNA could be detected by PCR up to 7 d after feeding using the mtDNA marker and up to 60 d using the STR marker. No difference was observed between male and female bed bugs. The stability of human DNA in bed bug bloodmeals for up to 60 d reveals the utility of bed bugs for forensic investigations and the time interval blood identification could be used to determine the time course of suspects in legal investigations.

KEY WORDS Bed bug, *Cimex lectularius*, forensic entomology, STR typing, hypervariable region

Bed bugs (*Cimex lectularius* L.) (Hemiptera: Cimicidae) have recently garnered significant attention because of their reemergence as a pest of possible medical concern in the United States. Population genetic studies of *C. lectularius* L. confirm the neoteric expansion of this pest across North America (Szalanski et al. 2008) with greater insight into their proposed acumen for broader establishment and coexistence with humans and concomitant animals. For this reason, the application of alternative studies that focus on ancillary applications of this re-emerging pest may prove to be useful in legal investigations where there is a need to link a suspect or a victim to a specific location and/or time frame; in this sense, bed bugs are ideal.

¹Accepted for publication 25 January 2008.

²Center for Urban & Structural Entomology, Department of Entomology, Texas A&M University, College Station, TX, USA.

³Department of Entomology, Virginia Tech, Blacksburg, VA, USA.

Identification of bloodmeal sources has been documented in a number of blood-feeding insects. The potential use of polymerase chain reaction (PCR) diagnostics of bloodmeals of hematophagous insects for forensics has been demonstrated for several insects, including mosquitoes (Kreike & Kampfer 1999), human crab louse, *Pthirus pubis* (L.) (Anoplura: Pthiridae) (Lord et al. 1998), fly larvae (Wells et al. 2001, Zehner et al. 2004), nitidulid beetles (DiZinno et al. 2002), triatomines (Bosseno et al. 2006, Pizzaro et al. 2007), and most recently bed bugs, *Cimex lectularius* (Szalanski et al. 2006). Time-course analyses of bloodmeals can provide insight into how long host DNA can be detected by PCR and have been conducted on mosquitoes (Ngo & Kramer 2003, Kent & Norris 2005), the black fly, *Simulium damnosum* s.l. (Diptera: Simuliidae) (Boakye et al. 1999), and human body lice, *Pediculus humanus* L. (Anoplura: Pediculidae) (Mumcuoglu et al. 2004).

The aim of this study was to conduct a time-course analysis of bed bug bloodmeals to determine how long human DNA can be detected from bed bug bloodmeals. This will provide insight into the time interval that bed bug bloodmeals could be used to determine the time course of suspects and/or victims in legal and/or forensic investigations.

Materials and Methods

The bed bugs evaluated for this study were taken from a laboratory strain maintained at Virginia Tech, Blacksburg, Virginia. Adult female and male bed bugs were allowed to feed on a human volunteer for 15 min and were then maintained at 22°C. Bed bug blood feeding was conducted as approved by the Virginia Tech Institutional Review Board (IRB 06-165). At 1-, 3-, 7-, 14-, 30-, 60-, and 90-d intervals, six bed bugs of each sex were removed from the experimental population, placed in glass vials and frozen at -20°C until DNA extraction.

DNA was extracted from individual bed bugs using a Puregene DNA isolation kit D-5000A (Gentra, Minneapolis, Minnesota). Extracted DNA was resuspended in 50 µL of Tris:EDTA and stored at -20°C. PCR was conducted with two primer sets, one amplifying a mtDNA marker and the other one amplifying a short tandem repeat (STR). PCR primers L15997 and R202 amplify a 381 bp region of the human mtDNA first hypervariable part (HRV1) of the control region (Cavelier et al. 2000) and primers 5'-CAAACCCGACTACCAGCAAC-3' and 5'-GAGCCATGTTTCATGCCACTG-3' amplify a 271 bp region of the human D18S51 locus (Barber & Parkin 1996). The PCR reactions were conducted with 2 µL of the extracted DNA following the method of Szalanski et al. (2003). PCR conditions for the D18S51 marker consisted of 35 cycles of denaturing at 94°C for 30 s, annealing at 55°C for 45 s, extension at 72°C for 45 s, and final extension at 72°C for 5 min (Egyed et al. 2006). The HRV1 thermocycler profile consisted of 28 cycles of denaturing at 94°C for 30 s, annealing at 60°C for 30 s, with extension at 72°C for 45 s, and final extension at 60°C for 30 min. The HVR1 PCR products were resolved on 2% agarose gels, whereas the STR PCR products were separated by 10% PAGE per Taylor et al. (1996).

Results and Discussion

The time-course experiment showed that from the 84 bed bugs subjected to DNA analysis, DNA was successfully isolated and amplified from the 1-, 3-, and

7-d-old samples for the mtDNA marker. For the STR marker, DNA was successfully amplified from 1-, 3-, 7-, 14-, 30-, and 60-d samples. One reason why the STR marker was more successful than the mtDNA marker for PCR is that the PCR product is smaller, 271 versus 381 bp, which may facilitate amplification of degraded DNA. No difference was observed between male and female bed bugs in terms of PCR success, this may indicate that blood digestion in females was not accelerated because of reproductive physiology.

The ability to detect host DNA for up to 2 mo from bed bugs is beyond most observations for other hematophagous insects. Blood meals from the blackfly *Simulium damnosum* s.l. are detectable via PCR up to 72 h after feeding (Boakye et al. 1999) and bloodmeals from the human body louse, *Pediculus humanus humanus* L. are detectable for up to 20 h using PCR (Mumguoglu et al. 2004). Studies on mosquito blood meals are able to detect host DNA via PCR up to 48–72 h after feeding (Lee et al. 2002, Ngo & Kramer 2003, Kent & Norris 2005). Reasons why bed bug bloodmeals are detectable for a longer time may be related to the biology of the insect. Cimicid feeding frequency depends on digestion rate, environmental temperature and host availability (Reinhardt & Siva-Jothy 2007). Bed bugs have a relatively long life span compared with mosquitoes and may survive for months without bloodmeals away from hosts, unlike lice. For example, the cliff swallow bug, *Oeciacus vicarius* (Horvath) cimicid, may survive up to 3 yr in a cliff swallow nest without hosts and have been found to survive for up to 1 year without feeding in a laboratory environment (Loye 1985). This ability is likely a function of the physiological synchronization that has been documented between arthropods and their preferred hosts, especially birds (Larimore 1987).

Whereas there is little debate about the ability of bed bugs to survive between bloodmeals for extended periods of time, the physiological ability to sustain bloodmeals within their guts for lengthy periods of time remains unknown. Understanding this capability would be important to establishing a reliable time estimate or expectation for the recovery of viable blood from bed bugs in legal investigations. Comparisons with other obligate blood feeding arthropods can provide some insight. Kirstein & Gray (1996) were able to detect host DNA in ticks up to 280 d after engorgement. It is likely that the vacuoles that frequently form around bloodmeals of engorged ticks contribute to their ability to maintain extended periods of time without feeding, and hence increases the duration between subsequent feeding intervals. It may also be that the sheer volume of blood consumed by undisturbed bed bugs, reportedly measuring 130–200% of the unfed adult body weight of *C. lectularius* (Johnson 1941, Wattal & Kalra 1961), may contribute to this ability. Isolation of blood meals from other arthropods to identify vertebrate hosts have been developed applying the 18S rRNA gene (Pinchon et al. 2003, 2005) and the 12S rDNA gene (Humair et al. 2007). Future investigations into the basic physiology of bed bugs should be directed to answer the questions of why blood meals appear to be sustained in a viable mode longer than reports from other hematophagous insects.

Bed bugs may have several advantages over other blood-feeding arthropods for human forensics. For example, bed bugs do not remain on the host after their blood feeding activity, and will remain in close proximity to host feeding sites, thus making them useful for corroborating the location of a suspect or a victim in legal and/or forensic investigations. Because they are obligate blood feeders, requiring a blood meal before eclosion, the bulk of any bed bug population provides

a proportionately greater number of individuals from which to sample when compared with other insects that might be used in studies and/or investigations of this type. These findings then may have forensic applications in the use of bed bug derived blood meals for host identification and warrants future research into host-specific identification using molecular approaches.

Acknowledgment

This research was supported in part by the University of Arkansas, Fayetteville, Arkansas, Agricultural Experiment Station, and the Center for Urban and Structural Entomology at Texas A&M University, College Station, Texas.

References Cited

- Barber, M. D. & B. H. Parkin. 1996.** Sequence analysis and allelic designation of the two short tandem repeat loci D18S51 and D8S1179. *Int. J. Legal Med.* 109: 62–65.
- Boakye, D. A., J. Tang, P. Truc, A. Merriweather & T. R. Unnasch. 1999.** Identification of bloodmeals in haematophagous Diptera by cytochrome B heteroduplex analysis. *Med. Veterin. Entomol.* 13: 282–287.
- Bosseno, M.-F., L. S. Garcia, F. Baunaure, E. M. Gastelú, M. S. Gutierrez, F. L. Kasten, E. Dumonteil & S. F. Brenière. 2006.** Short report: Identification in triatomine vectors of feeding sources and *Trypanosoma cruzi* variants by heteroduplex assay and a multiplex miniexon polymerase chain reaction. *Am. J. Trop. Med. Hyg.* 74: 303–305.
- Cavelier, L., E. Jazin, P. Jalonen & U. Gyllensten. 2000.** MtDNA substitution rate and segregation of heteroplasmy in coding and noncoding regions. *Hum. Genet.* 107: 45–50.
- DiZinno, J. A., W. D. Lord, M. B. Collins-Morton, M. R. Wilson & M. L. Goff. 2002.** Mitochondrial DNA sequencing of beetle larvae (Nitidulidae: Omosita) recovered from human bone. *J. Foren. Sci.* 47: 1337–1339.
- Egyed, B., S. Furedi, M. Angyal, I. Balogh, L. Kalmar & Z. Padar. 2006.** Analysis of the population heterogeneity in Hungary using fifteen forensically informative STR markers. *Forensic Sci. Int.* 158: 244–249.
- Humair, P.-F., V. Douet, F. M. Cadenas, L. M. Schouls, I. Van de Pol & L. Gern. 2007.** Molecular identification of bloodmeal source in *Ixodes ricinus* Ticks using 12S rDNA as a genetic marker. *J. Med. Entomol.* 44: 869–880.
- Johnson, C. G. 1941.** The ecology of the bed-bug *Cimex letularius* L., in Britain. *J. Hyg. (Lond.)* 41: 345–361.
- Kent, R. J. & D. E. Norris. 2005.** Identification of mammalian blood meals in mosquitoes by a multiplex polymerase chain reaction targeting cytochrome B. *Am. J. Trop. Med. Hyg.* 73: 336–342.
- Kirstein, F. & J. S. Gray. 1996.** A molecular marker for the identification of the zoonotic reservoirs of Lyme borreliosis by analysis of the blood meal in its European vector *Ixodes ricinus*. *Appl. Environ. Microbiol.* 62: 4060–4065.
- Kreike, J. & S. Kampfer. 1999.** Isolation and characterization of human DNA from mosquitoes (Culicidae). *Int. J. Legal Med.* 112: 380–382.
- Larimore, R. W. 1987.** Synchrony of cliff swallow nesting and development of the tick, *Ixodes baergi*. *Southwest. Nat.* 32: 121–126.
- Lee, J. H., H. Hassan, G. Hill, E. W. Cupp, T. B. Higazi, C. J. Mitchell, M. S. Godsey Jr. & T. R. Unnasch. 2002.** Identification of mosquito avian-derived bloodmeals by polymerase chain reaction-heteroduplex analysis. *Am. J. Trop. Med. Hyg.* 66: 599–604.
- Lord, W. D., J. A. DiZinno, M. R. Wilson, B. Budowle, D. Taplin & T. L. Meinking. 1998.** Isolation, amplification, and sequencing of human mitochondrial DNA obtained from human crab louse, *Phthirus pubis* (L.) blood meals. *J. Forensic Sci.* 43: 1097–1100.

- Loye, J. E. 1985.** The life history and ecology of the cliff swallow bug, *Oeciacus vicarius* (Hemiptera: Cimicidae). Cah. ORSTOM, ser. Ent. Med. et Parasitol. 23: 133–139.
- Mumcuoglu, K. Y., N. Gallili, A. Reshef, P. Brauner & H. Grant. 2004.** Use of human lice in forensic entomology. J. Med. Entomol. 41: 803–806.
- Ngo, K. A. & L. D. Kramer. 2003.** Identification of mosquito bloodmeals using polymerase chain reaction (PCR) with order-specific primers. J. Med. Entomol. 40: 215–222.
- Pichon, B., D. Egan, M. Rogers & J. Gray. 2003.** Detection and identification of pathogens and host DNA in unfed host-seeking *Ixodes ricinus* L. (Acari: Ixodidae). J. Med. Entomol. 40: 723–731.
- Pinchon, B., M. Rogers, D. Egan & J. Gray. 2005.** Blood-meal analysis for the identification of reservoir hosts of tick-borne pathogens in Ireland. Vector Borne Zoonotic Dis. 5: 172–180.
- Pizzaro, J. C., D. Lucero & L. Stevens. 2007.** A method for the identification of guinea pig blood meal in the Chagas disease vector, *Triatoma infestans*. Kinetoplastid Biol. Dis. 6: 1–6.
- Reinhardt, K. & M. T. Siva-Jothy. 2007.** Biology of the bed bugs (Cimicidae). Annu. Rev. Entomol. 52: 351–374.
- Szalanski, A. L., J. W. Austin & C. B. Owens. 2003.** Identification of *Reticulitermes* spp. (Isoptera: Rhinotermitidae) from South Central United States PCR-RFLP. J. Econ. Entomol. 96: 1514–1519.
- Szalanski, A. L., J. W. Austin, J. A. McKern, D. Miller, C. D. Steelman & R. E. Gold. 2006.** Isolation and characterization of human DNA from the bed bug, *Cimex lectularius* L. (Heteroptera: Cimicidae). J. Agric. Urban Entomol. 23: 189–194.
- Szalanski, A. L., J. W. Austin, J. A. McKern, D. Miller, T. McCoy & R. E. Gold. 2008.** Population genetics of the bed bug *Cimex lectularius* (Heteroptera: Cimicidae). J. Med. Entomol. in press.
- Taylor, D. B., A. L. Szalanski & R. D. Peterson II. 1996.** Identification of screwworm species (Diptera: Calliphoridae) by polymerase chain reaction – restriction fragment length polymorphism. Med. Vet. Entomol. 10: 63–70.
- Wattal, B. L. & N. L. Kalra. 1961.** New methods for the maintenance of a laboratory colony of the bed-bug *Cimex hemipterus* Fabricius, with observations on its biology. Indian J. Malariol. 15: 157–171.
- Wells, J. D., F. Introna Jr., G. Di Vella, C. P. Campobasso, J. Hayes & F. A. H. Sperling. 2001.** Human and insect mitochondrial DNA analysis from maggots. J. Forensic Sci. 46: 685–687.
- Zehner, R., J. Amendt & R. Krettek. 2004.** STR typing of human DNA from fly larvae fed on decomposing bodies. J. Forensic Sci. 49: 1–4.
-

The Gypsy Moth (Lepidoptera: Lymantriidae) Nucleopolyhedrovirus as a Synergist for Baculoviruses Against Beet Armyworm, Fall Armyworm and Corn Earworm (Lepidoptera: Noctuidae)¹

Martin Shapiro and B. Merle Shepard²

Insect Biocontrol Laboratory, USDA-ARS, Henry A. Wallace Agricultural Research Center,
Beltsville, Maryland 20705 USA

J. Agric. Urban Entomol. 23(4): 243–251 (October 2006)

ABSTRACT The gypsy moth, *Lymantria dispar* (L.) (Lepidoptera: Lymantriidae) nucleopolyhedrovirus LdMNPV was tested as a synergist for nucleopolyhedroviruses against *Helicoverpa zea* (Boddie), *Spodoptera exigua* (Hübner), and *Spodoptera frugiperda* (J.E. Smith) (Lepidoptera: Noctuidae) larvae. The addition of the gypsy moth NPV significantly decreased the LC₅₀s of AcMNPV, AfMNPV, GmMNPV, HaMNPV, HzSNPV, PxMNPV, RoMNPV, SeMNPV, and SfMNPV. In the case of the corn earworm, *H. zea*, the most active virus was the homologous virus HzSNPV. The addition of LdMNPV increased the activity of HzSNPV by 8.7-fold. Whereas LdMNPV increased the activities of each NPV against *H. zea*, none of the combinations were as active as HzSNPV alone. The most active NPV against the beet armyworm, *S. exigua*, was HaMNPV. AcMNPV and RoMNPV were comparable in activity with that of the homologous virus SeMNPV. The most active combinations against the beet armyworm were AfMNPV/LdMNPV and HaMNPV/LdMNPV, which were 7.5- and 4.4-fold more active than the SeMNPV/LdMNPV combination. The homologous NPV SfMNPV was the most active against fall armyworm, *S. frugiperda*, larvae, and the most active combination was SfMNPV/LdMNPV. These data indicate that the gypsy moth NPV could be used in combination with the homologous NPVs to increase the potencies of HzSNPV, SeMNPV, and SfMNPV against the corn earworm, beet armyworm, and fall armyworm, respectively.

KEY WORDS Insecta, nucleopolyhedroviruses, biological activity, synergism

An insect may be infected with more than one virus. Depending upon the viruses and the host, one virus can inhibit, enhance, or have no effect upon the other (Whitlock 1977, Derksen & Granados 1988, Hackett et al. 2000). Much of the research on virus–virus interactions has involved granuloviruses (GVs) and was stimulated by the research of Y. Tanada and colleagues (1956, 1971, 1973). Tanada (1956) reported synergism between a nucleopolyhedrovirus (NPV) and a

¹Accepted for publication 28 January 2008.

²Current address: Clemson University, Coastal Research and Education Center, Charleston, South Carolina 29414, USA.

GV of the armyworm, *Pseudaletia unipuncta* (Haworth) (Lepidoptera: Noctuidae) against armyworm larvae. In a subsequent publication, Tanada (1959) observed that the greatest effect occurred when both viruses were fed together and concluded that the granulovirus (PuGV) acted as the synergist. Derksen & Granados (1988) reported that a factor present in the cabbage looper, *Trichoplusia ni* (Hübner) (Lepidoptera: Noctuidae), GV (TnGV) enhanced the biological activity of the alfalfa looper, *Autographa californica* (Speyer) (Lepidoptera: Noctuidae), NPV (AcMNPV) for cabbage looper larvae. Further research by Granados and colleagues (Gallo et al. 1991, Lepore et al. 1996, Wang & Granados 1997, Peng et al. 1998) dealt with the molecular biology, site of action, and mode of action of the enhancer, now called *enhancin*. On the other hand, Hackett et al. (2000) reported that a GV from *Helicoverpa armigera* (Hübner) (Lepidoptera: Noctuidae) interfered with an NPV from *Helicoverpa zea* (Boddie) (Lepidoptera: Noctuidae) in *H. zea* larvae.

Whereas *enhancin* genes have been reported in several GVs (Roelvink et al. 1995, Hayakawa et al. 1999, Ou et al. 2000), these genes have been reported for few nucleopolyhedroviruses (e.g., the gypsy moth, *Lymantria dispar* (L.) (Lepidoptera: Lymantriidae), NPV (Bishoff & Slavicek 1997, Kuzio et al. 1999, Slavicek & Popham 2005), the bertha armyworm, *Mamestra configurata* Wlk.) (Lepidoptera: Noctuidae), NPV (Maco NPV) (Li et al. 2002, Li et al. 2003), and the spruce budworm, *Choristoneura fumiferana* Clemens, (Lepidoptera: Tortricidae), NPV (CfMNPV) (de Jong et al. 2005).

Because LdMNPV was tested only against gypsy moth larvae (Popham et al. 2001), a study was initiated to determine whether LdMNPV had any effect upon the biological activities of other baculoviruses against agriculturally important insects as the corn earworm, *H. zea*, the beet armyworm, *S. exigua*, and the fall armyworm, *Spodoptera frugiperda* (J. E. Smith).

Materials and Methods

Insects, virus inocula. The colonized strains of the beet armyworm (BAW), corn earworm (CEW), and the fall armyworm (FAW) were provided by the USDA-ARS (Tifton, Georgia) and had been in culture for more than 5 years. For testing, larvae were reared on a wheat germ diet developed for the gypsy moth (Bell et al. 1981), but a carrageenan DG611 (FMC, Rockland, Maine) was substituted for agar.

Nucleopolyhedroviruses from the beet armyworm (SeMNPV), and the fall armyworm (SfMNPV) were obtained from the USDA-ARS. The corn earworm NPV (HzSNPV) was obtained from Certis U.S.A. (Columbia, Maryland). The gypsy moth NPV (LdMNPV) was obtained from the USDA-FS (Hamden, Connecticut). These viruses have been maintained for many years in the Insect Biocontrol laboratory, USDA-ARS (Beltsville, Maryland) by passage in their natural hosts (Shapiro et al. 1981) and have not been cloned. The other NPVs were obtained from the following sources: *A. californica* NPV (AcMNPV) and *Rachiplusia ou* Guené (Lepidoptera: Noctuidae) NPV (RoMNPV) were obtained from the USDA-ARS (Beltsville, Maryland). *Anagrapha falcifera* (Kirby) (Lepidoptera: Noctuidae), NPV (AfMNPV), and *Plutella xylostella* (L.) (Lepidoptera: Plutellidae), NPV (PxMNPV) were obtained from the USDA-ARS (Columbia, Missouri); *Galleria mellonella* (L.) (Lepidoptera: Pyralidae) NPV (GmMNPV) from The Ohio State

University, Columbus, Ohio; and *H. armigera* NPV (HaMNPV) from the USDA-ARS (Tifton, Georgia). The wild-type NPVs were passed through BAW larvae before being assayed. For production of LdMNPV, the virus was diluted to a concentration of 1×10^6 OBs/mL per cup (= LC_{95} ; = 209.64 OB/mm²) and 100 fourth-stage larvae were infected (1 mL of virus per cup; 10 larvae per 180-mL cup; W.L. Enterprises, Newark, New Jersey). Larvae were reared at 28°C, 50% relative humidity, and a photoperiod of 12:12 (L:D) for 21 d. All virus-killed larvae were collected and frozen (-20°C) until used.

For bioassays involving HzSNPV, SeMNPV, and SfMNPV alone, the NPVs were extracted from virus-killed larvae (Shapiro et al. 1981) and were diluted in distilled water. Decimal dilutions were made, and OB suspensions were applied to the diet (0.1 mL per 30-mL cup; W.L. Enterprises, Newark, New Jersey) at final concentrations ranging from 10^1 to 10^5 OB per cup; = 0.01–74.74 OB/mm². Second instars (4 d old) were placed individually in each container and were reared for 14 d at 28°C, 50% relative humidity, and a photoperiod of 12:12 (L:D). In all tests, untreated larvae, as well as larvae exposed to LdMNPV alone (9×10^5 OB/30 mL cup; = 672.65 OB/mm²) were used. For bioassays involving both the gypsy moth NPV and NPVs from CEW, BAW, and FAW, HzSNPV, SeMNPV, and SfMNPV were diluted with suspensions of LdMNPV to final concentrations for HzSNPV, SeMNPV, and SfMNPV and a final concentration of 672.65 OB/mm² for LdMNPV. Tests were repeated eight times with 10 larvae per virus concentration per treatment per replicate, 10 LdMNPV-treated larvae, and 10 untreated control larvae per replicate. Mortality was initially assessed at day 3, then every 2–3 d thereafter until day 14.

Statistical methods. Concentration–mortality regressions were estimated by probit analysis (LeOra Software 1987) to monitor the biological activities of all baculoviruses, with and without the addition of LdMNPV. Failure of 95% confidence limits (CLs) at LC_{50} values to overlap was used as a measure of significance. Means were separated by the Fisher protected least significant difference test (Steel & Torrie 1980).

Results

The most active NPV tested against the CEW were the NPV isolated from and produced in CEW (HzSNPV) (Table 1). LC_{50} values for AcMNPV, AfMNPV, HaMNPV, and RoMNPV were 125.9-, 4.2-, 6.1-, and 12.7-fold greater than that of HzSNPV. In the case of GmMNPV and PxMNPV, however, only about 30% of the test larvae were killed by these NPVs at the highest concentration tested (= 74.7 OB/mm²) and LC_{50} s could not be obtained. Whereas the gypsy moth NPV LdMNPV caused no mortality and larval growth and development of CEW larvae were not affected, the addition of LdMNPV resulted in decreases in LC_{50} s for the NPVs tested (Table 1). The LC_{50} for the SzSNPV/LdMNPV combination was 8.7-fold lower than the value for HzSNPV only. LC_{50} values for the AcMNPV/LdMNPV, AfMNPV/LdMNPV, HaMNPV/LdMNPV and RoMNPV/LdMNPV combinations were 86.5-, 62.6-, 4.7-, and 10.2-fold lower, respectfully, that the LC_{50} s viruses without LdMNPV and these combinations were as active as the homologous NPV (= HzSNPV) for *H. zea*. After the addition of LdMNPV to GmMNPV and PxMNPV, LC_{50} values could be obtained, but the LC_{50} values of these combinations were significantly greater than that of HzSNPV. The most active combina-

Table 1. LC₅₀s of several NPVs against corn earworm larvae with and without the addition of LdMNPV.

Treatment ^a	LC ₅₀ (95% CL) ^b	Slope (± SE)	χ ²	AR ^c
H _z SNPV	0.23 (0.16–0.31)	1.22 (0.01)	2.91	1.00
H _z SNPV/LdMNPV	0.03 (0.02–0.04)	1.09 (0.01)	2.41	8.73
AcMNPV	28.58 (18.43–48.94)	0.92 (0.01)	1.87	0.01
AcMNPV/LdMNPV	0.59 (0.36–1.02)	0.89 (0.01)	2.44	0.39
AfMNPV	9.43 (5.87–16.25)	0.92 (0.01)	1.54	0.02
AfMNPV/LdMNPV	0.19 (0.13–0.28)	0.92 (0.01)	1.90	1.22
GmMNPV	Only 31.1% kill at the highest concentration used			
GmMNPV/LdMNPV	6.80 (4.32–11.18)	0.86 (0.01)	1.28	0.03
HaMNPV	1.39 (0.95–2.02)	0.99 (0.01)	4.45	0.16
HaMNPV/LdMNPV	0.22 (0.12–0.60)	0.95 (0.01)	1.78	1.06
PxMNPV	Only 33.3% kill at the highest concentration used			
PxMNPV/LdMNPV	1.00 (0.66–1.51)	0.80 (0.01)	4.10	0.23
RoMNPV	2.88 (1.96–4.25)	0.94 (0.01)	1.90	0.08
RoMNPV/LdMNPV	0.34 (0.23–0.49)	0.99 (0.01)	1.03	0.67

^aNPVs used were H_zSNPV, AcMNPV, AfMNPV, GmMNPV, HaMNPV, PxMNPV, RoMNPV, and LdMNPV against *H. zea*.

^bLC₅₀s were expressed as OBs/mm² of diet surface. Eight replicates; 10 larvae per NPV concentrations per treatment per replicate; five virus concentration per treatment per replicate; 10 untreated larvae per replicate; 10 LdMNPV-only-treated larvae per replicate. No mortality was found among any untreated or LdMNPV only-treated larvae.

^cActivity Ratio (AR) of LC₅₀ values, where activity of H_zSNPV equals 1.00.

tion against CEW was H_zSNPV/LdMNPV, followed by AfMNPV/LdMNPV, HaMNPV/LdMNPV, and AcMNPV/LdMNPV (Table 1).

In the case of the BAW, all NPVs tested were active (Table 2). LC₅₀ values for AcMNPV and RoMNPV were comparable with that of the homologous NPV SeMNPV ($P > 0.05$), whereas LC₅₀ values for GmMNPV and PxMNPV were 8.7- and 6.6-fold greater, respectively, than that for SeMNPV. On the other hand, LC₅₀s for AfMNPV and HaMNPV were 2.7- and 7.9-fold lower, respectively, than that of SeMNPV (Table 2). LdMNPV alone had no detrimental effects upon BAW larvae (i.e., growth and development were normal), and the addition of LdMNPV to the other NPVs significantly decreased LC₅₀ values for every NPV/LdMNPV combination (Table 2). The LC₅₀ value for SeMNPV/LdMNPV was reduced 3.8-fold to 0.2 OB/mm². LC₅₀ values were 4.3-fold (PxMNPV/LdMNPV) to 17.5-fold (GmMNPV/LdMNPV) lower when compared with the LC₅₀ values for the NPVs alone (Table 2). The most active combinations were HaMNPV/LdMNPV (LC₅₀ = 0.02 OB/mm²), AfMNPV/LdMNPV (= 0.02 OB/mm²), and AcMNPV/LdMNPV (= 0.09 OB/mm²), which were 12.6-, 10.9-, and 2.5-fold more active, respectively, than SeMNPV/LdMNPV (LC₅₀ = 0.2 OB/mm²) against *S. exigua*.

Table 2. LC₅₀s of several NPVs against beet armyworm larvae with and without the addition of LdMNPV.

Treatment ^a	LC ₅₀ (95% CL) ^b	Slope (± SE)	χ ²	AR ^c
SeMNPV	0.85 (0.61–1.13)	1.39 (0.02)	3.35	1.00
SeMNPV/LdMNPV	0.23 (0.08–0.63)	1.13 (0.03)	1.13	3.77
AcMNPV	1.02 (0.69–1.51)	0.99 (0.01)	3.50	0.84
AcMNPV/LdMNPV	0.09 (0.06–0.14)	0.92 (0.01)	1.43	9.37
AfMNPV	0.32 (0.23–0.46)	1.25 (0.01)	2.22	2.65
AfMNPV/LdMNPV	0.02 (0.01–0.03)	1.10 (0.01)	1.57	40.62
GmMNPV	7.40 (5.83–11.16)	1.03 (0.01)	1.88	0.12
GmMNPV/LdMNPV	0.42 (0.27–0.62)	0.96 (0.010)	1.64	2.02
HaMNPV	0.11 (0.08–0.16)	1.17 (0.01)	1.00	7.90
HaMNPV/LdMNPV	0.02 (0.01–0.03)	0.96 (0.01)	1.83	47.38
PxMNPV	5.64 (4.03–7.95)	1.29 (0.01)	1.48	0.15
PxMNPV/LdMNPV	1.32 (0.68–2.43)	1.22 (0.01)	3.29	0.65
RoMNPV	0.93 (0.61–1.38)	0.99 (0.01)	2.29	0.92
RoMNPV/LdMNPV	0.11 (0.07–0.16)	1.02 (0.01)	2.22	8.05

^aNPVs used SeMNPV, AcMNPV, AfMNPV, HaMNPV, PxMNPV, RoMNPV, and LdMNPV against *S. exigua*..

^bLC₅₀ values were expressed as OBs per mm² of diet surface; 10 larvae per NPV concentration per treatment per replicate; five virus concentrations per treatment per replicate; 10 untreated larvae per replicate. No mortality was found among any untreated of LdMNPV-only treated larvae.

^cRatio of LC₅₀ values where activity of SeMNPV equals 1.00.

In the case of the FAW, the most active NPV was SfMNPV (LC₅₀ = 3.1 OB/mm²). The only other NPV where LC₅₀ values could be obtained were HaMNPV and RoMNPV, which were 11.4- and 16.4-fold less active against FAW than the homologous NPV SfMNPV (Table 3). LC₅₀ values could not be obtained for AcMNPV, AfMNPV, GmMNPV, and PxMNPV, even at the highest concentration tested (= 74.74 OB/mm²; = 100,000 OB/cup). The addition of LdMNPV increased the activity for SfMNPV/LdMNPV, HaMNPV/LdMNPV, and RoMNPV/LdMNPV by 3.5-, 4.7-, and 3.6- fold, respectively. The most active combination was SfMNPV/LdMNPV (LC₅₀ = 0.7 OB/mm²), which was 10.3-, and 19.5-fold more active than the HaMNPV/LdMNPV and RoMNPV/LdMNPV combinations, respectively. Although the addition of LdMNPV to AcMNPV, GmMNPV, and PxMNPV resulted in increased virus-caused mortality, LC₅₀ values could not be obtained. AfMNPV alone, at 74.74% OB/mm², caused only 26% larval mortality but the addition of LdMNPV produced an LC₅₀ value of 35.0 OB/mm² (Table 3). Larval growth and development were normal among FAW larvae exposed to LdMNPV alone (672.65 OB/mm²), and no larvae died.

Discussion

The impetuses for this study were the discoveries that GVs could act as synergists for nucleopolyhedroviruses (Tanada 1956, Derksen & Granados 1988). Bishoff & Slavicek (1997) demonstrated that a nucleopolyhedrovirus from the

Table 3. LC₅₀ values of several NPVs against fall armyworm larvae with and without the addition of LdMNPV.

Treatment ^a	LC ₅₀ (95% CL) ^b	Slope (± SE)	χ ²	AR ^c
SfMNPV	3.1 (1.7–9.8)	1.27 (0.02)	3.45	1.00
SfMNPV/LdMNPV	0.7 (0.3–1.2)	1.10 (0.01)	1.78	4.29
AcMNPV	21.7% kill at the highest concentration used			
AcMNPV/LdMNPV	40.0% kill at the highest concentration used			
AfMNPV	26.0% kill at the highest concentration used			
AfMNPV/LdMNPV	35.0 (18.6–84.6)	1.08 (0.01)	2.46	0.09
GmMNPV	10.0% kill at the highest concentration used			
GmMNPV	40.0% kill at the highest concentration used			
LdMNPV				
HaMNPV	35.0 (15.9–113.6)	1.06 (0.01)	3.30	0.09
HaMNPV/LdMNPV	7.4 (3.8–16.2)	1.06 (0.01)	2.28	0.42
PxMNPV	0.0% kill at the highest concentration used			
PxMNPV/LdMNPV	25.0% kill at the highest concentration used			
RoMNPV	50.6 (35.6–80.8)	1.29 (0.03)	3.32	0.06
RoMNPV/LdMNPV	14.1 (9.1–22.7)	1.18 (0.01)	1.67	0.22

^aNPVs used were SfMNPV, AcMNPV, AfMNPV, GmMNPV, HaMNPV, PxMNPV, RoMNPV, and LdMNPV against *S. frugiperda*.

^bLC₅₀ values were expressed as OBs per mm² of diet surface. Seven replicates; 10 larvae per NPV concentration per treatment per replicate; five virus concentrations per treatment per replicate; 10 untreated larvae per replicate; 10 No mortality was found among any untreated larvae and LdMNPV-only treated larvae.

^cRatio of LC₅₀s where activity of SfMNPV equals 1.00.

gypsy moth (LdMNPV) contained a homolog to the GV *enhancin* genes and exhibited 29% amino acid identity to the *enhancin* proteins of TnGV, PuGV, and HaGV. Subsequently, Popham et al. (2001) identified a second *enhancin* gene from LdMNPV and determined that both genes were involved in viral pathogenicity. When both genes were deleted from LdMNPV, however, some gypsy moth larvae died of a viral infection, which infers that *enhancins* are not the sole determinants for pathogenicity. Li et al. (2002) reported that the bertha armyworm, *Mamestra configurata* Walker, NPV (MacoNPV) contained an *enhancin*, which exhibited 20% amino acid identity with other baculovirus *enhancins*. These NPVs were the first two NPVs that were demonstrated to contain *enhancin* genes.

For more than 20 years, our research on the gypsy moth NPV has been focused on improving the biological activity (Shapiro et al. 1992, Shapiro 2002, Stanley & Shapiro 2006) and improving UV stability (Shapiro et al. 1983, Shapiro 1989,

1992) to optimize efficacy of the virus against its natural host, *L. dispar*. During the course of this research, we investigated the effects of an *enhancin*-containing GV HaGV (Roelvink et al. 1995, Liu et al. 2001) upon the activity of the gypsy moth NPV. The addition of HaGV to LdMNPV adversely affected the growth and development of *L. dispar* larvae and reduced both the LC₅₀ and the LT₅₀ for the NPV (Shapiro 2000). When it was reported that the gypsy moth NPV also contained *enhancins* (Bishoff & Slavicek 1997, Popham et al. 2001, Slavicek & Popham 2005) we set up a series of experiments to determine whether LdMNPV could act as a synergist for three NPVs isolated from the CEW, *H. zea*, the BAW, *S. exigua*, and the FAW, *S. frugiperda*. Because the gypsy moth NPV was isolated from *L. dispar* and is specific to certain lymantriids (Lewis 1981, Riegel et al. 1994, Solomon 1999), we felt that LdMNPV would not be infective to the noctuids (i.e., CEW, BAW, FAW). Even at the high NPV concentration chosen for our study (i.e., 672.65 OB/mm²; $\approx$ 900,000 OB/cup), LdMNPV had no detrimental effects upon the growth and development of CEW, BAW, and FAW larvae and no LdMNPV-treated larvae died. Regardless of the virus tested (i.e., HzMNPV, SeMNPV, SfMNPV) or the insect used (i.e., CEW, BAW, FAW), the addition of LdMNPV to a given virus resulted in an increase in the activity of that virus, as measured by LC₅₀ values (Tables 1–3). In cases where LC₅₀ values could not be obtained against *H. zea* and *S. frugiperda*, virus-caused mortalities were increased by the addition of LdMNPV.

The mechanism for the synergistic activity of LdMNPV is not as clear-cut as it may seem. A reasonable explanation would be that the two *enhancins* of LdMNPV were responsible (Bishoff & Slavicek 1997, Popham et al. 2001). LdMNPV has also been shown to contain other identifiable genes (e.g., *host range factor-1*) that can increase host range (Thiem et al. 1996, Du & Thiem 1997, Chen et al. 1998). In vivo assays were conducted against CEW, BAW, and *Plutella xylostella* L. (Lepidoptera: Plutellidae) larvae using wild-type AcMNPV and a recombinant AcMNPV containing the *hrf-1* gene from LdMNPV. In the case of the CEW, the addition of *hrf-1* resulted in a 5.8-fold increase in activity. The recombinant NPV, however, was no more active against BAW and *P. xylostella* than was the wild-type NPV (Chen et al. 1998). In subsequent studies, Ishikawa et al. (2004) and Ikeda et al. (2005) determined that the *hrf-1* gene promoted replication of *A. californica* NPV AcMNPV, *Hyphantria cunea* Drury (Lepidoptera: Arctiidae) m NPV HycuNPV, *Bombyx mori* L. (Lepidoptera: Bombycidae), and *S. exigua* NPV SeNPV in nonpermissive cells of the gypsy moth IPL-652Y. The authors concluded that *hrf-1* was essential for these viruses to infect normally nonpermissive cells.

At present, we do not know whether the synergistic activity of LdMNPV was caused by the LdMNPV *enhancins* (Popham et al. 2001), the LdMNPV *hrf-1* gene (Thiem et al. 1996), or unidentified gene or combinations of genes. Research on LdMNPV/virus combinations should be continued and tested to determine whether the use of the LdMNPV synergist can represent a new strategy for effective insect pest control against agriculturally important pest insects.

Acknowledgments

The research was conducted (2002–2003) when Martin Shapiro was a Research Entomologist, Insect Biocontrol Laboratory, USDA-Agricultural Research Service, Beltsville, Maryland. Technical Contribution No. 5326 of the Clemson University Experiment Station.

References Cited

- Bell, R. A., C. D. Owens, M. Shapiro & J. G. R. Tardif. 1981. Development of mass rearing technology, pp. 599–633. In C. C. Doane and M. L. McManus [Eds.], The gypsy moth: research toward integrated pest management. U.S. Dept. Agric. Tech. Bull. 1584.
- Bishoff, D. S. & J. M. Slavicek. 1997. Molecular analysis of an enhancing gene in the *Lymantria dispar* nuclear polyhedrosis virus. J. Virol. 71: 8133–8140.
- Chen, C. J., M. E. Quentin, L. A. Brennan, C. Kunkel & S. M. Thiem. 1998. *Lymantria dispar* hrf-1 expands the larval host range of *Autographa californica* nucleopolyhedrovirus. J. Virol. 72: 2526–2531.
- de Jong, J. G., H. A. M. Lauzon, C. Dominy, A. Poloumienko, E. B. Carstens, B. M. Arif & P. J. Krell. 2005. Analysis of the *Choristoneura fumiferana* nucleopolyhedrovirus gene. J. Gen. Virol. 86: 929–943.
- Derksen, A. C. G. & R. R. Granados. 1988. Alteration of a lepidopteran peritrophic membrane by baculoviruses and enhancement of viral activity. Virology 167: 242–250.
- Du, X. & S. M. Thiem. 1997. Characterization of host range factor 1 (hrf-1) expression in *Lymantria dispar* M nucleopolyhedrovirus and recombinant *Autographa californica* M nucleopolyhedrovirus-infected IPLB-Ld652Y cells. Virology 227: 420–430.
- Gallo, L. G., B. G. Corsaro, P. R. Hughes & R. R. Granados. 1991. In vivo enhancement of baculovirus infection by the viral enhancing factor of a granulosis virus of the cabbage looper, *Trichoplusia ni* (Lepidoptera: Noctuidae). J. Invertebr. Pathol. 58: 203–210.
- Hackett, K. J., A. Boore, C. Deming, E. Buckley, M. Camp & M. Shapiro. 2000. *Helicoverpa armigera* granulovirus interference with progression of *H. zea* nucleopolyhedrovirus disease in *H. zea* larvae. J. Invertebr. Pathol. 75: 99–106.
- Hayakawa, T., R. Ko, K. Okano, S. Seong, C. Goto & S. Maeda. 1999. Sequence analysis of the *Plutella xylostella* granulovirus genome. Virology 262: 277–297.
- Ikeda, M., E. A. Reimbold & S. M. Thiem. 2005. Functional analysis of the baculovirus host range gene, hrf-1. Virology 332: 602–613.
- Ishikawa, H., M. Ikeda, C. A. Alves, S. M. Thiem & M. Kobayashi. 2004. Host range factor 1 from *Lymantria dispar* nucleopolyhedrovirus (NPV) is an essential viral factor required for productive infection of NPVs in IPLB-652Y cells derived from *L. dispar*. J. Virol. 48: 12703–12708.
- Kuzio, J., M. N. Pearson, S. H. Harwood, C. J. Funk, J. R. Evans, J. M. Slavicek & G. F. Rohrmann. 1999. Sequence and analysis of the genome of a baculovirus pathogenic for *Lymantria dispar*. Virology 253: 17–34.
- LeOra Software. 1987. POLO-PC. A users' guide to probit or Logit analysis. Berkeley, CA.
- Lepore, L. S., P. R. Roelvink & R. R. Granados. 1996. Enhancing the granulosis virus protein that facilitates nucleopolyhedrovirus (NPV) infections, is a metalloprotease. J. Invertebr. Pathol. 68: 131–140.
- Lewis, F. B. 1981. Gypsy moth nucleopolyhedrosis virus, pp. 454–461. In C. C. Doane and M. L. McManus [Eds.], The gypsy moth: research toward integrated pest management. U.S. Dept. Agric. Tech. Bull. 1584.
- Li, L., C. Donly, Q. Li, L. G. Willis, B. A. Keddie, M. A. Erlandson & D. A. Theilman. 2002. Identification and genomic analysis of a second species of nucleopolyhedrovirus isolated from *Mamestra configurata* nucleopolyhedrovirus. Virology 297: 226–244.
- Li, Q., L. Li, K. Moore, C. Donly, D. A. Theilmann & M. Erlandson. 2003. Characterization of *Mamestra configurata* nucleopolyhedrovirus enhancing and its functional analysis via expression in an *Autographa californica* M nucleopolyhedrovirus recombinant. J. Gen. Virol. 84: 125–132.
- Liu, X., G. Yang, B. Oui, K. Zhang & P. Tian. 2001. Expression of two truncated enhancing genes from *Helicoverpa armigera* granulosis virus in *E. coli* and its preliminary bioassay. Wei Sheng Wu Xue Bao 41: 167–172.
- Ou, Y., X. L. Meng & J. P. Xu. 2000. Expression of 2.1 kb enhancing gene fragment from *Helicoverpa armigera* granulosis in *Escherichia coli*. Sheng Wu Gong Xue Bao 16: 595–598.

- Peng, J., J. Zhong & R. R. Granados. 1998.** A baculovirus enhancing alters the permeability of a mucosal peritrophic membrane matrix from lepidopteran larvae. *J. Insect Physiol.* 45: 159–166.
- Popham, H. J., D. S. Bishoff & J. M. Slavicek. 2001.** Both *Lymantria dispar* nucleopolyhedrovirus enhancing genes contribute to viral potency. *J. Virol.* 75: 8639–8648.
- Riegel, C. I., C. Lanner & J. M. Slavicek. 1994.** Identification and characterization of the ecdysteroid UDP-glucosyltransferase gene of the *Lymantria dispar multinucleocapsid* nuclear polyhedrosis virus. *J. Gen. Virol.* 74: 829–838.
- Roelvink, P. W., B. G. Corsaro & R. R. Granados. 1995.** Characterization of the *Helicoverpa armigera* and *Pseudaletia unipuncta* granulovirus enhancin gene. *J. Gen. Virol.* 76: 2693–2705.
- Shapiro, M. 1989.** Congo red as an ultraviolet protectant for the gypsy moth (Lepidoptera: Lymantriidae) nuclear polyhedrosis virus. *J. Econ. Entomol.* 82: 548–550.
- Shapiro, M. 1992.** Use of optical brighteners as radiation protectants for the gypsy moth (Lepidoptera: Lymantriidae) nuclear polyhedrosis virus. *J. Econ. Entomol.* 85: 1682–1686.
- Shapiro, M. 2000.** Effect of two granulosis viruses on the activity of the gypsy moth (Lepidoptera: Lymantriidae) nuclear polyhedrosis virus. *J. Econ. Entomol.* 93: 1633–1637.
- Shapiro, M. 2002.** Effects of anion and cation inhibitors and carbonic anhydrase inhibitors upon the activity of the gypsy moth (Lepidoptera: Lymantriidae) nucleopolyhedrovirus. *J. Econ. Entomol.* 95: 237–242.
- Shapiro, M., P. P. Agin & R. A. Bell. 1983.** Ultraviolet protectants of the gypsy moth (Lepidoptera: Lymantriidae) nucleopolyhedrosis virus. *Environ. Entomol.* 12: 982–985.
- Shapiro, M., J. J. Hamm & E. M. Dougherty. 1992.** Composition and methods for bio-control using fluorescent brighteners. U.S. Patent No. 5,124,149.
- Shapiro, M., M. E. Martignoni, J. C. Cunningham & R. H. Goodwin. 1982.** Potential use of the saltmarsh caterpillar as a production host of the nucleopolyhedrosis viruses. *J. Econ. Entomol.* 75: 69–71.
- Slavicek, J. M. & H. J. R. Popham. 2005.** The *Lymantria dispar* nucleopolyhedrovirus enhancins are components of occlusion-derived virus. *J. Virol.* 79: 10578–10588.
- Solomon, D. 1999.** Gypsy moth in southwest Michigan. Michigan State University. W. K. Kellogg Biological Station, 3 pp.
- Stanley, D. & M. Shapiro. 2006.** Eicosanoid biosynthesis inhibitors increase the susceptibility of *Lymantria dispar* nucleopolyhedrovirus LdMNPV. *J. Invertebr. Pathol.* 95: 119–124.
- Steel, R. D. G. & J. H. Torrie. 1980.** Principles and procedures of statistics. McGraw-Hill, New York, NY, 633 pp.
- Tanada, Y. 1956.** Some factors affecting the susceptibility of the armyworm to virus infections. *J. Econ. Entomol.* 49: 52–57.
- Tanada, Y. 1959.** Synergism between two viruses of the armyworm, *Pseudaletia unipuncta* (Haworth) (Lepidoptera: Noctuidae). *J. Insect Pathol.* 45: 215–231.
- Tanada, Y. & H. Watanabe. 1971.** Disc electroporetic and serological studies of the capsule proteins obtained from two strains of a granulosis virus of the armyworm, *Pseudaletia unipuncta*. *J. Invertebr. Pathol.* 18: 307–312.
- Tanada, Y., M. Himeno & E. M. Omi. 1973.** Isolation of a factor, from the capsule of a granulosis virus, synergistic for a nuclear polyhedrosis virus of the armyworm. *J. Invertebr. Pathol.* 21: 31–40.
- Thiem, S. M., X. Du, M. E. Quentin & M. M. Berner. 1996.** Identification of baculovirus gene that promotes *Autographa californica* nuclear polyhedrosis virus replication in a nonpermissive cell line. *J. Virol.* 70: 2221–2229.
- Wang, P. & R. R. Granados. 1997.** An intestinal mucin is the target substrate for a baculovirus enhancing. *Proc. Natl. Acad. Sci. USA* 94: 6977–6982.
- Whitlock, V. H. 1977.** Simultaneous treatments of *Heliothis armigera* with a nuclear polyhedrosis and granulosis virus. *J. Invertebr. Pathol.* 29: 297–303.
-

NOTICE TO CONTRIBUTORS

The Editorial Committee of *Journal of Agricultural Entomology* considers the timely publication and documentation of research endeavors critical to the advancement of agricultural and urban entomology. As such, JAUE places a high priority on prompt processing of submitted works, as well as prompt publication of accepted works. However, the overall turnaround time is to a large extent in the hands of authors making revisions.

In light of this, the Committee requests that authors return revised manuscripts to their subject editor within three months of receipt. Manuscripts held for revisions for six months or more will, at the discretion of the editor, be subject to a 10% surcharge over the normal cost of publication.

The Committee recognizes that extenuating circumstances arise, and will consider special arrangements for authors in such situations.

***South Carolina
Entomological Society, Inc.***

**3517 Flowering Oak Way
Mt. Pleasant, SC 29466**

***MEMBERSHIP
APPLICATION
FORM***

NAME: _____ DATE: _____

MAILING ADDRESS: street _____

city: _____ state: _____ zip code: _____

HOME ADDRESS: street _____

city: _____ state: _____ zip code: _____

WORK PHONE: _____ HOME PHONE: _____

FAX # _____ E-MAIL _____

EDUCATION:

	DEGREE	INSTITUTION	YEAR
1.	_____	_____	_____
2.	_____	_____	_____
3.	_____	_____	_____
4.	_____	_____	_____

EMPLOYER: _____ SINCE: _____

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS:

1. _____
2. _____
3. _____
4. _____
5. _____

I hereby make application for membership in the South Carolina Entomological Society, Inc. with the status of active (\$30), student (\$15), emeritus (\$10) membership.

Signature: _____ Date: _____

Return form with membership dues to address above.

- ALVERSON, D.R. 1
ANDERSON, P.A. 17
ÁNGEL-SAHAGÚN, C.A. 41
AUSTIN, J.W. 189, 237
- BOHAC, J.R. 77, 141
BRUCK, D.J. 111
BUCKLEY, P.M. 87, 97
- CANHILAL, R. 1, 27, 159
CARNER, G.R. 1, 27, 159
CATANGUI, M.A. 225
COATS, J. 209
- DARA, S.K. 65
DAVES, C.A. 87, 97
DEACUTIS, J.M. 105
- GALINDO-VELASCO, E. 41
GEDEN, C.J. 105
GENTZ, M.C. 201
GERRY, A.C. 105, 195
GOLD, R.E. 189
GONZALEZ-ANDUJAR, J.L. 51
GONZÁLEZ-REYES, E. 41
GRACE, J.K. 201
GRIFFIN, R.P. 1
GUNNARSON, R.D. 111
- HANSEN, J.D. 17
HEIDT, M.L. 17
HOOD, W.M. 167
- JACKSON, D.M. 1, 77, 141
- KAUFMAN, P.E. 195
KERING, M.K. 7
- LEICHTER, C.A. 105
LEWIS, L.C. 111
LEZAMA-GUTIÉRREZ, R. 41
LOPEZ, M.A. 51
LOPEZ, M.D. 111
- MCCOY, T. 237
MCKERN, J.A. 189, 237
MICHEL-ACEVES, A.C. 41
MILLER, D.M. 189, 237
MILLER, F. 209
MO, J. 57
MOLINA-OCHOA, J. 41
- PALUCH, G. 209
PESCADOR-RUBIO, A. 41
PRASIFKA, J.R. 111
- RAMIREZ-DAVILA, J.F. 51
RIEDEL, W.E. 225
RUTZ, D.A. 105, 195
- SCOTT, J.G. 105, 195
SEMTNER, P.J. 65
SHAPIRO, M. 243
SHEPARD, B.M. 243
STEELMAN, C.D. 189, 237
SZALANSKI, A.L. 189, 237
- WALDSTEIN, D.E. 7
WANG, Z. 57
WATSON, W.D. 105
WILLIAMS, W.P. 87, 97
WINDHAM, G.L. 97
- YEE, W.L. 125
- ZHANG, S. 57
ZHU, J. 209
- OCETE, R. 51

- activation 57
Aedes albopictus 57
Aflatoxin 97
Aphis glycines Matsumura 225
apple 7
apple bins 17
arthropods 111
Aspergillus flavus 97
attraction 57

Bacillus thuringiensis 87
bed bug 189, 237
bioassay 77
biological activity 243
biological control 27, 41
boric acid 201
Brentidae 141
Bt corn 111

cherry 7
Cimex lectularius 189, 237
codling moth 17
coefficient of variation 111
corn 87
Crambidae 87, 97
cross-resistance 195
Cydia pomonella 17
Cylas formicarius 141

dairy 105
Diabrotica 77
Diatraea grandiosella 87, 97

Elm 209
entomology 167
entomopathogenic nematodes 27, 41, 159

flue-cured tobacco 65
forensic entomology 189, 237

Glycine max L. Merrill 225
Grapholita molesta 7
green peach aphid 65
gypsy moth 209

heat 17
Helicoverpa zea 87
Heterorhabditidae 27
Heterorhabditis 159
history 167
host plant resistance 77, 97
human skin volatile 57
hypervariable region 189, 237

imidacloprid 195
Insecta 195, 243
insecticide resistance monitoring 105
Ipomoea 77, 141
Isoptera 201

Japanese beetle 209

Lepidoptera 87, 97
life history 1
life table 1

mango 41
Melittia cucurbitae 1, 27
Mexican fruit fly 41
Mexico 41
microencapsulation 7
Musca domestica 105, 195
Myzus nicotianae 65

natural occurrence 159
Neonicotinoid resistance 195
Nicotiana tabacum 65
Noctuidae 87
nontargets 111
nucleopolyhedroviruses 243

organization 167

peach 7
pear 7
pheromones 7, 141
plant insect interactions 225
plum 7
population dynamics 141
poultry 105
protein baits 125

Rhagoletis indifferens 125
Rhinotermitidae 201

seed composition 225
seed yield 225
South Carolina 159
South Carolina Entomological Society 167
soybean 225
Soybean aphid 225
spinosad 125
Spodoptera frugiperda 87
Squash 1

squash vine borer 1, 27

Steinernema 159

Steinernema carpocapsae 41

Steinernema riobrave 41

Steinernematidae 27

STR typing 189, 237

sweetpotato 77, 141

synergism 243

tobacco aphid 65

trap 57, 141

water baths 17

within-plant distribution 65

wood preservation 201

Zea mays 87, 97